How Many Evaluator Scores Certify Output Retention Under a Churn Budget?

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Abstract

Replacing an evaluator can change which language outputs pass an ensemble cutoff. We study the score evidence needed to certify that a returned subset meets an average normalized churn budget and retains a count within a specified tolerance of the full-information optimum under that budget. The normalization, cutoff and a finite family of single-evaluator replacements are fixed, and the guarantees hold for every score completion consistent with the observed entries. We characterize minimum certificates chosen with full-score knowledge and verified from revealed entries only. For odd median pools, the evidence for feasibility and the evidence excluding larger retained sets are disjoint, which yields closed-form minima. Mean and even-median pools allow one observation to serve both purposes. Under exact count certification from an empty history, we construct mean-pooling families whose joint certificate approaches half the size of the best union of separately minimum certificates. Across 1,920 mean targets from cached RewardBench 2 and five WMT panels, numerical optimization finds modest joint gains in 89 targets, all of which require exclusion evidence, and the corresponding whole-column targets show no gain. Discovering a two-entry certificate can require queries linear in batch size, even with positive count tolerance. We also extend the analysis to weighted and per-replacement churn objectives and to verifiable cross-entry score constraints. Certificate size, discovery cost and inference time require separate evaluation, and retention stability does not certify output quality.

1 Introduction

Reward models, translation metrics and LLM judges score language outputs (Malik et al., 2026; Freitag et al., 2023; Zheng et al., 2023), and ensembles of such evaluators are often used to filter a batch (Coste et al., 2024; Eisenstein et al., 2024). Evaluators are replaced over time. Before rescoring a selected batch after replacing one ensemble member, an analyst may want evidence that many outputs would keep their selection status within a declared churn allowance. Returning nothing is always feasible, so a useful certificate must also bound how many outputs are lost relative to the best budget-feasible retention.

We study this problem for a fixed batch. A base pool of evaluators selects outputs whose median or mean normalized score reaches a cutoff, and a finite family of candidate evaluators defines every single-member replacement. Candidate scores on the batch are initially unknown and can be revealed entry by entry. A *certificate* is a set of revealed entries from which a verifier can conclude, for every completion of the unrevealed scores, that a returned subset meets the churn budget and that its count is within ε outputs of the optimum. This guarantee is deterministic and completion-wise. It complements conformal and selective risk control, which bound expected risk under exchangeability (Angelopoulos et al., 2024; Bai and Jin, 2026).

Our contributions are as follows. (i) We define safety evidence, which certifies feasibility, and exclusion evidence, which rules out larger retained sets, and we show that both bounds are sharp under the declared completion model (Section 4). (ii) For odd medians the two kinds of evidence are disjoint, giving closed-form minimum certificates. For mean and even-median pools a single entry can serve both, and we give an exact joint program together with a tight factor-two separation from the best union of separately minimum witnesses (Section 5). (iii) We separate certificate size from discovery: two-entry certificates can require queries linear in batch size even at positive tolerance (Section 6). (iv) On RewardBench 2, WMT and SummEval panels we measure inspection, minimum evidence, and joint gains, and we identify when sharing evidence helps (Section 7). (v) We extend the theory to weighted and per-replacement churn

objectives and to verifiable cross-entry constraints (Appendices R–S). Stability differs from quality: evaluator agreement can coexist with shared errors and reward hacking (Eisenstein et al., 2024).

2 Related Work

Risk control and judge stability. Conformal risk control (CRC) bounds an expected monotone loss (Angelopoulos et al., 2024; Yeh et al., 2025), and risk-controlling prediction sets give high-probability bounds (Bates et al., 2021). SCoRE, SCOPE and multi-expert CRC address selective risk, pairwise acceptance and multiple judges (Bai and Jin, 2026; Badshah et al., 2026; Cheng et al., 2026). Replicable calibration stabilizes thresholds across samples (Papamichalis et al., 2026). LLM judges are sensitive to position and prompt variations (Wang et al., 2024; Thakur et al., 2024), and ensemble agreement need not exclude common errors (Afrin and Shihab, 2026). Predictive churn has been studied for model updates and training-sample changes (Milani Fard et al., 2016; Watson-Daniels et al., 2025; Prastalo and Jablonka, 2026), and fixed-threshold routing can change even with matching class predictions (Kim, 2026). Our guarantee instead concerns one fixed batch, holds for every consistent score completion, and is stated relative to a declared replacement family.

Verification, discovery and robust selection. Priced information and matroid verification distinguish proof size from discovery cost (Charikar et al., 2002; Diwan et al., 2026), and Boolean evaluation studies adaptive queries (Deshpande et al., 2016; Gkenosis et al., 2018). Robust selection and knapsack problems choose queries before uncertain parameters are resolved (Chen et al., 2026; Goerigk et al., 2015), often with budgeted uncertainty sets (Bertsimas and Sim, 2004); uncertain-profit knapsack certifies good packings and excludes better ones under resource augmentation (Schlöter, 2025). Recoverable selection permits later exchanges (Chassein et al., 2018), and covering and query-competitive models study sharing and discovery (Megow and Schlöter, 2023; Erlebach and Hoffmann, 2015; Erlebach et al., 2021). These models leave costs or profits uncertain. Here the uncertainty is a matrix of candidate scores under a known pooling rule, and we verify feasibility and optimality for a fixed scenario family rather than solve a min–max design problem. Unlike black-box certification (Blanc et al., 2022; Gupta and Manoj, 2023), we query a fixed matrix.

Top-k with partial information. Sorted-list aggregation supports sorted and random accesses (Fagin et al., 2003); our certificates assume no sorted oracle. Top-k queries over uncertain or incomplete data compute possible and certain answers (Imieliński and Lipski, 1984; Soliman et al., 2007; Amarilli et al., 2017), and partial voting profiles admit necessary and possible winners (Imber and Kimelfeld, 2020). Our safe and optimistic counts play the role of certain and possible retention, but the object certified is a budget-constrained optimum over a replacement family, which leads to different certificate combinatorics. Ranking robustness covers weight changes and word attacks (Asudeh et al., 2018; Wu et al., 2022); PAC selection, Col-Bandit and budget-adaptive ranking acquire scores adaptively (Kalyanakrishnan et al., 2012; Pony et al., 2026; Oettershagen, 2026). Selective ensembles, pruning and early exit reduce scoring cost (Black et al., 2022; Li et al., 2022; Inoue, 2019; Busolin et al., 2021), and Best-of-Evidence queries factors within a fixed pool (Zhang et al., 2026). MT filtering and metric biases motivate separate quality assessment (Perrella et al., 2024; Shayegh et al., 2025).

3 Selection under Evaluator Replacements

Rows index outputs and columns index evaluators. Pools P and R contain $M \geq 1$ base and $T \geq 1$ candidate evaluators. Normalization maps and a cutoff $0 < t_0 \leq 1$ are fitted on a reference split of scored outputs (here, *reference* denotes the fitting split, not a human reference translation). Write $s_e(i) \in [0, 1]$ for normalized scores. Replacing base member j by candidate r gives the pool $P^{j \to r}$; the $H = MT$ replacements have uniform weight. The maps and cutoff remain fixed. Base scores are visible; queries reveal candidate test scores.

The base list is $S_0(P) = \{i : c_P(i) \geq t_0\}$, where c_P is median or mean consensus; an even-sized median averages the two central scores, and cutoff ties are included. For a batch of N outputs, fix a nominal selection fraction $q \in (0, 1]$ and the nominal count $k = \max(1, \mathrm{round}(qN))$ (half-integers round to even). The count k scales the risk and imposes no return quota. With $\sigma_Q(i) = \mathbf{1}[i \in S_0(Q)]$, base scores and one candidate score determine all M replacement outcomes of a row.

Quantity	Definition	Example
Churn budget	$R(C) \le \alpha$, Eq. (2)	.10
Changes/replacement	$\rho = 2k\alpha$	0.4
Flip budget	$b_H = \lfloor \rho H \rfloor$	8
Oppositions (odd M)	$b = \lfloor b_H/(h+1) \rfloor$	2
Count tolerance	$V^+ - V^- \le \varepsilon$	$\varepsilon \ge 0$
Active inspection	$100\,Q/(mT)$	–

Table 1: Units and conversions. The example uses $M = 7$ ($h = 3$), $T = 3$ and $k = 2$, so $H = 21$ (Appendix A.7). Because $2k$ is a fixed nominal scale, $R(C) > 1$ is possible when many outputs are returned; only $2k\alpha$ matters for feasibility (Appendix L).

Define the flip count and row total

$$f_{ir} = \sum_{j=1}^{M} \mathbf{1}[\sigma_P(i) \neq \sigma_{P^{j\to r}}(i)], \quad F_i = \sum_{r=1}^{T} f_{ir}. \tag{1}$$

A commitment mask C fixes baseline decisions without review and returns $S_0(P) \cap C$. Its average normalized churn over the declared replacements is

$$R(C) = \frac{1}{H} \sum_{j,r} \frac{|(S_0(P) \triangle S_0(P^{j\to r})) \cap C|}{2k} = \frac{\sum_{i\in C} F_i}{2kH}. \tag{2}$$

Decisions outside C are referred for review. We maximize the returned count subject to $R(C) \le \alpha$. Nonselected outputs do not increase the returned count and receive no dedicated entry queries, and an exclusion is committed only when base scores rule out every membership flip. With all costs observed, the integer flip budget is $b_H = \lfloor 2k\alpha H \rfloor$; sorting selected outputs by increasing F_i and taking the longest prefix within b_H gives the full-information optimum V^*.

Units. Table 1 collects the conversions. Allowing ρ average membership changes per replacement gives $\alpha = \rho/(2k)$. This is an average replacement allowance, not a per-replacement limit or a review-capacity quota.

Choice of risk objective. The uniform average in Eq. (2) makes the full-information problem a sorted prefix and yields the closed forms below. Three alternatives are natural. *Fixed nonnegative replacement weights* preserve every bound and the sorted-prefix argument; Theorem 3 extends with column-dependent caps, and odd-median evidence remains disjoint (Appendix R.1). *Certifying each single replacement*, i.e., a per-replacement cap $\max_{j,r} |\cdot|/(2k) \le \alpha$, keeps verification of a given mask linear-time and sharp, but computing the optimal retained count becomes NP-hard even with full information, so count gaps require integer programs (Appendix R.2). This matters in practice: 6,985 of 15,360 average-certified masks exceed .10 for at least one individual replacement (Appendix I). *A known single replacement* reduces to one candidate column with Boolean row costs. Uniform weights correspond to a replacement chosen uniformly from the declared family.

4 Certificates from Partial Scores

A certificate uses visible base scores and queried candidate scores to verify churn feasibility and bound the count shortfall. One entry query determines all M replacement outcomes of its row. Let $f_i(x)$ count flips when a candidate assigns score x to output i, so $f_{ir} = f_i(s_r(i))$ and $p_i = \max_{x\in[0,1]} f_i(x)$. Under either pooling rule, replacement consensus is nondecreasing in x; for selected outputs f_i is nonincreasing, with maximum p_i at $x = 0$ and zero at $x = 1$. Visible base scores determine p_i. After t_i queries on row i, let ℓ_i sum their observed contributions. Bounding each unobserved count between zero and p_i gives

$$\ell_i \le F_i \le u_i = \ell_i + (T - t_i)p_i. \tag{3}$$

For nonnegative costs w, define the retention capacity $V(w) = \max\{|C \cap S_0(P)| : \sum_{i\in C} w_i \le b_H\}$, computed by sorting. Maximizing retention under u yields a mask returning $V^- = V(u)$ outputs that is feasible for every consistent completion; the optimistic count $V^+ = V(\ell)$ is attained in some completion.

The completion model. Unobserved scores range freely over $[0, 1]$. This is a set of possible matrices, not a statistical assumption: assigning every unobserved entry its zero-flip endpoint realizes ℓ, and assigning its maximum-flip endpoint realizes u. We use this worst-case set because correlations among evaluators, however strong on reference data, do not by themselves guarantee where an unseen test score lies; predictions therefore guide query order but never tighten bounds. Valid cross-entry constraints can only shrink the completion set: every certificate in this paper remains valid under them and minimum certificates can only become smaller. A verifiable class stays tractable. If the constrained completion set is a lattice, with

coordinatewise greatest and least elements, then the same two completions give sharp bounds. This covers per-entry score supports, monotonicity constraints and Lipschitz-with-slack constraints across outputs for each evaluator, whose envelopes are computed in linear time per query (Propositions 18–19, Appendix S). In a controlled study with valid Lipschitz-with-slack constraints, primary-stop inspection falls from 72–96% to 4–68% of active entries, depending on the slack and depth; a misspecified constant produced invalid certificates in 12 of 80 exact-stop runs, so such constraints must be justified externally.

Proposition 1 (Safe retention and stopping)**.** *At any adaptive query history, let C^- maximize retained count subject to the upper-cost budget. Then $R(C^-) \le \alpha$ and $V^- \le V^* \le V^+$. If $V^+ - V^- \le \varepsilon$, its returned count is at most ε outputs below V^*. Under the declared replacement family and completion model, these guarantees hold for every acquisition order and stopping time.*

Proof. Equation (3) bounds each true cost. Upper-cost feasibility implies true feasibility, which implies lower-cost feasibility; maximizing cardinality gives the count bounds. The returned count is V^-, so its shortfall is at most the gap. □

Integer $\varepsilon \ge 0$ bounds count shortfall, and zero requires count optimality. Full revelation gives $V^- = V^+ = V^*$. An output is structurally invariant ($p_i = 0$) if no candidate score can change its membership; noninvariant outputs can still have zero true cost. Zero-query retention uses upper costs Tp_i and can admit noninvariant outputs within budget. Let $m_0 = |\{i \in S_0 : p_i = 0\}|$ count invariant selected rows; only the $m = |\{i \in S_0 : p_i > 0\}|$ active selected rows are eligible for entry queries, and active inspection is $Q/(mT)$ for Q entry queries (zero when $m = 0$). The safe count cannot decrease, although retained sets need not be nested.

5 Minimum Certificate Complexity

Odd medians: disjoint evidence. For $M = 2h + 1$, median selection requires $h + 1$ votes at or above t_0, and a single replacement changes at most one vote. Selected rows with at least $h + 2$ votes are invariant; those with exactly $h + 1$ form the boundary set B, and a below-cutoff candidate excludes a boundary row exactly when it replaces one of its $h + 1$ voting members. Thus $p_i = h + 1$ on B. For $i \in B$ define opposition bits $y_{ir} = \mathbf{1}[s_r(i) < t_0]$ and row sums $z_i = \sum_r y_{ir}$, so $F_i = (h + 1)z_i$. In opposition units, feasibility is exactly $\sum_{i \in C \cap B} z_i \le b$ with $b = \lfloor b_H/(h + 1) \rfloor$ (Table 1). On a boundary row, an *agreement* (candidate score at or above the cutoff) lowers the upper bound by one opposition, and an *opposition* raises the lower bound by one: with a_i agreements and o_i oppositions observed, $o_i \le z_i \le T - a_i$. An offline constructor uses the true bit matrix to choose entries; the verifier sees only revealed entries and checks every consistent Boolean completion.

Theorem 1 (Safety witness)**.** *If the true matrix admits a feasible set of L boundary rows, the minimum number of observations certifying that an L-row set is feasible under every consistent completion is $Q_{\text{safe}}(L) = \max\{0, LT - b\}$. If no L-row set is truly feasible, no certificate exists.*

An L-row set starts with upper cost LT, and only agreements reduce it, one unit each; any truly feasible set contains enough agreements (Appendix A gives proofs and a worked example). Feasibility need not establish optimality. Let $z_{(1)} \le \cdots \le z_{(m)}$ and let L be the largest prefix size with sum at most b; the full retained optimum is $m_0 + L$.

Theorem 2 (Exact optimality witness)**.** *If $L = m$, the minimum certificate proving exact optimality is $\max(0, mT - b)$. Otherwise set $p = L + 1$, $B_0 = b + 1$, $D(t) = \sum_{i=1}^{p} \min(z_{(i)}, t)$ and $t^* = \min\{t \in \{0, \ldots, T\} : D(t) \ge B_0\}$. The exact offline certificate size is*

$$Q_{\text{exact}} = \max(0, LT - b) + B_0 + (m - p)t^*. \quad (4)$$

The exclusion part reveals $b + 1$ oppositions across the p cheapest rows, capped at $\min(z_{(i)}, t^*)$, and t^* oppositions in every remaining row. These cells are disjoint from the safety agreements, so the two costs add. With certified safe boundary count s, tolerance ε requires every $(s+\varepsilon+1)$-row boundary set to be excluded unless that size exceeds m.

Corollary 1 (Minimum gap witness)**.** *Let L^* be the true optimal boundary count and $\varepsilon \ge 0$ an integer. For $L^* < p \le m$, let $D_p(t) = \sum_{i=1}^{p} \min(z_{(i)}, t)$, $t_p = \min\{t : D_p(t) > b\}$ and $E(p) = b + 1 + (m - p)t_p$; for $p > m$ set $E(p) = 0$. Then*

$$Q_\varepsilon = \min_{s \in \mathcal{S}_\varepsilon} \big[\max(0, sT - b) + E(s + \varepsilon + 1) \big], \quad (5)$$

with $\mathcal{S}_\varepsilon = \{\max(0, L^* - \varepsilon), \ldots, L^*\}$. In particular $Q_0 = Q_{\text{exact}}$.

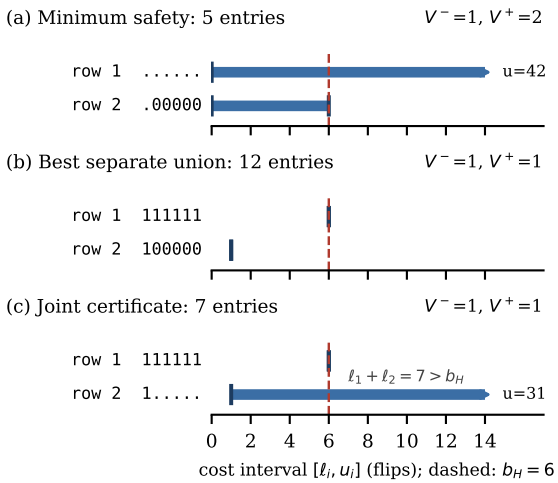


Figure 1: One mean-pooling entry can serve safety and exclusion (Proposition 9 at $T = 6$; caps $p = (7, 6)$, $b_H = 6$, exact stopping). Digits are revealed flip counts; dots are unqueried. Bars show $[\ell_i, u_i]$. (a) The shortest safety proof certifies row 2. (b) Its best union with the unique minimum exclusion witness reveals all 12 entries. (c) Retaining row 1 instead uses the six entries that exclusion already needs: they set $u_1 = 6 \le b_H$ (safety) and, with one entry of row 2, make the two smallest lower costs sum to $7 \ge b_H + 1$ (exclusion).

These minima lower-bound entry queries for the same instance, budget, model and tolerance from an empty history (Appendix A.6); residual witnesses account for existing observations (Appendix M.4).

Joint evidence: mean and even medians. An entry with true contribution f_{ir} raises its row's lower cost by f_{ir} and reduces its upper cost by $p_i - f_{ir}$. Mean pooling can have integer contributions $0 < f_{ir} < p_i$, so one query contributes to both proofs. Figure 1 shows the mechanism. Exclusion of every p-row set is equivalent to the p smallest lower costs summing to at least $b_H + 1$, which we write with a single threshold t as $\sum_i (t - \ell_i)_+ \le pt - b_H - 1$ (Lemma 1, Appendix M.5). In the figure, the six entries of row 1 are needed for exclusion; choosing row 1 as the retained row turns them into a safety proof as well, so the longer safety proof yields the shorter complete certificate.

For a history $\mathcal{O}$, let ℓ_i^0, u_i^0 be its row bounds and U_i its unqueried candidates. Using true contributions offline, define $\mathcal{D}_i = \{(|J|, \sum_{r \in J} f_{ir}) : J \subseteq U_i\}$. A state (a_i, v_i) gives $\ell_i = \ell_i^0 + v_i$ and $u_i = u_i^0 - a_i p_i + v_i$. Let L^* be the true optimal active count.

Theorem 3 (Joint entry witness)**.** *For the fixed true matrix, history $\mathcal{O}$ and integer $\varepsilon \ge 0$, the minimum additional entry count Q_{joint} is $\min \sum_i a_i$ over $s \in \{\max(0, L^* - \varepsilon), \dots, L^*\}$, row states in $\mathcal{D}_i$ and a safe set C of s active rows with $\sum_{i \in C} u_i \le b_H$. If $p = s + \varepsilon + 1 \le m$, also require an integer $0 \le t \le TM$ such that $\sum_i (t - \ell_i)_+ \le pt - b_H - 1$, where $(x)_+ = \max(0, x)$. For $p > m$, exclusion is automatic.*

Every successful query set induces reachable row states and a safe count, and every feasible state choice is attained by a query set of the same size (Appendix M.5). The offline dynamic program is polynomial in explicit m, T, M but has a budget-indexed grid; an equivalent integer program groups equal within-row outcomes into G row–outcome groups with $O(G+m)$ variables and constraints per safe-count target, without guaranteeing polynomial solver time. On the 1,920 full mean targets, this program closes 1,876 within a nominal 60 seconds, and constructed-score grids up to $N = 8192$, $T = 65$ close in four of eight cases (Appendix M.7). Its optima are identical across HiGHS settings and CBC on our solver-sensitivity grid (Appendix W).

What joint optimization adds. Let Q_{split} minimize the sum of separately minimum safety and exclusion sizes over admissible safe counts, and let $Q_{\cup,\text{sep}}$ minimize the union of tied minimum safety and exclusion witnesses at the same count, then over counts. For the same instance, history and tolerance,

$$Q_{\text{joint}} \le Q_{\cup,\text{sep}} \le Q_{\text{split}} \le 2Q_{\text{joint}}. \qquad (6)$$

Realizable mean families attain $Q_{\cup,\text{sep}}/Q_{\text{joint}} \to 2$, or $53/30$ at fixed $M = 7$, $\alpha = .10$, with empty history and exact stopping (Appendix M.8). The sandwich holds for any fixed nonnegative replacement weights, and the factor-two family persists for every weight vector within relative distance $1/(2MT)$ of uniform (Proposition 14, Appendix R.1).

When can sharing help? Three conditions are checkable before any candidate-test score is seen. A strict gain $Q_{\cup,\text{sep}} > Q_{\text{joint}}$ requires (a) a nonzero initial gap, $|S_0| - V^-(0) > \varepsilon$, and (b) an active row with two distinct replacement thresholds in $(0, 1]$, since otherwise every contribution is 0 or p_i and Corollary 3 gives equality; odd medians never satisfy (b). Ex post, (c) a gain requires a witness that cannot certify exclusion automatically, and all 1,831 no-gain NLP targets have a witness with $V^- + \varepsilon \ge |S_0|$ (Appendix M.9). A reference-predicted version of (c), $\hat{V}^* + \varepsilon < |S_0|$, gives

a conservative pre-scoring screen: on controlled panels it flagged every target with a gain, and no unflagged target had one (Appendix U). A history alone can never certify a strictly positive remaining advantage, because a completion with zero unseen contributions makes safety alone sufficient (Proposition 6).

6 Acquiring a Certificate

Fixed order uses seeded row priority followed by candidate index; random order permutes active entries; largest width maximizes $u_i - \ell_i$ over eligible rows. Reference priority fits one ridge model per candidate to reference-row flip counts f_{ir} (features: pool scores, cutoff margin, score range and p_i/M; predictions clipped to $[0, p_i]$, Appendix B). It selects the eligible row with the smallest estimated total cost and, within it, the unqueried candidate with the largest predicted upper-cost reduction $p_i - \hat{f}_{ir}$. Frontier priority (mean pooling) clips predicted active capacity to $[V^- - m_0 + 1, V^+ - m_0]$, obtaining a target s, and scores predicted progress towards safety for s and exclusion of $s+1$ (Appendix P.3). For odd medians, the closed forms suggest a targeted policy, OM-CF, that pursues agreements in the predicted safe set and capped oppositions at the Theorem 2 level (Appendix T). Predictions guide order, not bounds.

These policies differ in what they can guarantee. With exact predictions, OM-CF stops after exactly Q_{exact} queries (Proposition 20), so the closed forms make a prediction-based policy consistent in the sense of learning-augmented algorithms (Lykouris and Vassilvitskii, 2018; Purohit et al., 2018). No policy, however, can be robust: Propositions 7 and 8 give instances with fixed $M = T = 7$, $q = 1/20$, $\alpha = 1/10$ where any always-valid policy needs queries linear in batch size, even at the primary positive tolerance, while the minimum certificate has one or two entries. Sorting computes each capacity bound in $O(m \log m)$ time.

7 NLP Score-Inspection Experiments

7.1 Panels and protocol

RewardBench 2 has 1,723 four-completion prompts and 14 reward models (Malik et al., 2026). WMT23 en-de has 557 source segments, 13 output streams and 24 reference-based metrics (Freitag et al., 2023); removing `synthetic_ref` gives a twelve-stream variant with refitted maps and cutoff. The frozen first-seven pools leave $T = 7$ RewardBench or $T = 17$ WMT23 candidates (Appendix B); Section 7.4 tests composition sensitivity. Forty seeded splits preserve prompt/source groups. Each evaluator uses a reference-fitted right empirical cumulative distribution function (ECDF), and the cutoff is the $\max(1, \text{round}(qn_{\text{ref}}))$-th highest reference consensus. We use both pooling rules, $q \in \{.05, .10, .20, .30\}$ and $\alpha \in \{.02, .05, .10, .15, .20\}$ (primary .10). Primary stopping uses $V^+ - V^- \le \varepsilon = \lceil .02|S_0(P)| \rceil$, returning $K = V^-$ so that $V^* - K \le \varepsilon$; exact stopping uses zero. Queries access cached scores; reference and base scoring are charged separately, and acquisition counts exclude panel construction and full-panel WMT23 preprocessing reads. These are logical access counts, not inference savings, and the overlapping splits support descriptive summaries.

7.2 Retention baselines and risk guarantees

Selected coverage is $K/|S_0|$. At $q = .05$, $\alpha = .10$ and median pooling, zero-query coverage spans 80.9–99.2% across the three original panels, versus 99.5–99.9% under full information (Appendix Table 8); invariant and budget-admissible outputs both contribute (Appendix G.1). Fixed-count, batch-feasible and calibrated baselines have different constraints and guarantees (Appendix C). Envelope CRC calibrates an output-cost threshold with the loss-envelope correction of Yeh et al. (2025); under group exchangeability its expectation guarantee does not ensure the budget on every batch. Grouped SCoRE controls marginal deployment risk through whole-group decisions; at $q = .05$, $\alpha = .10$, RewardBench coverage is 96.4% (median) and 98.8% (mean) without candidate test scoring for selection, but unequal-size WMT24 documents are excluded. Tables 26–31 compare returned counts and churn with reference, calibration and test costs charged in the same units; the guarantees remain distinct.

7.3 Inspection and certificate size

At $q = .05$ (Table 2), reference priority averages 61.4% active-entry inspection with median pooling on RewardBench and 45.1% on thirteen-stream WMT23; mean pooling averages 85.4–86.5% across the three panels. Twelve-stream WMT23 under median pooling stops at zero on 37/40 splits.

Panel	Pooling	Z (%)	Fixed (%)	Random (%)	Width (%)	Ref. (%)	Cov. (%)
RewardBench 2	median	0.0	63.8	64.0	63.0	61.4	97.3
WMT23, 13 streams	median	10.0	57.9	58.4	65.0	45.1	97.7
WMT23, 12 streams	median	92.5	3.9	3.8	3.9	3.8	99.6
RewardBench 2	mean	0.0	90.9	91.5	91.7	86.5	97.5
WMT23, 13 streams	mean	0.0	91.3	92.2	92.6	85.7	97.5
WMT23, 12 streams	mean	0.0	89.5	90.2	89.9	85.4	97.0

Table 2: $q = .05$, $\alpha = .10$, gap $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Z: percentage of 40 splits stopping without queries. Order columns: mean active inspection ($100Q/(mT)$), zero for empty active sets. Ref.: reference priority; Cov.: its mean selected coverage ($100K/|S_0|$).

Matching minima. Comparing exact odd-median stopping with its matching minimum (Theorem 2), the median splitwise reference-query/Q_{exact} ratio is 1.968 for RewardBench and 1.662 for thirteen-stream WMT23, conditional on $Q_{\text{exact}} > 0$ (Appendix Table 36, M.2). Positive-gap comparisons use Q_ε (Appendix N). RewardBench 2, two WMT23 variants and three WMT24 directions give six panels; four depths and 40 splits yield 960 mean instances and 1,920 targets across the two stops. Each joint and optimal-union witness matches a numerical global lower bound (Appendix M.9; Table 3); this is distinct from an exact rational proof certificate.

Beyond deduplication: when do gains occur? Joint optimization improves on $Q_{\cup,\text{sep}}$ in 89 of the 1,920 targets, saving 6,269 entries in total (Appendix M.9). Median savings are zero; maximum reductions relative to $Q_{\cup,\text{sep}}$ are 2.61% (primary) and .87% (exact), with mean savings of 5.34 and 1.19 entries (Table 56). The gains concentrate where exclusion evidence is necessary. Full retention is feasible in 903 of the 960 instances, which makes exclusion automatic in their 1,806 targets. Of the 57 instances that cannot retain the full baseline list, all 57 exact targets and 32 primary targets show strict gains, and the remaining 25 primary targets have an optimal witness that certifies at least $|S_0| - \varepsilon$ outputs safe (Table 57). WMT24 ja-zh contributes 63 of the 89 gains. The prevalence of intermediate contributions $0 < f_{ir} < p_i$ does not explain gains on its own: in 13 strata containing both outcomes, gain targets have the larger intermediate fraction in eight and the smaller in five. Multiplying the budget by .75 and 1.25 moves the number of non-full instances from 57 to 174 and 19 (Table 60), so tighter budgets enlarge the region where sharing can matter. All 8,640 whole-column tasks on this grid have equal joint and best separate-union minima.

Online discovery. Across the 960 primary mean targets, frontier priority averages 5210.6 queries versus 5286.5 for reference and 5349.7 for width. It beats reference in 403 instances and loses in 554, with three ties. The lower mean despite more losses has a simple diagnosis: the exclusion term helps when exclusion is necessary (it saves 20.28 entries on average over safety-only on the 32 such targets) and costs a few entries elsewhere (3.72 on the 903 full-retention targets), so frontier's wins are larger than its losses (Appendix P.3, Table 71). Safety-only frontier has a slightly lower overall mean, and the ablations do not establish that both terms are needed.

7.4 Budget and evaluator composition

Inspection need not decrease with the churn budget: safety can become easier while exclusion becomes harder (Appendix G). Composition changes both the replacement family and the selected set; Appendix G reports twenty random pools per original panel and WMT23 pools with a fixed mix of surface-overlap and other metrics.

7.5 External panels and quality diagnostics

WMT24 en-de, en-es and ja-zh use document splits, exclude human-reference outputs and retain four depths, 40 splits and $\alpha = .10$ (Freitag et al., 2024). At $q = .05$, median zero-query coverage is 98.1–99.8%. The objective and panel criteria preceded WMT24 result inspection, but numerical priorities were repaired after a replay failure (a partial freeze; Appendix H). Quality is assessed after selection. RewardBench precision is the preferred-label fraction among returned outputs; its mean differences from label-blind, same-consensus top-K are -0.46 to -0.03 percentage points across depths under median reference priority. Released MQM differences have either sign; missing labels are not imputed (Appendices E and H).

Panel	Num. closed	Primary Q	Primary Ref./Q	Exact Ref./Q	Gain/num. closed
RewardBench 2	320/320	1216.9	1.205	1.186	0/320
WMT23, 13	320/320	2012.1	1.394	1.318	7/320
WMT23, 12	320/320	2024.3	1.291	1.258	13/320
WMT24 en-de	320/320	6922.7	1.185	1.169	3/320
WMT24 en-es	320/320	5968.4	1.187	1.174	3/320
WMT24 ja-zh	320/320	8717.0	1.119	1.107	63/320

Table 3: Full-test mean evidence at $\alpha = .10$: four depths, 40 splits and two count-gap targets per panel. Primary uses $\varepsilon = \lceil .02|S_0| \rceil$; exact uses zero. Primary Q averages the minimum joint entry count over 160 primary targets per panel; each ratio uses its matching target. All joint and union minima are numerically closed. Ratios average splitwise reference queries divided by positive matching minima. Gain compares joint evidence with the best union of separately optimal witnesses. Appendix M.9 reports all depths.

7.6 Reference mapping and query granularity

On thirteen-stream WMT23 with median pooling at $q = .05$, switching from full-reference right to midrank ECDF raises zero-query selected coverage from 84.9% to 99.8% and lowers mean active inspection at primary stopping from 45.1% to 0.3%. Test groups stay fixed, but baseline-set Jaccard similarity is 0.77: midrank reduces the share of selected outputs exactly at the cutoff from 28.8% to 7.3%, so the saving concerns a changed selection task (Appendix K). Difficulty is therefore not an intrinsic property of the data. We recommend fixing the mapping, reference-size rule and tie convention as part of the selection protocol before any candidate-test score is seen, choosing them for selection validity (e.g., the score resolution of each metric, Deutsch et al. 2023) rather than to minimize queries, and reporting reference-bootstrap sensitivity (Appendix K.2) together with the share of selected outputs at the cutoff. Replicable calibration can stabilize the cutoff, but not a difficulty that depends on the test batch (Appendix V).

At $q = .05$ and $\alpha = .10$, median RewardBench reference priority reveals 1.77% of all NT candidate-test entries through entry queries versus 71.8% through columns, which charge all N entries per column. Every column is required on all 114 non-full mean targets; a sufficient range condition covers 101 (Appendix M.12), while other realizable instances permit strict column gains.

7.7 Summarization and real scoring

A frozen SummEval protocol (Fabbri et al., 2021) tests all ten three-base/two-candidate compositions. Under mean pooling, joint frontier saves two queries on average when $V^* + \varepsilon < |S_0|$ but adds queries in full-retention cases. On a fresh five-metric panel, 192 A40 runs cover eight methods, four depths, two pools and three repeats; primary reference priority averages 89.97 s versus 108.18 s for full scoring, including common setup (Appendix Q).

7.8 Discussion and practical guidance

Joint optimization couples retention to evidence, extending the verification–discovery distinction (Charikar et al., 2002; Diwan et al., 2026) to a setting where the retained set is also chosen. Its empirical gains are modest, and discovery is the larger cost: on the same 960 primary targets, reference queries exceed Q_{joint} by 809.58 entries on average, whereas $Q_{\cup,\mathrm{sep}} - Q_{\mathrm{joint}}$ averages 5.34. Even true-score priorities fail to yield query optimality (Appendix P.5). For practitioners we suggest: (1) fix the replacement family, weights, mapping and cutoff before scoring, and choose the average, weighted or per-replacement objective to match how replacements will be executed; (2) check the initial gap $|S_0| - V^-(0)$ first, since it decides whether any query is needed; (3) invoke joint optimization only when exclusion is likely necessary ($\hat{V}^* + \varepsilon < |S_0|$), otherwise a minimum safety witness suffices; (4) report evidence size, queries and scoring time separately, together with externally labeled quality assessments.

8 Conclusion

Retention evidence depends on safety, exclusion and query granularity. Odd medians separate the two proof tasks; mean pools permit sharing beyond optimal separate unions, with modest empirical gains confined to targets that need exclusion evidence. The analysis extends to weighted and per-replacement objectives and to lattice-structured score constraints. Small certificates ensure neither inexpensive discovery nor inference.

Limitations

The main certificates cover uniformly weighted single-member replacements from the declared finite family, with fixed maps and cutoff. They control average churn without requiring each replacement to satisfy α and do not cover evaluators outside the family; drift beyond the family requires extending the family and its budget. Per-replacement caps make the full-information optimum NP-hard (Appendix R.2), and weighted odd-median minima no longer have closed forms. Adding original-batch outputs to the returned set requires rechecking its budget, and newly generated outputs change the batch being certified. Nominal $2k$ normalization does not constrain the returned count to k. The odd-median closed forms assume unit-cost entry queries and Boolean completions without cross-entry constraints. Constrained completions (Appendix S) are valid only if the constraints hold for the true scores; learned correlations do not establish this, and minimum-certificate formulas do not transfer to them. The near-full criterion (Proposition 11) assumes empty history, exact stopping and a restricted family of minimum exclusion witnesses. The joint dynamic program has a large count-indexed state space, and outcome grouping reduces model size without guaranteeing polynomial solver time. Forty splits overlap, so aggregate comparisons are descriptive. True-score and oracle priorities are diagnostics, not deployable policies, and OM-CF with learned predictions did not improve on reference priority in our controlled study. The revision experiments in Appendices R–W use controlled synthetic panels and were not rerun on the NLP panels.

RewardBench and WMT23 were used during method development. WMT24 results had not been inspected when the risk objective and panel-inclusion criteria were fixed; after an exact-replay failure, numerical query priorities were corrected, which affected query order but not the certificate definition. This partial freeze is not a fully preregistered evaluation. Independence between evaluator training data and the benchmarks is not established, and some released meta-metrics use human-score calibration. The rationale for the original frozen evaluator ordering is not established by the records, and sampled pools do not cover all future evaluators. Reference priority has no online competitive guarantee. Incomplete MQM coverage restricts quality diagnostics to annotated outputs. The SummEval timing study measures five fixed scorers on one A40 with batch-eight full scoring and sequential entry queries; its three repetitions measure runtime variability rather than independent task replication.

Ethical Considerations

The experiments reuse evaluator panels and existing labels or generate controlled numerical scores; they collect no new human labels or language responses. Source reconstruction matches existing benchmark text to cached scores and output identities locally. Available records do not document a systematic audit for personally identifiable information or offensive content. Stable selection can preserve evaluator errors, so a certificate does not endorse retained content. Appendices B and H document data sources and acquisition. We have not established redistribution permission for every separate RewardBench result or contributed WMT score; benchmark and toolkit licenses do not resolve the status of every asset. The data archive remains local. Intended use is research on retention certification. Generative AI tools supported problem framing, proof discussion, coding, analysis and manuscript drafting. Human authors remain responsible for claims, code, results, citations and final text. Historical compute accounting is incomplete; `RESPONSIBLE_NLP_CURRENT.md` records available details.

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Appendix guide. Repeated baseline, budget and grouped-SCoRE grids are provided in `supplement/result_grids.csv`. Full-depth mapping, column, tolerance and discovery grids are in `supplement/extended_result_grids.csv`. Companion JSON files preserve headers, captions and table sources. Matched minimum-evidence and query costs for both pooling rules are in `supplement/matched_evidence_summary.csv`, with per-instance records in `supplement/matched_evidence_records.csv`. Full proofs, replay specifications and additional diagnostics are provided below. Proofs begin in Appendix A (p. 13); replay specification begins in Appendix B (p. 15). Appendix C specifies baselines; D–J collect retention, quality and risk comparisons; K–L examine selection protocols and loss definitions. Minimum-evidence extensions are in Appendix M (p. 45), with a compact joint mean formulation in Appendix M.5 (p. 55) and full-batch evidence in Appendix M.9 (p. 72). N–O discuss tolerance and operational scope. Prediction interventions and frontier acquisition appear in Appendix P (p. 88). Appendix Q reports SummEval and measured scoring costs.

A Proofs and Completion Model

A.1 Notation and units

Table 4 collects the main quantities; Section 5 and Appendix A.7 give worked examples.

Symbol	Meaning and unit
N, M, T	Outputs, base evaluators, candidate evaluators.
k, S_0	Nominal count and baseline-selected set; $\lvert S_0\rvert$ need not equal k.
$H = MT$	Number of single-member replacement scenarios.
f_{ir}, p_i	Entry's flip count and its row's maximum flip count.
F_i	Full row cost $\sum_r f_{ir}$, in flips.
b_H	Integer flip budget $\lfloor 2k\alpha H \rfloor$.
ℓ_i, u_i	Lower and upper row costs, in flips.
V^-, V^+, V^*	Safe, optimistic, and true optimal retained counts.
ε	Allowed certified retained-count gap.
Q_{joint}	Minimum joint evidence under the stated query unit.
$Q_{\cup,\text{sep}}$	Minimum union over separately minimum proofs.
Q_{split}	Minimum sum of separate proof sizes.

Table 4: Core notation. In odd-median Boolean formulas, dividing by the nonzero flip cap gives opposition-count costs and budget b. A column query reveals and is charged for all N score entries.

A.2 Flip geometry and sharp intervals

Our cutoff rule includes scores equal to the threshold. Under the nonempty reference ECDF conventions used here, every score in the reference split maps into $(0, 1]$, so a cutoff fitted from reference consensus satisfies $0 < t_0 \le 1$. For median pooling with $M = 2h + 1$, let v_i be the number of base scores at least t_0. If $v_i \ge h + 2$, removing one vote leaves at least $h + 1$ votes. If $v_i \le h - 1$, adding one vote leaves at most h votes. These rows are invariant. When $v_i = h + 1$, a below-cutoff candidate changes membership exactly when it replaces one of the $h+1$ voting members. When $v_i = h$, an at-or-above-cutoff candidate changes membership exactly when it replaces one of the $h+1$ nonvoting members. Thus the maximum flip count is $h + 1$ on either boundary and zero elsewhere.

For mean pooling, each replacement mean is nondecreasing in candidate score x. On a selected row, increasing x cannot create an exclusion, so $f_i(x)$ is nonincreasing, with maximum $f_i(0)$. At $x = 1$, the candidate score is at least every base score it can replace; all replacement means remain at or above the cutoff, giving $f_i(1) = 0$. On an unselected row, $f_i(x)$ is nondecreasing, with maximum $f_i(1)$ and $f_i(0) = 0$. The same argument applies to the average of the two central scores in an even-sized median pool: increasing a candidate score cannot decrease either order statistic. It also covers ties because membership always uses $\ge$.

Each row's cost sums observed flip counts and unobserved counts bounded by zero and p_i. With no cross-entry constraints, joint endpoint assignments realize both ℓ and u. Consequently, $V(\ell)$ is the largest full-information retention capacity among consistent completions, and $V(u)$ the smallest. These extrema are sharp in the stated model. The declared completion set does not encode additional constraints from evaluator functions, score correlations or raw-score support. Sharpness is relative to this set; it does not claim every endpoint assignment is realizable by a deployed evaluator.

A.3 Safety witness

Suppose a certificate proves some boundary mask C of cardinality L safe. For each $i \in C$, let a_i count its observed agreements. In the worst-case completion its opposition cost is $\sum_{i \in C}(T - a_i)$.

Feasibility requires

$$\sum_{i \in C} a_i \geq LT - b.$$

Each query reveals at most one agreement, establishing the lower bound $\max(0, LT - b)$.

A truly feasible set C of L boundary rows contains $LT - \sum_{i \in C} z_i \geq LT - b$ agreement cells. Revealing any $\max(0, LT - b)$ of them makes its upper cost at most b, attaining the lower bound. If no truly feasible L-row set exists, no observations can certify one; this infeasible target is represented by infinite certificate cost. For $L = 0$ the empty mask requires no information. When $b \geq LT$, every L-mask is safe before querying.

A.4 Exact optimality witness

Let L be the true optimal boundary count. When $L = m$, certifying the full boundary safe certifies optimality automatically; the safety witness gives the claimed value. Consider $L < m$, so $p = L + 1$ and the sum of the p smallest true row costs exceeds b. Therefore the level t^* in Equation 5 exists.

A certificate establishing that no p-row mask is feasible under any consistent completion must reveal opposition counts x_i such that

$$0 \leq x_i \leq z_i, \qquad \sum_{i=1}^{p} x_{(i)} \geq B_0 = b + 1.$$

Otherwise, choose the p rows with the smallest observed opposition counts and complete their unknown cells with agreements. This makes that set feasible in a consistent completion. Rowwise inequalities $x_i \leq z_i$ imply order-statistic inequalities $x_{(j)} \leq z_{(j)}$ for every j. Let $t = x_{(p)}$, the p-th smallest observed count. Then

$$B_0 \leq \sum_{i=1}^{p} x_{(i)} \leq \sum_{i=1}^{p} \min(z_{(i)}, t),$$

so $t \geq t^*$. Every one of the remaining $m - p$ observed row counts is at least t. The number of observed opposition cells is consequently at least $B_0 + (m - p)t^*$.

For attainment, order the rows by their true costs. Assign integer observed counts to the first p rows between zero and $\min(z_{(i)}, t^*)$, with sum exactly B_0. This is possible because the sum of the caps is at least B_0. If all were below t^*, their sum would be at most $D(t^* - 1) < B_0$; at least one must therefore equal t^*. The level also satisfies $t^* \leq z_{(p)}$, since the cap at $z_{(p)}$ leaves the first p true row counts unchanged. Assign observed count t^* to every subsequent row. Each is within that row's true opposition capacity. The p smallest observed counts then sum to B_0, and the total number of revealed opposition cells is exactly $B_0 + (m-p)t^*$.

Choose safety agreements from any truly optimal set of L rows. Their cells are disjoint from the exclusion witness's opposition cells. Revealing the union certifies feasibility and rules out any feasible p-row set under every completion, at the sum of the two costs. Every exact certificate must meet both lower bounds, proving Equation 6.

For $L = 0 < m$, $p = 1$ and $t^* = b + 1$; exact exclusion requires $b + 1$ oppositions on every row. For $m = 0$, full optimality is immediate and the certificate size is zero. The formula is not an online competitive bound and does not apply as a lower bound to a positive-ε stop. Nonuniform replacement weights and unequal entry costs require a different characterization. Appendix M treats whole-column revelation separately.

A.5 Online difficulty and acquisition objectives

For one row and $0 \leq b < T$, feasibility asks whether its opposition count is at most b. An adversary can arrange the first $T - 1$ replies to contain exactly b oppositions in total. The final reply then distinguishes cost b from cost $b + 1$. Any deterministic exact strategy has a worst case of T queries. An offline feasible certificate needs only $T - b$ agreements; an infeasible certificate needs $b + 1$ oppositions.

At $b = T - 1$, a feasible row with one agreement has a one-entry offline certificate, whereas an adversary can force T queries. Proposition 7 strengthens this obstruction: with $M = T = 7$, $q = .05$, $\alpha = .10$ and $|S_0| = k$, exact discovery cost can grow with batch size while the offline minimum remains one.

Reduction in the integer retention gap need not have diminishing returns. For one row with $T = 2$ and $b = 0$, the first agreement leaves the upper cost one and the gap one. A second agreement lowers the gap to zero. The second agreement has a larger marginal gap reduction after the first is observed. This prevents transferring adaptive-submodular guarantees to the retention-gap objective without further assumptions. Stochastic Boolean query methods analyze objectives and distributions with their own conditions (Deshpande

et al., 2016; Gkenosis et al., 2018).

A.6 Minimum witnesses at positive retention tolerance

The same agreement and opposition geometry characterizes the smallest certificate at any integer tolerance. The implemented target is the observable frontier gap $V^+ - V^- \le \varepsilon$. The realized regret satisfies $0 \le V^* - V^- \le V^+ - V^-$ and can be smaller than the gap. The gap criterion certifies shortfall for every consistent completion; checking realized regret alone on the complete matrix would not supply that guarantee. Appendix A.7 gives a state with regret zero and gap one.

Corollary 1 in Section 5 states the minimum witness size.

Proof. Selected interior rows add the same m_0 to both count bounds and cancel from the gap. Consider any valid certificate and let s be its safe boundary count, so $V^- = m_0 + s$. Since $V^- \le m_0 + L^* \le V^+$ and $V^+ - V^- \le \varepsilon$, this count lies between $\max(0, L^* - \varepsilon)$ and L^*. Safety requires at least $\max(0, sT - b)$ agreements by Theorem 1. If $p = s + \varepsilon + 1 \le m$, observed opposition counts must rule out every p-row mask, making the optimistic boundary count at most $p-1$. The order-statistic argument for Theorem 2 applies to this p, yielding at least $E(p)$ oppositions. Their cells are disjoint, so every certificate has at least the corresponding sum in Equation 7.

Conversely, choose any s in that range and a true-feasible s-row mask. Reveal a minimum safety witness in it. When $p \le m$, reveal the capped opposition witness of size $E(p)$; it exists because $p > L^*$, so the p smallest true costs exceed b. The union certifies at least s safe rows and at most $s+\varepsilon$ feasible rows in any completion. When $p > m$, the upper count is already at most $m \le s + \varepsilon$, and no opposition witness is needed. Thus the minimum is attained. The empty boundary and tolerances at least m give zero observations. □

Increasing ε enlarges the set of permitted certificates, so Q_ε is nonincreasing. The offline constructor uses the true matrix to locate the witness; it does not prescribe an online query order. Mean costs need not be Boolean opposition sums, and whole-column queries couple the available cells, so neither model directly inherits this median entry formula. Theorem 3 treats joint mean evidence, while Appendix M.3 treats exact median columns.

Queries	$\ell/4$	$u/4$	V^-	V^+
0	$(0,0,0)$	$(3,3,3)$	0	3
4	$(0,0,0)$	$(0,2,3)$	2	3
7	$(0,1,2)$	$(0,2,3)$	2	2

Table 5: Certificate updates for the three-row example with opposition budget $b = 2$. The first four observations are row 1's three agreements and row 2's second entry. The final three are row 2's first opposition and row 3's first two oppositions.

A.7 A complete certificate example

Consider median pooling with $N = 20$, $M = 7$, $T = 3$, $k = 2$, $\alpha = .10$ and cutoff $t_0 = .5$ fixed by construction. Each of the three selected rows has four base scores .8 and three .2; the remaining seventeen rows have all seven scores .2 and remain unselected under a single replacement. Each candidate–output entry assigns .1 for opposition and .9 for agreement. Their opposition matrix is

$$Y = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

Each opposition reverses membership under four member replacements. Hence true flip costs are $(0, 4, 8)$ and the integer flip budget is $b_H = \lfloor 2k\alpha H \rfloor = 8$, equivalent to the bit budget $b = 2$. Table 5 reports $\ell/4$ and $u/4$ in opposition-count units.

After the first four queries, returning rows 1 and 2 has certified risk at most $8/84 = 2/21 \le .10$, while its actual risk is $4/84 = 1/21$. After four queries the frontier gap is one, although realized regret is already zero. Enumerating all $2^9 = 512$ subsets of the nine observable entries in this fixed matrix gives $Q_0 = 7$ and $Q_1 = 4$. Thus a lower query count at positive tolerance can reflect a smaller required witness as well as the acquisition order.

B Reproducible Experimental Specification

Panel construction. The frozen panels contain numerical scores and identifiers. For each RewardBench evaluator, we exclude result records with subset `Ties` or without four eligible numeric scores. Eligibility unwraps singleton score lists and requires four numeric values. The constructed panel subsequently passes a finite-value assertion; this is distinct from the initial type and length filter. We retain prompt IDs eligible for every evaluator; each prompt forms a four-output group. For

each evaluator, alignment checks match the native prompt and every completion to the corresponding model-formatted input, preserving evaluator-specific formatting. Aligned records retain native order: the chosen completion followed by three rejected completions. Selection ties use the separate label-blind priorities below.

The allenai/reward-bench-2-results snapshot is pinned at `b19c7033e964187d12e74a43a07f2d727a3d37e5`. Within each model JSON, the builder traverses the parallel `id`, `subset` and `scores` arrays in their native order and retains the last eligible occurrence of a repeated identifier. Common prompt IDs are then sorted numerically. The model/file mapping is `evidence/source-model-files.json`; `ACQUISITION_INDEX.md` gives pinned-source acquisition and reconstruction commands.

For WMT23, we retain reference-based metrics under the common reference-A configuration and remove duplicate raw `refA.seg.score` files by SHA-256 of their byte contents before parsing scores and aligning systems.

Before assigning splits, the WMT23 builder checks the Pearson correlation of each retained metric with BLEURT-20 (Sellam et al., 2020; Pu et al., 2021) over all 557 source segments and 13 output streams. All correlations are positive, with a minimum of 0.18425. A nonpositive correlation would reject the build. Since all are positive, this check changes neither score signs nor metric inclusion. It provides no human-quality labels and lies outside the reported post-freeze acquisition counts. A post hoc reference-only check over the same 40 splits also finds positive correlations for all 24 metrics: the minima are 0.09446 for thirteen streams and 0.07931 after removing `synthetic_ref`. This post hoc diagnostic leaves scores unchanged; the original preprocessing used the full panel.

RewardBench splits use seed $20260815 + r$ and WMT23 uses $87000 + r$, for $r = 0, \ldots, 39$. Each seed shuffles group indices with Python `random.Random(seed).shuffle`. RewardBench reference and calibration sets each contain 574 prompts, leaving 575 for testing. WMT23 reference and calibration sets each contain 185 source segments, leaving 187 for testing. Let n_{ref} count reference outputs, including all outputs within each reference group. For each evaluator e, we separately fit

$$\widehat{F}_e(s) = \frac{1}{n_{\text{ref}}} \sum_{\ell=1}^{n_{\text{ref}}} \mathbf{1}[s_{e,\ell}^{\text{ref}} \le s].$$

All reference scores tied at s enter the cumulative count. The cutoff is the reference consensus at position $n_{\text{ref}} - \max(1, \text{round}(q n_{\text{ref}}))$ in ascending order, with zero-based indexing. The implementation uses Python's nearest-integer `round`, which resolves exact half-integer ties to the even integer, for both this index and the nominal test count k. Cutoff membership includes scores equal to the boundary. Throughout the tables, a top-$100q\%$ setting denotes the nominal selection fraction q, rather than an exact returned fraction.

The WMT twelve-stream panel excludes the entry named `synthetic_ref` before CDF fitting, reference consensus and all subsequent calculations. It has 12 outputs per segment instead of 13. RewardBench has 2,300 test outputs, thirteen-stream WMT23 has 2,431, and the twelve-stream panel has 2,244. The family contains 49 replacements for RewardBench and 119 for each WMT23 panel.

The ordered RewardBench evaluators are Tulu-3-8B, QRM-Gemma-27B, WorldPM-72B, Skywork-V2-8B, DeBERTa-v3, InternLM2-20B, Starling-34B, UltraRM-13B, URM-LLaMa-8B, OffsetBias-8B, GRM-Llama3-8B, Skywork-V2-3B, LDL-Gemma-27B and BTRM-Qwen2-7B. The first seven constitute the pool and the last seven are candidates. This is the frozen original ordering; its selection rationale is not established by the available records. Analysis files record the source-model identity mapping. A results-dataset revision identifies the cached-score snapshot, not necessarily the model-weight checkpoints. Replaying scores requires the former; reproducing scorer execution additionally requires verified model revisions and interfaces.

The ordered WMT metrics are BERTscore, BLEU, BLEURT-20, COMET, Calibri-COMET22, MEE4, MaTESe, MetricX-23, MetricX-23-b, MetricX-23-c, XCOMET-Ensemble, XCOMET-XL, XCOMET-XXL, XLsim, YiSi-1, chrF, docWMT22CometDA, eBLEU, embed_llama, f200spBLEU, instructscore, mre-score-labse-regular, prismRef and tokengram_F. The first seven form the pool, and the remainder are candidates in both WMT populations.

Implementation and replay checks. The replay rebuilds replacement score vectors and uses the same left-to-right mean reduction for complete-matrix evaluation and individual queries. It does not subtract a member from a previously rounded sum, keeping observed and endpoint flip calculations numerically consistent. For rational budget $\alpha = a/d$, it computes $b_H = \lfloor 2kMTa/d \rfloor$ by exact integer division.

Safe retention and query priorities use a label-blind permutation of test-item indices generated by NumPy `default_rng(...).permutation` with seed split seed + 700001. The priorities remain fixed throughout each path. Random acquisition uses the same NumPy permutation API with seed split seed + 99173. Reindexing carries each output's original tie priority with it, preserving features and zero-query masks. For each candidate, ridge uses every reference output and its flip count for the current pool and cutoff. Features are the M ECDF-normalized base scores, signed cutoff margin $c_P(i) - t_0$, base-pool score range and p_i/M, with no additional standardization. The objective is the sum of squared residuals plus a squared ℓ_2 penalty of coefficient 1, with an unpenalized intercept. The normal equations are solved with `numpy.linalg.solve`; the recorded replay uses NumPy 2.0.2. The ridge penalty is fixed at one; available records do not document held-out tuning or identify the original BLAS library and platform versions. Test labels and candidate test values are not fitting inputs. Width chooses candidate index order within its chosen row. Reference priority chooses the maximum predicted upper-cost reduction, resolving candidate ties by increasing index.

An active-row entry query reveals one candidate score; known base scores determine its member-replacement outcomes and flip count. At each stopping or checkpoint state, the implementation stores the mask, retained-count bounds, query count and upper-cost sum. Certificate computation uses revealed scores and endpoint bounds only. A separate audit uses the complete cached matrix to recompute replies, reject duplicate queries and reconstruct recorded intervals and masks. For each recorded stopping target, it checks actual and upper-budget feasibility, $V^- \le V^* \le V^+$ and the corresponding stopping condition. Its access to hidden scores and V^* is for evaluation, not acquisition or certification.

The accounting records Tn_{ref} candidate-reference entries, Mn_{ref} base-reference entries and MN base-test entries. The full-information controls inspect TN candidate-test entries; CRC additionally uses candidate calibration scores. Active query fractions divide by mT, with m the number of selected rows having $p_i > 0$, whereas panel fractions divide by NT. These entry counts are not GPU-call counts: reward models can batch completions, and document-level metrics can share computation across segments.

For each policy, commitment coverage is $|C|/N$ (including committed exclusions) and selected coverage is $|S_0 \cap C|/|S_0|$, using that policy's own base list S_0. The automatically returned count is $K = |S_0 \cap C|$; K/k is its ratio to the nominal count before review. RewardBench precision is the preferred-response count among returned outputs divided by K. Reported ratios and paired differences are computed within each split before aggregation. Active inspection is defined as zero when $m = 0$ (no eligible work), and this value enters the 40-split mean. Other ratios with empty denominators, such as returned precision, are undefined and omitted from their respective means. Display rounding is applied only after aggregation. Cross-table returned and queried counts use exact integer sums divided by the number of splits, with decimal half-up rounding to one place. The selected-coverage summaries have 40 defined splits per configuration; RewardBench precision comparisons and the WMT24 reference-priority quality tables likewise have 40.

No active set is empty in the original-pool budget experiments or primary WMT24 experiments. In composition sensitivity, thirteen-stream WMT23 with random pool 04, median pooling and $q = .05$ has an empty active set in all 40 splits. In mapping sensitivity, WMT24 ja-zh, quarter reference, midrank ECDF, median, $q = .05$ has one empty-active split (index 33). Its reference-priority positive-gap mean is 24.283% over all 40 splits, versus 24.905% conditional on the 39 nonempty splits. The tables use the former convention.

The original WMT23 score panels contain no quality labels, so their original precision fields remain missing. Appendix E separately aligns official MQM after freezing the masks and paths.

C Comparison Methods

Full-information controls use exact observed costs and b_H; the zero-query control uses endpoint upper costs. Table 6 summarizes their different decision

objects and guarantees. The four acquisition orders share the selected-retention solver, information boundary, risk budget and stopping tolerance; their query counts form the same-task algorithm comparison. The controls below compare selection objectives or statistical guarantees and have different scoring requirements.

- Zero-query retention maximizes returned baseline-selected count using upper costs Tp_i, with no candidate-test queries.
- Batch uniform threshold chooses the largest feasible threshold among zero and the observed costs F_i. The total cost of all selected and unselected outputs at or below it must not exceed b_H; cost ties are included together.
- Batch cardinality sorts all items by cost, then by label-blind tie priority, and takes the maximum feasible prefix. It maximizes committed-decision count, including exclusions.
- Batch selected retention maximizes returned baseline-selected count using true costs F_i, applying the sort only to selected outputs. Nonselected outputs with $p_i = 0$ are also committed, at zero churn cost. Its returned count is V^*.
- Fixed-count retention recomputes the complete top-k list for every replacement. It sums membership flips for this different rule and maximizes retention within its base list under the same nominal risk normalization.

Native and matched fixed-count controls. The native-tie fixed-count control selects the k largest consensus–item-index keys in the baseline and every replacement. Its matched-tie counterpart uses the same seeded label-blind priority as the retention solver. Both are evaluated with full information on six panels, two pooling rules, four depths and 40 splits: 1,920 paired split instances at $\alpha = .10$.

The matched control orders every baseline and complete-batch replacement by decreasing consensus, then increasing the same seeded label-blind priority used by the retention solver. It refits neither maps nor cutoff and changes no query path. We recompute membership flips, maximum-retention masks, own-baseline coverage, average and replacement-specific churn. Labels are not loaded. Table 7 summarizes all 48 panel–pooling–depth settings; `outputs/matched-tie-control/` supplies masks and per-split records.

Relative to native ties, 452 baseline lists and 453 retained sets change; replacement memberships change in 1,869 configurations. Returned count changes in only one configuration, by one output, but average churn changes in 1,435. The sign of fixed-count-minus-frozen returned count is unchanged in all 1,920 configurations. Thus count ordering is robust to this tie correction on these panels, while memberships and risk are not generally invariant. This removes a tie-priority difference, but fixed-count and frozen-cutoff rules still define different baseline tasks. We do not apply the row-local median entry formula to globally coupled top-k membership or claim a new online certificate.

All 3,840 control records (1,920 native and 1,920 frozen) reconcile with the original results. An independent sort and cardinality implementation checks matched masks, costs and priority transport under reindexing on 96 configurations across all panels, depths and pooling rules. All 5,760 records across native, matched and frozen controls meet their integer churn budgets. These are cached-score checks, not evaluator execution or independent population replications.

The CRC expectation control uses fully observed candidate scores to compute $d_i = F_i/H$ for every calibration and test item. Its group loss is $L_\lambda(U) = B|U|^{-1}\sum_{i\in U} d_i \mathbf{1}[d_i \le \lambda]$, with $B = N/(2k)$ fixed to the test batch's nominal scale. For calibration groups $U_1, \ldots, U_n$, define

$$\hat{R}_n(\lambda) = \frac{1}{n}\sum_{a=1}^{n} L_\lambda(U_a).$$

The mathematical control chooses the maximum threshold from zero and observed calibration d_i values satisfying

$$\frac{n\hat{R}_n(\lambda) + B\lambda}{n+1} \le \alpha.$$

Zero is always feasible and includes zero-disagreement items. The following finite-grid argument supplies the expectation guarantee for this threshold convention, using the envelope correction of Yeh et al. (2025) and the exchangeability principle of Angelopoulos et al. (2024).

Finite-grid expectation guarantee. Condition on the reference-fitted maps, cutoff, frozen evaluator family and design quantities H, N, k, hence

Control	Decision object	Guarantee or objective	Candidate scores
Zero-query retention	Baseline-selected outputs	Endpoint-safe retained count	None on test
Batch selected retention	Baseline-selected outputs	Optimal count under churn	Full test matrix
Batch cardinality	Selected and excluded decisions	Optimal committed count	Full test matrix
Batch uniform threshold	Outputs at or below a cost threshold	Feasible tied cost level	Full test matrix
Fixed-count retention	Replacement-specific top-k lists	Retention under a different rule	Full test matrix
Envelope CRC	Output-cost threshold	Expected churn	Calibration + test
Grouped SCoRE	Group acceptance or abstention	Expected group loss	Calibration

Table 6: Objects and information used by the controls. The acquisition policies alone share the same selected-retention target and stopping criterion; the other rows compare objectives or guarantees.

$B = N/(2k)$. Because F_i is an integer, all item losses lie on the fixed grid $\Lambda = \{0, 1/H, \ldots, 1\}$. Assume that the loss vectors $(L_\lambda(U_a))_{\lambda\in\Lambda}$ for the n calibration groups and one future group are conditionally exchangeable. Each group loss is non-decreasing in λ, is zero at zero, and is at most $B\lambda$. Let $\hat{\lambda}$ be any searched grid point satisfying the correction in exact arithmetic. Define the auxiliary, permutation-invariant threshold

$$\tilde{\lambda} = \max\left\{\lambda \in \Lambda : \frac{\sum_{a=1}^{n+1} L_\lambda(U_a)}{n+1} \le \alpha\right\}.$$

The calibration correction and $L_{\hat{\lambda}}(U_{n+1}) \le B\hat{\lambda}$ make $\hat{\lambda}$ feasible for this auxiliary problem, so $\hat{\lambda} \le \tilde{\lambda}$. Monotonicity and conditional exchangeability give

$$\begin{aligned}\mathbb{E}[L_{\hat{\lambda}}(U_{n+1})] &\le \mathbb{E}[L_{\tilde{\lambda}}(U_{n+1})] \\ &= \mathbb{E}\left[\frac{1}{n+1}\sum_{a=1}^{n+1} L_{\tilde{\lambda}}(U_a)\right] \le \alpha,\end{aligned}$$

where expectations are conditional on the fixed quantities above. Searching only zero and observed calibration values suffices: any grid point satisfying the correction is feasible for the auxiliary problem. Finiteness avoids a continuity assumption at the inclusive threshold's jumps.

RewardBench groups have four outputs; the two WMT23 panels have thirteen and twelve. With their fixed group counts, N is 2,300, 2,431 and 2,244, respectively; q fixes k before test scores are used. For equal-size test groups sharing $\hat{\lambda}$, the batch churn is their average group loss. Applying the bound to each future group and using linearity of expectation gives expected batch churn at most α. This assumes exchangeability; it does not prove it for the NLP population or bound every realized batch. Unequal-size WMT24 documents do not admit this unweighted reduction. The deterministic query certificate instead enforces the current batch budget under every declared score completion.

Numerical calibration. Historical runs evaluate the correction in floating point. An integer-arithmetic audit of all 4,800 calibrations in the original-pool budget grid confirms that every chosen threshold satisfies the exact inequality. One RewardBench mean instance (split 6, $q = .10$, $\alpha = .15$) selects $29/49$ instead of the exact maximum $30/49$. Both return 245 selected outputs, but their committed exclusions and churn differ. All 960 primary thresholds match the exact maxima. The tables retain the audited historical runs. The code package also provides an integer implementation that enforces the displayed correction exactly; the expectation argument above requires that exact feasibility condition.

Panel	Pooling	q	Frozen Cov.	Matched Cov.	ΔK_{tie} Matched–Native	Changed sets/40	Changed signs/40
RewardBench 2	median	0.05	99.6	99.4	0.025	10	0
RewardBench 2	median	0.10	100.0	100.0	0.000	13	0
RewardBench 2	median	0.20	100.0	100.0	0.000	11	0
RewardBench 2	median	0.30	100.0	100.0	0.000	11	0
RewardBench 2	mean	0.05	100.0	100.0	0.000	1	0
RewardBench 2	mean	0.10	100.0	100.0	0.000	1	0
RewardBench 2	mean	0.20	100.0	100.0	0.000	0	0
RewardBench 2	mean	0.30	100.0	100.0	0.000	0	0
WMT23, 13	median	0.05	99.5	100.0	0.000	32	0
WMT23, 13	median	0.10	100.0	100.0	0.000	15	0
WMT23, 13	median	0.20	100.0	100.0	0.000	14	0
WMT23, 13	median	0.30	100.0	100.0	0.000	10	0
WMT23, 13	mean	0.05	99.7	100.0	0.000	1	0
WMT23, 13	mean	0.10	100.0	100.0	0.000	1	0
WMT23, 13	mean	0.20	100.0	100.0	0.000	2	0
WMT23, 13	mean	0.30	100.0	100.0	0.000	1	0
WMT23, 12	median	0.05	99.9	100.0	0.000	19	0
WMT23, 12	median	0.10	100.0	100.0	0.000	13	0
WMT23, 12	median	0.20	100.0	100.0	0.000	10	0
WMT23, 12	median	0.30	100.0	100.0	0.000	13	0
WMT23, 12	mean	0.05	99.1	100.0	0.000	1	0
WMT23, 12	mean	0.10	100.0	100.0	0.000	3	0
WMT23, 12	mean	0.20	100.0	100.0	0.000	0	0
WMT23, 12	mean	0.30	100.0	100.0	0.000	0	0
WMT24 en-de	median	0.05	100.0	100.0	0.000	22	0
WMT24 en-de	median	0.10	100.0	100.0	0.000	20	0
WMT24 en-de	median	0.20	100.0	100.0	0.000	13	0
WMT24 en-de	median	0.30	100.0	100.0	0.000	13	0
WMT24 en-de	mean	0.05	99.9	100.0	0.000	15	0
WMT24 en-de	mean	0.10	100.0	100.0	0.000	5	0
WMT24 en-de	mean	0.20	100.0	100.0	0.000	2	0
WMT24 en-de	mean	0.30	100.0	100.0	0.000	2	0
WMT24 en-es	median	0.05	100.0	100.0	0.000	21	0
WMT24 en-es	median	0.10	100.0	100.0	0.000	18	0
WMT24 en-es	median	0.20	100.0	100.0	0.000	16	0
WMT24 en-es	median	0.30	100.0	100.0	0.000	16	0
WMT24 en-es	mean	0.05	99.9	100.0	0.000	19	0
WMT24 en-es	mean	0.10	100.0	100.0	0.000	4	0
WMT24 en-es	mean	0.20	100.0	100.0	0.000	6	0
WMT24 en-es	mean	0.30	100.0	100.0	0.000	3	0
WMT24 ja-zh	median	0.05	99.8	100.0	0.000	15	0
WMT24 ja-zh	median	0.10	100.0	100.0	0.000	16	0
WMT24 ja-zh	median	0.20	100.0	100.0	0.000	20	0
WMT24 ja-zh	median	0.30	100.0	100.0	0.000	11	0
WMT24 ja-zh	mean	0.05	97.5	99.8	0.000	2	0
WMT24 ja-zh	mean	0.10	98.5	100.0	0.000	5	0
WMT24 ja-zh	mean	0.20	99.7	100.0	0.000	5	0
WMT24 ja-zh	mean	0.30	100.0	100.0	0.000	2	0

Table 7: Matched label-blind top-k control, $\alpha = .10$, full information. Cov. averages splitwise selected coverage (%) using each rule's own baseline. ΔK is matched-minus-native returned count, averaged over 40 splits. Changed sets/40 counts changed retained sets under that tie change. Changed signs/40 counts changes in the sign of fixed-count-minus-frozen returned count; a zero returned-count difference has sign zero. This compares selection tasks, not online query orders. All baseline and replacement rankings use decreasing consensus and increasing seeded priority.

Panel	Policy	Risk	Sel. cov. (%)	Returned (outputs)	Precision
RewardBench 2	No queries	0.053	80.9	95.5	0.844
RewardBench 2	Envelope CRC	0.079	72.2	85.6	0.868
RewardBench 2	Batch threshold	0.081	72.9	86.4	0.868
RewardBench 2	Batch cardinality	0.099	77.3	91.5	0.865
RewardBench 2	Batch retention	0.082	99.6	118.4	0.823
RewardBench 2	Fixed-count retention	0.096	99.3	114.3	0.828
WMT23, 13 streams	No queries	0.056	84.9	117.9	–
WMT23, 13 streams	Envelope CRC	0.067	77.0	107.9	–
WMT23, 13 streams	Batch threshold	0.085	86.0	119.8	–
WMT23, 13 streams	Batch cardinality	0.096	91.1	126.5	–
WMT23, 13 streams	Batch retention	0.082	99.5	137.3	–
WMT23, 13 streams	Fixed-count retention	0.056	100.0	122.0	–
WMT23, 12 streams	No queries	0.042	99.2	112.4	–
WMT23, 12 streams	Envelope CRC	0.062	95.2	106.9	–
WMT23, 12 streams	Batch threshold	0.071	98.6	111.5	–
WMT23, 12 streams	Batch cardinality	0.072	98.9	112.1	–
WMT23, 12 streams	Batch retention	0.044	99.9	113.7	–
WMT23, 12 streams	Fixed-count retention	0.043	100.0	112.0	–

Table 8: Risk is the mean of replacement-average churn $R(C)$ over 40 splits; Returned is the mean output count K. Median pooling, $q = .05$, $\alpha = .10$: full-information controls and zero-query retention. Selected coverage is $K/|S_0|$ using each policy's base list. Counts and splitwise ratios are separately averaged over 40 splits. These controls use native ties; the matched-tie comparison is reported separately in Appendix C. RewardBench precision is the preferred fraction among returned outputs; these WMT23 controls omit human-quality labels (see Appendix E for the separate MQM diagnostic). CRC controls expectation; batch controls constrain realized churn. Count means use decimal half-up rounding after aggregating integer records.

D All Depths and Pooling Rules

Tables 9 and 10 report acquisition at positive-gap and exact stopping. The complete baseline grids in `supplement/result_grids.csv` use each control's own baseline for selected coverage. Their lists can differ, so these values are not retention gains on an identical output set; preference precision is separate. Across depths, the amount inspected must be interpreted jointly with initial coverage and the stopping target, as discussed in Section 7.

E Post Hoc Quality Diagnostics at Matched Count

The stability policy optimizes selected retention under a churn budget. It does not optimize preference precision. RewardBench labels identify the preferred completion among each prompt's four outputs. For a returned set of size K, we compare its preference precision with that of the K outputs having the highest base-consensus scores. The same seeded label-blind secondary key resolves ties. The comparator uses the same consensus and $K \leq |S_0|$, so its top-K outputs lie within the baseline cutoff list.

This post hoc diagnostic leaves frozen fits, acquisition paths, stopping states and protocol unchanged. RewardBench and WMT23 informed development; this statement does not assert label-blind development history. For each split, the comparator matches returned count but is not constrained by the churn budget. Averaging and rounding follow Appendix B. All RewardBench comparisons have 40 defined splits (Table 11).

E.1 Official WMT23 MQM alignment

All 173,784 metric-score entries match the frozen 24-metric, 557-segment, thirteen-stream panel using strict equality of parsed Python float values, without a tolerance. This is numerical equality, not raw-file byte identity. The 44 official-file acquisition identities are in `review-revision/acceptance-review-implementation/wmt23-official-files.json`. We align MQM by system name and original segment identifier, separately from the original score builder. Twelve MT systems have labels on 460 of 557 segments each, giving 5,520 observed labels among 7,241 outputs. The `synthetic_ref` output has no MQM label. Higher released MQM scores are better; missing annotations remain missing.

We preserve all original splits, mappings, masks and acquisition paths. The analysis covers both WMT23 populations, both pooling rules, four depths and 40 splits, with all four entry orders at exact and positive-gap stops, plus zero-query and full-information masks: 6,400 records. Labels do not choose the acquisition order, stopping point or protocol. For each returned set, we report total count K, annotated count K_{lab} and annotated coverage. We compare the returned annotated subset's mean MQM with the mean for the K_{lab} highest-consensus outputs among all annotated test outputs, using the original label-blind tie key. This matches observed-label count; the comparator need not satisfy the churn budget. It does not estimate quality on unannotated outputs.

Table 12 gives reference-priority positive-gap results at every depth; complete records retain all orders and stops. At median nominal $q = .05$, mean returned-label coverage is 28.3% on thirteen streams and 77.3% on twelve streams. The corresponding returned-minus-comparator mean MQM differences are approximately $-.0020$ and $-.0002$, respectively. Other configurations have differences of either sign. Low coverage, especially when `synthetic_ref` is admitted, prevents extending these MQM comparisons to all returned outputs. The 40 overlapping splits provide descriptive averages, not independent samples for a significance test. A separate implementation rebuilds labels and both baseline masks and recomputes every diagnostic record.

Panel	Pooling	q	Z (%)	Active inspection (%) Fixed	Random	Width	Ref.	Cov. (%)	Gap (outputs)
RewardBench 2	median	0.05	0.0	63.8	64.0	63.0	61.4	97.3	3.0
RewardBench 2	median	0.10	2.5	49.6	49.4	47.6	45.1	97.8	5.2
RewardBench 2	median	0.20	7.5	14.4	14.2	14.5	14.1	97.9	9.5
RewardBench 2	median	0.30	97.5	0.0	0.1	0.0	0.0	99.9	1.1
RewardBench 2	mean	0.05	0.0	90.9	91.5	91.7	86.5	97.5	3.0
RewardBench 2	mean	0.10	0.0	87.5	88.3	87.9	83.5	97.8	5.2
RewardBench 2	mean	0.20	0.0	78.4	79.2	76.3	76.0	97.9	9.7
RewardBench 2	mean	0.30	0.0	64.7	65.5	58.8	67.0	97.9	14.3
WMT23, 13 streams	median	0.05	10.0	57.9	58.4	65.0	45.1	97.7	2.9
WMT23, 13 streams	median	0.10	100.0	0.0	0.0	0.0	0.0	100.0	0.1
WMT23, 13 streams	median	0.20	100.0	0.0	0.0	0.0	0.0	100.0	0.0
WMT23, 13 streams	median	0.30	100.0	0.0	0.0	0.0	0.0	100.0	0.0
WMT23, 13 streams	mean	0.05	0.0	91.3	92.2	92.6	85.7	97.5	3.0
WMT23, 13 streams	mean	0.10	0.0	88.4	89.3	89.7	83.2	97.8	5.5
WMT23, 13 streams	mean	0.20	0.0	56.4	57.3	51.3	57.4	97.9	10.1
WMT23, 13 streams	mean	0.30	0.0	39.2	40.0	34.1	45.1	97.9	15.2
WMT23, 12 streams	median	0.05	92.5	3.9	3.8	3.9	3.8	99.6	0.4
WMT23, 12 streams	median	0.10	87.5	2.2	2.2	2.3	1.8	99.6	1.1
WMT23, 12 streams	median	0.20	97.5	0.6	0.5	0.5	0.5	99.9	0.3
WMT23, 12 streams	median	0.30	97.5	0.3	0.3	0.3	0.4	99.9	0.8
WMT23, 12 streams	mean	0.05	0.0	89.5	90.2	89.9	85.4	97.0	2.8
WMT23, 12 streams	mean	0.10	0.0	86.0	86.8	86.5	81.3	97.8	5.0
WMT23, 12 streams	mean	0.20	0.0	60.3	61.3	56.1	59.9	97.9	9.5
WMT23, 12 streams	mean	0.30	0.0	51.0	52.2	45.9	54.5	97.9	14.1

Table 9: All depths and pooling rules at $\alpha = .10$, $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Order columns are mean queried fractions of mT in percent; empty active sets contribute zero. Z is the zero-query stopping percentage over 40 splits. Cov. is reference-priority $K/|S_0|$ in percent; Gap is its retained-count gap in outputs. These overlapping splits provide descriptive summaries.

Panel	Pooling	q	Z (%)	Active inspection (%) Fixed	Random	Width	Ref.	Cov. (%)	Gap (outputs)
RewardBench 2	median	0.05	0.0	73.8	73.9	72.0	71.1	99.6	0.0
RewardBench 2	median	0.10	0.0	58.3	58.3	55.5	53.4	100.0	0.0
RewardBench 2	median	0.20	2.5	23.5	23.2	23.4	22.0	100.0	0.0
RewardBench 2	median	0.30	85.0	0.8	0.8	0.8	0.8	100.0	0.0
RewardBench 2	mean	0.05	0.0	93.7	93.7	93.8	90.2	100.0	0.0
RewardBench 2	mean	0.10	0.0	90.5	90.4	90.0	86.9	100.0	0.0
RewardBench 2	mean	0.20	0.0	82.0	82.0	79.5	79.5	100.0	0.0
RewardBench 2	mean	0.30	0.0	69.7	69.8	63.0	71.0	100.0	0.0
WMT23, 13 streams	median	0.05	10.0	68.4	67.6	71.2	54.3	99.5	0.0
WMT23, 13 streams	median	0.10	97.5	0.2	0.2	0.2	0.2	100.0	0.0
WMT23, 13 streams	median	0.20	100.0	0.0	0.0	0.0	0.0	100.0	0.0
WMT23, 13 streams	median	0.30	100.0	0.0	0.0	0.0	0.0	100.0	0.0
WMT23, 13 streams	mean	0.05	0.0	94.1	94.0	94.6	88.8	99.7	0.0
WMT23, 13 streams	mean	0.10	0.0	90.9	91.0	91.4	86.3	100.0	0.0
WMT23, 13 streams	mean	0.20	0.0	62.2	62.1	55.7	62.1	100.0	0.0
WMT23, 13 streams	mean	0.30	0.0	47.4	47.2	40.2	51.6	100.0	0.0
WMT23, 12 streams	median	0.05	90.0	5.7	5.5	5.5	5.3	99.9	0.0
WMT23, 12 streams	median	0.10	80.0	5.2	5.1	5.9	4.8	100.0	0.0
WMT23, 12 streams	median	0.20	97.5	0.9	0.8	0.8	0.8	100.0	0.0
WMT23, 12 streams	median	0.30	95.0	0.9	0.9	0.8	0.9	100.0	0.0
WMT23, 12 streams	mean	0.05	0.0	92.2	92.2	92.0	88.7	99.1	0.0
WMT23, 12 streams	mean	0.10	0.0	88.6	88.6	88.4	84.3	100.0	0.0
WMT23, 12 streams	mean	0.20	0.0	66.0	66.0	60.6	64.8	100.0	0.0
WMT23, 12 streams	mean	0.30	0.0	58.1	58.2	51.4	60.0	100.0	0.0

Table 10: All depths and pooling rules at $\alpha = .10$, $V^+ - V^- = 0$. Order columns are mean queried fractions of mT in percent; empty active sets contribute zero. Z is the zero-query stopping percentage over 40 splits. Cov. is reference-priority $K/|S_0|$ in percent; Gap is its retained-count gap in outputs. These overlapping splits provide descriptive summaries.

Pooling	q	Strategy	K	Precision	Top-K	Difference
median	0.05	fixed	115.7	0.823	0.828	-0.0048
median	0.05	random	115.6	0.825	0.828	-0.0033
median	0.05	largest_width	115.6	0.827	0.828	-0.0008
median	0.05	reference_priority	115.6	0.827	0.828	-0.0009
median	0.10	fixed	229.1	0.702	0.706	-0.0045
median	0.10	random	229.1	0.705	0.706	-0.0015
median	0.10	largest_width	229.1	0.705	0.706	-0.0006
median	0.10	reference_priority	229.1	0.701	0.706	-0.0046
median	0.20	fixed	451.4	0.586	0.588	-0.0020
median	0.20	random	451.4	0.587	0.588	-0.0013
median	0.20	largest_width	451.4	0.585	0.588	-0.0030
median	0.20	reference_priority	451.4	0.586	0.588	-0.0027
median	0.30	fixed	686.0	0.511	0.511	-0.0003
median	0.30	random	686.0	0.511	0.511	-0.0003
median	0.30	largest_width	686.0	0.511	0.511	-0.0003
median	0.30	reference_priority	686.0	0.511	0.511	-0.0003
mean	0.05	fixed	117.3	0.825	0.827	-0.0027
mean	0.05	random	117.3	0.827	0.827	0.0001
mean	0.05	largest_width	117.3	0.829	0.827	0.0012
mean	0.05	reference_priority	117.3	0.826	0.827	-0.0009
mean	0.10	fixed	231.8	0.705	0.707	-0.0019
mean	0.10	random	231.8	0.710	0.707	0.0028
mean	0.10	largest_width	231.8	0.710	0.707	0.0032
mean	0.10	reference_priority	231.8	0.709	0.707	0.0014
mean	0.20	fixed	452.8	0.591	0.593	-0.0017
mean	0.20	random	452.8	0.594	0.593	0.0004
mean	0.20	largest_width	452.8	0.595	0.593	0.0023
mean	0.20	reference_priority	452.8	0.588	0.593	-0.0048
mean	0.30	fixed	675.1	0.530	0.530	-0.0004
mean	0.30	random	675.1	0.530	0.530	-0.0003
mean	0.30	largest_width	675.1	0.533	0.530	0.0034
mean	0.30	reference_priority	675.1	0.529	0.530	-0.0015

Table 11: RewardBench preference diagnostic at matched returned count. All results use $\alpha = .10$ and the primary stopping criterion $V^+ - V^- \leq \lceil .02|S_0(P)| \rceil$. Within each of 40 grouped splits, the returned count K determines the same-consensus top-K comparator with label-blind seeded tie priorities. Entries are splitwise means; differences are computed before display rounding. The comparator is constructed without a churn-budget constraint. This descriptive analysis was added after the inspection protocol; labels never enter acquisition or stopping. Count means use decimal half-up rounding after aggregating integer records.

Panel	Pooling	q	K	K_{lab}	Obs. (%)	MQM	Δ
WMT23 (13)	median	0.05	134.8	38.5	28.3	-0.3402	-0.0020
WMT23 (13)	median	0.10	241.3	101.0	41.1	-0.4141	0.0000
WMT23 (13)	median	0.20	489.1	297.8	60.7	-0.7860	0.0000
WMT23 (13)	median	0.30	730.9	490.4	67.0	-1.3198	0.0000
WMT23 (13)	mean	0.05	123.4	43.6	35.4	-0.3046	0.0092
WMT23 (13)	mean	0.10	241.9	112.2	45.8	-0.5335	0.0016
WMT23 (13)	mean	0.20	476.4	281.6	58.8	-0.8979	-0.0163
WMT23 (13)	mean	0.30	714.8	472.0	65.9	-1.3320	0.0031
WMT23 (12)	median	0.05	113.3	87.6	77.3	-0.3996	-0.0002
WMT23 (12)	median	0.10	227.4	185.1	81.5	-0.6429	-0.0013
WMT23 (12)	median	0.20	457.3	374.4	81.9	-1.0051	0.0013
WMT23 (12)	median	0.30	680.8	559.3	82.1	-1.5591	0.0013
WMT23 (12)	mean	0.05	113.7	88.2	77.3	-0.4973	0.0063
WMT23 (12)	mean	0.10	222.2	175.2	79.1	-0.6437	-0.0042
WMT23 (12)	mean	0.20	442.1	358.4	81.0	-1.1279	-0.0090
WMT23 (12)	mean	0.30	666.9	542.7	81.3	-1.5970	-0.0293

Table 12: Post hoc WMT23 MQM diagnostic for the original reference-priority positive-gap stop, $\alpha = .10$, full-reference right ECDF. K counts all returned outputs and K_{lab} those with labels. Obs. averages splitwise labeled fractions. MQM is higher-is-better on labeled outputs; Δ subtracts the consensus-ranked comparator with the same labeled count. Each row averages 40 overlapping splits; no population inference or missing-label imputation is made. All four acquisition methods, both stops and the two controls are included in the reproducibility results.

F Safe Retention Frontiers

Figure 1 shows the three development panels under both pooling rules at $q = .05$. The single-split trajectory below uses a different depth to illustrate how equal query counts can yield different safe sets.

Table 13 follows the first seeded split of thirteen-stream WMT23 with mean pooling at $q = .30$, whose aggregate comparison favors largest width. Both orders start with 646 safe outputs and an optimistic count of 715. After 1,122 inspections, largest width raises the safe count to 694, compared with 674 for reference priority. It reaches the 15-output tolerance at 1,249 queries rather than 1,715. Both certify the same full-information count of 715 at exact stopping. The batch, risk budget and certificate criterion stay fixed; the trace illustrates the ordering difference without identifying its cause.

G Budget and Composition Evidence

The budget grid contains $3 \times 2 \times 4 \times 40 \times 5 = 4{,}800$ risk-budget configurations. The 960 budget-independent panel–pooling–depth–split settings each use four strategies, yielding 3,840 orders. The four original orders (fixed, random, width and reference) depend on base geometry, reference fits, revealed entries and label-blind priorities, but not the risk budget. For each budget, we locate its first positive-gap and exact stopping prefixes on the same budget-independent acquisition path. The saved $\alpha = .10$ records, masks, checkpoints and trace prefixes permit independent replay. Five separate stopping problems do not count as five independent data samples.

For composition, the random-pool RNG seeds are 620271 (RewardBench) and 620272 (WMT). We sample 20 distinct seven-member pools other than the original pool. Its individual evaluators remain eligible for the sampled pools. Full and twelve-stream WMT use the same evaluator subsets. The stored evaluator indices in `review-revision/composition-protocol.json` determine the replay; the historical pool-generation API is not established. The metadata-defined WMT pool uses seed 620273: three members from BLEU, chrF, eBLEU, f200spBLEU and tokengram_F, plus four from the remaining names. This category is based on evaluator identity; the remaining group is not restricted to learned metrics. All 62 added panel–pool units run both aggregators, four depths and 40 splits, yielding 19,840 configurations and 79,360 acquisition paths. Candidate sets are the full remaining evaluator set, and the reference cutoff and ridge are refitted for each pool. For a fixed evaluator and reference-output population, ECDF maps can be reused across pools and budgets. Changing the base pool still requires refitting the consensus cutoff and ridge targets.

The companion grids report reference priority at all budgets and depths; the composition tables give descriptive pool-level distributions. Complete result paths and field meanings for acquisition orders, masks, stopping targets, quality diagnostics, witness decompositions and replacement-specific losses are listed in `RESULT_DATA_DICTIONARY.md`. The prepared local code bundle includes the listed summaries and numerical query traces. Original third-party text and full score arrays remain outside it; redistribution clearance is still incomplete. Pool-level quantiles summarize sensitivity across sampled pools, not population confidence intervals.

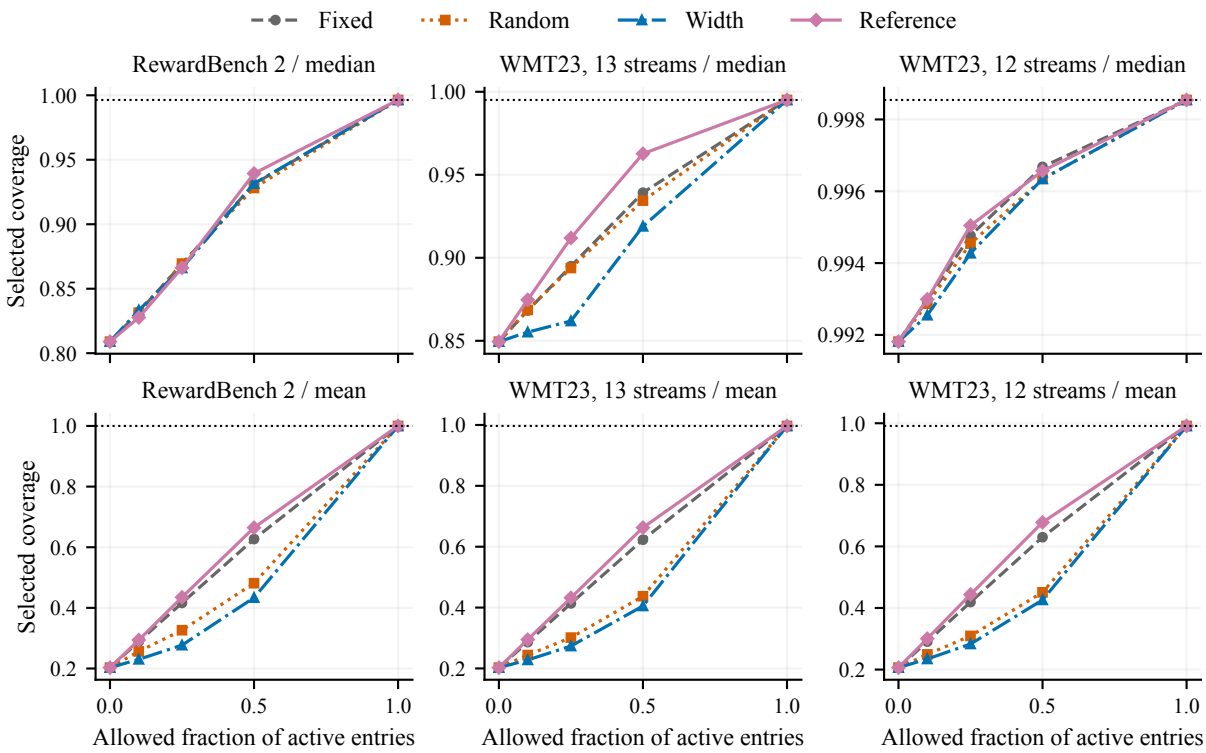


Figure 1: At $q = .05$, $\alpha = .10$, colored curves show safe selected coverage $V^-/|S_0|$ and dotted lines show $V^*/|S_0|$. No optimistic frontier is plotted. The horizontal axis is the allowed fraction of mT active entries; actual queries can be fewer after early exact stopping. Every plotted mask satisfies the risk budget. Curves and dotted lines are arithmetic means of splitwise coverage over 40 splits. Vertical ranges differ across panels.

G.1 Invariant and budget-admissible zero-query retention

Let $I = \{i \in S_0 : p_i = 0\}$ and let C_0 be the zero-query mask computed with Tp_i. Every invariant selected output is retained for free. For each split,

$$\frac{|S_0 \cap C_0|}{|S_0|} = \frac{|I|}{|S_0|} + \frac{|(S_0 \cap C_0) \setminus I|}{|S_0|}.$$

Both terms divide by $|S_0|$. The second counts retained noninvariant outputs whose combined upper cost fits the budget. Zero-query stopping additionally requires the initial retained-count gap to meet the tolerance. Tables 14 and 15 report all six panels at four depths under median and mean pooling, respectively, using each panel's original evaluator pool and averaging splitwise fractions. Removing the `synthetic_ref` output stream changes the output population as well as the refitted maps and cutoff. The twelve- and thirteen-stream comparison does not isolate deletion from these protocol changes.

G.2 Complete composition and budget results

Tables 16, 17 and 18 summarize evaluator-pool sensitivity across depths. Full five-budget grids for all three original panels and both pooling rules are retained as `review-budget-*` in `supplement/result_grids.csv`; the companion JSON preserves captions and units.

Stage	Reference priority Queries	V^-	V^+	Largest width Queries	V^-	V^+
Initial	0	646	715	0	646	715
10% allowance	449	652	715	449	662	715
25% allowance	1122	674	715	1122	694	715
First positive-gap stop	1715	700	715	1249	700	715
Exact stop	2054	715	715	1558	715	715

Table 13: Query-prefix illustration from the first seeded split of thirteen-stream WMT23, mean pooling, $q = .30$ and $\alpha = .10$. Both orders begin with 646 safe outputs and 4,488 active entries. Allowance rows compare the two strategies after the same number of queries; stopping rows report each strategy's own first stopping state. The allowed gap is 15 outputs. Counts come from the existing traces, not a new evaluation sample.

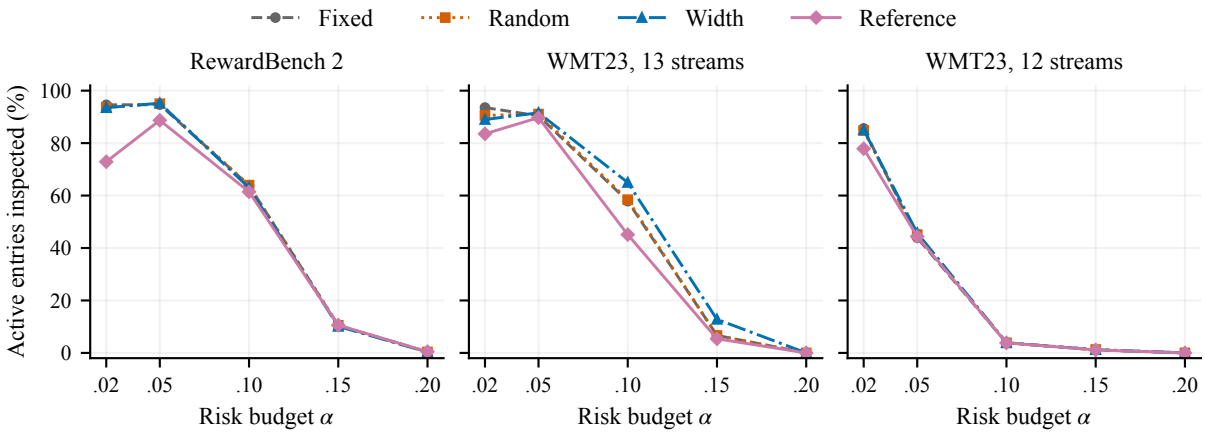


Figure 2: Median pooling at $q = .05$, all five budgets. Points average queried fractions of mT active selected entries over 40 splits, stopping at $V^+ - V^- \leq \lceil .02|S_0| \rceil$ outputs. Empty active sets contribute zero inspection (Appendix B). Orders are shared across budgets; first stopping states are recorded separately.

Panel	q	Invariant	Noninv.	Zero	Z
RewardBench 2	0.05	46.6	34.3	80.9	0.0
RewardBench 2	0.1	50.5	34.4	84.9	2.5
RewardBench 2	0.2	59.6	35.1	94.6	7.5
RewardBench 2	0.3	66.5	33.3	99.8	97.5
WMT23, 13 streams	0.05	54.4	30.6	84.9	10.0
WMT23, 13 streams	0.1	78.8	21.2	100.0	100.0
WMT23, 13 streams	0.2	77.3	22.7	100.0	100.0
WMT23, 13 streams	0.3	76.4	23.6	100.0	100.0
WMT23, 12 streams	0.05	75.1	24.1	99.2	92.5
WMT23, 12 streams	0.1	71.7	27.7	99.4	87.5
WMT23, 12 streams	0.2	73.5	26.3	99.9	97.5
WMT23, 12 streams	0.3	72.0	27.9	99.9	97.5
WMT24 en-de	0.05	85.3	14.2	99.5	95.0
WMT24 en-de	0.1	77.9	21.8	99.7	87.5
WMT24 en-de	0.2	76.2	23.8	100.0	100.0
WMT24 en-de	0.3	76.2	23.8	100.0	100.0
WMT24 en-es	0.05	83.7	16.1	99.8	97.5
WMT24 en-es	0.1	78.2	21.8	99.9	100.0
WMT24 en-es	0.2	76.6	23.4	100.0	100.0
WMT24 en-es	0.3	78.1	21.9	100.0	100.0
WMT24 ja-zh	0.05	77.8	20.3	98.1	77.5
WMT24 ja-zh	0.1	77.1	22.2	99.3	87.5
WMT24 ja-zh	0.2	75.5	24.4	99.9	97.5
WMT24 ja-zh	0.3	77.9	22.1	100.0	100.0

Table 14: Median pooling, $\alpha = .10$, full-reference right ECDF, original evaluator pools. Invariant and Noninv. are selected-output shares retained through $p_i = 0$ and budget-admissible $p_i > 0$, respectively. Before rounding, their sum is zero-query selected coverage (Zero). Shares divide by $|S_0|$ within each split, then average over 40 splits. Z is the percentage of splits stopping at zero queries when $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Invariant, Noninv., Zero and Z are percentages, rounded separately; q is the nominal selection fraction.

Panel	q	Invariant	Noninv.	Zero	Z
RewardBench 2	0.05	0.6	19.7	20.3	0.0
RewardBench 2	0.1	7.2	24.5	31.7	0.0
RewardBench 2	0.2	22.2	29.5	51.7	0.0
RewardBench 2	0.3	37.3	33.4	70.8	0.0
WMT23, 13 streams	0.05	0.0	20.3	20.3	0.0
WMT23, 13 streams	0.1	0.0	19.9	19.9	0.0
WMT23, 13 streams	0.2	49.2	28.5	77.6	0.0
WMT23, 13 streams	0.3	59.0	28.7	87.7	0.0
WMT23, 12 streams	0.05	0.0	20.6	20.6	0.0
WMT23, 12 streams	0.1	3.2	21.1	24.3	0.0
WMT23, 12 streams	0.2	47.9	28.0	75.8	0.0
WMT23, 12 streams	0.3	53.3	29.6	82.9	0.0
WMT24 en-de	0.05	0.0	22.0	22.0	0.0
WMT24 en-de	0.1	29.9	24.8	54.7	0.0
WMT24 en-de	0.2	41.6	25.7	67.3	0.0
WMT24 en-de	0.3	48.4	28.0	76.4	0.0
WMT24 en-es	0.05	0.0	21.2	21.2	0.0
WMT24 en-es	0.1	27.8	23.8	51.6	0.0
WMT24 en-es	0.2	40.7	25.6	66.3	0.0
WMT24 en-es	0.3	51.4	27.7	79.2	0.0
WMT24 ja-zh	0.05	0.0	21.4	21.4	0.0
WMT24 ja-zh	0.1	0.0	21.8	21.8	0.0
WMT24 ja-zh	0.2	11.7	26.4	38.1	0.0
WMT24 ja-zh	0.3	24.1	29.2	53.3	0.0

Table 15: Mean pooling, $\alpha = .10$, full-reference right ECDF, original evaluator pools. Invariant and Noninv. are selected-output shares retained through $p_i = 0$ and budget-admissible $p_i > 0$, respectively. Before rounding, their sum is zero-query selected coverage (Zero). Shares divide by $|S_0|$ within each split, then average over 40 splits. Z is the percentage of splits stopping at zero queries when $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Invariant, Noninv., Zero and Z are percentages, rounded separately; q is the nominal selection fraction.

Panel	Pooling	q	Zero (%)	Ref. (%)	10–90%	Δ Width
RewardBench 2	median	0.05	81.6	55.8	43.7–68.9	-6.3
RewardBench 2	median	0.10	87.2	39.0	20.8–49.3	-6.7
RewardBench 2	median	0.20	96.2	9.0	0.9–22.0	-1.0
RewardBench 2	median	0.30	100.0	0.0	0.0–0.7	0.0
WMT23, 13 streams	median	0.05	98.6	2.5	0.0–67.8	0.0
WMT23, 13 streams	median	0.10	99.4	1.4	0.0–18.5	0.0
WMT23, 13 streams	median	0.20	100.0	0.0	0.0–3.1	0.0
WMT23, 13 streams	median	0.30	100.0	0.0	0.0–1.1	0.0
WMT23, 12 streams	median	0.05	98.1	6.4	2.8–29.3	-0.4
WMT23, 12 streams	median	0.10	98.6	3.6	0.6–13.2	-0.1
WMT23, 12 streams	median	0.20	99.6	0.9	0.0–6.6	-0.1
WMT23, 12 streams	median	0.30	99.9	0.1	0.0–4.6	0.0

Table 16: Primary stopping uses $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Median pooling: Random evaluator composition at $\alpha = .10$, 20 pools with 40 splits each. Zero is the median of pool-wise mean zero-query selected coverage. Ref. is the median pool-wise mean queried fraction of mT in percent, with descriptive 10th–90th percentiles; empty active sets contribute zero. Within each pool we average splitwise Reference minus Width active-inspection differences. Δ is the median of these 20 differences in percentage points; positive values mean more Reference queries.

Panel	Pooling	q	Zero (%)	Ref. (%)	10–90%	Δ Width
RewardBench 2	mean	0.05	20.9	86.9	84.1–94.4	-4.8
RewardBench 2	mean	0.10	29.9	83.8	81.3–88.5	-4.4
RewardBench 2	mean	0.20	52.2	75.4	74.6–77.3	-0.3
RewardBench 2	mean	0.30	70.9	66.2	62.6–67.3	5.6
WMT23, 13 streams	mean	0.05	20.8	81.6	79.0–84.4	-4.3
WMT23, 13 streams	mean	0.10	31.8	78.7	74.6–81.4	-4.3
WMT23, 13 streams	mean	0.20	72.9	60.8	55.0–68.3	2.2
WMT23, 13 streams	mean	0.30	82.8	51.9	48.4–59.8	6.6
WMT23, 12 streams	mean	0.05	21.1	81.2	78.5–87.0	-3.9
WMT23, 12 streams	mean	0.10	40.9	76.3	71.3–79.7	-3.5
WMT23, 12 streams	mean	0.20	69.4	63.9	61.1–69.3	0.1
WMT23, 12 streams	mean	0.30	79.7	57.4	54.0–63.5	5.8

Table 17: Primary stopping uses $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Mean pooling: Random evaluator composition at $\alpha = .10$, 20 pools with 40 splits each. Zero is the median of pool-wise mean zero-query selected coverage. Ref. is the median pool-wise mean queried fraction of mT in percent, with descriptive 10th–90th percentiles; empty active sets contribute zero. Within each pool we average splitwise Reference minus Width active-inspection differences. Δ is the median of these 20 differences in percentage points; positive values mean more Reference queries.

Panel	Pooling	q	Zero	Fixed	Random	Width	Ref.
WMT23, 13 streams	median	0.05	79.1	89.6	90.0	89.9	85.5
WMT23, 13 streams	median	0.10	98.8	3.7	3.7	3.0	3.2
WMT23, 13 streams	median	0.20	100.0	0.0	0.0	0.0	0.0
WMT23, 13 streams	median	0.30	99.9	0.2	0.2	0.1	0.1
WMT23, 13 streams	mean	0.05	20.4	85.4	86.5	85.6	81.5
WMT23, 13 streams	mean	0.10	28.1	83.3	84.3	84.1	78.5
WMT23, 13 streams	mean	0.20	81.3	49.4	50.5	47.3	51.1
WMT23, 13 streams	mean	0.30	85.6	44.0	44.8	41.2	48.0
WMT23, 12 streams	median	0.05	97.8	10.4	10.8	10.6	10.1
WMT23, 12 streams	median	0.10	99.5	1.7	1.8	1.6	1.5
WMT23, 12 streams	median	0.20	99.2	2.0	2.1	1.7	1.8
WMT23, 12 streams	median	0.30	99.3	1.5	1.5	1.2	1.3
WMT23, 12 streams	mean	0.05	21.2	85.7	86.7	86.3	81.7
WMT23, 12 streams	mean	0.10	52.0	74.7	75.8	73.6	70.9
WMT23, 12 streams	mean	0.20	74.1	61.2	62.3	59.5	59.7
WMT23, 12 streams	mean	0.30	80.0	56.7	57.9	54.4	57.3

Table 18: Primary stopping uses $V^+ - V^- \leq \lceil .02|S_0| \rceil$. WMT pool with three surface-overlap metrics and four remaining metrics (not necessarily learned). All depths, $\alpha = .10$, 40 split means. Zero is zero-query $K/|S_0|$ in percent; order columns are queried mT percentages, with zero for empty active sets.

H WMT24 Admission and Quality Diagnostics

We obtained the official MTME archive from https://data.statmt.org/wmt26/mt-metrics-eval-v2.tgz, the replacement host identified by the maintainer in https://github.com/google-research/mt-metrics-eval/issues/27. The prepared source/code packages contain review-revision/acquisition/official-hosting.json and review-revision/external-admission.json, identifying the archive and used files. The analysis uses the WMT24 files inside that archive; the wmt26 path is the hosting location.

In those files, paragraph granularity is labeled seg; these units are paragraphs, not individual sentences. Standard references are refB for en-de and refA for en-es/ja-zh. Using system-name metadata, we exclude human references as outputs and retain other MT systems common to the admitted metrics, including benchmark outliers. Every retained paragraph has finite scores for every retained MT system and admitted default-reference metric. Score files are parsed into floating-point arrays, with systems sorted by name and paragraphs in source order. On the common finite paragraphs, deduplication compares SHA-256 signatures of the array byte representations, keeping the alphabetically first metric for each signature. It does not compare raw text files. One duplicate is removed in each English-source panel; no paragraph or MT system is removed on the basis of MQM, churn or query performance.

The en-de, en-es and ja-zh panels contain 171, 171 and 197 source documents, respectively. For split $r \in \{0, \ldots, 39\}$, the document-shuffle seed is $924000 + r$. Let D be the number of documents. Unique document identifiers are collected in first-appearance order and shuffled with Python `random.Random(seed).shuffle`, and partitioned using $d_{\text{third}} = \lfloor D/3 \rfloor$. The first two partitions each contain d_{third} documents and the remainder forms test: 171 documents give 57/57/57 and 197 give 65/65/67. Documents remain intact; output counts depend on document lengths. The second partition is reserved for calibration and is not used by these WMT24 query experiments. Unequal document sizes make test count N and nominal k vary across splits. ECDF maps and cutoff use all outputs in reference documents. Tie and random-query seeds add 700001 and 99173, respectively, to the current document-shuffle seed. Each panel uses $M = 7$; en-de and en-es have $T = 14$, and ja-zh has $T = 15$. Ridge features and the unnormalized squared-error objective with penalty one follow Appendix B. We use both $\varepsilon = \lceil .02|S_0| \rceil$ and exact stopping. The exact ordered base and candidate names and removed duplicates appear in supplement/wmt24_evaluator_manifest.csv, derived from the admission record. We evaluate all three admitted directions. The pre-results freeze covered the risk objective and panel-inclusion criteria; the later numerical repair is described in the numerical-query-priorities paragraph below. Evaluator-training independence is not claimed.

Numerical query priorities. For WMT24 entry acquisition, each predicted flip count is converted separately to an integer fixed-point value $e_{ir} = \text{rint}(10^{12} \hat{f}_{ir})$, with nearest-even rounding. Active rows are ranked in ascending order

Language	Pooling	Selected coverage (%) Zero	Opt.	Ref.	Active (%)
en-de	median	99.5	100.0	99.9	1.3
en-de	mean	22.0	99.9	97.9	83.3
en-es	median	99.8	100.0	99.9	1.1
en-es	mean	21.2	99.9	97.8	82.1
ja-zh	median	98.1	99.8	99.3	11.3
ja-zh	mean	21.4	97.5	96.0	85.9

Table 19: Ref. stops at $V^+ - V^- \leq \lceil .02|S_0| \rceil$; Opt. uses full information. WMT24, $q = .05$, $\alpha = .10$: 40 document-grouped split means. Zero, Opt. and Ref.: zero-query, full-information and reference-priority coverage. Active: reference inspection ($100Q/(mT)$), zero for empty active sets. Admission uses metadata and finite scores.

of $10^{12}\ell_i + \sum_{r \text{ unqueried}} e_{ir}$. Within the chosen row, candidates are ordered by decreasing predicted upper-cost reduction from the rounded per-cell values; candidate index breaks ties. Row ties use the frozen, seeded label-blind item priority defined in Appendix B. Each value is rounded before summation.

These fixed-point values affect query order only; observed flip costs, budgets and certificates retain integer arithmetic. RewardBench and WMT23 entry acquisition retain their original priority arithmetic; only WMT24 entry acquisition uses this fixed-point rule. Column acquisition follows the separate rule in Appendix K. The fixed-point rule for WMT24 entry acquisition was introduced after an exact-replay failure and was outside the pre-results freeze. The first fresh replay differed in 22 result fields for en-de, mean pooling, split 37, $q = .10$, reference priority: 3,063 versus 3,059 queries, both with count bounds 477/487, but different masks and quality diagnostics. The recorded first query-order difference was at query 71. Old prediction arrays were not retained, so a specific floating-point backend is not an established cause. After introducing fixed-point priorities, all 960 WMT24 configurations were rerun and checked against an independent replay; the unaffected original-panel computations were reused.

We align native MQM by exact system name and original zero-based paragraph index. Annotations are available for 8,279 of 25,948 en-de outputs, 8,722 of 22,954 en-es outputs and 7,840 of 15,884 ja-zh outputs. Values missing in the official release stay missing. Selection uses the complete metric panel; it is not filtered to annotated systems or paragraphs. For a returned list of size K, the original matched-K comparator uses the same consensus and label-blind tie priorities. Its observed-label mean and that of the certified list may involve different numbers of annotated outputs. For a descriptive diagnostic, we extract the K_{lab} annotated returned outputs without changing the certified mask. We compare their mean with the top K_{lab} consensus outputs among all test outputs having MQM. This comparator matches annotated count and concerns only the observed-label subpopulation. MQM means use only annotated outputs within each list. A difference is defined only when both compared means are defined; splitwise differences are then averaged without imputing missing values. All reference-priority quality comparisons reported here have 40 defined splits. Neither comparator is constructed under the churn-budget constraint. These diagnostics do not estimate quality for unannotated outputs.

Tables 20 and 21 report the median and mean diagnostics separately.

Official source-only and reference-only sentinel score streams (Perrella et al., 2024b) are evaluated descriptively on the same observed MQM entries. Pooled Pearson correlation uses all matching entries; within-paragraph pair accuracy uses annotated system pairs with unequal MQM scores and gives one half to metric ties. The probes contain no candidate translation information and tie all systems within a paragraph. These streams are probes; their diagnostic performance does not trigger evaluator removal, cutoff refitting or candidate-family changes. This differs from removing WMT23's `synthetic_ref` output stream, which changes the output axis rather than the evaluator axis. Table 22 reports the two sentinel diagnostics without changing the certified selection protocol.

Language	Pooling	q	Obs. (%)	MQM	$\Delta\mathrm{MQM}_{K}$	$\Delta\mathrm{MQM}_{K_{\mathrm{lab}}}$
en-de	median	0.05	37.2	-0.0695	-0.0010	-0.0010
en-de	median	0.10	36.7	-0.2688	0.0014	0.0009
en-de	median	0.20	38.2	-0.7180	0.0000	0.0000
en-de	median	0.30	39.0	-1.1547	0.0000	0.0000
en-es	median	0.05	30.9	-0.0668	0.0000	0.0000
en-es	median	0.10	34.0	-0.1181	-0.0001	-0.0001
en-es	median	0.20	38.3	-0.1791	-0.0001	-0.0001
en-es	median	0.30	40.1	-0.2218	0.0000	0.0000
ja-zh	median	0.05	54.8	-0.1467	0.0018	0.0021
ja-zh	median	0.10	56.2	-0.3327	-0.0003	0.0001
ja-zh	median	0.20	59.8	-0.6149	-0.0007	-0.0008
ja-zh	median	0.30	61.7	-0.9359	0.0000	0.0000

Table 20: Median pooling: native MQM after reference-priority stopping. All results use $\alpha = .10$ and the primary stopping criterion $V^+ - V^- \leq \lceil .02|S_0(P)| \rceil$. Entries average 40 split values; differences precede rounding. Obs. is the percentage of returned outputs with MQM annotations. MQM is higher-is-better. $\Delta\mathrm{MQM}_{K}$ is the certified set's observed-label mean MQM minus the matched-K comparator's mean. $\Delta\mathrm{MQM}_{K_{\mathrm{lab}}}$ uses the top-K_{lab} annotated-output comparator, matching the number of annotated returned outputs. The matched-K comparator matches total count; each MQM mean uses the annotated outputs in its own list. Positive differences favor the certified set. Missing labels are not imputed; both diagnostics concern annotated outputs.

Language	Pooling	q	Obs. (%)	MQM	$\Delta\mathrm{MQM}_{K}$	$\Delta\mathrm{MQM}_{K_{\mathrm{lab}}}$
en-de	mean	0.05	36.1	-0.0636	0.0003	0.0040
en-de	mean	0.10	36.7	-0.2389	-0.0068	-0.0065
en-de	mean	0.20	37.8	-0.6264	-0.0030	-0.0034
en-de	mean	0.30	39.2	-1.1102	-0.0113	-0.0102
en-es	mean	0.05	29.7	-0.0577	-0.0016	-0.0002
en-es	mean	0.10	34.8	-0.1010	0.0006	-0.0005
en-es	mean	0.20	38.8	-0.1609	0.0011	0.0030
en-es	mean	0.30	40.5	-0.2081	-0.0017	-0.0018
ja-zh	mean	0.05	55.8	-0.3203	-0.0015	-0.0009
ja-zh	mean	0.10	60.7	-0.4475	0.0057	0.0053
ja-zh	mean	0.20	64.0	-0.6797	0.0049	0.0063
ja-zh	mean	0.30	62.4	-0.9202	0.0012	0.0039

Table 21: Mean pooling: native MQM after reference-priority stopping. All results use $\alpha = .10$ and the primary stopping criterion $V^+ - V^- \leq \lceil .02|S_0(P)| \rceil$. Entries average 40 split values; differences precede rounding. Obs. is the percentage of returned outputs with MQM annotations. MQM is higher-is-better. $\Delta\mathrm{MQM}_{K}$ is the certified set's observed-label mean MQM minus the matched-K comparator's mean. $\Delta\mathrm{MQM}_{K_{\mathrm{lab}}}$ uses the top-K_{lab} annotated-output comparator, matching the number of annotated returned outputs. The matched-K comparator matches total count; each MQM mean uses the annotated outputs in its own list. Positive differences favor the certified set. Missing labels are not imputed; both diagnostics concern annotated outputs.

Language	Probe	Pearson	Pairs	Metric ties (%)	Accuracy
en-de	Source	0.257	39819	100.0	0.500
en-de	Reference	0.245	39819	100.0	0.500
en-es	Source	0.207	21736	100.0	0.500
en-es	Reference	0.195	21736	100.0	0.500
ja-zh	Source	0.244	29850	100.0	0.500
ja-zh	Reference	0.252	29850	100.0	0.500

Table 22: Official missing-information sentinel probes on native annotated WMT24 entries. Pooled Pearson can be positive despite complete within-paragraph ties across MT systems. Pair accuracy assigns one half to metric ties and excludes human-score ties. This descriptive diagnostic changes no candidate family and does not identify invalid real metrics.

I Variation across Individual Replacements

Only replacement-average churn is constrained. For example, thirteen-stream WMT23 at mean $q = .05$ has mean churn .072, mean within-split maximum .327, and a mean 27.9% of replacements above $\alpha = .10$ (Table 24). For each returned mask, we descriptively report the largest replacement-specific churn and the fraction exceeding α, averaging each statistic across splits. The mean of split-specific maxima is not the maximum over all splits; neither diagnostic is a certificate constraint. Table 23 gives the corresponding odd-median results.

Risk across individual replacements. For each of the 15,360 saved masks from six panels, two pooling rules, four depths, 40 splits, four acquisition orders and two stopping targets, we reconstruct all MT member-replacement outcomes. Each individual risk divides committed membership flips by the same $2k$; their average matches $R(C)$. All masks satisfy the average budget .10, while 6,985 have at least one replacement above .10. Among 960 reference-priority primary masks per pooling rule, this occurs in 308 median and 541 mean cases. Their averages of per-mask maximum risk are .0945 and .1404, respectively; their mean replacement-average risks are .0427 and .0496. The largest individual risk across all masks is .50. These observations illustrate the distinction between average and per-replacement constraints, rather than violations of the stated certificate. The accompanying per-panel, depth, policy and stopping summaries include the 95th percentile and fraction of replacements above budget. Splits overlap, so these are descriptive frequencies. Scripts and all summaries are in `review-revision/granularity-risk-implementation/`.

Panel	q	Avg.	Mean max	Over (%)
RewardBench 2	0.05	0.077	0.164	17.3
RewardBench 2	0.10	0.066	0.128	9.3
RewardBench 2	0.20	0.047	0.088	0.3
RewardBench 2	0.30	0.041	0.068	0.0
WMT23, 13 streams	0.05	0.078	0.234	38.0
WMT23, 13 streams	0.10	0.035	0.072	0.3
WMT23, 13 streams	0.20	0.035	0.073	0.0
WMT23, 13 streams	0.30	0.033	0.069	0.1
WMT23, 12 streams	0.05	0.043	0.114	6.1
WMT23, 12 streams	0.10	0.048	0.112	5.5
WMT23, 12 streams	0.20	0.042	0.085	1.6
WMT23, 12 streams	0.30	0.040	0.082	1.1
WMT24 en-de	0.05	0.022	0.063	2.4
WMT24 en-de	0.10	0.036	0.081	1.1
WMT24 en-de	0.20	0.034	0.073	0.0
WMT24 en-de	0.30	0.032	0.065	0.0
WMT24 en-es	0.05	0.029	0.070	2.4
WMT24 en-es	0.10	0.036	0.081	0.9
WMT24 en-es	0.20	0.036	0.075	0.6
WMT24 en-es	0.30	0.031	0.064	0.0
WMT24 ja-zh	0.05	0.046	0.106	9.3
WMT24 ja-zh	0.10	0.048	0.110	8.3
WMT24 ja-zh	0.20	0.047	0.103	4.9
WMT24 ja-zh	0.30	0.041	0.088	1.5

Table 23: Primary stopping uses $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Median pooling: reference-priority returns at $\alpha = .10$, all depths. Avg. averages the splitwise replacement-average churn $R(C)$ over 40 splits. Mean max averages the maximum replacement-specific churn within each split; it is not the maximum across all splits. Over is the mean percentage of replacements exceeding .10. Each split separately satisfies $R(C) \leq \alpha$; individual replacements need not.

Panel	q	Avg.	Mean max	Over (%)
RewardBench 2	0.05	0.068	0.193	14.6
RewardBench 2	0.10	0.059	0.144	12.3
RewardBench 2	0.20	0.045	0.104	4.1
RewardBench 2	0.30	0.036	0.073	0.0
WMT23, 13 streams	0.05	0.072	0.327	27.9
WMT23, 13 streams	0.10	0.056	0.211	14.2
WMT23, 13 streams	0.20	0.031	0.093	0.3
WMT23, 13 streams	0.30	0.031	0.072	0.0
WMT23, 12 streams	0.05	0.069	0.290	20.3
WMT23, 12 streams	0.10	0.049	0.185	8.8
WMT23, 12 streams	0.20	0.037	0.094	0.6
WMT23, 12 streams	0.30	0.036	0.080	0.0
WMT24 en-de	0.05	0.063	0.175	23.7
WMT24 en-de	0.10	0.032	0.075	0.3
WMT24 en-de	0.20	0.035	0.087	1.1
WMT24 en-de	0.30	0.030	0.072	0.2
WMT24 en-es	0.05	0.056	0.177	17.0
WMT24 en-es	0.10	0.037	0.087	1.2
WMT24 en-es	0.20	0.035	0.081	0.8
WMT24 en-es	0.30	0.029	0.068	0.2
WMT24 ja-zh	0.05	0.078	0.203	24.5
WMT24 ja-zh	0.10	0.077	0.177	23.6
WMT24 ja-zh	0.20	0.071	0.161	19.8
WMT24 ja-zh	0.30	0.061	0.141	14.0

Table 24: Primary stopping uses $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Mean pooling: reference-priority returns at $\alpha = .10$, all depths. Avg. averages the splitwise replacement-average churn $R(C)$ over 40 splits. Mean max averages the maximum replacement-specific churn within each split; it is not the maximum across all splits. Over is the mean percentage of replacements exceeding .10. Each split separately satisfies $R(C) \leq \alpha$; individual replacements need not.

J A Same-Loss Statistical Comparison

SCoRE provides marginal deployment risk (MDR) and selective deployment risk (SDR) control for general bounded losses (Bai and Jin, 2026). We use its marginal criterion because churn has a fixed batch denominator. A selected-output denominator would control a different quantity. For a prompt/source group U of g outputs, let

$$L(U) = \frac{\sum_{i \in U \cap S_0} F_i}{gH} \in [0, 1].$$

The group decision $\psi(U)$ accepts or rejects its baseline-selected outputs jointly. Structurally invariant decisions may additionally be committed at zero churn cost. For n_{test} equal-size groups, $N = gn_{\text{test}}$ and the original risk satisfies

$$R(C) = \frac{N}{2k} \frac{1}{n_{\text{test}}} \sum_U L(U)\psi(U).$$

For equal-size groups, the identity converts churn into $B = N/(2k)$ times average group loss. SCoRE-MDR at target α/B therefore bounds expected churn by α, assuming conditional whole-group exchangeability after independent reference fitting. This algebraic reduction does not establish that a fixed benchmark satisfies the assumption. The method accepts a group's selected outputs together, unlike output-level retention.

For a group containing selected outputs, its reference-ridge priority is

$$\hat{v}(U) = \frac{\sum_{i \in U \cap S_0} \sum_r \hat{f}_{ir}}{H|U \cap S_0|}.$$

When $|U \cap S_0| = 0$, we set $\hat{v}(U) = 1$. Only selected outputs enter this prediction; the loss still divides by the full group size g. The same candidate-specific ridge model and feature definitions predict calibration and test flip costs. Calibration losses use the fully scored candidate calibration matrix. For calibration groups, set $v_a = \hat{v}(U_a)$; for the test group, set $v = \hat{v}(U)$. With n calibration groups, SCoRE's Proposition 4.4 at nominal level α/B and $\gamma = \alpha/B$ accepts a test group with priority v when

$$\frac{1 + \sum_{a=1}^{n} L(U_a)\mathbf{1}[v_a \le v]}{n+1} \le \frac{\alpha}{B}.$$

We implement the inequality by exact integer cross-multiplication of group flip sums and the rational budget. Candidate test scores are unavailable until decisions are fixed. Each of the 960 original-pool configurations runs all five budgets. A separate implementation recomputes calibration costs, all 1,518,400 group decisions and test churn; 300 leave-one-out arithmetic checks additionally verify the symmetric construction on finite instances. These finite checks test implementation consistency; they neither prove the published risk theorem nor establish exchangeability of the benchmark groups.

Of 4,800 panel-and-budget records, 70 have realized batch churn above their specified budget. Outputs recur across overlapping splits and budgets, so 70/4,800 is descriptive rather than an estimate from independent batches. An expectation guarantee does not require every batch to meet its budget. These exceedances therefore do not contradict the marginal expectation guarantee under the stated exchangeability assumptions. High coverage without candidate test queries likewise does not establish the deterministic completion certificate. Grouped SCoRE requires fully scored base and candidate calibration data; query certification uses reference and base-test information plus the candidate test scores it inspects. These costs are recorded separately.

WMT24 documents contain unequal numbers of outputs. An unweighted document-level MDR objective would average document risks rather than reproduce the output-average loss, so we do not apply this equal-size-group reduction there. A weighted adaptation requires a separate derivation.

The complete grouped-SCoRE grids, including every budget-exceedance count, are retained as `review-score-*` in `supplement/result_grids.csv`.

SCOPE uses bidirectional judgment probabilities to compute Bidirectional Preference Entropy (BPE), which these scalar score panels do not provide (Badshah et al., 2026). Transductive full-ranking inference observes the true relative ranks of a calibration subset and assumes tie-free, exchangeable rankings (Fermanian et al., 2025). Its rank-coverage guarantees would require a separate derivation for churn and retained-count certification. We do not evaluate such an adaptation.

Setup and amortization. Each configuration records Mn_{ref} base-reference entries, Tn_{ref} candidate-reference entries and MN base-test entries. CRC and grouped SCoRE also record fully scored base and candidate calibration entries. The earlier WMT23 correlation check reads all $24 \times 557 \times 13 = 173{,}784$ retained scores, including outputs later assigned to test splits. It is excluded from post-freeze counts, so replay does not establish acquisition savings from the start of panel construction. Reference scoring could be reused across batches only while fitted maps, cutoff protocol and declared family remain applicable. The 40 overlapping, separately refitted splits do not evaluate batches sharing one frozen setup and therefore do not measure deployment amortization.

Removing a candidate changes the certified family. A claim about the original family still needs conservative treatment of its unobserved scores. Endpoint bounds avoid queries while retaining the candidate in the family.

Table 25 reports the corresponding information-access accounting.

Common information accounting. Tables 26–31 summarize four methods across three original panels, two pooling rules, four depths and 40 splits: 3,840 method–configuration records at $\alpha = .10$. Candidate reference, calibration and test observations share the denominator NT; the records separately list pool-score stages. Setup is charged as recorded, not optimized for deployment. Zero-query retention and reference priority certify the current batch; CRC and SCoRE retain their expectation guarantees and calibration assumptions. SCoRE decides whole groups. A common numerical loss and cost unit do not make these guarantees identical. Panel preprocessing and full-matrix posthoc verification are outside the selection-visible counts.

K Reference Mapping and Query Granularity

K.1 Reference-only mapping sensitivity

We retain the original 40 group or document splits, all test outputs, both pooling rules, four selection depths and all four acquisition methods. Within each reference split, nested prefixes of the original shuffled group order retain all, half or one quarter of the reference groups. Fractional group counts are rounded down, with at least one group retained. The calibration and test assignments remain unchanged. For each evaluator, the right empirical CDF is the original protocol. For a raw score s, the midrank variant is $(n_{<s} + n_{\leq s})/(2n_{\text{ref}})$, where both counts and n_{ref} use the current reference subset. The reference subset supplies both mapping and cutoff fitting, and each cutoff remains fixed during certification. Ridge feature definitions, regularization, priority rules, integer risk arithmetic and stopping criteria are unchanged. Fitted predictions and query order can change with the reference subset and mapping. WMT24 entry acquisition retains the integer-tick comparisons and ties specified in Appendix H.

The six mapping protocols run at $\alpha = .10$ on RewardBench, both WMT23 variants and the three WMT24 directions, yielding 11,520 configurations and 46,080 entry-acquisition paths. These are reference-protocol ablations on existing data, rather than new independent samples. Table 32 summarizes nominal $q = .05$; the supplement's `mapping-all-*` grids cover every depth. Each range spans the six protocol-specific means over 40 splits, not a confidence interval. Machine-readable summaries also retain each mapping and reference size separately, including exact stopping, setup costs and the agreement/opposition witness decomposition.

Mapping affects baseline selection and, under median pooling, its one-vote boundary. In thirteen-stream WMT23 with median pooling at nominal $q = .05$, changing full-reference right ECDF to

Panel	Method	Setup	Cal.	Cand. test	Accounted total
RewardBench 2	Reference	48244	0	285.0	48529.0
RewardBench 2	Full batch	48244	0	16100.0	64344.0
RewardBench 2	CRC	48244	32144	16100.0	96488.0
RewardBench 2	SCoRE	48244	32144	0.0	80388.0
WMT23, 13 streams	Reference	74737	0	523.2	75260.2
WMT23, 13 streams	Full batch	74737	0	41327.0	116064.0
WMT23, 13 streams	CRC	74737	57720	41327.0	173784.0
WMT23, 13 streams	SCoRE	74737	57720	0.0	132457.0
WMT23, 12 streams	Reference	68988	0	43.3	69031.3
WMT23, 12 streams	Full batch	68988	0	38148.0	107136.0
WMT23, 12 streams	CRC	68988	53280	38148.0	160416.0
WMT23, 12 streams	SCoRE	68988	53280	0.0	122268.0

Table 25: Full batch denotes the full-information benchmark, not an online acquisition policy. At $q = .05$, median pooling, $\alpha = .10$: logical score-entry accounting. Setup includes pool/candidate reference and pool test entries; Cal. includes pool/candidate calibration entries; Cand. test excludes base-test scores already counted in Setup. Accounted total sums these three columns after panel construction; it omits preprocessing reads of the full panel. Values average 40 splits and do not measure runtime. Endpoint zero-query certification can omit candidate-reference scoring; no setup-optimality claim is made.

midrank reduces the mean fraction of baseline-selected outputs exactly at the cutoff from 28.8% to 7.3%; mean splitwise base-list Jaccard similarity is 0.772. This diagnostic was computed after execution without selecting a preferred mapping. The same change reduces the mean minimum certificate size for exact stopping from 415.9 to 5.1 entries. Conversely, WMT24 ja-zh under right ECDF needs 11.3% of active entries with full reference groups and 34.0% with a quarter. Its zero-query selected coverage falls from 98.1% to 92.9%, while $|S_0|/k$ rises from 1.27 to 1.66. These are different selection protocols; smaller inspection counts do not by themselves identify a preferable returned set. The original right ECDF remains the primary protocol; midrank is a sensitivity variant. A deployment must fix its mapping and reference-size rule before interpreting candidate-test outcomes. These analyses were conducted on previously inspected panels and were not prospectively preregistered.

K.2 Reference-cutoff bootstrap

A paired group bootstrap compares cutoff-only sensitivity with joint map-and-cutoff sensitivity. For each of six panels and splits 0–7, we draw 50 reference samples with replacement at the original prompt/source/document unit. At $q = .05$, both pooling rules use two arms: retaining original right ECDF maps and resampling the reference-consensus cutoff, or refitting maps and cutoff on the same sampled groups. Test groups and all candidates stay fixed. The generator is NumPy `default_rng`, with seed $620275 + 1000h + r$, where h is the zero-based panel index and r the split index. The panel order is RewardBench 2, thirteen-stream WMT23, twelve-stream WMT23, WMT24 en-de, en-es and ja-zh. Reference groups retain their first-occurrence order; each draw samples the original number of groups with replacement, using the same sampled groups for both arms. This yields 9,600 protocol instances. We recompute initial bounds, the full-information count and, for median pooling, matching exact and primary minima; no labels, reference-priority refit or online acquisition enter this analysis.

For all 9,600 records, saved accounting checks verify unique configuration keys, positive sample counts, integer budgets, cutoff and Jaccard ranges, ordered capacity bounds, gap identities, active-plus-invariant selected counts and the zero-query stopping condition. Mean-pooling records carry no median-only witness value. A separate bisect-based ECDF and explicit replacement replay reconstructs 24 fixed full-test cases (split zero, draw zero, every panel, arm and pooling rule), not every bootstrap record. Table 33 summarizes selection Jaccard, zero-query primary stops and median exact minima. On thirteen-stream WMT23 median, cutoff-only and map-plus-cutoff Jaccard medians are .997 and .959; zero-query primary-stop fractions are 21.0% and 25.5%. We also compute the sample cutoff standard deviation within each 50-draw split and average these deviations over eight splits. On thirteen-stream WMT23 median, the cutoff-only and map-plus-cutoff averages are .0115 and .0078 on the normalized score scale; all 24 cells accompany the records. A draw changes the fixed selection protocol, so a smaller certificate

q	Method	K	Cov.	Risk	Over	Ref.	Cal.	Test	Total
0.05	Zero	95.5	80.9	0.053	0	0.998	0.000	0.000	0.998
0.05	Reference	115.6	97.3	0.077	0	0.998	0.000	0.018	1.016
0.05	CRC	85.6	72.2	0.079	2	0.998	0.998	1.000	2.997
0.05	SCoRE	114.1	96.4	0.078	1	0.998	0.998	0.000	1.997
0.10	Zero	198.2	84.9	0.048	0	0.998	0.000	0.000	0.998
0.10	Reference	229.1	97.8	0.066	0	0.998	0.000	0.024	1.022
0.10	CRC	186.1	79.5	0.079	6	0.998	0.998	1.000	2.997
0.10	SCoRE	234.3	100.0	0.070	0	0.998	0.998	0.000	1.997
0.20	Zero	435.7	94.6	0.043	0	0.998	0.000	0.000	0.998
0.20	Reference	451.4	97.9	0.047	0	0.998	0.000	0.012	1.010
0.20	CRC	436.5	94.7	0.089	9	0.998	0.998	1.000	2.997
0.20	SCoRE	460.9	100.0	0.050	0	0.998	0.998	0.000	1.997
0.30	Zero	685.9	99.8	0.041	0	0.998	0.000	0.000	0.998
0.30	Reference	686.0	99.9	0.041	0	0.998	0.000	0.000	0.998
0.30	CRC	687.0	100.0	0.091	0	0.998	0.998	1.000	2.997
0.30	SCoRE	687.1	100.0	0.041	0	0.998	0.998	0.000	1.997

Table 26: RewardBench 2, median pooling, $\alpha = .10$: common recorded churn and candidate-score accounting. Risk is replacement-average churn, averaged across the 40 splits. K is returned count; Cov. is selected coverage in percent; Over counts batches above budget out of 40. Ref., Cal., Test and Total are logical candidate entries divided by the same test NT; Total includes the three stages and can exceed one. K, coverage, risk and stage-cost ratios are splitwise means; Over is a count of splits. Reference priority stops at $V^+ - V^- \le \lceil .02|S_0| \rceil$. Zero and Reference give deterministic batch feasibility; CRC and SCoRE give expectation guarantees under group exchangeability, so Over does not refute those guarantees. SCoRE decides whole groups. Reference setup is charged as recorded, not minimized. Pool scores and panel-preprocessing reads are separate; full matrices used only for posthoc validation are outside selection-visible information. Full pool-stage counts accompany the records.

concerns a changed task. The saved percentiles describe these draws, not population confidence intervals or independent replications.

K.3 Whole-evaluator query controls

A column oracle reveals all N scores from one candidate evaluator for the current test batch. It is charged N logical score entries, including invariant and nonselected outputs. The same certificate solver updates the active selected rows with the newly observed flips. Fixed column order follows candidate index, random column order uses the original query seed plus 571921, and reference column order sorts candidates by their summed predicted upper-cost reduction across active selected rows. Let $e_{ir} = \mathrm{round}_{\mathrm{even}}(10^{12}\hat{f}_{ir})$ be prediction ticks. Columns sort by decreasing $\sum_{i \in S_0 : p_i > 0}(10^{12} p_i - e_{ir})$, with candidate-index ties. This static column priority uses reference and visible base information only, not hidden candidate test scores. It aggregates the entry method's per-cell predictions at the column level.

The original full-reference right-ECDF configuration runs all three column orders at every depth and pooling rule on 40 splits, yielding 5,760 paths. The positive-gap and exact stops are recorded separately. Table 34 uses the positive-gap stop, and the supplement's `column-all-*` grids give every depth. Entry and column protocols share the risk budget and stopping criterion, although their positive-gap returned sets can differ. All-column percentages divide by NT; dividing them by mT would mix scores outside the active selected rows with active-entry inspection. The entry witness Q_{exact} is not an unscaled bound on column count. Under the same odd-median completion model and exact stopping criterion, any successful certificate using c columns satisfies $Q_{\mathrm{exact}} \le mc \le Nc$; the entry witness therefore lower-bounds its score-entry cost. This exact bound does not apply directly to the positive-gap results. Appendix M.3 gives the converted bound $\lceil Q_{\mathrm{exact}}/m \rceil$ for $m > 0$ and exact column witnesses; $m = 0$ needs no queries.

At $q = .05$, $\alpha = .10$ and primary stopping, RewardBench reference-priority entry paths access all seven candidate evaluators on average under both pooling rules. Reference column paths access 5.0 under median and 6.7 under mean. The queried-score fractions are reported in Table 34. These results separate revealed entries from the number of candidate test columns involved. Reference and base setup costs remain recorded for each protocol. Cached-score replay establishes neither model-loading savings nor end-to-end inference savings.

q	Method	K	Cov.	Risk	Over	Ref.	Cal.	Test	Total
0.05	Zero	24.0	20.3	0.012	0	0.998	0.000	0.000	0.998
0.05	Reference	117.3	97.5	0.068	0	0.998	0.000	0.045	1.043
0.05	CRC	97.3	81.1	0.095	15	0.998	0.998	1.000	2.997
0.05	SCoRE	118.7	98.8	0.072	0	0.998	0.998	0.000	1.997
0.10	Zero	74.9	31.7	0.005	0	0.998	0.000	0.000	0.998
0.10	Reference	231.8	97.8	0.059	0	0.998	0.000	0.080	1.078
0.10	CRC	207.3	87.4	0.093	12	0.998	0.998	1.000	2.997
0.10	SCoRE	237.0	100.0	0.063	0	0.998	0.998	0.000	1.997
0.20	Zero	238.7	51.7	0.003	0	0.998	0.000	0.000	0.998
0.20	Reference	452.8	97.9	0.045	0	0.998	0.000	0.119	1.117
0.20	CRC	454.6	98.3	0.097	15	0.998	0.998	1.000	2.997
0.20	SCoRE	462.5	100.0	0.048	0	0.998	0.998	0.000	1.997
0.30	Zero	487.7	70.8	0.006	0	0.998	0.000	0.000	0.998
0.30	Reference	675.1	97.9	0.036	0	0.998	0.000	0.126	1.124
0.30	CRC	689.0	100.0	0.084	0	0.998	0.998	1.000	2.997
0.30	SCoRE	689.3	100.0	0.039	0	0.998	0.998	0.000	1.997

Table 27: RewardBench 2, mean pooling, $\alpha = .10$: common recorded churn and candidate-score accounting. Risk is replacement-average churn, averaged across the 40 splits. K is returned count; Cov. is selected coverage in percent; Over counts batches above budget out of 40. Ref., Cal., Test and Total are logical candidate entries divided by the same test NT; Total includes the three stages and can exceed one. K, coverage, risk and stage-cost ratios are splitwise means; Over is a count of splits. Reference priority stops at $V^+ - V^- \le \lceil .02|S_0| \rceil$. Zero and Reference give deterministic batch feasibility; CRC and SCoRE give expectation guarantees under group exchangeability, so Over does not refute those guarantees. SCoRE decides whole groups. Reference setup is charged as recorded, not minimized. Pool scores and panel-preprocessing reads are separate; full matrices used only for posthoc validation are outside selection-visible information. Full pool-stage counts accompany the records.

K.4 Scope of the completion and query model

Observed score correlations may guide query order without changing the endpoint certificate. Using relations among missing scores to tighten bounds requires guarantees that those relations hold for the true scores; learned low-rank approximations or concentrated posteriors alone provide no such guarantee. Point predictions do not remove any consistent completion. Both joint endpoint vectors remain attainable, so predictions cannot tighten the frontiers in this model. A margin ruling out every single-member flip is already captured by $p_i = 0$; base-pool variance alone does not restrict missing candidate scores. Fixed replacement weights are treated below; the odd-median unit-count formulas assume uniform weights and Boolean entry outcomes y_{ir}. Mean pooling can have several flip-count levels and different p_i across rows; Theorem 3 handles their shared evidence without the median binary decomposition. Simultaneous replacements change median boundary geometry; continuous mixtures also require a specified risk measure. These are different certification models.

Fixed replacement weights. The weights apply to replacement scenarios, not to scores within the pooling rule. For specified $w_{jr} \ge 0$ with $\sum_{j,r} w_{jr} = 1$, define the candidate contribution

$$g_{ir}(x) = \sum_j w_{jr} \mathbf{1}[\sigma_P(i) \neq \sigma_{P^{j\to r}(x)}(i)].$$

The weighted risk is $R_w(C) = \sum_{i \in C} \sum_r g_{ir}(s_r(i))/(2k)$. For selected rows, let $a_{ir} = g_{ir}(0)$. The observed sum is a valid lower bound, and adding $\sum_{r \in U_i} a_{ir}$ gives its upper bound; unselected rows use the opposite endpoint. Nonnegative weights preserve joint endpoint attainment and the sorted-prefix retained-count argument. The budget here is the real-valued $2k\alpha$, without an integer floor unless contributions are first rescaled to a common integer unit. Nonuniform weights can give candidate-specific endpoint caps; Theorem 3 and the median closed forms are stated for the original uniform family. No weighted-acquisition advantage is claimed.

Restricting only unobserved completions to a discrete set retaining both joint endpoint assignments leaves these coordinate bounds and capacities unchanged, with observed scores, base scores and cutoff fixed. An arbitrary grid restriction is not itself evidence that the true missing scores belong to that set. Reference-sample minima and maxima alone do not establish hard support for unseen test scores.

A known hard constraint $\mathcal{K}$ satisfied by the true

q	Method	K	Cov.	Risk	Over	Ref.	Cal.	Test	Total
0.05	Zero	117.9	84.9	0.056	0	0.989	0.000	0.000	0.989
0.05	Reference	134.8	97.7	0.078	0	0.989	0.000	0.013	1.002
0.05	CRC	107.9	77.0	0.067	1	0.989	0.989	1.000	2.979
0.05	SCoRE	114.9	82.2	0.049	0	0.989	0.989	0.000	1.979
0.10	Zero	241.3	100.0	0.035	0	0.989	0.000	0.000	0.989
0.10	Reference	241.3	100.0	0.035	0	0.989	0.000	0.000	0.989
0.10	CRC	239.1	99.1	0.064	0	0.989	0.989	1.000	2.979
0.10	SCoRE	241.4	100.0	0.035	0	0.989	0.989	0.000	1.979
0.20	Zero	489.1	100.0	0.035	0	0.989	0.000	0.000	0.989
0.20	Reference	489.1	100.0	0.035	0	0.989	0.000	0.000	0.989
0.20	CRC	487.2	99.6	0.060	0	0.989	0.989	1.000	2.979
0.20	SCoRE	489.1	100.0	0.035	0	0.989	0.989	0.000	1.979
0.30	Zero	730.9	100.0	0.033	0	0.989	0.000	0.000	0.989
0.30	Reference	730.9	100.0	0.033	0	0.989	0.000	0.000	0.989
0.30	CRC	730.3	99.9	0.066	0	0.989	0.989	1.000	2.979
0.30	SCoRE	730.9	100.0	0.033	0	0.989	0.989	0.000	1.979

Table 28: WMT23, 13, median pooling, $\alpha = .10$: common recorded churn and candidate-score accounting. Risk is replacement-average churn, averaged across the 40 splits. K is returned count; Cov. is selected coverage in percent; Over counts batches above budget out of 40. Ref., Cal., Test and Total are logical candidate entries divided by the same test NT; Total includes the three stages and can exceed one. K, coverage, risk and stage-cost ratios are splitwise means; Over is a count of splits. Reference priority stops at $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Zero and Reference give deterministic batch feasibility; CRC and SCoRE give expectation guarantees under group exchangeability, so Over does not refute those guarantees. SCoRE decides whole groups. Reference setup is charged as recorded, not minimized. Pool scores and panel-preprocessing reads are separate; full matrices used only for posthoc validation are outside selection-visible information. Full pool-stage counts accompany the records.

matrix restricts completions to $\Omega_{\mathcal{K}} \subseteq \Omega$. Every original safe certificate remains valid, but endpoint bounds can cease to be tight and the minimum sufficient evidence can change. An observed correlation alone does not establish $\mathcal{K}$. Original minimum-query lower bounds need not remain lower bounds for this restricted model. If a learned set contains the true matrix only with a stated probability, certification within that set inherits the containment event; marginal coverage for individual entries does not establish simultaneous validity under adaptive acquisition. Such a change requires its own information and guarantee specification.

Transformations that preserve the task. For odd-sized median pooling, apply a common strictly increasing bijection $h : [0, 1] \to [0, 1]$ to every score and replace the cutoff by $h(t_0)$. Median order statistics commute with h, and cutoff membership is unchanged for every replacement. Transporting each unobserved score by h gives a bijection of completion matrices, so flip costs and certificate minima are unchanged. Separate evaluator-wise ECDF refits are not such a common transformation. For mean pooling, the corresponding property holds for a common positive affine map $h(x) = ax + b$, $a > 0$, with the cutoff and score-completion range transported together. A nonlinear common map need not commute with averaging. These transformations give sufficient conditions for task equivalence. Outside them, equivalence must be checked rather than assumed; the ECDF-refitting experiments above change baseline selected sets.

Evidence requirements under changed protocols. Table 35 connects each changed median protocol to its initial count gap, invariant and budget-admissible zero-query retention, and matching exact entry minimum. We reaggregate 5,760 median configurations (six reference protocols, six panels, four depths and 40 splits) without new query runs. A lower minimum under a different reference map is evidence about a different selected set, not a same-instance acquisition improvement.

q	Method	K	Cov.	Risk	Over	Ref.	Cal.	Test	Total
0.05	Zero	24.0	20.3	0.015	0	0.989	0.000	0.000	0.989
0.05	Reference	123.4	97.5	0.072	0	0.989	0.000	0.045	1.035
0.05	CRC	116.2	91.8	0.074	0	0.989	0.989	1.000	2.979
0.05	SCoRE	96.5	75.7	0.051	2	0.989	0.989	0.000	1.979
0.10	Zero	48.0	19.9	0.012	0	0.989	0.000	0.000	0.989
0.10	Reference	241.9	97.8	0.056	0	0.989	0.000	0.085	1.074
0.10	CRC	242.6	98.1	0.080	2	0.989	0.989	1.000	2.979
0.10	SCoRE	246.9	99.8	0.061	0	0.989	0.989	0.000	1.979
0.20	Zero	377.6	77.6	0.012	0	0.989	0.000	0.000	0.989
0.20	Reference	476.4	97.9	0.031	0	0.989	0.000	0.059	1.048
0.20	CRC	484.4	99.6	0.058	0	0.989	0.989	1.000	2.979
0.20	SCoRE	486.5	100.0	0.034	0	0.989	0.989	0.000	1.979
0.30	Zero	638.5	87.7	0.017	0	0.989	0.000	0.000	0.989
0.30	Reference	714.8	97.9	0.031	0	0.989	0.000	0.057	1.047
0.30	CRC	729.3	99.9	0.065	0	0.989	0.989	1.000	2.979
0.30	SCoRE	730.0	100.0	0.034	0	0.989	0.989	0.000	1.979

Table 29: WMT23, 13, mean pooling, $\alpha = .10$: common recorded churn and candidate-score accounting. Risk is replacement-average churn, averaged across the 40 splits. K is returned count; Cov. is selected coverage in percent; Over counts batches above budget out of 40. Ref., Cal., Test and Total are logical candidate entries divided by the same test NT; Total includes the three stages and can exceed one. K, coverage, risk and stage-cost ratios are splitwise means; Over is a count of splits. Reference priority stops at $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Zero and Reference give deterministic batch feasibility; CRC and SCoRE give expectation guarantees under group exchangeability, so Over does not refute those guarantees. SCoRE decides whole groups. Reference setup is charged as recorded, not minimized. Pool scores and panel-preprocessing reads are separate; full matrices used only for posthoc validation are outside selection-visible information. Full pool-stage counts accompany the records.

q	Method	K	Cov.	Risk	Over	Ref.	Cal.	Test	Total
0.05	Zero	112.4	99.2	0.042	0	0.989	0.000	0.000	0.989
0.05	Reference	113.3	99.6	0.043	0	0.989	0.000	0.001	0.990
0.05	CRC	106.9	95.2	0.062	1	0.989	0.989	1.000	2.979
0.05	SCoRE	109.6	97.2	0.037	0	0.989	0.989	0.000	1.979
0.10	Zero	226.6	99.4	0.047	0	0.989	0.000	0.000	0.989
0.10	Reference	227.4	99.6	0.048	0	0.989	0.000	0.001	0.990
0.10	CRC	225.3	98.8	0.075	3	0.989	0.989	1.000	2.979
0.10	SCoRE	228.5	100.0	0.048	0	0.989	0.989	0.000	1.979
0.20	Zero	456.8	99.9	0.041	0	0.989	0.000	0.000	0.989
0.20	Reference	457.3	99.9	0.042	0	0.989	0.000	0.000	0.990
0.20	CRC	456.4	99.7	0.076	1	0.989	0.989	1.000	2.979
0.20	SCoRE	457.6	100.0	0.042	0	0.989	0.989	0.000	1.979
0.30	Zero	680.3	99.9	0.040	0	0.989	0.000	0.000	0.989
0.30	Reference	680.8	99.9	0.040	0	0.989	0.000	0.000	0.990
0.30	CRC	681.3	100.0	0.079	0	0.989	0.989	1.000	2.979
0.30	SCoRE	681.6	100.0	0.040	0	0.989	0.989	0.000	1.979

Table 30: WMT23, 12, median pooling, $\alpha = .10$: common recorded churn and candidate-score accounting. Risk is replacement-average churn, averaged across the 40 splits. K is returned count; Cov. is selected coverage in percent; Over counts batches above budget out of 40. Ref., Cal., Test and Total are logical candidate entries divided by the same test NT; Total includes the three stages and can exceed one. K, coverage, risk and stage-cost ratios are splitwise means; Over is a count of splits. Reference priority stops at $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Zero and Reference give deterministic batch feasibility; CRC and SCoRE give expectation guarantees under group exchangeability, so Over does not refute those guarantees. SCoRE decides whole groups. Reference setup is charged as recorded, not minimized. Pool scores and panel-preprocessing reads are separate; full matrices used only for posthoc validation are outside selection-visible information. Full pool-stage counts accompany the records.

q	Method	K	Cov.	Risk	Over	Ref.	Cal.	Test	Total
0.05	Zero	22.0	20.6	0.015	0	0.989	0.000	0.000	0.989
0.05	Reference	113.7	97.0	0.069	0	0.989	0.000	0.046	1.035
0.05	CRC	103.8	89.0	0.077	4	0.989	0.989	1.000	2.979
0.05	SCoRE	89.2	77.6	0.048	1	0.989	0.989	0.000	1.979
0.10	Zero	55.6	24.3	0.010	0	0.989	0.000	0.000	0.989
0.10	Reference	222.2	97.8	0.049	0	0.989	0.000	0.080	1.069
0.10	CRC	222.6	98.2	0.076	2	0.989	0.989	1.000	2.979
0.10	SCoRE	226.3	99.5	0.054	0	0.989	0.989	0.000	1.979
0.20	Zero	342.0	75.8	0.011	0	0.989	0.000	0.000	0.989
0.20	Reference	442.1	97.9	0.037	0	0.989	0.000	0.063	1.053
0.20	CRC	450.3	99.7	0.075	0	0.989	0.989	1.000	2.979
0.20	SCoRE	451.6	100.0	0.040	0	0.989	0.989	0.000	1.979
0.30	Zero	563.3	82.9	0.015	0	0.989	0.000	0.000	0.989
0.30	Reference	666.9	97.9	0.036	0	0.989	0.000	0.078	1.067
0.30	CRC	680.3	99.9	0.076	0	0.989	0.989	1.000	2.979
0.30	SCoRE	681.0	100.0	0.039	0	0.989	0.989	0.000	1.979

Table 31: WMT23, 12, mean pooling, $\alpha = .10$: common recorded churn and candidate-score accounting. Risk is replacement-average churn, averaged across the 40 splits. K is returned count; Cov. is selected coverage in percent; Over counts batches above budget out of 40. Ref., Cal., Test and Total are logical candidate entries divided by the same test NT; Total includes the three stages and can exceed one. K, coverage, risk and stage-cost ratios are splitwise means; Over is a count of splits. Reference priority stops at $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Zero and Reference give deterministic batch feasibility; CRC and SCoRE give expectation guarantees under group exchangeability, so Over does not refute those guarantees. SCoRE decides whole groups. Reference setup is charged as recorded, not minimized. Pool scores and panel-preprocessing reads are separate; full matrices used only for posthoc validation are outside selection-visible information. Full pool-stage counts accompany the records.

Panel	Pooling	$\lvert S_0\rvert/k$	Zero coverage (%)	Active (%)	Q_{exact}	Jaccard
RewardBench 2	median	0.99–1.03	80.9–82.6	53.5–61.4	178.9–196.6	0.84–1.00
RewardBench 2	mean	1.01–1.05	20.3–21.5	85.3–86.6	–	0.90–1.00
WMT23, 13 streams	median	1.04–1.15	84.9–99.8	0.3–45.1	5.1–415.9	0.72–1.00
WMT23, 13 streams	mean	1.02–1.12	20.3–21.3	84.6–87.5	–	0.78–1.00
WMT23, 12 streams	median	1.01–1.20	95.5–99.2	3.8–24.1	41.1–220.6	0.66–1.00
WMT23, 12 streams	mean	1.04–1.14	20.4–22.1	83.3–85.4	–	0.73–1.00
WMT24 en-de	median	1.09–1.37	96.7–99.5	1.3–13.1	37.1–408.2	0.65–1.00
WMT24 en-de	mean	1.08–1.35	22.0–23.6	80.8–83.9	–	0.64–1.00
WMT24 en-es	median	1.11–1.35	96.9–99.8	1.1–14.4	18.1–343.9	0.71–1.00
WMT24 en-es	mean	1.11–1.29	20.9–22.0	81.6–83.4	–	0.71–1.00
WMT24 ja-zh	median	1.25–1.66	92.9–98.1	10.8–34.0	194.1–1143.5	0.61–1.00
WMT24 ja-zh	mean	1.20–1.48	21.4–25.4	82.2–85.9	–	0.43–1.00

Table 32: Mapping sensitivity at $q = .05$, $\alpha = .10$. Ranges span six protocol-specific 40-split means (full/half/quarter reference, right/midrank ECDF), keeping test groups fixed. Zero is zero-query selected coverage in percent. Active is reference-priority queried mT percentage when $V^+ - V^- \leq \lceil .02|S_0| \rceil$, with zero for empty active sets. Q_{exact} is the 40-split mean entry witness under median pooling, an exact target. Jaccard compares base selected sets with full-reference right ECDF. Active and witness columns use different stopping targets and do not form an online-to-lower-bound ratio. Ranges are not confidence intervals.

Panel	Pool	J_t	$J_{\mathrm{map+t}}$	Zero stops t (%)	Zero stops map+t (%)	$Q_{0,t}$	$Q_{0,\mathrm{map+t}}$
RewardBench 2	median	0.951	0.909	0.0	0.0	153.0	153.0
RewardBench 2	mean	0.933	0.933	0.0	0.0	–	–
WMT23, 13 streams	median	0.997	0.959	21.0	25.5	312.0	244.0
WMT23, 13 streams	mean	0.892	0.890	0.0	0.0	–	–
WMT23, 12 streams	median	0.847	0.842	82.2	80.8	0.0	0.0
WMT23, 12 streams	mean	0.862	0.855	0.0	0.0	–	–
WMT24 en-de	median	0.855	0.856	92.0	90.0	0.0	0.0
WMT24 en-de	mean	0.853	0.846	1.0	1.5	–	–
WMT24 en-es	median	0.851	0.854	98.0	95.2	0.0	0.0
WMT24 en-es	mean	0.877	0.877	0.5	0.0	–	–
WMT24 ja-zh	median	0.841	0.831	77.5	74.5	0.0	0.0
WMT24 ja-zh	mean	0.829	0.836	0.0	0.0	–	–

Table 33: Reference-group bootstrap at $q = .05$, $\alpha = .10$: 50 paired draws in each of eight splits, 400 instances per cell. t resamples the reference-consensus cutoff while retaining original maps; map+t refits ECDFs and cutoff on the same resampled prompt/source/document groups. Test groups and the evaluator family stay fixed. J is median baseline-selection Jaccard against the original split. Zero reports the fraction meeting the primary count-gap stop without queries; Q_0 is the median matching exact entry minimum, reported only for median pooling. Every draw defines a changed fixed protocol. Percentiles in the saved records are descriptive bootstrap summaries, not population confidence intervals. No label-based tuning or new online acquisition is performed.

Panel	Pooling	Entry (%)	Touched	Fixed (%)	Random (%)	Ref. (%)	Cols.	T
RewardBench 2	median	1.77	7.0	69.3	71.4	71.8	5.0	7
RewardBench 2	mean	4.52	7.0	99.6	99.6	96.1	6.7	7
WMT23, 13 streams	median	1.27	15.1	67.5	62.9	40.1	6.8	17
WMT23, 13 streams	mean	4.53	17.0	95.4	94.7	90.1	15.3	17
WMT23, 12 streams	median	0.11	1.3	4.1	4.3	4.1	0.7	17
WMT23, 12 streams	mean	4.61	17.0	92.5	92.6	89.9	15.3	17
WMT24 en-de	median	0.03	0.7	2.0	2.3	1.2	0.2	14
WMT24 en-de	mean	4.69	14.0	92.1	92.1	89.6	12.6	14
WMT24 en-es	median	0.03	0.3	1.2	1.4	1.4	0.2	14
WMT24 en-es	mean	4.76	14.0	90.5	90.5	88.6	12.4	14
WMT24 ja-zh	median	0.36	3.3	12.7	12.3	12.3	1.9	15
WMT24 ja-zh	mean	5.46	15.0	93.0	93.5	92.8	13.9	15

Table 34: Whole-evaluator queries at $q = .05$, $\alpha = .10$, full-reference right ECDF, stopping when $V^+ - V^- \leq \lceil .02|S_0| \rceil$. Entry is reference-priority entry inspection; Touched counts candidate columns accessed by entry queries; it is not a scoring-call count. Fixed, Random and Ref. are column orders, charged N scores per evaluator. Percentages divide by NT; Cols. counts reference-order evaluators. T is the candidate count; the replacement family has $H = MT$ members. Counts and ratios average 40 splits separately. No inference time is measured.

Panel	Ref.	ECDF	$\lvert S_0 \rvert$	Initial gap	Inv.	Noninv.	Q_0	Jaccard
RewardBench 2	1	Right	118.9	23.4	55.5	40.0	196.6	1.000
RewardBench 2	1	Mid	118.9	23.3	55.7	40.0	189.1	0.994
RewardBench 2	1/2	Right	114.3	20.8	53.5	40.0	178.9	0.896
RewardBench 2	1/2	Mid	114.5	21.1	53.5	40.0	187.5	0.897
RewardBench 2	1/4	Right	114.0	21.0	53.0	40.0	180.2	0.836
RewardBench 2	1/4	Mid	114.2	21.0	53.3	40.0	179.9	0.836
WMT23, 13	1	Right	137.9	20.0	76.7	41.2	415.9	1.000
WMT23, 13	1	Mid	126.4	0.4	93.6	32.5	5.1	0.772
WMT23, 13	1/2	Right	139.3	14.8	83.7	40.9	326.7	0.906
WMT23, 13	1/2	Mid	128.9	2.1	92.0	34.8	33.0	0.739
WMT23, 13	1/4	Right	140.9	12.6	88.3	40.0	239.4	0.823
WMT23, 13	1/4	Mid	133.5	3.4	95.1	35.0	71.8	0.724
WMT23, 12	1	Right	114.0	1.6	85.5	26.9	41.1	1.000
WMT23, 12	1	Mid	117.6	3.8	84.9	29.0	63.2	0.762
WMT23, 12	1/2	Right	113.5	2.7	85.9	24.9	60.7	0.821
WMT23, 12	1/2	Mid	116.0	3.9	84.5	27.6	64.5	0.718
WMT23, 12	1/4	Right	133.9	8.6	96.8	28.5	220.6	0.724
WMT23, 12	1/4	Mid	131.3	5.9	95.2	30.2	98.1	0.659
WMT24 en-de	1	Right	464.6	2.7	400.3	61.6	37.1	1.000
WMT24 en-de	1	Mid	466.1	2.9	401.4	61.8	40.2	0.992
WMT24 en-de	1/2	Right	506.7	13.0	407.5	86.2	245.5	0.774
WMT24 en-de	1/2	Mid	510.2	12.5	414.3	83.3	238.9	0.775
WMT24 en-de	1/4	Right	581.5	21.0	451.9	108.7	408.2	0.666
WMT24 en-de	1/4	Mid	578.0	16.2	457.2	104.7	341.6	0.651
WMT24 en-es	1	Right	411.8	1.4	347.7	62.7	19.0	1.000
WMT24 en-es	1	Mid	413.1	1.3	350.2	61.6	18.1	0.997
WMT24 en-es	1/2	Right	441.2	9.0	353.0	79.2	124.2	0.785
WMT24 en-es	1/2	Mid	440.9	9.1	356.0	75.9	125.3	0.778
WMT24 en-es	1/4	Right	493.0	18.2	385.3	89.5	334.8	0.714
WMT24 en-es	1/4	Mid	499.4	18.8	388.9	91.6	343.9	0.710
WMT24 ja-zh	1	Right	345.5	11.8	266.0	67.7	271.6	1.000
WMT24 ja-zh	1	Mid	338.5	13.1	257.6	67.8	194.1	0.789
WMT24 ja-zh	1/2	Right	396.8	25.8	302.1	68.9	625.0	0.778
WMT24 ja-zh	1/2	Mid	364.3	21.7	275.3	67.3	348.7	0.707
WMT24 ja-zh	1/4	Right	448.4	52.8	325.9	69.7	1143.5	0.661
WMT24 ja-zh	1/4	Mid	398.1	36.7	291.7	69.7	606.0	0.614

Table 35: Median protocol sensitivity at $q = .05$, $\alpha = .10$, means over 40 fixed test splits. Ref. is the fraction of reference groups. Initial gap is $|S_0| - V(u^{(0)})$; Inv. and Noninv. are output counts decomposing zero-query retention. Q_0 counts score entries in the matching exact minimum for each changed protocol. Jaccard compares its baseline selected set with full-reference right ECDF on the same split. These joint descriptions connect changed selection tasks to evidence requirements; they are not same-instance acquisition interventions or an endorsement of a mapping.

L Risk Scaling and Selection Objects

For a batch-fixed positive denominator d, define $R_d(C) = \sum_{i \in C} F_i/(dH)$. Then $R_d = (2k/d)R(C)$, and using $\alpha_d = 2k\alpha/d$ preserves the integer budget and every feasible mask. Choosing $d = |S_0|$ requires a nonempty base list; choosing $d = N$ scales by all NH possible membership flips. Keeping the same numerical α instead changes the allowed flip count. For example, with $k = 10$, $|S_0| = 30$ and $\alpha = .10$, the original replacement-average allowance is two flips. Base-list normalization preserves it at $\alpha_d = 1/15$; using $.10$ would allow three. A denominator equal to the returned count depends on the mask and has no such fixed rescaling.

For two k-element subsets of an N-output batch, the largest symmetric difference is $2\min(k, N-k)$. The nominal $2k$ scale is a general upper bound, attained when $2k \leq N$. Frozen-cutoff lists can have other sizes, so it is not a universal bound on their churn. An allowance of ρ average membership changes corresponds to $\alpha = \rho/(2k)$. Requiring each replacement to satisfy its own budget instead assigns outputs a vector of replacement costs. Combining those constraints with the average budget no longer justifies sorting a single scalar cost and taking its longest feasible prefix; the optimization and certificate complexity need separate analysis.

The all-mask replacement-risk diagnostics are reported together in Appendix I.

Fixed-count and frozen-cutoff stability can differ even without ties. Consider consensus scores $(.90, .80, .10)$ with $k = 2$ and cutoff $.75$. If a replacement changes the third score to $.85$, frozen-cutoff selection adds that output and preserves the first two memberships. Top-two selection instead adds the third and excludes the second. This example compares selection objects under the same score change; it is not a comparison of query algorithms.

The grouped SCoRE reduction in Appendix J targets the same nonnegative loss with a fixed batch denominator. Further pruning an accepted set cannot increase that loss, so a subsequent certification step that only prunes the accepted set preserves the marginal expectation bound under the original calibration assumptions. Selected-risk guarantees have a changing denominator and need a separate argument. Re-certifying each future batch already bounds its realized churn within the declared family; calibration does not extend that family or establish semantic correctness.

For any certified mask C, a subsequent content policy can restrict commitments to $C' \subseteq C$. Nonnegative F_i and the fixed denominator give $R(C') \leq R(C) \leq \alpha$. The pruned returned count $K' = |S_0(P) \cap C'|$ has certified regret at most $V^+ - K'$ relative to the original optimum. If $C = C^-$ is returned at gap tolerance ε, then $K = V^-$ and this bound is at most $\varepsilon + (K - K')$. The realized regret $V^* - K'$ can be evaluated when V^* is known, through exact certification or full-information evaluation. Improved preference precision still requires label evidence.

M Exact Information Costs Across Query Granularities

This appendix separates certificate characterizations, computation and empirical comparisons. Column and residual evidence lead to joint entry certificates and resource measurements. Appendix M.8 compares joint evidence with optimal deduplication and studies discovery limits; Appendix M.9 reports full-batch numerical comparisons. Mixed prices, query granularity and controlled panels then test how the access model changes evidence requirements. The following subsections connect certificate models to their computational and em-

pirical consequences. Joint entry formulations and their validation appear in Appendices M.5 and M.6; shared-evidence structure and complete-batch results follow in Appendices M.8 and M.9. The later column and pricing analyses change the query actions while retaining the certification objective.

M.1 Acquisition when the initial gap requires queries

Before any candidate-test query, all lower costs are zero, so $V^+(0) = |S_0|$. The zero-query mask has count $V^-(0)$. We stratify configurations by whether $|S_0| - V^-(0) > \varepsilon$, using the same initial state and tolerance for every acquisition policy. This condition, rather than high initial coverage alone, determines whether queries are needed. For each panel, depth, pooling rule and stopping target, we report the subset size and conditional query counts for all four strategies. Exact and positive-gap targets can give different subsets. When the subset is empty, its means are undefined.

At $q = .05$, $\alpha = .10$ and the primary two-percent stop, thirteen-stream WMT23 median requires queries on 36/40 splits; reference priority inspects 50.1% of active entries on those splits, versus 45.1% across all 40. Twelve-stream WMT23 requires queries on 3/40, with conditional inspection 51.0% versus 3.8% overall. RewardBench median requires queries on all 40, so its conditional and overall inspection both equal 61.4%. Mean pooling requires queries on every split of these original panels. Tables 37–40 cover all six panels and four depths, including empty subsets. These summaries describe the existing acquisition records; they do not tune protocols or provide independent-population inference.

Panel	q	Safety	Exclusion	Offline	Fixed	Ref.
RewardBench 2	0.05	158.8	37.9	196.6	338.8	327.6
RewardBench 2	0.10	249.2	0.0	249.2	487.3	446.9
WMT23, 13 streams	0.05	319.5	96.5	415.9	774.0	622.6
WMT23, 13 streams	0.10	1.3	0.0	1.3	2.6	3.4
WMT23, 12 streams	0.05	20.5	20.6	41.1	59.0	56.0
WMT23, 12 streams	0.10	31.1	0.0	31.1	81.0	74.6

Table 36: Count means use decimal half-up rounding after aggregating integer records. Exact median stopping, $\alpha = .10$: mean entry counts over 40 splits. Safety counts agreement entries; Exclusion counts opposition entries. They sum to the offline minimum before separate rounding. For every instance, Fixed and Ref. online counts are no smaller than this minimum.

Panel	q	n	Fixed	Random	Width	Ref.	Ref. active	K	Gap
RewardBench 2	0.05	40	296.3	296.8	292.2	285.0	61.4	115.6	3.0
RewardBench 2	0.10	39	426.6	425.4	409.0	388.1	46.2	230.1	5.3
RewardBench 2	0.20	37	210.8	207.7	211.2	204.8	15.2	453.8	9.8
RewardBench 2	0.30	1	27.0	38.0	35.0	33.0	1.8	735.0	15.0
WMT23, 13	0.05	36	738.3	746.0	817.2	581.4	50.1	132.5	3.2
WMT23, 13	0.10	0	–	–	–	–	–	–	–
WMT23, 13	0.20	0	–	–	–	–	–	–	–
WMT23, 13	0.30	0	–	–	–	–	–	–	–
WMT23, 12	0.05	3	588.0	584.7	585.7	577.7	51.0	167.0	4.0
WMT23, 12	0.10	5	283.2	282.6	303.0	238.2	14.8	300.6	6.6
WMT23, 12	0.20	1	712.0	658.0	605.0	641.0	20.1	551.0	12.0
WMT23, 12	0.30	1	637.0	588.0	563.0	683.0	14.8	821.0	17.0
WMT24 en-de	0.05	2	1066.0	1094.5	1247.0	881.0	25.8	592.0	12.5
WMT24 en-de	0.10	5	256.4	263.2	313.8	205.2	3.6	1563.2	32.6
WMT24 en-de	0.20	0	–	–	–	–	–	–	–
WMT24 en-de	0.30	0	–	–	–	–	–	–	–
WMT24 en-es	0.05	1	1484.0	1439.0	1293.0	1329.0	45.4	733.0	15.0
WMT24 en-es	0.10	0	–	–	–	–	–	–	–
WMT24 en-es	0.20	0	–	–	–	–	–	–	–
WMT24 en-es	0.30	0	–	–	–	–	–	–	–
WMT24 ja-zh	0.05	9	1278.1	1275.8	1288.1	1228.4	50.2	558.2	12.1
WMT24 ja-zh	0.10	5	1374.0	1352.6	1466.2	1273.0	33.7	1001.2	21.0
WMT24 ja-zh	0.20	1	336.0	329.0	595.0	359.0	9.7	765.0	16.0
WMT24 ja-zh	0.30	0	–	–	–	–	–	–	–

Table 37: Median entry acquisition with $\varepsilon = \lceil .02|S_0| \rceil$ and $\alpha = .10$. Each panel–depth setting has 40 splits. The shared subset has initial gap greater than ε; n gives its size. Query counts are conditional means for fixed, random, largest-width and reference-priority acquisition. Ref. active is the mean $100Q/(mT)$. For reference priority, $K = V^-$ is returned count and Gap is $V^+ - V^-$; both average the same initially non-stopping splits. Empty subsets are marked –. Integer count means use decimal half-up rounding; ratios are averaged separately. These overlapping splits give descriptive results.

Panel	q	n	Fixed	Random	Width	Ref.	Ref. active	K	Gap
RewardBench 2	0.05	40	338.8	339.2	331.6	327.6	71.1	118.4	0.0
RewardBench 2	0.10	40	487.3	487.1	464.3	446.9	53.4	234.3	0.0
RewardBench 2	0.20	39	321.6	316.9	320.5	300.5	22.6	461.8	0.0
RewardBench 2	0.30	6	98.5	92.5	95.0	89.7	5.1	730.5	0.0
WMT23, 13	0.05	36	859.9	851.0	891.1	691.8	60.3	135.3	0.0
WMT23, 13	0.10	1	105.0	129.0	125.0	137.0	9.2	368.0	0.0
WMT23, 13	0.20	0	–	–	–	–	–	–	–
WMT23, 13	0.30	0	–	–	–	–	–	–	–
WMT23, 12	0.05	4	590.3	579.3	578.0	560.3	53.1	165.3	0.0
WMT23, 12	0.10	8	404.8	399.6	454.5	373.1	23.9	286.9	0.0
WMT23, 12	0.20	1	1191.0	1074.0	985.0	1017.0	31.8	563.0	0.0
WMT23, 12	0.30	2	829.5	806.0	738.0	776.0	17.2	839.5	0.0
WMT24 en-de	0.05	3	969.7	1009.3	1048.3	802.3	24.3	639.7	0.0
WMT24 en-de	0.10	5	1245.0	1259.0	1389.2	1011.6	18.6	1595.8	0.0
WMT24 en-de	0.20	0	–	–	–	–	–	–	–
WMT24 en-de	0.30	0	–	–	–	–	–	–	–
WMT24 en-es	0.05	2	1066.0	1014.0	950.0	927.5	31.9	716.0	0.0
WMT24 en-es	0.10	3	339.3	350.0	456.0	298.3	8.7	890.3	0.0
WMT24 en-es	0.20	1	440.0	429.0	724.0	333.0	6.3	1423.0	0.0
WMT24 en-es	0.30	0	–	–	–	–	–	–	–
WMT24 ja-zh	0.05	13	1194.5	1203.3	1207.6	1144.0	49.9	528.4	0.0
WMT24 ja-zh	0.10	9	1412.2	1409.3	1515.4	1290.3	36.7	929.8	0.0
WMT24 ja-zh	0.20	2	664.5	668.5	873.0	688.5	17.8	1243.0	0.0
WMT24 ja-zh	0.30	0	–	–	–	–	–	–	–

Table 38: Median entry acquisition with $\varepsilon = 0$ and $\alpha = .10$. Each panel–depth setting has 40 splits. The shared subset has initial gap greater than ε; n gives its size. Query counts are conditional means for fixed, random, largest-width and reference-priority acquisition. Ref. active is the mean $100Q/(mT)$. For reference priority, $K = V^-$ is returned count and Gap is $V^+ - V^-$; both average the same initially non-stopping splits. Empty subsets are marked –. Integer count means use decimal half-up rounding; ratios are averaged separately. These overlapping splits give descriptive results.

Panel	q	n	Fixed	Random	Width	Ref.	Ref. active	K	Gap
RewardBench 2	0.05	40	763.6	768.5	770.6	727.9	86.5	117.3	3.0
RewardBench 2	0.10	40	1349.4	1362.4	1356.2	1289.6	83.5	231.8	5.2
RewardBench 2	0.20	40	1978.4	1998.9	1925.8	1919.6	76.0	452.8	9.7
RewardBench 2	0.30	40	1957.3	1982.9	1781.6	2028.7	67.0	675.1	14.3
WMT23, 13	0.05	40	1984.8	2004.0	2013.2	1873.7	85.7	123.4	3.0
WMT23, 13	0.10	40	3729.6	3768.0	3782.5	3518.8	83.2	241.9	5.5
WMT23, 13	0.20	40	2381.8	2423.1	2167.6	2419.2	57.4	476.4	10.1
WMT23, 13	0.30	40	2067.3	2111.6	1800.6	2363.8	45.1	714.8	15.2
WMT23, 12	0.05	40	1827.4	1843.5	1841.1	1757.7	85.4	113.7	2.8
WMT23, 12	0.10	40	3219.9	3246.8	3239.9	3047.4	81.3	222.2	5.0
WMT23, 12	0.20	40	2439.8	2481.9	2273.8	2413.8	59.9	442.1	9.5
WMT23, 12	0.30	40	2797.3	2863.3	2520.6	2979.8	54.5	666.9	14.1
WMT24 en-de	0.05	40	5848.3	5904.3	5920.0	5619.0	83.3	451.6	9.7
WMT24 en-de	0.10	40	6809.8	6896.9	6785.0	6538.0	69.7	900.5	19.0
WMT24 en-de	0.20	40	10060.3	10220.6	9911.6	9591.7	63.0	1743.6	36.0
WMT24 en-de	0.30	40	11020.1	11247.0	10569.2	11135.4	58.0	2591.7	53.4
WMT24 en-es	0.05	40	5173.8	5222.9	5228.2	4962.1	82.1	404.5	8.8
WMT24 en-es	0.10	40	6335.6	6414.1	6272.6	6038.1	72.4	789.1	16.6
WMT24 en-es	0.20	40	8977.9	9121.1	8836.9	8509.9	64.4	1535.9	31.9
WMT24 en-es	0.30	40	8750.2	8914.4	8309.6	8924.9	55.6	2301.2	47.5
WMT24 ja-zh	0.05	40	4560.0	4599.3	4590.0	4402.5	85.9	306.7	6.9
WMT24 ja-zh	0.10	40	8142.3	8189.5	8182.0	7834.2	86.0	562.3	12.1
WMT24 ja-zh	0.20	40	13303.9	13385.2	13133.2	12544.0	82.5	1103.9	23.1
WMT24 ja-zh	0.30	40	15658.6	15797.3	15180.5	14436.1	76.1	1633.2	33.9

Table 39: Mean entry acquisition with $\varepsilon = \lceil .02|S_0| \rceil$ and $\alpha = .10$. Each panel–depth setting has 40 splits. The shared subset has initial gap greater than ε; n gives its size. Query counts are conditional means for fixed, random, largest-width and reference-priority acquisition. Ref. active is the mean $100Q/(mT)$. For reference priority, $K = V^-$ is returned count and Gap is $V^+ - V^-$; both average the same initially non-stopping splits. Empty subsets are marked –. Integer count means use decimal half-up rounding; ratios are averaged separately. These overlapping splits give descriptive results.

Panel	q	n	Fixed	Random	Width	Ref.	Ref. active	K	Gap
RewardBench 2	0.05	40	786.6	786.5	787.8	759.2	90.2	120.2	0.0
RewardBench 2	0.10	40	1395.2	1394.8	1387.1	1341.4	86.9	237.0	0.0
RewardBench 2	0.20	40	2069.8	2068.5	2006.4	2006.6	79.5	462.5	0.0
RewardBench 2	0.30	40	2111.0	2111.6	1906.5	2148.8	71.0	689.3	0.0
WMT23, 13	0.05	40	2042.8	2041.7	2053.1	1940.4	88.8	126.1	0.0
WMT23, 13	0.10	40	3833.5	3838.8	3856.5	3647.6	86.3	247.3	0.0
WMT23, 13	0.20	40	2622.2	2621.9	2352.5	2615.6	62.1	486.5	0.0
WMT23, 13	0.30	40	2486.0	2473.2	2108.1	2685.8	51.6	730.0	0.0
WMT23, 12	0.05	40	1879.9	1880.6	1879.4	1823.0	88.7	116.1	0.0
WMT23, 12	0.10	40	3315.4	3314.0	3311.7	3160.6	84.3	227.2	0.0
WMT23, 12	0.20	40	2668.5	2666.6	2447.8	2611.1	64.8	451.6	0.0
WMT23, 12	0.30	40	3180.4	3181.9	2816.3	3274.4	60.0	681.0	0.0
WMT24 en-de	0.05	40	5999.6	6004.5	6017.4	5809.8	86.2	461.0	0.0
WMT24 en-de	0.10	40	7121.7	7122.1	7007.2	6847.4	73.1	919.4	0.0
WMT24 en-de	0.20	40	10716.2	10718.8	10388.9	10201.6	67.2	1779.6	0.0
WMT24 en-de	0.30	40	12079.3	12092.0	11327.7	11985.3	62.5	2645.1	0.0
WMT24 en-es	0.05	40	5310.6	5312.2	5316.6	5126.9	84.9	413.0	0.0
WMT24 en-es	0.10	40	6611.1	6612.9	6471.5	6320.6	75.9	805.7	0.0
WMT24 en-es	0.20	40	9562.9	9563.1	9263.8	9054.4	68.5	1567.7	0.0
WMT24 en-es	0.30	40	9699.3	9684.2	9008.4	9703.0	60.5	2348.7	0.0
WMT24 ja-zh	0.05	40	4674.0	4678.6	4672.9	4554.2	89.1	310.5	0.0
WMT24 ja-zh	0.10	40	8339.0	8340.0	8332.5	8093.6	89.1	570.1	0.0
WMT24 ja-zh	0.20	40	13731.9	13735.6	13549.6	13115.0	86.4	1123.6	0.0
WMT24 ja-zh	0.30	40	16379.6	16383.9	15846.2	15245.9	80.3	1667.1	0.0

Table 40: Mean entry acquisition with $\varepsilon = 0$ and $\alpha = .10$. Each panel–depth setting has 40 splits. The shared subset has initial gap greater than ε; n gives its size. Query counts are conditional means for fixed, random, largest-width and reference-priority acquisition. Ref. active is the mean $100Q/(mT)$. For reference priority, $K = V^-$ is returned count and Gap is $V^+ - V^-$; both average the same initially non-stopping splits. Empty subsets are marked –. Integer count means use decimal half-up rounding; ratios are averaged separately. These overlapping splits give descriptive results.

M.2 Multiplicative entry discovery ratios

An offline constructor selects certificate entries using the complete score matrix. We define the multiplicative discovery ratio as $Q_{\mathrm{online}}/Q_{\mathrm{exact}}$ for exact median entry certification, conditional on $Q_{\mathrm{exact}} > 0$. Splits with $Q_{\mathrm{exact}} = 0$ have undefined ratios and are reported separately; all acquisition policies stop without queries on those splits. Ratios are computed per split before aggregation. Table 41 reports the median, interpolated 90th percentile and maximum ratios for reference priority at $q = .05$. The supplement's `entry-overhead-all-*` grids include every original depth and order. Across six panels, four depths, six reference protocols, four acquisition orders and 40 splits, this exact median analysis gives 23,040 records and 576 panel–depth–protocol–order summary cells. Overlapping splits give descriptive distributions, not population confidence intervals or a competitive guarantee.

RewardBench 2 has a mean exact-certificate size of 196.625 entries; reference priority queries 327.575 entries on average. Their ratio, about 1.666, differs from the mean per-instance ratio 1.895 and the median per-instance ratio 1.968. WMT24 en-de and en-es have only three and two positive witnesses at this depth, limiting interpretation of their conditional ratios. Positive-gap, mean and whole-column paths have different witness targets and are excluded from this entry-ratio denominator.

Panel	$Q_{\text{exact}} > 0$	$Q_{\text{exact}} = 0$	Median	P90	Max
RewardBench 2	40	0	1.968	2.195	2.878
WMT23, 13 streams	36	4	1.662	2.156	2.554
WMT23, 12 streams	4	36	2.058	2.519	2.672
WMT24 en-de	3	37	1.824	2.110	2.182
WMT24 en-es	2	38	2.172	2.390	2.445
WMT24 ja-zh	13	27	1.873	2.533	3.061

Table 41: Exact median entry-query discovery ratios at $q = .05$, $\alpha = .10$, full-reference right ECDF and reference priority. Ratios are $Q_{\text{online}}/Q_{\text{exact}}$ on positive-witness splits; zero-witness splits are separate and use no online queries. $Q_{\text{exact}} > 0$ and $Q_{\text{exact}} = 0$ report split counts; each panel has 40 splits. P90 uses linear interpolation; overlapping splits supply descriptive distributions, not population intervals.

M.3 Odd-median column certificate characterization

A column query reveals one candidate's scores for all N outputs and costs N score entries. Selected interior rows have zero churn cost; their scores remain included in the N entries charged for each column query. Let $Y \in \{0,1\}^{m\times T}$ contain the boundary rows' opposition bits. Under bit budget b, define

$$V_B(w) = \max\Big\{|C| : C \subseteq \mathcal{B},\ \sum_{i\in C} w_i \le b\Big\},$$

and true optimum $L = V_B(z)$. For a queried column subset A, write $o_i(A) = \sum_{r\in A} y_{ir}$. Every consistent completion has row cost between $o_i(A)$ and $o_i(A) + T - |A|$. The minimum number of column queries for exact certification is

$$K_{\text{col}} = \min_{A\subseteq\{1,\dots,T\}} |A|$$
$$\text{subject to}$$
$$V_B(o(A) + T - |A|) = V_B(o(A)) = L.$$

Both capacities here concern boundary rows, with the same bit budget. Adding the known interior count gives the full selected-retention optimum.

Proposition 2 (Exact column condition)**.** *Order the observed counts as $o_{(1)}(A) \le \cdots \le o_{(m)}(A)$. A subset A certifies exact optimality if and only if*

$$\sum_{i=1}^{L} o_{(i)}(A) + L(T - |A|) \le b,$$
$$\sum_{i=1}^{L+1} o_{(i)}(A) > b \quad \text{if } L < m.$$

The first condition uses an empty sum when $L = 0$; the second is omitted when $L = m$. For $m > 0$,

$$\left\lceil \frac{Q_{\text{exact}}}{m} \right\rceil \le K_{\text{col}} \le \min(T, Q_{\text{exact}}).$$

If $m = 0$ or $Q_{\text{exact}} = 0$, $K_{\text{col}} = 0$.

Proof. The upper and lower costs share their order because the unobserved-column offset is common to every row. Their cheapest prefixes therefore have the displayed costs. The true optimum lies between their capacities. Since upper-cost capacity cannot exceed the true optimum L, it equals L exactly when the cheapest L-row upper-cost prefix is feasible. The capacity under lower costs is at most L exactly when the cost sum of its cheapest $(L+1)$-row prefix exceeds the budget. This proves necessity and sufficiency, including both endpoints.

A c-column witness reveals at most mc relevant boundary entries. Keeping only the boundary entries yields an entry certificate, so $Q_{\text{exact}} \le mc$. Conversely, reveal every column touched by a minimum entry witness. Extra observations cannot decrease safe capacity or increase optimistic capacity, so they cannot enlarge the gap; at most Q_{exact} columns are touched. Revealing all T columns always suffices. A zero entry witness certifies the same problem before any column query; with no boundary rows optimality is already known. □

For these candidate counts, we compute exact column minima by enumerating subsets and testing Proposition 2. Encode each row's opposition support as a T-bit mask and merge equal supports with their multiplicities. For each column subset, counting opposition bits in its intersection with each row mask gives a histogram of observed costs. Histogram prefixes test both conditions without enumerating returned masks. With D_{sup} distinct row supports, worst-case time is $O(mT + 2^T(D_{\text{sup}} + T))$ and space is $O(m + 2^T + D_{\text{sup}} + T)$ beyond the input. This bound treats a T-bit mask intersection and population count as unit-cost operations, as when masks fit a machine word with a population-count primitive. Multiword implementations must include their bit-operation costs; the original seven-member-pool panels considered

here have $T \le 17$. The entry lower bound and feasible online column counts restrict the subset sizes requiring enumeration. Every reported optimum is additionally checked using raw-row sorted costs and a separate enumeration of every $(K_{\mathrm{col}} - 1)$-subset. Revealing more columns cannot invalidate a certificate, so failure of every subset at that size rules out every smaller subset. When $K_{\mathrm{col}} = 0$ there is no smaller level.

Evidence shared across rows. Take $m = T = 2$, $b = 0$ and row costs $(1, 1)$. The matrices

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

share $L = 0$ and $Q_{\mathrm{exact}} = 2$, but have minimum column counts one and two. They are realizable with seven-member pools having four scores above a .5 cutoff and three below. For these selected median boundary rows, candidate–output scores below the cutoff implement opposition bits; scores at or above it implement agreement bits. Row sums determine entry certificate size; column certificate size also depends on whether the needed evidence occurs in the same columns. More generally, for any $m \ge T \ge 1$, give each row one opposition and set $b = 0$. Shared support on one candidate gives $K_{\mathrm{col}} = 1$; assigning the opposition cyclically across all T candidates gives $K_{\mathrm{col}} = T$. Both have $Q_{\mathrm{exact}} = m$ and identical row costs. They attain the lower and upper bounds of Proposition 2, respectively.

When $b = 0$ and every boundary row has a true opposition, $L = 0$ and the exact column condition requires at least one observed opposition in every row. Associate each candidate column with the rows where it has an opposition. The minimum column certificate is a minimum set cover of the boundary rows. Covering formulations and queries revealing coefficients across constraints already occur in uncertainty exploration (Megow and Schlöter, 2023; Schlöter, 2025). This specialization identifies the coupling in the present median model; it does not supply an online approximation or competitive guarantee.

M.4 Minimum residual entry evidence for odd medians

An existing observation can supply agreement or opposition evidence before further entries are queried. This matters when a column is bought once and the remaining evidence is acquired cell by cell. We condition the minimum entry witness on arbitrary revealed cells, keeping the original odd-median completion model and bit budget b.

Let $\mathcal{O}$ be the already observed cells, with agreement counts a_i, opposition counts o_i and upper costs $u_i = T - a_i$. The true row costs are z_i, and $L^* = V_B(z)$. Interior outputs add the same known count to both count bounds and cancel from the gap. For $0 \le s \le L^*$, define

$$G_s(\mathcal{O}) = \max\left\{0, \min_{\substack{|C|=s \\ \sum_{i\in C} z_i \le b}} \sum_{i \in C} u_i - b\right\}.$$

The minimum ranges over boundary rows; for $s = 0$ its sum is zero. It counts the fewest new agreements needed to make an s-row set safe. In this offline minimization, true costs restrict the sets to those actually feasible. Existing agreements determine the additional evidence needed for each such set. A dynamic program indexed by retained-row count and total true cost computes the minimum. Let $D_j(s, c)$ be the minimum upper-cost sum for s boundary rows among the first j whose total true cost is exactly c. Initialize $D_0(0, 0) = 0$ and all other states to $+\infty$. Then

$$\begin{aligned} D_j(s, c) = \min\{&D_{j-1}(s, c), \\ &D_{j-1}(s - 1, c - z_j) + u_j\}. \end{aligned}$$

Infeasible indices are omitted. The resulting minimum is

$$G_s(\mathcal{O}) = \max\left\{0, \min_{0 \le c \le b} D_m(s, c) - b\right\}.$$

The cost dimension can be capped at $\min(b, mT)$. Writing $B_{\mathrm{DP}} = \min(b, mT) + 1$, a dense implementation takes $O(m(L^* + 1)B_{\mathrm{DP}})$ time; the supplied implementation stores only reachable states. This is an offline computation using true costs z_i, not an acquisition policy.

For an exclusion target $L^* < p \le m$, put

$$v_i(t) = \max\{o_i, \min(z_i, t)\}.$$

If the p smallest o_i already sum above b, set $E_p(\mathcal{O}) = 0$. Otherwise, over integer levels $0 \le$

$t \le T$, define

$$\lambda_p = \min\left\{ t : \sum_{j=1}^{p} v_{(j)}(t) > b \right\},$$

$$E_p(\mathcal{O}) = \sum_i \big(v_i(\lambda_p) - o_i\big) - \left(\sum_{j=1}^{p} v_{(j)}(\lambda_p) - b - 1\right).$$

At each level t, the values $v_i(t)$ are sorted anew; this order need not match the true-cost order. For $p > m$, set $E_p(\mathcal{O}) = 0$. Thus existing opposition evidence is retained rather than charged again.

Proposition 3 (Minimum residual entry witness)**.** *For integer $\varepsilon \ge 0$, the minimum additional entry count certifying a gap at most ε is*

$$Q_\varepsilon(\mathcal{O}) = \min_{s=\max(0,L^*-\varepsilon)}^{L^*} \left[G_s(\mathcal{O}) + E_{s+\varepsilon+1}(\mathcal{O})\right].$$

With no initial observations this reduces to Corollary 1.

Proof. For a truly feasible s-row set C, each additional agreement reduces its upper-cost sum by one. At least $\max(0, \sum_{i\in C} u_i - b)$ additional agreements are necessary. The available agreements total $\sum_{i\in C}(u_i - z_i)$, enough to attain this number because $\sum_{i\in C} z_i \le b$. Minimizing over C gives G_s.

For exclusion, write $B = b + 1$ and assume the observed prefix does not already exclude p rows. The critical level exists since $p > L^*$. Moreover $\lambda_p \le z_{(p)}$. Rowwise, $v_i(t) \ge \min(z_i, t)$, so their ordered values satisfy the same inequality. At $t = z_{(p)}$, the p smallest truncated true costs sum to $\sum_{j=1}^{p} z_{(j)} \ge B$; hence the p smallest $v_i(t)$ also sum to at least B.

Consider any sufficient final opposition counts $w_i \in [o_i, z_i]$, and let $t = w_{(p)}$. For $w_i \le t$, we have $v_i(t) \ge w_i$ because $o_i \le w_i \le z_i$; for $w_i > t$, $v_i(t) \ge t$. Thus $v_i(t) \ge \min(w_i, t)$ rowwise. Since $t = w_{(p)}$, truncating at t preserves the sum of the p smallest counts. Their sum cannot decrease when $\min(w, t)$ is replaced by $v(t)$, so it remains at least B, implying $t \ge \lambda_p$. Every row with $o_i > t$ lies outside that prefix. Consequently the total new opposition count is at least

$$B + \sum_{j=1}^{m-p} \big[\max(o, t)\big]_{[j]} - \sum_i o_i,$$

where brackets order values from largest to smallest. This expression is nondecreasing in t. At $t = \lambda_p$ it equals E_p: because $\lambda_p \le z_{(p)}$, the largest $m - p$ values of $v(\lambda_p)$ equal those of $\max(o, \lambda_p)$.

To attain the bound, start with $v(\lambda_p)$ and remove new opposition counts one at a time among the current p smallest row counts until the sum of those counts is B, reordering after each removal. A removable count has $w_i > o_i$ and $w_i \le w_{(p)}$; reducing it decreases that prefix sum by exactly one. Such a count exists whenever the sum remains above B. Otherwise the cheapest p counts are unchanged observations, and all positive increments lie above their threshold. Every row with a positive increment has final count at most λ_p. Since all such rows lie above the prefix threshold, that threshold is below λ_p. The vector $v(\lambda_p - 1)$ would then have a prefix sum at least as large, contradicting minimality of λ_p. The construction removes exactly the excess subtracted in E_p.

Any successful gap certificate has safe count $s \in [\max(0, L^* - \varepsilon), L^*]$. If $p = s + \varepsilon + 1 \le m$, it must prove no p-row set feasible under any consistent completion. If $p > m$, no exclusion evidence is needed. New agreement and opposition cells are disjoint, so their lower bounds add. Conversely, the two constructions can be united: agreements change upper costs and oppositions change lower costs. They certify the required gap and attain the displayed minimum. □

M.5 Joint entry evidence beyond Boolean costs

Mean pooling changes how observations serve a certificate. For a selected row with base-score sum S_i, replacement of member j flips membership when $x < Mt_0 - (S_i - s_j(i))$. Thus $f_i(x)$ is an integer staircase, although scores are continuous. The two endpoint costs remain attainable, as required by Proposition 1; intermediate query outcomes can tighten both bounds, raising the lower cost and lowering the upper cost.

Consider two selected rows with identical base scores $(.1, .7, .9)$, cutoff .5 and one candidate scoring .6 on each row. Take $M = 3$, $N = 2$, $q = .5$, $k = 1$ and $\alpha = 1/6$, so $b_H = 1$. Each row has $p_i = 2$ and true cost one. No queries give $(V^-, V^+) = (0, 2)$, one query gives $(1, 2)$, and two give $(1, 1)$. Safety requires one query and exclusion requires two, but their union requires two rather than three. This example shows the evidence overlap absent from odd-median boundary rows.

Attainable row states. For initial observations $\mathcal{O}$, let U_i be the unobserved candidates on active row i. Define

$$\mathcal{D}_i = \left\{ \left(|J|, \sum_{r \in J} f_{ir} \right) : J \subseteq U_i \right\}.$$

Initialize the row-state set with $(0, 0)$. For each remaining candidate, retain existing states and add $(a + 1, v + f_{ir})$ from each (a, v). Store one witnessing subset for each pair. There are at most $(T + 1)(Tp_i + 1)$ pairs. The offline constructor uses the true contributions to build these states; the verifier subsequently sees only the chosen observations.

Lemma 1 (Excluding a cardinality by one threshold)**.** *For nonnegative integer lower costs ℓ and $1 \le p \le m$,*

$$\sum_{j=1}^{p} \ell_{(j)} = \max_{t \in \{0,\dots,TM\}} \left[pt - \sum_i (t - \ell_i)_+ \right],$$

provided every $\ell_i \le TM$.

Proof. For any p-row set A, the bracket is at most $\sum_{i \in A} \ell_i$: bounding each $(t - \ell_i)_+$ below by $t - \ell_i$ on A proves this inequality. At $t = \ell_{(p)}$, all values below t belong to the cheapest p rows, while any remaining rows in that prefix equal t. The bracket therefore equals their sum. Integer costs make exceeding b_H equivalent to reaching $b_H + 1$. □

Proof of Theorem 3. Any successful observation set has active safe count $s = V^- - m_0$ in $[\max(0, L^* - \varepsilon), L^*]$, where m_0 is the invariant selected count. Its queried cells define row states, and a safe prefix supplies C. If $p = s + \varepsilon + 1 \le m$, optimistic active count $V^+ - m_0$ is at most $s + \varepsilon$, so the cheapest p lower costs exceed b_H. Lemma 1 supplies the required threshold. Thus every successful certificate appears in the minimization.

Conversely, witnessing row subsets produce the stated bounds. The safe-set constraint gives at least s active retained outputs. The threshold condition gives optimistic active count at most $s + \varepsilon$; for $p > m$, that bound is automatic. Hence the count gap is at most ε. Each additional cell belongs to exactly one row, so its cost is counted once even when it improves both bounds. Minimizing constructs a certificate of exactly the minimum size. The proof uses joint endpoint attainability, not attainability of every integer between those endpoints.

Row-state dynamic program. For fixed s, t, put $D = pt - b_H - 1$ and discard negative D. For each processed row prefix, a state (n, u, d) stores the minimum additional query count. Here n counts safe rows, u is their upper-cost sum, and d is the accumulated threshold deficit. Initialize $(0, 0, 0)$ with cost zero. For an attainable row state (a, v) and a safe-row indicator $y \in \{0, 1\}$, update

$$\begin{aligned} n' &= n + y, \\ u' &= u + y(u_i^0 - ap_i + v), \\ d' &= d + (t - \ell_i^0 - v)_+, \end{aligned}$$

adding query cost a. Discard states with more than s safe rows, upper cost above b_H or deficit above D; read the minimum at $n = s$ and reconstruct its row subsets and safe set. When $p > m$, omit the deficit coordinate and threshold. Minimizing over the permitted s and thresholds gives Theorem 3, with safety and exclusion coupled through the same row states.

Writing $R = \max_i |\mathcal{D}_i|$, the dense computation for fixed s, t takes $O(mR(s+1)(b_H+1)(D+1))$ time and $O((s+1)(b_H+1)(D+1))$ working memory, apart from witness storage. The implementation stores reachable states. In the single-member family, $p_i \le M$ and row costs are at most TM. If $b_H \ge \sum_i Tp_i$, zero queries suffice; otherwise $b_H < mTM$ and $D \le mTM$. Together with $R \le (T + 1)(TM + 1)$, these bounds make the algorithm polynomial in the explicit counts m, T, M, although the dense state space can be large. For generic integer contributions with binary-encoded large caps, the same budget-indexed algorithm is pseudo-polynomial. The odd-median formulas provide a much smaller specialization.

Complete search cost. Row states can be cached once in $O(\sum_i T|\mathcal{D}_i|)$ time and $O(\sum_i |\mathcal{D}_i|)$ space, apart from stored subsets. There are at most $\min(\varepsilon, L^*) + 1$ safe targets and $TM + 1$ thresholds per target. Multiplying the fixed-s, t bound by these counts bounds total search time; the working state grid is reused. Thus the explicit single-member model is tractable in polynomial time even when its dense implementation is impractical. Binary-encoded generic costs need not satisfy these polynomial bounds. An incumbent query count q_0 permits discarding partial states already costing at least q_0, since remaining query costs are nonnegative. This pruning seeks strict improvement and preserves a previously constructed witness when no improvement exists.

Algorithm 2 Offline minimum joint witness by row states

Require: True $m \times T$ contributions $0 \le f_{ir} \le p_i \le M$, base-pool size M, consistent observations, integer budget $b_H \ge 0$ and integer tolerance $\varepsilon \ge 0$, under Theorem 3.

Construct and cache each $\mathcal{D}_i$; compute L^* from true row costs.
Initialize an incumbent with all remaining active entries.
for $s = \max(0, L^* - \varepsilon), \dots, L^*$ **do**
  Set $p = s + \varepsilon + 1$.
  for each integer $t \in \{0, \dots, TM\}$ with $pt \ge b_H + 1$ (or once without t if $p > m$) **do**
    Initialize $(n, u, d) = (0, 0, 0)$ with cost zero; omit d when $p > m$.
    Apply the row transitions above; keep the cheapest witness for each reachable state.
    Compare feasible terminal states ($n = s$, $u \le b_H$ and, if $p \le m$, $d \le pt - b_H - 1$) against the incumbent.
  end for
end for
return The minimum query count and its reconstructed entries.

Even-sized median pools. The separation between Boolean and shared evidence concerns replacement outcomes, not only the name of the pooling rule.

Corollary 2 (Joint evidence for even medians)**.** *Let M be even and define median consensus as the average of the two central scores. Under the same fixed cutoff, uniformly weighted single-member replacements and completion model without cross-entry constraints, Theorem 3 characterizes the minimum additional entry certificate size. Its row-state dynamic program and outcome-count integer formulation remain valid.*

Proof. Median consensus is coordinatewise nondecreasing. For a base-selected row, replacing any member by 1 preserves selection, while replacing it by 0 maximizes the exclusion indicator for that replacement. Thus $f_i(1) = 0$, $f_i(0) = p_i$, and $f_i(x)$ is a nonincreasing integer-valued function bounded by $p_i \le M$. The Cartesian product of the entry domains makes both endpoint cost vectors jointly attainable. Querying (i, r) raises the lower cost by f_{ir} and lowers the upper cost by $p_i - f_{ir}$. These are precisely the bounds and reachable row states used in Theorem 3; its safety, threshold-exclusion and attainability arguments therefore apply unchanged. □

For example, take two rows with base scores $(.4, .8)$, cutoff $.5$ and one candidate scoring $.4$ on each row. With $k = 1$ and $\alpha = 1/4$, the flip budget is one. Both rows have endpoint cap two and true cost one. The successive count bounds are $(0, 2)$, $(1, 2)$ and $(1, 1)$ after zero, one and two queries. Safety needs one observation and exclusion needs two, while their union needs two. Even medians therefore need not have the disjoint evidence underlying the odd-median closed forms.

Solver implementation and resources. The integer programs use Python 3.9.6, NumPy 2.0.2 and SciPy 1.13.1 on macOS 15.6 arm64, through `scipy.optimize.milp` on an Apple M4 Pro. The solver runtime banner reports HiGHS 1.2.0. The original SciPy calls explicitly set the remaining time limit, zero relative MIP gap and presolve. Later direct-HiGHS calls also set one thread and random seed zero. These call families do not fully record effective feasibility or integrality tolerances for the historical targets; present-day defaults cannot recover them. The explicit settings for the new secondary analysis are reported in Appendix M.8; they do not reconstruct omitted historical settings. A numerical lower bound L_B is converted to $\max(0, \lceil L_B - 10^{-6} \rceil)$; reconstructed witnesses are checked in integer arithmetic. This combines numerical optimality bounds with exact feasibility checks, rather than rational proof certification. Worker processes set BLAS/OpenMP thread limits to one. The environment and attempt ledger report machine details, time limits and retained failures; process memory measurements are not isolated solver peaks.

A compact outcome-count formulation. Invariant selected rows cancel from the count gap. For unit-cost entry witnesses, unqueried entries within the same active row are interchangeable when their true flip contributions are equal. For active row i, let d_{iv} count remaining candidates with contribution $v \in \{0, \dots, p_i\}$. Querying a_{iv} of them gives the same row bounds regardless of which candidates are chosen. This groups integer outcomes without discretizing scores.

For a fixed safe target s, write $p = s + \varepsilon + 1$. Use

integer counts $0 \le a_{iv} \le d_{iv}$, binary safe-row indicators y_i, and continuous products z_{iv} constrained by

$$\begin{aligned} 0 \le z_{iv} &\le a_{iv}, \\ z_{iv} &\le d_{iv} y_i, \\ z_{iv} &\ge a_{iv} - d_{iv}(1 - y_i). \end{aligned}$$

These inequalities make $z_{iv} = a_{iv} y_i$ exactly. Minimize $\sum_{i,v} a_{iv}$ subject to

$$\sum_i y_i = s,$$

$$\sum_i u_i^0 y_i - \sum_{i,v} (p_i - v) z_{iv} \le b_H.$$

If $p \le m$, also use an integer threshold $0 \le t \le TM$ and nonnegative continuous deficits w_i satisfying

$$w_i \ge t - \ell_i^0 - \sum_v v a_{iv},$$

$$pt - \sum_i w_i \ge b_H + 1.$$

For $p > m$, omit the threshold and deficit constraints.

Proposition 4 (Outcome-count equivalence)**.** *Minimizing this program over the same safe targets as Theorem 3 gives the same minimum additional entry count. With $G = \sum_i |\{v : d_{iv} > 0\}| \le m \min(T, M+1)$, each target has $O(G+m)$ variables and constraints.*

Proof. Every entry witness determines counts a_{iv} and a safe set of s rows. The product constraints encode its safe upper-cost sum. Lemma 1 supplies a threshold and deficits satisfying the exclusion constraint. Conversely, choose any a_{iv} distinct remaining entries from each outcome group. They realize lower costs $\ell_i^0 + \sum_v v a_{iv}$ and upper costs $u_i^0 - \sum_v (p_i - v) a_{iv}$. The indicated s rows are safe, and the deficits exclude every p-row set. Each query is counted once. Both mappings preserve objective value, proving equivalence. There are at most $M+1$ outcome groups per row and at most T nonempty groups; the variable and constraint bounds follow. These size bounds do not bound integer-program search time. □

The ungrouped formulation uses one binary variable per unqueried entry, treating each entry as a separate group of capacity one. Both formulations use the true matrix offline. Outcome grouping is specific to unit-cost entries; whole-column actions still depend on candidate identity.

Offline threshold screening. For exact stopping, $s = L^*$ and $p = L^* + 1$. Every partial lower cost satisfies $\ell_i \le F_i$, so a threshold is impossible if

$$pt - \sum_i (t - F_i)_+ < b_H + 1.$$

Discarding these thresholds and solving the compact program at every remaining integer $t \in \{0, \dots, TM\}$ preserves the global minimum. If $p > m$, no exclusion threshold is needed. This decomposition uses true costs offline; it does not expose them to the online policy or assert that a partially solved threshold gives a global optimum.

Separate witnesses as a comparison. For each safe target s, independently minimize safety queries and exclusion queries for $p = s + \varepsilon + 1$, then minimize their sum over the same target range. Call this additive upper bound Q_{split}. It may count shared cells twice and is not the size of an optimally deduplicated union. The union of these separately constructed witnesses is feasible because both bounds improve monotonically under additional observations. Therefore $Q_{\text{joint}} \le Q_{\text{split}}$. Equality holds for median boundary evidence under Proposition 3; mean intermediate contributions can make the inequality strict.

M.6 Joint-certificate validation and NLP subbatches

Independent query-subset enumeration checks 6,316 capped-integer cases, 768 scaled median cases and 400 realizable mean cases with exact rational scores. Histories, endpoint outcomes, intermediate flips, ties, budgets and count tolerances are included. The 768 scaled odd-median cases use empty histories and match Corollary 1. The generator `scripts/check_joint_certificates.py` specifies the enumerated domains and the seeded random instances; subset enumeration is exhaustive within each generated instance, not over every possible input.

For each of six evaluated panels, both pooling rules, four depths and forty splits, take the first four baseline-selected outputs in the original label-blind tie order, or all of them if fewer exist. Reference maps, the cutoff and complete candidate family stay fixed. Set N to this subbatch size, reset $k = \max(1, \text{round}(qN))$ and use $\alpha = .10$. The original q still determines the original reference cutoff; this construction defines a separate batch, not a subset certificate for the original budget. Invariant rows are included and require no queries.

All 1,920 realized subbatches have four outputs; the smaller-batch fallback is not exercised by these records (Table 42).

We compute minima at $\varepsilon \in \{0, 1\}$ from empty history and from a fixed quarter-history: reveal the first $\lfloor mT/4 \rfloor$ eligible cells in seeded row then candidate-index order. These 1,920 batch configurations give 7,680 joint minima and matching Q_{split} comparisons. A separate binary integer optimization enumerates all safe and excluded output subsets, instead of using the threshold identity or row states. All 7,680 small-batch optimization checks report zero optimizer optimality gap; their witnesses are checked in integer arithmetic. This solver gap is distinct from the retained-count gap. Both independent formulations return the same minimum additional joint query count on all 7,680 tasks, with no unresolved instances. Complete records include reconstructed query sets and safe masks.

Under exact mean certification, $Q_{\text{joint}} < Q_{\text{split}}$ in 186/960 empty-history instances; their differences total 306 entries, or .319 per instance. With fixed quarter-history, 105 instances have a total difference of 157 additional entries. At $\varepsilon = 1$, these counts are 74 and 23, respectively. All 3,840 median history/tolerance comparisons have zero difference. These reductions from an additive upper bound are modest and instance-dependent. The sum of separate minima remains valid but need not equal the minimum joint certificate size.

The four original complete acquisition policies are also run on these same batches from empty history through exact stopping, retaining their original reference-fitted predictions and tie conventions. Their first stops at $\varepsilon = 0$ and one give 15,360 records. Table 43 reports mean-pooling discovery ratios at $q = .05$, averaged over splits with positive matching subbatch minima. Full-panel mean acquisition is not divided by these small-batch minima. These analyses use previously exposed panels and provide descriptive evidence.

Even-median validation. For pools of size two and four, exact rational scores on the quarter-step grid generate 3,690 two-row cases with two candidates, cutoff ties, endpoint and intermediate outcomes, empty and partial histories, and exact and positive-gap targets. A separate reconstruction enumerates all 38,887 eligible query subsets across these cases and agrees with the saved joint minima. These finite checks complement Corollary 2. The scripts `check_even_median.py` and `verify_rational_cases.py` in `review-revision/even-median-resource-20261007/` specify case generation and independent exhaustive reconstruction, respectively.

We also fix even pools of size six and eight on all six NLP panels, using the first M evaluators in the existing order and all remaining evaluators as candidates. Six panels, two even pool sizes, 40 splits, four depths and two stopping targets give 3,840 full-batch tasks. Each task refits its consensus cutoff on the same reference outputs, retains right-ECDF normalization and $\alpha = .10$, and uses either the primary gap or exact stopping. Integer ECDF rank numerators make comparisons of the two central scores exact, including cutoff ties. Per-task records retain lower and feasible upper query bounds after a nominal 60-second outcome-MILP budget. Table 44 averages those bounds over 160 targets per row. These exposed-panel, offline results neither evaluate a new acquisition policy nor compare $M = 6, 7, 8$ as the same selection task.

M.7 Resource scaling and full-test mean evidence

Matched-target resource comparison. Table 45 compares row-state DP, entry MILP and outcome-count MILP on the same 1,920 original full-mean targets. Methods run sequentially in fresh single-thread processes on the same workstation, once per target, with their order rotated deterministically across sorted targets. The nominal 60-second algorithm budget includes row-state or model construction; imports and input loading precede timing. Construction time is deducted before the MILP call receives the remaining budget. No external wall-clock deadline terminates the process. All 111 attempts exceeding 60 seconds remain unresolved; measured elapsed times, including the maximum 66.91 seconds, are retained. DP additionally permits at most 100,000 states and 10 million transitions. We record sampled whole-process RSS, final query bounds, model dimensions and status for all 5,760 attempts, including unresolved cases. The all-entry witness supplies the initial feasible bound; time to the first improving solver incumbent was not measured. Previously obtained numerical minima check every reported interval and new numerical closure in this resource study.

An interrupted attempt has no recoverable elapsed-time measurement and is excluded from

Panel	q	Q_{joint}	Q_{split}	$Q_{\mathrm{joint}} < Q_{\mathrm{split}}$ splits/40	$Q_{\mathrm{joint}}(\mathcal{O})$	$Q_{\mathrm{split}}(\mathcal{O})$	$Q_{\mathrm{joint}} < Q_{\mathrm{split}}$ splits/40
RewardBench 2	0.05	19.62	19.95	13	13.85	13.95	4
RewardBench 2	0.10	19.05	19.30	6	13.30	13.45	3
RewardBench 2	0.20	13.95	14.47	11	10.72	11.05	8
RewardBench 2	0.30	12.93	13.07	5	9.55	9.70	5
WMT23, 13 streams	0.05	39.17	39.30	4	25.48	25.57	4
WMT23, 13 streams	0.10	49.33	49.65	8	35.92	36.05	3
WMT23, 13 streams	0.20	21.05	21.18	5	15.25	15.32	3
WMT23, 13 streams	0.30	15.90	16.40	8	10.88	11.10	3
WMT23, 12 streams	0.05	43.25	43.25	0	29.52	29.52	0
WMT23, 12 streams	0.10	46.38	46.42	2	33.08	33.08	0
WMT23, 12 streams	0.20	19.30	19.62	5	13.82	14.00	3
WMT23, 12 streams	0.30	19.07	19.60	13	13.18	13.35	7
WMT24 en-de	0.05	37.85	38.08	7	27.35	27.50	5
WMT24 en-de	0.10	32.55	32.73	5	23.48	23.50	1
WMT24 en-de	0.20	21.32	21.73	12	14.72	14.97	8
WMT24 en-de	0.30	18.82	19.10	6	13.07	13.25	5
WMT24 en-es	0.05	40.67	40.95	10	28.98	29.12	5
WMT24 en-es	0.10	29.98	30.05	3	21.88	21.90	1
WMT24 en-es	0.20	23.02	23.23	6	16.23	16.38	4
WMT24 en-es	0.30	18.55	18.70	3	13.20	13.30	2
WMT24 ja-zh	0.05	40.12	40.70	13	28.40	28.75	8
WMT24 ja-zh	0.10	40.25	40.83	12	29.65	29.82	6
WMT24 ja-zh	0.20	36.83	37.35	11	28.50	28.85	7
WMT24 ja-zh	0.30	28.57	29.55	18	20.77	21.20	10

Table 42: Exact mean entry certificates on the predefined subbatches of at most four baseline-selected outputs. Each row averages 40 splits, retaining original maps, cutoff and all candidates while resetting N, k and b_H for this subbatch. The first group starts empty; the second reports minimum additional entries after observing the first quarter of eligible cells in seeded row and candidate-index order. Q_{split} adds independently minimized safety and exclusion sizes over the same safe-target range. The $Q_{\mathrm{joint}} < Q_{\mathrm{split}}$ columns count splits below the additive bound; they do not compare against the best deduplicated union. An independent subset formulation gives matching numerical optima; reconstructed witnesses pass integer feasibility checks. These are not minima for the original full test batches.

completed-attempt timing summaries. The run ledger preserves the interruption and subsequent continuation. Pauses between attempts permitted separate validation and compilation; the timings do not establish performance on an otherwise idle machine. Complete task-level records include m, T, G, model sizes and resource-limit reasons.

Resource tails and instance structure. Table 46 reaggregates those same outcome-count attempts by G. Within $2000 \le G < 5000$, 205/218 primary attempts close, and the 95th percentile reaches 60.03 seconds. The three primary instances with $G \ge 5000$ all close within 0.41 seconds. Larger group counts do not consistently require longer solution times on this grid. Panel, active count, budget and exclusion structure vary together; these are descriptive strata rather than controlled interventions. Saved summaries include m, T, G, maximum formulation sizes, status counts and timing tails, including every unresolved attempt.

The cached row-state implementation constructs each row table once, removes invariant query actions and prunes states whose query count cannot improve a feasible full-remaining-entry incumbent. We fixed six panels, splits 0–7, $q \in \{.05, .30\}$ and selected sizes 4, 8, 16, 32, 64 and full before examining these new runs. Empty and fixed quarter-histories with exact and primary gaps give 2,304 cases. All saved incumbents satisfy their integer certificate conditions. Independent subset MILP checks 396 predefined cases: every case of size at most four and split-zero empty exact eight-row cases. Larger completed DP minima are not all independently subset-enumerated. Table 47 reports completed pairs and resource-limited attempts, including the limits.

For the matching 576 empty-history exact tasks, entry and outcome-count MILPs each numerically close 573 joint/separate pairs within the primary 60-second budget (Table 48). The two formulations agree on every pair both numerically close; they also agree with all 264 completed joint DP minima; 263 of these tasks also have a completed separate optimum and enter the paired summary. Both addi-

Panel	ε	Mean Q_{joint}	n_+	Fixed	Random	Width	Reference
RewardBench 2	0	19.62	40	1.405	1.447	1.456	1.288
RewardBench 2	1	12.72	40	1.741	2.029	2.180	1.710
WMT23, 13 streams	0	39.17	40	1.795	1.844	1.836	1.774
WMT23, 13 streams	1	23.18	40	2.666	3.179	3.312	3.061
WMT23, 12 streams	0	43.25	40	1.587	1.658	1.642	1.406
WMT23, 12 streams	1	26.75	40	2.090	2.431	2.481	2.080
WMT24 en-de	0	37.85	40	1.562	1.591	1.573	1.474
WMT24 en-de	1	24.48	40	1.956	2.362	2.452	2.130
WMT24 en-es	0	40.67	40	1.384	1.421	1.422	1.324
WMT24 en-es	1	26.82	40	1.689	2.127	2.215	1.858
WMT24 ja-zh	0	40.12	40	1.563	1.563	1.561	1.361
WMT24 ja-zh	1	25.68	40	1.954	2.156	2.217	1.818

Table 43: Mean-pooling discovery ratios from empty history on the same predefined subbatches at $q = .05$, $\alpha = .10$. The four policies use their original full definitions, including WMT24 priorities. Policy columns average $Q_{\text{online}}/Q_{\text{joint}}$ over the n_+ splits with positive matching minima; zero minima have undefined ratios. Mean Q_{joint} counts entry observations; policy columns are dimensionless mean splitwise ratios. Counts and certificate gaps concern this separate small-batch task, not the full-panel acquisition records.

tionally match all 7,680 earlier small-batch minima. Append-only 600-second retries close two more pairs. All 576 outcome-count pairs are now closed. The last 64-row ja-zh task has joint minimum 555 and separate minimum 569. Its variable-threshold run and a diagnostic 600-second retry had retained bounds 554–555. Offline screening left exactly $t = 8, \ldots, 16$; each of the nine fixed-threshold programs attained matching numerical primal and dual objectives; the minimum across their nine optimal values was 555. The resource-limited records are retained.

The 96 full instances preserve original test N, nominal k, maps, cutoff, candidate family and budget. Their active-row counts range from 53 to 1,860; the largest outcome formulation has 11,116 variables and 13,448 constraints. Twelve additional split-zero full-instance probes sample process RSS every 10 ms: peaks range from 88.1 to 115.4 MiB, including Python, inputs, solver and resident caches. Concurrent load and sampled allocation preclude a controlled backend speedup or exact solver-memory claim.

To vary dimensions beyond the cached panels, we fix $N \in \{2048, 8192\}$, $M \in \{7, 15\}$ and $T \in \{17, 65\}$ before running the complete outcome-count joint model. Base and candidate scores are independently drawn from $U(0, 1)$ in a declared PCG64 order, with seed $731191 + N + 100M + T$. Mean pooling uses a construction-fixed cutoff .5, $q = .30$, $\alpha = .10$, empty history and exact stopping. Every selected row remains in the problem. Each target runs in its own process, with one HiGHS thread, zero relative MIP gap and a nominal 60-second budget including model work. Protocol and version records accompany `review-revision/discovery-bound-implementation/scale/`.

Four of eight targets numerically close; the other four retain their bounds in Table 49. Independent explicit replacements reconstruct all costs and validate every feasible witness. The largest model has 39,097 variables and 52,154 constraints; measured process peaks span 156.0–635.1 MiB. At the largest grid point, the true optimum retains every selected output, making exclusion automatic; this instance closes quickly. Thus dimensions alone do not determine difficulty. These constructed-score, exact-stop probes measure computational scope, not NLP generalization, evaluator inference time or a uniform scalability guarantee.

All 96 full mean minima are positive. The four original exact-stop policies give 384 same-instance discovery counts, all at least the matching minimum (Table 50). In 4/96 instances, Q_{joint} is below the additive Q_{split}, by 2,645 of 435,876 separately counted entries (0.607%). Appendix M.8 separates double counting from improvements beyond optimal deduplication. A compact exact formulation therefore extends the evidence analysis to full tasks; these records do not establish broadly large gains from sharing or an optimal online policy. The eight overlapping splits and previously exposed panels remain descriptive.

Panel	M	T	Stop	Closed	Zero	Mean LB	Mean UB
RewardBench	6	8	primary	160	0	355.8	355.8
RewardBench	6	8	exact	160	0	434.2	434.2
RewardBench	8	6	primary	160	62	55.8	55.8
RewardBench	8	6	exact	160	47	78.5	78.5
WMT23 (13 streams)	6	18	primary	160	143	9.2	9.2
WMT23 (13 streams)	6	18	exact	160	123	23.0	23.0
WMT23 (13 streams)	8	16	primary	160	159	0.019	0.019
WMT23 (13 streams)	8	16	exact	160	155	1.6	1.6
WMT23 (12 streams)	6	18	primary	160	141	20.5	20.5
WMT23 (12 streams)	6	18	exact	160	126	41.7	41.7
WMT23 (12 streams)	8	16	primary	160	156	3.5	3.5
WMT23 (12 streams)	8	16	exact	160	150	8.0	8.0
WMT24 en–de	6	15	primary	160	115	298.8	298.8
WMT24 en–de	6	15	exact	160	92	522.3	522.3
WMT24 en–de	8	13	primary	160	157	4.0	4.0
WMT24 en–de	8	13	exact	160	151	17.4	17.4
WMT24 en–es	6	15	primary	160	113	241.4	241.4
WMT24 en–es	6	15	exact	160	86	420.6	420.6
WMT24 en–es	8	13	primary	160	158	4.3	4.3
WMT24 en–es	8	13	exact	160	155	8.5	8.5
WMT24 ja–zh	6	16	primary	160	93	447.0	447.0
WMT24 ja–zh	6	16	exact	160	82	628.4	628.4
WMT24 ja–zh	8	14	primary	160	147	48.5	48.5
WMT24 ja–zh	8	14	exact	160	140	71.2	71.2

Table 44: Even-median joint evidence on complete test batches, averaging all four depths and 40 splits (160 targets) per row. Closed counts numerically closed minima; Zero counts zero-query certificates. Each target uses $\alpha = .10$ and either $\varepsilon = \lceil .02|S_0| \rceil$ (primary) or zero (exact). Mean lower and upper bounds include every target. The positive mean below 0.1 is shown to three decimals; an unresolved target is not replaced by a claimed minimum. Pools of size six and eight use the first members in the frozen evaluator order. Changing M also changes the candidate family and cutoff, so these are separate certification tasks.

Method	Stop	Closed	Mean s	Median s	Max s	RSS MiB	Gap
Row DP	primary	4	8.82	9.04	12.34	138.6	7371.6
Row DP	exact	82	6.83	7.20	10.57	137.7	7279.6
Entry MILP	primary	920	4.09	0.46	62.50	1116.9	169.2
Entry MILP	exact	933	2.26	0.07	66.91	1081.8	0.7
Outcome MILP	primary	931	2.21	0.05	60.12	618.9	132.7
Outcome MILP	exact	945	1.49	0.02	60.34	420.4	0.1

Table 45: Sequential single-thread measurements on the same 1,920 full mean targets, with 960 targets per stopping criterion. RSS is the maximum sampled whole-process memory across attempts in the row; Gap averages final feasible query counts minus numerical lower bounds over all 960 attempts in each row, each retaining a feasible incumbent. Each attempt has a nominal 60-second algorithm budget, not an external wall-clock cutoff; DP also retains its 100,000-state and 10-million-transition caps. All 5,760 saved attempts with resource measurements enter timing and memory summaries, including attempts with unresolved targets; the interrupted attempt without recoverable timing is excluded. RSS is sampled whole-process memory, not solver-only allocation.

G	Stop	Closed / attempts	m range	Max vars	Median s	95th s	Max s
0–499	exact	173/174	53–276	1227	0.008	0.07	60.01
0–499	primary	173/174	53–276	1233	0.014	0.10	60.01
500–1999	exact	557/565	170–921	5181	0.018	1.63	60.12
500–1999	primary	550/565	170–921	5183	0.042	1.80	60.11
2000–4999	exact	212/218	706–1860	11116	0.082	15.32	60.34
2000–4999	primary	205/218	706–1860	11116	0.179	60.03	60.12
5000+	exact	3/3	1907–2025	12381	0.170	0.17	0.17
5000+	primary	3/3	1907–2025	12381	0.360	0.40	0.40

Table 46: Outcome-count MILP resource strata from the existing 1,920 full-mean attempts in Table 45. G counts row–outcome groups and m active rows. All attempts, including unresolved ones, enter the time summaries; quantiles use linear interpolation. Closure uses the original nominal 60-second algorithm budget. These are posthoc descriptive strata: G, panel, active count and budget co-vary, so differences are not controlled scaling effects. Machine-readable summaries also stratify by T.

Selected size	Both closed	Median s	90th s	90th MiB
4	384/384	0.010	0.276	207.5
8	318/384	0.273	4.157	195.5
16	202/384	3.287	5.976	174.4
32	60/384	4.990	7.538	155.5
64	62/384	5.733	8.384	158.0
full	16/384	6.587	12.915	187.5

Table 47: Cached row-state DP resource study: six panels, splits 0–7, $q \in \{.05, .30\}$, empty/quarter histories and exact/primary gaps give 384 instances per size. Subbatches reset N, k, b_H; full preserves the original test task. Both closed counts completed joint and separate minima. Limits are 100,000 reachable states, ten million transitions and 60 seconds shared by the two optimizations. Resource-limited rows retain valid witnesses without claiming minima. Times include unresolved attempts; RSS is sampled every 10 ms and includes Python, inputs and caches. Six concurrent workers share a workstation; these are not controlled speedup measurements.

Size	Row DP	Entry MILP	Outcome MILP	Outcome vars	Median s
4	96/96	96/96	96/96	45	0.017
8	81/96	96/96	96/96	76	0.033
16	48/96	96/96	96/96	132	0.059
32	14/96	95/96	95/96	281	0.107
64	16/96	95/96	95/96	497	0.154
full	8/96	95/96	95/96	11116	0.128

Table 48: Matching empty-history exact mean tasks at $\alpha = .10$: 96 cases per size. Counts require both joint and separate optima within the primary resource budget. Each MILP backend receives 60 seconds shared between these optimizations; DP additionally has state/transition limits. Outcome MILP groups equal integer flip contributions (Proposition 4); its median time includes unresolved attempts. Outcome MILP numerically closes 573/576 joint-and-separate pairs in the primary run. Append-only retries numerically close two more pairs; exhaustive feasible-threshold decomposition numerically closes the last joint minimum, completing all 576 pairs. Initial censoring remains in this table. Outcome vars counts the largest outcome-count formulation represented in each row, not a bound on solve time.

N	M	T	m	G	Joint bounds	Seconds	Peak MiB	Closed
2048	7	17	790	3182	12778–12778	24.24	158.2	yes
2048	7	65	759	3883	46602–46673	60.18	305.8	no
2048	15	17	634	3483	10648–10648	21.08	226.6	yes
2048	15	65	619	5122	39767–39767	22.92	311.6	yes
8192	7	17	3178	13219	0–54026	61.56	309.5	no
8192	7	65	3142	16079	192553–194598	63.20	635.1	no
8192	15	17	2529	13747	42795–42830	61.67	595.2	no
8192	15	65	2419	19294	132536–132536	2.01	156.0	yes

Table 49: Controlled mean-score resource grid: all eight predefined exact joint targets. Bounds are a numerical lower bound and an independently checked feasible entry count; zero can be an uninformative lower bound. Seconds include model construction, optimization and witness validation, which can exceed the nominal 60-second budget. Peak process memory includes inputs and Python. One instance per point and concurrent background work preclude a calibrated runtime scaling law.

Panel	q	Mean Q_{joint}	Fixed	Random	Width	Reference
RewardBench 2	0.05	668.4	1.141	1.141	1.145	1.095
RewardBench 2	0.30	1560.6	1.336	1.334	1.204	1.359
WMT23, 13 streams	0.05	1689.9	1.174	1.173	1.176	1.102
WMT23, 13 streams	0.30	1851.8	1.504	1.516	1.289	1.631
WMT23, 12 streams	0.05	1735.3	1.125	1.129	1.128	1.082
WMT23, 12 streams	0.30	2370.9	1.476	1.476	1.295	1.524
WMT24 en-de	0.05	4698.9	1.132	1.132	1.133	1.065
WMT24 en-de	0.30	9857.6	1.295	1.295	1.212	1.283
WMT24 en-es	0.05	3920.0	1.116	1.117	1.117	1.060
WMT24 en-es	0.30	7553.6	1.322	1.322	1.222	1.345
WMT24 ja-zh	0.05	4289.8	1.142	1.150	1.147	1.090
WMT24 ja-zh	0.30	13957.3	1.271	1.271	1.219	1.174

Table 50: Full-test mean entry minima and online discovery from empty history at exact stopping, $\alpha = .10$, splits 0–7. Every row averages eight original full tasks with positive minima; ratio columns average the splitwise $Q_{\mathrm{online}}/Q_{\mathrm{joint}}$, not a ratio of means. All four original query paths are unchanged. Across 96 minima and 384 paths, every online count is at least its matching minimum. Mean Q_{joint} counts entry observations; policy columns are dimensionless. Joint minima are below the additive sum in 4/96 tasks, by 2,645 of 435,876 separately counted entries (0.607%). The best deduplicated comparison is in Table 52. These denominators are not used for positive-tolerance stops.

M.8 Joint evidence beyond deduplicating separate optima

For each safe target s in Theorem 3, let $\mathcal{A}_s$ be the set of all minimum-size additional safety witnesses, and let $\mathcal{B}_{s+\varepsilon+1}$ be the set of all minimum-size additional exclusion witnesses. Exclusion is automatic beyond the active-row count, with the empty witness. Define

$$Q_{\cup,\mathrm{sep}} = \min_{s} \min_{A\in\mathcal{A}_s,\, B\in\mathcal{B}_{s+\varepsilon+1}} |A \cup B|.$$

The history is fixed and not charged again. This comparison ranges over all tied component optima, rather than one pair returned by a solver.

Proposition 5 (Deduplicated separate evidence)**.** *For unit-cost entry certification under the same instance, history and tolerance,*

$$Q_{\mathrm{joint}} \le Q_{\cup,\mathrm{sep}} \le Q_{\mathrm{split}} \le 2Q_{\mathrm{joint}}.$$

Within each active row, group unqueried entries with equal true contribution. If two witnesses choose a_g, b_g entries from a group, their minimum possible union size is $\max(a_g, b_g)$. Minimizing $\sum_g \max(a_g, b_g)$ at separately optimal component cardinalities exactly characterizes $Q_{\cup,\mathrm{sep}}$.

Proof. The union preserves safety and exclusion by monotonicity, proving the first inequality. Choosing the target and witnesses attaining the additive bound gives the second. For the last inequality, take a minimum joint witness X and its safe target. The same X is feasible for each component, so each component minimum is at most $|X|$. Their sum at this target is at most $2|X|$, and minimizing over targets cannot increase it. This also covers $Q_{\mathrm{joint}} = 0$, automatic exclusion and fixed prior observations. Within a contribution group, identities do not affect either row bound. Nested choices attain union size $\max(a_g, b_g)$; no union can contain fewer than either component. Summing and optimizing over targets proves the characterization. This exchange argument uses unit-cost entry access; column identities need not be interchangeable. □

What a query history can establish about joint advantage. Instance-wise savings concern the realized hidden scores. A stronger question asks whether a history alone establishes strict savings for every consistent completion. Here each Q measures the cost of additional reveal actions after that history; past observations are free. Both component minima use the same action family.

Proposition 6 (No uniform residual advantage from a history)**.** *Fix any consistent history $\mathcal{O}$ under the completion model of Section 4 with no cross-entry constraints, an integer count tolerance $\varepsilon \ge 0$, and either entry or whole-column queries with fixed nonnegative additive action costs. There exists a completion consistent with $\mathcal{O}$ for which*

$$Q_{\mathrm{joint}}(\mathcal{O}) = Q_{\cup,\mathrm{sep}}(\mathcal{O}) = Q_{\mathrm{split}}(\mathcal{O}).$$

Consequently, no history can certify a strictly positive residual joint advantage uniformly over all its consistent completions.

Proof. Complete every unobserved selected-row score at its zero-flip endpoint. This is admissible because no cross-entry constraints exclude that simultaneous assignment. The completed row costs equal their current lower costs ℓ^0. Let $L_0 = V(\ell^0) - m_0$ be the optimal active count in this completion. Every additional query reveals zero contributions, so the optimistic active count remains L_0. A gap certificate must therefore establish safety for at least $s_0 = \max(0, L_0 - \varepsilon)$ active rows. Conversely, any such safety witness is a joint certificate: the history already excludes every count above L_0, and $s_0 + \varepsilon \ge L_0$. The

minimum safety witness for s_0 attains the joint minimum. Pairing it with the empty additional exclusion witness gives a separate union and additive cost of the same value. The inequalities in Proposition 5, or the fixed-action version below, give equality throughout. This also covers an already satisfied gap and zero-cost actions. □

The proposition concerns residual minima, not the original empty-history minima of an instance eventually revealed by queries. It does not rule out predicting realized savings or designing efficient acquisition. A uniformly valid positive residual test requires an additional constraint that excludes the zero-contribution completion. A statistical prediction and a deterministic certificate then have different roles.

Discovery at fixed pool size and risk budget. The separation between a short witness and finding it persists with the numerical parameters of the primary study. It concerns exact stopping, not the primary positive tolerance.

Proposition 7 (Exact discovery with fixed parameters)**.** *For every integer $h \geq 0$, there is a median instance with $M = T = 7$, $q = 1/20$, $\alpha = 1/10$, $k = 20h + 17$, $N = 20k$ and $|S_0| = k$ whose minimum exact entry certificate has size one. Every deterministic policy correct under the full completion model has an instance in this family requiring at least $49h + 42$ queries. Every randomized always-valid policy has a fixed instance with expected cost at least $(49h + 43)/2$. Thus no batch-size-independent competitive factor holds for exact discovery even with these parameters fixed.*

Proof. Set $m = 7h + 6$. The m selected boundary rows have four base scores .75 and three .25. A further $k - m$ selected rows have all base scores .75, and the remaining $N - k$ rows have all base scores .25. At cutoff $t_0 = 3/5$, only the boundary rows can change membership. Candidate scores are .25 everywhere except one active candidate–output entry with score .75. Each active opposition contributes four flips; the unique agreement contributes zero.

These normalized values and cutoff can be realized by the reference procedure. Use 20 reference outputs indexed $a = 0, \ldots, 19$. For evaluator $j = 0, \ldots, 13$, set its raw reference scores to $1 + ((a + 3j) \bmod 20)$. Each evaluator's right ECDF maps raw scores 5 and 15 to .25 and .75. The first seven reference columns have maximum median consensus $3/5$. Since $k_{\text{ref}} = 1$, the fitted cutoff is $3/5$. The reference panel is fixed across all instances and supplies no information about the agreement's location.

The flip budget and opposition budget are

$$b_H = \lfloor 2k\alpha MT \rfloor = 196h + 166,$$
$$b = \lfloor b_H/4 \rfloor = 49h + 41 = mT - 1.$$

All k selected outputs are truly feasible. Initially, however, the upper costs allow only $k - 1$ and the lower costs allow k. Observing the unique agreement reduces the total upper cost from $4mT$ to $4(mT - 1) \leq b_H$, certifying all k. Hence the exact offline minimum is one.

Before that entry is observed, the all-opposition completion remains consistent and permits only $k - 1$ selected outputs, while a completion with an agreement at any unseen active entry permits k. These two completions prevent exact stopping. The policy must be correct on the full completion set; it is not given a promise that exactly one agreement exists. Place the agreement last on a deterministic policy's all-opposition path to force mT active queries. For a uniformly chosen agreement location, any fixed random seed induces an order with expected discovery position at least $(mT + 1)/2$. Averaging first over locations and then over seeds gives a fixed location with at least that expected cost. Queries outside the active entries can only add cost. □

The initial gap is one. Therefore every $\varepsilon \geq 1$, including the primary $\lceil .02|S_0| \rceil$ criterion, allows stopping without queries on this family. The lower bound neither explains the measured positive-tolerance overhead nor establishes how often hard instances occur in NLP. It shows why a short exact offline certificate alone does not imply an efficient exact online policy. The reference construction, explicit replacement checks for all 273 agreement positions at $h = 0, 1, 2$, and exact-arithmetic checks of 1,003 values of h are provided in `review-revision/discovery-bound-implementation/`; finite checks supplement the proof.

Discovery at the primary positive tolerance. A different construction keeps the primary gap nontrivial as the batch grows.

Proposition 8 (Positive-tolerance discovery with fixed parameters)**.** *For every integer $m \geq 2$, put $e = \lceil m/50 \rceil$, $s = m - e - 1$, $k = 5s + 4$ and*

$N = 20k$. *There is a family of mean-pooling instances with $M = T = 7$, $q = 1/20$, $\alpha = 1/10$ and $|S_0| = m$, using the same reference-fitted maps and cutoff, whose minimum entry certificate at the primary tolerance $\varepsilon = e$ has size two. Every deterministic policy correct under the full completion model requires at least m queries on some instance in this family. Every randomized always-valid policy has a fixed instance with expected cost at least $(m + 1)/2$. Consequently, no competitive factor independent of batch size holds even at the primary positive tolerance.*

Proof. Use 20 reference outputs and raw scores $1 + ((a + 3j) \bmod 20)$ for output $a = 0, \ldots, 19$ and evaluator $j = 0, \ldots, 13$. Each right ECDF maps raw rank v to $v/20$. Since $k_{\mathrm{ref}} = 1$, the fitted mean cutoff is the largest reference consensus, $t_0 = 83/140$. Give every selected test row the base scores

$$(1, 3/20, 3/10, 9/20, 3/5, 3/4, 9/10),$$

whose mean equals t_0. Every base score is positive, so a candidate score zero causes seven flips, whereas score one causes none. Set all other base rows to zero; they remain unselected even after replacing one base score by one, since $1/7 < t_0$. These test scores are realized by the same reference ECDFs, using raw ranks or zero.

All candidate scores on selected rows are zero except two entries in one hidden row, in fixed candidate columns, whose scores are one. Thus every selected row has cap seven and upper cost 49 initially. The hidden row has true cost 35; every other selected row has true cost 49. The common reference data reveal neither the hidden row nor its test scores. The budget is

$$b_H = \lfloor 2k\alpha MT \rfloor = 49s + 39.$$

The initial safe and optimistic counts are s and m, with gap $e + 1$. Revealing both exceptional entries lowers one upper cost to 35. The safe count becomes $s + 1$, the optimistic count remains m, and the gap becomes e. No single entry suffices. A zero-flip reply lowers one upper cost only to 42, leaving the safe count s because $42 > 39$; a seven-flip reply leaves all upper costs unchanged. In either case the total lower cost is at most seven and all m rows remain optimistically feasible. Hence the minimum is exactly two.

Consider any history of at most $m - 1$ queries with only seven-flip replies on selected rows. Its lower-cost sum is at most $7(m-1) \le b_H$, while its upper costs remain 49, so the gap stays $e+1$. Here $e \le m/2$ for $m \ge 2$, which verifies the displayed budget inequality. At least one selected row has not been queried. Placing the two exceptional entries in that row gives a family member consistent with the history. A deterministic policy therefore cannot finish before m queries in the worst case. Queries on invariant rows cannot shorten this history.

For randomized policies, choose the hidden row uniformly among the m selected rows, independently of the policy randomness. After $t < m$ queries, at most t rows have been touched along the all-seven-flip path. With probability at least $1 - t/m$, the hidden row is untouched and the policy cannot stop. If Q is its query count, then

$$\begin{aligned} \mathbb{E}Q &\ge \sum_{t=0}^{m-1} \Pr(Q > t) \\ &\ge \sum_{t=0}^{m-1} (1 - t/m) = (m + 1)/2. \end{aligned}$$

Averaging over rows gives a fixed instance with at least this expected cost. Policies must be valid on the full completion set; the exceptional-row pattern is not supplied as a promise. Since $N = 20(5s+4)$ grows linearly with m, the cost ratio to the two-entry minimum is unbounded in batch size. □

This family uses the primary tolerance and fixed numerical parameters but changes the score geometry and selected fraction as specified above. It establishes a worst-case discovery obstruction, not the frequency or cause of overhead on NLP panels. Exact score reconstruction checks the reference cutoff and replacement outcomes. Independent enumeration checks all 16,384 and 2,097,152 entry subsets for $m = 2, 3$; further finite checks cover 2,002 parameter values and 63,466 adversarial histories. These checks supplement the proof (`review-revision/stable-batch-implementation/check_positive_discovery.py`).

Fixed query actions. For a fixed family of reveal actions with nonnegative additive costs, define each component optimum using the same actions, history and safe-count target. Define the best separate union by minimizing its action cost over all tied component optima and admissible targets. Then

$$Q_{\mathrm{joint}} \le Q_{\cup,\mathrm{sep}} \le Q_{\mathrm{split}} \le 2Q_{\mathrm{joint}}.$$

A union preserves both component certificates and costs no more than their separately charged sum. A minimum joint action set is feasible for each component at its attained safe target, so each component minimum costs at most that joint set. These two observations prove the inequalities. The bound applies to fixed-price column actions; the within-row exchange formula for entries does not apply to columns. History-dependent batching and loading costs require a different cost model.

Strict column advantage is possible even though the entry tight family does not establish column tightness. Take mean pooling with $M = T = 3$, $N = 2$, $k = 1$, cutoff $1/2$ and flip budget five, corresponding to $\alpha = 5/18$. The two base-score rows are $(2/5, 3/5, 4/5)$ and $(1/10, 7/10, 9/10)$. Candidate-score rows $(2/5, 2/5, 2/5)$ and $(2/5, 3/5, 2/5)$ give caps $(3, 2)$ and flip contributions

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \end{pmatrix}.$$

Exactly one row can be retained. Column two alone is the unique one-column safety witness. Columns one and three form the unique minimum exclusion witness: their lower costs sum to six, while either other pair sums to five. That pair also certifies row one safe, with upper cost five. Thus $Q_{\text{joint}} = 2$ and $Q_{\cup,\text{sep}} = Q_{\text{split}} = 3$ columns. This realizes a ratio of $3/2$, without establishing a tight factor-two column separation.

Corollary 3 (Intermediate contributions bound joint advantage)**.** *Under Theorem 3, let $\mathcal{X}^*$ be the set of all minimum additional joint witnesses for the fixed history and tolerance. For $X \in \mathcal{X}^*$, define*

$$I(X) = \{(i, r) \in X : 0 < f_{ir} < p_i\},$$
$$d_* = \min_{X \in \mathcal{X}^*} |I(X)|.$$

Then

$$Q_{\cup,\text{sep}} \le Q_{\text{split}} \le Q_{\text{joint}} + d_*.$$

In particular, an optimal witness with no intermediate contributions implies $Q_{\cup,\text{sep}} = Q_{\text{split}} = Q_{\text{joint}}$. This includes endpoint-only unobserved contributions, without requiring equal row caps.

Proof. Take any $X \in \mathcal{X}^*$ and its certified safe target s. Retain only entries with $p_i - f_{ir} > 0$ to form A_X. Removing the other entries leaves every upper cost unchanged, so A_X still certifies safety at s. Retain only entries with $f_{ir} > 0$ to form B_X. Every lower cost is unchanged, so B_X still certifies exclusion at $s + \varepsilon + 1$. When exclusion is automatic, the empty witness also suffices. Previously observed entries are held fixed throughout.

The component minima at this target are at most $|A_X|$ and $|B_X|$. Their sum is at most

$$|A_X \cup B_X| + |A_X \cap B_X| \le |X| + |I(X)|.$$

Minimizing over safe targets and then over $X \in \mathcal{X}^*$ proves the bound. The equality conclusion follows from Proposition 5. □

Strict joint advantage requires $d_* > 0$, but $d_* > 0$ does not imply strict advantage. It holds for arbitrary row counts, prior observations and integer count-gap tolerances. It does not provide a general structural characterization of strict gain. For a particular optimal witness X, $|I(X)|$ is a valid, possibly looser upper bound on the gain; obtaining d_* requires minimizing over all tied joint optima.

Direct query-subset enumeration checks 14,626 empty-history instances with caps one or two and at most three rows and two candidates, plus 1,200 seeded instances with arbitrary histories, zero caps and up to four rows. Every count tolerance from zero to the row count is checked. Separately optimal witnesses include every tie. These are finite formula checks; they neither certify the proof independently nor constitute NLP experiments.

Minimizing intermediate evidence. We compute d_* on the existing 1,920 full mean targets by fixing the previously numerically closed cardinality Q_{joint} and minimizing $\sum_{i,v} \mathbf{1}\{0 < v < p_i\} a_{iv}$ in the outcome-count formulation, subject to $\sum_{i,v} a_{iv} = Q_{\text{joint}}$. Here a_{iv} is the number of queried entries from outcome group (i, v). Every admissible safe-count target remains eligible; within-row groups preserve all contribution multiplicities. This is a secondary objective over the full joint-optimal set, rather than a count from one saved witness. When the true optimal active count is m, exclusion is automatic and the smallest admissible safe count suffices by monotonicity.

Exact full retention has a simpler integer solution. Let $j = Q_{\text{joint}}$ and let $E(a)$ and $I(a)$ sum the a largest reductions $p_i - f_{ir}$ among unobserved endpoint and intermediate cells, respectively. Let $D = \sum_i u_i^{(0)} - b_H$. Then

$$d_* = \min\{d : E(j - d) + I(d) \ge D\},$$

where the two class sizes restrict admissible d. Exact retention of every active row makes the sum of upper costs the sole remaining constraint. For a fixed d, the two sorted prefixes maximize its reduction among all j-entry sets with d intermediate cells; therefore the first feasible d is minimal. Selecting those prefixes reconstructs a witness. This argument fixes j and does not establish its primary optimality.

For the remaining cases we retain all numerical bounds and integer-check each reconstructed query set. Three append-only rounds use nominal wall-time budgets of 5, 60 and 120 seconds per unresolved target. The remaining budget is computed before each target model is assembled; model assembly and solver termination can make elapsed time exceed that budget. Maximum recorded elapsed times are 8.06, 61.82 and 121.20 seconds, respectively. The secondary runs explicitly request zero relative gap and 10^{-7} primal, dual and MIP feasibility tolerances. Independent query-subset enumeration checks 240 seeded cases with histories, varying caps and tolerances, plus 80 cases restricted to nonzero secondary objectives and nonfull retention. Both entry and grouped formulations match all 320 cases (640 comparisons). Full-grid checks validate each saved witness; numerical lower bounds are not rational proof certificates.

Table 51 reports 1,879 closed secondary objectives and 41 remaining intervals. Across both stops, 1,024 targets have $d_* = 0$, compared with 457 zero counts in the original saved witnesses. All have no strict gain. At least 807 no-gain targets have $d_* > 0$. Although 41 objective values retain intervals, every target is classified as zero or positive. Every known union gain respects the bound, and 1,207 new witnesses improve the original intermediate-count upper bound. Minimizing over ties certifies more zero values, but these do not add empirical equality coverage beyond the simpler full-retention condition: all 1,024 zero cases lie among its 1,806 targets (Table 57). A positive value remains insufficient for advantage. The 89 strict gains, including 63 on ja-zh, retain their original values. These overlapping-split summaries neither estimate population frequencies nor identify a causal effect.

Matched multirow counterexample. Intermediate counts also fail to determine the gain when basic instance summaries are fixed. Consider the two contribution matrices

$$F^A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 3 \\ 1 & 2 & 2 \end{pmatrix}, \qquad F^B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 1 & 1 & 3 \end{pmatrix}.$$

Both have three rows, three candidates, caps $p_i = 3$, row totals $(0, 4, 5)$, empty history, budget $b_H = 8$ and exact stopping. Their optimal retained count is two; their safety and exclusion minima are four and five, respectively. Both have $Q_{\text{joint}} = 7$ and $d_* = 4$, with four intermediate cells in the complete matrix. Yet $Q_{\cup,\text{sep}}$ is eight for F^A and seven for F^B. Thus even these matched summaries do not determine compatibility of separately optimal witnesses.

These are realizable mean scores. Use base scores $(.50, .51, .52)$ on each row and $t_0 = .5$. Candidate scores $1, .485, .475, 0$ realize contributions $0, 1, 2, 3$, respectively. Set $M = T = N = 3$, $q = 1/3$, $k = 1$ and $\alpha = 4/9$. Exact rational replacement calculations recover both matrices. Independent enumeration of all 2^9 query subsets per matrix verifies every minimum and all component ties. The search uses three rows, three candidates and common cap three. It enumerates nondecreasing row triples from $\{0, 1, 2, 3\}$, then unordered multisets of three such rows, and every integer budget with true retained count one or two. This removes within-row entry permutations and row permutations from the entry-query search, yielding 16,720 instances. The pair refutes sufficiency of these summaries, not a population hypothesis or every multirow criterion.

The two-row example in Appendix M.5 has $Q_{\text{split}} = 3$ but $Q_{\cup,\text{sep}} = Q_{\text{joint}} = 2$: its difference is entirely double counting. We therefore compare both reductions on the existing 3,840 mean small-batch tasks and 96 complete exact tasks, retaining their original histories, budgets and targets. In 3,544 tasks, equality follows from the previously numerically closed joint and additive optima together with the theoretical bound sandwich. For the 392 remaining tasks, an outcome-count program fixes both component cardinalities before minimizing their union. A separate binary-entry model enumerates row subsets on all 388 small tasks; entry-variable threshold programs check the four full tasks. An exhaustive query-subset implementation also checks 160 fixed-seed numerical cases with at most nine entries, including histories and positive tolerances. These are finite implementation checks, not a human proof review.

Panel	$d_* = 0$	$d_* > 0$	Unresolved sign	Closed objective
RewardBench 2	13 / 4	147 / 156	0 / 0	160 / 160
WMT23, 13	132 / 108	28 / 52	0 / 0	159 / 160
WMT23, 12	98 / 65	62 / 95	0 / 0	159 / 160
WMT24 en-de	147 / 121	13 / 39	0 / 0	160 / 160
WMT24 en-es	152 / 132	8 / 28	0 / 0	160 / 160
WMT24 ja-zh	28 / 24	132 / 136	0 / 0	139 / 142

Table 51: Secondary minimization of intermediate entries over all minimum joint witnesses in the full mean grid. Cells give primary / exact counts out of 160 targets (four depths and 40 overlapping splits) per panel, using full-reference right ECDF and $\alpha = .10$. Primary uses $\varepsilon = \lceil .02|S_0| \rceil$; exact uses zero. The first three columns classify targets using integer-checked witnesses and bounds from Corollary 3 or numerical optimization. Solver-based classifications depend on the validity of the numerical bounds. Closed objective counts equality of lower and upper bounds, conditional on the previously numerically closed joint minimum. The remaining objectives retain intervals; a positive sign does not imply an exact objective value.

Table 53 identifies the four full-batch cases with gains beyond optimal deduplication.

Small gains beyond deduplication. On the 96 full tasks, 2,573 of the 2,645-entry additive difference disappears by deduplication. The remaining joint reduction totals 72 entries in four instances, or 0.0166% of 433,303 optimally deduplicated entries. Thus joint selection can improve on the best union of separately optimal proofs, but the full-batch improvements here are small. The comparison concerns offline evidence, not a gain achieved by an online policy. Small-task results remain a separate target with reset batch size and budget. Saved records retain query sets, component minima, all-target checks and solver states. One primary outcome run reported infeasibility despite an independently feasible witness; disabling presolve closed the same model. One full entry check reached its initial time limit and was resolved by an append-only retry. Failed and unresolved primary attempts are preserved.

Proposition 9 (Strict gain beyond optimal deduplication)**.** *For every even $T \geq 2$, a realizable $M = T + 1$ mean-pooling instance with two selected outputs, empty history and exact stopping satisfies*

$$Q_{\text{joint}} = T + 1, \qquad Q_{\cup,\text{sep}} = Q_{\text{split}} = 2T.$$

Thus the factor-two upper bound in Proposition 5 is asymptotically tight even after optimal deduplication. It concerns offline unit-cost entry certificates, not online query competitiveness.

Proof. Use endpoint caps $(T + 1, T)$, flip rows $(1, \ldots, 1)$ and $(1, 0, \ldots, 0)$, and budget $b_H = T$. Their true costs are T and one, so the optimal retained count is one. After a queries, the first row's upper cost is $T(T + 1) - aT$; certifying it requires all T entries. On the second row, querying a_2 entries gives upper cost $T(T - a_2) + e$, where $e \in \{0, 1\}$ records whether its positive entry was queried. Minimum safety therefore uses its $T - 1$ zero entries. Excluding the two-row set requires observed total cost at least $T + 1$, hence all $T + 1$ positive entries across both rows. These separately minimum witnesses are disjoint, giving union and additive counts $2T$.

A joint witness queries the first row completely and the positive entry of the second row. It retains the first row and excludes retaining both, using $T + 1$ entries. Every exact certificate must exclude retaining both and therefore reveal these $T + 1$ positive entries. The joint minimum follows.

For a score realization, set cutoff $1/2$ and define the first base row by $a_j = 1/2 + j/(4M^2)$ for $j = 0, \ldots, M - 1$. Set the second row to $b_0 = 0$ and $b_{j+1} = M/(2(M-1)) + j/(4M^2)$ for $j = 0, \ldots, M - 2$. Both means exceed the cutoff. Replacing base member j flips the first row's membership precisely when the candidate score is below $\tau_j = M/2 - \sum_v a_v + a_j$; replace a by b for the second row. The first row has M positive distinct thresholds; the second has $M - 1$. In each row, the midpoint of the largest two positive thresholds realizes exactly one flip. Use that score in every first-row candidate and in one second-row candidate; set all other second-row candidate scores to one. All scores lie in $[0, 1]$. With $N = 2$, $q = 1/2$, $k = 1$ and $\alpha = 1/(2M)$, the budget is T. □

Table 1 instantiates the construction at $T = 6$. The separately shortest safety proof retains row 2; joint optimization retains row 1 using evidence already needed for exclusion. Exact rational score checks and independent query-subset enumeration

Task	History	ε	n	$Q_{\text{joint}} < Q_{\cup,\text{sep}}$	Double count	Joint reduction
Full	empty	0	96	4	2573	72
Small	empty	0	960	84	190	116
Small	empty	1	960	43	37	71
Small	quarter	0	960	38	106	51
Small	quarter	1	960	9	17	18

Table 52: Mean entry evidence: separating deduplication from joint optimization. The strict-gain column counts instances. Double count sums $Q_{\text{split}} - Q_{\cup,\text{sep}}$; Joint reduction sums $Q_{\cup,\text{sep}} - Q_{\text{joint}}$ in additional entry observations. Each small row covers six panels, four depths and 40 splits, with the original reset subbatch budget. Full tasks preserve the original test batch and budget at two depths and eight splits per panel. All component ties are optimized; equal joint/additive cases use the proved bound sandwich. Histories are not charged again.

Panel	Split	q	Q_{joint}	$Q_{\cup,\text{sep}}$	Q_{split}
WMT23, 12 streams	6	0.05	2273	2293	2521
WMT23, 12 streams	7	0.05	2347	2361	2602
WMT24 ja-zh	1	0.05	5149	5167	6023
WMT24 ja-zh	4	0.05	10103	10123	11371

Table 53: The four full mean instances with joint minima below optimally deduplicated separate witnesses. All use empty history, exact stopping and $\alpha = .10$. The other 92 full tasks have equality throughout. Both outcome-count and binary-entry programs numerically close these union optima; original time-limited attempts remain in the records. All costs count entry observations, not scorer calls or execution time.

verify the construction; NLP reductions above remain measured on their own fixed panels.

Separation with a fixed evaluator pool. The preceding family increases the pool size with the candidate count. The same mechanism operates with a fixed pool; the integer budget determines how much evidence the separate minima can share.

Proposition 10 (Fixed-pool separation)**.** *Fix integers $M \geq 3$, $T \geq 1$ and flip budget $b = b_H$ with $T \leq b < (M-1)T$. Write $b = r(M-1) + t$ with $0 \leq t < M-1$. There is a realizable mean-pooling instance with two selected outputs, $k = 1$, $\alpha = b/(2MT)$, empty history and exact stopping such that*

$$Q_{\text{joint}} = T + 1, \qquad Q_{\text{split}} = 2T + 1 - r,$$

$$Q_{\cup,\text{sep}} = \begin{cases} T + 1, & r = 0, \\ 2T + 1 - r - \mathbf{1}[t > 0], & r \geq 1. \end{cases}$$

Strict joint advantage holds precisely when $r \geq 1$ and $T - r > \mathbf{1}[t > 0]$ within this family.

Proof. Use caps $(M, M-1)$. Give the first row T positive integer contributions at most $M-1$, summing to b; such a row exists because $T \leq b < (M-1)T$. Give the second row one contribution of one and $T-1$ zeros. The true costs are b and one, so the optimum retains one row. Excluding both requires all $T+1$ positive entries, since their total is $b+1$. This unique minimum exclusion witness also certifies the first row, proving $Q_{\text{joint}} = T + 1$.

Certifying the first row requires all its entries: every unqueried entry has strictly positive upper-cost slack. With a queries on the second row, its upper cost is $(T-a)(M-1) + e$, where e is one if its positive entry was queried and zero otherwise. If $r \geq 1$, minimum safety uses $a = T - r < T$ queries on this row. With $e = 0$ the upper cost is $b - t$, whereas $e = 1$ is feasible precisely when $t > 0$. A minimum safety witness can therefore share at most $\mathbf{1}[t > 0]$ entries with the unique minimum exclusion witness, and that overlap is attainable. This proves both the additive and best-union formulas, including all ties. If $r = 0$, both rows require T queries for safety. Choosing the first row makes its safety witness a subset of the minimum exclusion witness, giving union size $T + 1$.

For a score realization, use cutoff $1/2$, first-row base scores $a_j = 1/2 + j/(4M^2)$ for $j = 0, \ldots, M-1$, and second-row scores $b_0 = 1/(16M^2)$ and $b_{j+1} = M/(2(M-1)) + j/(4M^2)$ for $j = 0, \ldots, M-2$. Both base means exceed the cutoff. Their replacement thresholds $M/2 - \sum_v a_v + a_j$ (and analogously for the second row) are distinct; respectively M and $M-1$ lie in $(0, 1)$. Candidate scores strictly between consecutive positive thresholds realize each needed flip count. A score strictly above the largest threshold

realizes zero. Taking $N = 2$, $q = 1/2$ and the stated α gives integer budget b. □

The same instance is valid for any α with $\lfloor 2MT\alpha \rfloor = b$. Fix M and $1/(2M) \le \alpha < (M-1)/(2M)$, and take this rounded budget as T grows. Then

$$\frac{Q_{\cup,\mathrm{sep}}}{Q_{\mathrm{joint}}} \longrightarrow 2 - \frac{2M\alpha}{M-1}.$$

At $M = 7$ and $\alpha = .10$, the limit is $53/30$ without a divisibility restriction on T. The original specialization $T = 30n$, $b = 42n$ gives costs $30n + 1$ and $53n+1$. These are attained ratios in this family, not tight upper bounds for every fixed-pool instance. The exact formula also identifies equality cases: at $r = 0$ the first row supplies a minimum safety witness, and at $r = T - 1$, $t > 0$ a one-entry safety witness lies in the exclusion witness.

At gap tolerance one, either safety for one row or exclusion of both suffices. Their minima are $T - r$ and $T + 1$, respectively, so joint and best separate-union costs both become $T - r$. Thus the two-row construction loses its strict advantage at the primary rounded tolerance.

Local score perturbations. The construction does not require cutoff ties. Choose each positive-count candidate at the midpoint between its adjacent thresholds, and each zero-count candidate midway between the largest threshold and one. Consider the distances of the constructed base and candidate scores from zero and one, together with the absolute cutoff margins of base means, observed replacement means and replacement means at candidate endpoints zero and one. Let δ be half the smallest of these distances and margins. All these distances are positive. Perturbing each score by less than δ, with cutoff and budget fixed, changes each mean by less than δ. Base memberships, endpoint caps and observed flip contributions therefore stay unchanged, preserving both certificate minima. This is local robustness of the normalized-score construction, not robustness to refitting ECDF maps.

Exact rational checks cover 595 constructed instances, including every integer budget $T \le b < (M-1)T$ for $M = 3, \dots, 9$ and $T = 1, \dots, 6$. Literal enumeration examines 882,928 query subsets across 589 cases. An independent outcome-count enumeration covers 545,800 states across all 595 cases, optimizing over all tied component minima. Both agree with the exact and gap-one formulas; 223 cases have no strict joint advantage. At $M = 7$, $\alpha = .10$ and $T = 7, 17$, the rounded budgets are 9 and 23. Joint costs are 8 and 18, versus best separate-union costs 13 and 31 and additive costs 14 and 32. At $T = 30, 60, 90$, joint costs are 31, 61 and 91, versus separate-union costs 54, 107 and 160. The constructed score margins permit $\delta = 1/5488$ in these three cases. These are theorem-instance checks, separate from the NLP panels.

When separate minima are incompatible. Changing the retained set can make a longer safety proof share more exclusion evidence. We give a prefix-sum criterion for any number of active rows when the true optimum retains all but one, including a class with tied minimum exclusion witnesses. Consider $m \ge 2$ selected active rows, empty history, exact stopping and integer budget $b = b_H$, with

$$\sum_{i=1}^{m-1} F_{(i)} \le b < \sum_{i=1}^{m} F_i.$$

Thus $L^* = m - 1$. For each $(m-1)$-row set C, define reductions $d_{ir} = p_i - f_{ir}$ and required reduction $D_C = T\sum_{i\in C} p_i - b$. Let a_C be the length of the shortest descending reduction prefix in C reaching D_C: zero if $D_C \le 0$, and infinity if no prefix suffices. Set $a = \min_C a_C$.

Order all mT contributions as $c_1 \ge \cdots \ge c_{mT}$, put $C_j = \sum_{v=1}^{j} c_v$, and let $e = \min\{j : C_j > b\}$. For the unique and tied cases specified below, we represent the family of minimum exclusion witnesses as

$$\mathcal{E} = \{H \cup B' : B' \subseteq B,\ |B'| = \kappa\}.$$

If $e = mT$, the only minimum exclusion witness is the full entry set: take it as H, with $B = \varnothing$ and $\kappa = 0$. For $e < mT$, the top-e set is unique when $C_e - c_e + c_{e+1} \le b$; use the same convention. In the remaining case, suppose $C_e = b + 1$; take H to contain entries with contribution greater than c_e, B those equal to c_e, and $\kappa = e - |H|$. All other entries form Z. For $X \in \{H, B, Z\}$, let $P_C^X(t)$ sum the t largest reductions among entries of X in rows C, with $P_C^X(0) = 0$ and value $-\infty$ when fewer than t entries exist.

Proposition 11 (Near-full compatibility with exclusion ties)**.** *For a row set C, write $h_C = \sum_{(i,r)\in H: i\in C} d_{ir}$ and $b_C = |B \cap (C \times [T])|$. Suppose*

$$h_C + P_C^B(\min\{\kappa, b_C\}) \ge D_C$$

for some $(m-1)$-row set C. Let $\mathcal{F}_C$ be the set of all nonnegative integer triples (u, v, w) satisfying

$$u + v + w = a,$$
$$P_C^H(u) + P_C^B(v) + P_C^Z(w) \geq D_C.$$

Define

$$\delta = \min_{C: a_C = a} \min_{(u,v,w) \in \mathcal{F}_C} \left[w + (v - \kappa)_+ \right].$$

Then

$$Q_{\text{joint}} = e, \qquad Q_{\cup,\text{sep}} = e + \delta.$$

Hence strict joint advantage is equivalent to $\delta > 0$ within this class. Sorting and prefix sums compute the tied case in $O(m^2 T \log(mT) + ma^2)$ time; the unique case reduces to $O(m^2 T \log(mT))$.

Proof. Exclusion requires total observed contribution greater than b, so every exclusion witness has at least e entries. If $e = mT$, there is only one size-e set. If $e < mT$ and the unique-case inequality holds, every other size-e set has sum at most $C_e - c_e + c_{e+1} \leq b$. In the remaining tied case with $C_e = b+1$, integer contributions force every size-e exclusion witness to attain this maximum sum. It must contain all entries above c_e and choose exactly κ entries tied at c_e. Thus $\mathcal{E}$ is the complete family in either case.

For fixed C, the mandatory entries in rows C contribute their reductions, while up to κ boundary entries in C can contribute the largest available reductions. Remaining boundary slots can be filled without reducing this sum. The assumed inequality therefore says that some $E \in \mathcal{E}$ is safe for $m-1$ rows. It attains the exclusion lower bound, proving $Q_{\text{joint}} = e$.

Every minimum safety witness has size a and lies in a set C with $a_C = a$; otherwise an entry outside a set made safe could be removed. Suppose such a witness takes u, v, w entries from H, B, Z. Its best possible reduction is the sum of the three prefixes, attained by taking those entries. A size-e exclusion witness contains all u mandatory entries and can share at most $\min(v, \kappa)$ chosen boundary entries. It shares no Z entries. Its union with the safety witness therefore has minimum size $e + w + (v - \kappa)_+$. Minimizing over the feasible counts and row sets proves the formula. For each of m row sets, sorting at most mT reductions and scanning u, v costs $O(mT \log(mT) + a^2)$. If B is empty, $v = 0$ and $w = a - u$, leaving only u to scan. □

The unique-exclusion case is Proposition 11 with a singleton minimum-exclusion family; in Proposition 9, $e = T + 1$, $a = T - 1$ and $\delta = T - 1$. The tied case optimizes over all minimum exclusion witnesses rather than arbitrarily breaking a tie. It still requires a minimum exclusion witness that itself certifies safety. It does not cover arbitrary exclusion families, nonzero tolerance or larger retained-count deficits.

A realizable multirow family. For any $m, T \geq 2$, use $M = T$, caps $p_i = T$, contribution rows $(0, \ldots, 0, 1)$ and $m - 1$ copies of $(1, \ldots, 1)$, and budget $b = (m-1)T$. All but one row can be retained. Exclusion requires every positive entry, so its unique minimum E has size $(m-1)T+1$. It certifies the $m-1$ all-one rows. For any $(m-1)$-row set, the required upper-cost reduction is $D = (m-1)T(T-1)$. A query reduces the upper cost by $T-1$, or by T at one of the $T-1$ zero entries. With a queries, reduction is at most $a(T-1) + (T-1)$. Reaching D requires $a \geq (m-1)T - 1$, with equality only if all $T-1$ zeros are queried. This bound is attained by taking those zeros and $(m-2)T$ positive entries within a set containing the first row and $m-2$ other rows. Thus

$$Q_{\text{joint}} = (m-1)T + 1, \qquad Q_{\cup,\text{sep}} = mT.$$

Here $\delta = T - 1$: allowing a longer safety proof avoids all zero entries and reduces the union.

For a mean-score realization, give every row base scores $a_j = 1/2 + j/(4T^2)$, $j = 0, \ldots, T-1$, and cutoff $1/2$. All T replacement thresholds $T/2 - \sum_v a_v + a_j$ are positive and distinct. A candidate midway between the largest two thresholds realizes contribution one; a candidate score of one realizes zero. With $N = m$, $q = 1/m$, $k = 1$ and $\alpha = (m-1)/(2T)$, the budget is exactly $(m-1)T$. Exact rational arithmetic checks all 25 members with $m, T \in \{2, \ldots, 6\}$; independent enumeration checks the 12 members with $mT \leq 12$, totaling 19,600 query subsets. Larger checked members use the explicit witnesses and criterion, not exhaustive subset enumeration.

Finite checks and empirical scope. Independent subset enumeration checks 16,149 canonical near-full instances: $(m, T) = (3, 2)$ with caps in $\{1, 2, 3\}$, $(4, 2)$ with caps in $\{1, 2\}$, and $(3, 3)$ with every cap equal to three. Within each grid we enumerate row contribution multisets, row multisets and every integer budget satisfying $L^* =$

$m-1$. Across 5,688,256 query subsets, the unique-exclusion formulas agree in all 2,220 applicable instances: 615 have strict gain and 1,605 have equality. Another 2,424 strict-gain instances fall outside that unique-exclusion criterion. These finite checks test the formulas, not a general proof or NLP prevalence.

The earlier two-row check remains a specialization: all 14,674 instances with $T \in \{1, 2, 3\}$, caps in $\{1, 2, 3\}$ and budgets satisfying $F_1, F_2 \leq b < F_1 + F_2$ and $b < Tp_1, Tp_2$ span 913,084 query subsets in total. Within this search, the criterion applies to 2,812 instances spanning 171,772 subsets; all agree with enumeration (742 gains and 2,070 equalities). Returning to the 16,149-instance multi-row grid, allowing the tied maximum-sum family adds 1,203 applicable instances: 237 gains and 966 equalities. The formulas agree with freshly enumerated minima in all 3,423 applicable instances, spanning 1,061,120 query subsets. The other instances are outside the sufficient conditions; they are not classified by this test. Among 960 exact NLP targets, five have active optimum $m - 1$. All five satisfy the tied-family condition, but none of their minimum exclusion witnesses can certify safety. The extended criterion therefore still has zero coverage on this NLP grid. Positive-tolerance targets are outside its scope.

Evidence position and entry discovery. Use the two-row score construction of Proposition 9, with caps $(T + 1, T)$, and vary only the position $j \in \{1, \ldots, T\}$ of the second-row positive entry. Every row retains its contribution multiset, so $Q_{\text{joint}} = T + 1$ and $Q_{\cup,\text{sep}} = 2T$ throughout. A fixed order querying the first row and then the second, each in candidate-index order, stops exactly after $T + j$ queries: the first row is safe after T observations, and exclusion becomes certified when the second-row positive entry is seen. Reversing the row order takes $2T$ queries.

We enumerate every position for $T \in \{2, 4, 6, 8, 10, 12\}$, using those two fixed orders and largest width (row-one ties first, candidate index ascending). All 42 score instances use exact rational realizations and exact stopping from empty history. The 126 paths are checked at every reply and first stop; independent subset enumeration verifies the joint minima for $T \leq 6$. At $T = 6$, the first fixed order takes 7–12 queries, the reverse order always 12, and width 11–12, while the joint minimum remains seven. These controlled positions separate evidence size from discoverability under entry access. They do not establish how often this mechanism explains the NLP results or evaluate learned reference priorities. Complete scores, positions and paths accompany the code.

M.9 Complete-batch mean evidence across depths and tolerances

We retain the original six panels, four depths and 40 splits, full-reference right ECDF, complete candidate families and $\alpha = .10$. For each of the 960 instances, we optimize both $\varepsilon = 0$ and $\varepsilon = \lceil .02|S_0| \rceil$, starting from empty candidate-test history. Batch size, cutoff, selected set and budget are unchanged. Outcome-count optimization ranges over every admissible safe target. Equivalent fixed-threshold formulations cover every level not excluded by true costs; continuous relaxations screen levels whose lower bound reaches an existing feasible witness. Strict-improvement searches retain that witness when no smaller one exists. All levels contribute to the global lower bound.

For all 1,920 targets, the integer-checked feasible joint witness matches its numerical global lower bound using the rounding convention in Appendix M.5. Historical solver feasibility and integrality settings were not fully recorded. For the hard cases left by the initial runs, complete integer safe-target partitions strengthen the fixed-threshold bounds; every branch contributes to the global lower bound. Each witness is reconstructed as actual score entries and checked against both capacities. The original 96 exact targets agree with the earlier full-test study. Comparisons include all four original query orders and the three frontier variants, each using the same instance and stopping target. Tables 54 and 55 report all depths and the number of positive minima in each ratio denominator.

For separately optimal safety and exclusion witnesses, the union program minimizes over tied component optima and safe targets. Equal joint and additive optima close the union by a bound sandwich. A feasible union of separately optimal components also closes the comparison when its size reaches the numerically closed joint minimum; other safe targets then need not be solved. More generally, a feasible union of separately optimal components certifies the global minimum when every admissible safe target has a lower bound at least as large as the cardinality of this union. This does not require solving every safe-target subproblem to optimality.

Exhaustive integer-threshold partitions, with LP bounds and integer witnesses, resolve the remaining comparisons. A minimum safety component contains no query with zero upper-cost reduction; a minimum exclusion component contains no query with zero lower-cost increase. The 1,920 closed union comparisons contain 89 strict joint improvements, totaling 6,269 entries. These gains concern offline evidence, not savings achieved by the acquisition policies. Numerical optimality uses solver bounds, not exact rational proof certificates.

Independent numerical-bound audit. An audit covers 73 continuation attempts on 40 distinct targets: 21 attempts reach numerical closure, 51 remain unresolved and one returns an inconsistent numerical bound that is rejected. Integer checks validate the returned witnesses in the 72 non-error attempts. In WMT24 ja-zh, split 16, $q = .10$, at the primary stop, a union subproblem for active safe count 913 reports an optimum of 14,463. The returned component witnesses instead have a feasible union of 14,462 entries, certifying safety for 913 and optimistic count at most 933. This contradicts that subproblem's reported numerical lower bound, which we discard. Disabling presolve yields matching numerical primal and dual values of 14,462 for this same safe target. The witness exceeds the historical all-target union value 14,423, so it does not refute that value or independently prove it. The implementation retains such witnesses as feasible upper bounds and rejects their conflicting optimality status. Historical merged numerical comparisons and exact minimality proofs remain distinct.

Where joint gains occur. Strict reductions beyond optimal deduplication occur in 32 primary and 57 exact targets out of 960 each; both gain medians are zero. Among targets with strict gains, mean savings are 160.34 and 19.96 entries, respectively. WMT24 ja-zh accounts for 24 of the 32 primary improvements and 39 of the 57 exact improvements; RewardBench has none at either stop. Table 56 reports relative and absolute magnitudes. These differences describe the recorded panels and do not identify a causal effect of language or evaluator composition.

Full retention is feasible in 903 of the 960 instances, hence in 1,806 targets across the two stops. This makes exclusion automatic at either tolerance. Let V_A denote capacity on the active selected rows, equivalently total capacity after appending zero invariant-row costs minus m_0. If $L^* = m$, every history satisfies $V_A(\ell) = m$. The count-gap condition is equivalent to $V_A(u) \geq \max(0, m - \varepsilon)$, so only safety evidence is needed and $Q_{\text{joint}} = Q_{\cup,\text{sep}} = Q_{\text{split}}$. Full-information feasibility does not imply zero-query certifiability.

Table 57 cross-classifies this condition with the minimized intermediate-entry count. All 1,024 targets with $d_* = 0$ are inside the full-retention region, adding zero no-gain coverage beyond that simpler condition. The remaining 25 no-gain targets are non-full primary targets with $d_* > 0$. Replaying their saved optimal witnesses shows $V^- + \varepsilon \geq |S_0|$ in every case, so no count-exclusion evidence is needed. More generally, if a minimum joint witness satisfies this inequality, it is a minimum safety witness for its attained count: a shorter safety witness would also be a joint certificate, contradicting minimality. Pairing it with the empty exclusion witness gives $Q_{\text{joint}} = Q_{\cup,\text{sep}} = Q_{\text{split}}$. This is sufficient, not necessary in general. For example, two rows with $T = p_1 = p_2 = 1$, true contributions $(1, 1)$, $b = 1$ and $\varepsilon = 0$ have zero-query safety for one row and require both entries to exclude two rows. Joint and separate costs equal two, although automatic exclusion is impossible. In this grid, all 1,831 no-gain targets have an optimal witness satisfying $V^- + \varepsilon \geq |S_0|$; none of the 89 strict-gain targets has such a witness. The classification uses saved optimal witnesses and is not a pre-query predictor or an independent confirmation sample.

To test sensitivity within these 25 exposed non-full, no-gain cases, we fix their matrices, maps, cutoff and candidate family and multiply the original integer budget by $.75, 1, 1.25$ (rounding down) and the original tolerance by $0, 1, 2$. Of 225 targets, 159 joint/union comparisons close numerically: 42 show strict gains and 117 equality; 66 remain unresolved. Fifty targets coincide with previously closed tasks. At the original budget, removing tolerance changes all 25 no-gain cases to strict gains, whereas increasing the budget by 25% yields equality for all three tested tolerances. This is a deterministic change of certification task on outcome-selected matrices, not evidence of a population effect or a general monotonicity law for joint advantage. Initial nominal limits were two seconds for each complete joint or additive-separate objective and four seconds for union optimization. Retries gave each complete joint or additive-separate objective 15 seconds; union retries allowed ten seconds per solver call within a 30-second total budget. Selected additive-separate rechecks used

Panel	q	Mean Q	Closed	n_+	Ref.	Width	Frontier	Gain/n
RewardBench 2	0.05	663.2	40	40	1.098	1.165	1.134	0/40
RewardBench 2	0.10	1157.8	40	40	1.114	1.173	1.139	0/40
RewardBench 2	0.20	1605.5	40	40	1.197	1.200	1.147	0/40
RewardBench 2	0.30	1441.1	40	40	1.410	1.237	1.192	0/40
WMT23, 13	0.05	1709.4	40	40	1.105	1.197	1.132	2/40
WMT23, 13	0.10	3289.3	40	40	1.070	1.153	1.108	0/40
WMT23, 13	0.20	1712.0	40	40	1.427	1.267	1.208	0/40
WMT23, 13	0.30	1337.7	40	40	1.976	1.350	1.297	0/40
WMT23, 12	0.05	1613.3	40	40	1.090	1.150	1.117	5/40
WMT23, 12	0.10	2834.3	40	40	1.073	1.141	1.106	0/40
WMT23, 12	0.20	1769.1	40	40	1.381	1.285	1.246	0/40
WMT23, 12	0.30	1880.7	40	40	1.618	1.345	1.302	0/40
WMT24 en-de	0.05	5144.4	40	40	1.097	1.170	1.133	0/40
WMT24 en-de	0.10	5886.1	40	40	1.120	1.157	1.123	0/40
WMT24 en-de	0.20	8277.0	40	40	1.181	1.208	1.183	0/40
WMT24 en-de	0.30	8383.2	40	40	1.343	1.264	1.223	0/40
WMT24 en-es	0.05	4641.5	40	40	1.067	1.132	1.102	1/40
WMT24 en-es	0.10	5401.0	40	40	1.124	1.165	1.127	0/40
WMT24 en-es	0.20	7306.3	40	40	1.169	1.215	1.191	0/40
WMT24 en-es	0.30	6525.0	40	40	1.387	1.279	1.241	0/40
WMT24 ja-zh	0.05	4115.2	40	40	1.087	1.156	1.142	12/40
WMT24 ja-zh	0.10	7347.5	40	40	1.080	1.150	1.126	11/40
WMT24 ja-zh	0.20	11132.9	40	40	1.131	1.196	1.178	1/40
WMT24 ja-zh	0.30	12272.4	40	40	1.179	1.242	1.206	0/40

Table 54: Full-batch mean evidence from empty history, $\alpha = .10$, $\varepsilon = \lceil .02|S_0| \rceil$. All four depths and 40 original splits are included. Q averages the matching minimum joint entry count over all 40 splits; all targets are numerically closed. Closed counts minima whose feasible value matches the numerical global lower bound; n_+ counts positive matching minima. Ref., Width and Frontier average Q_{online}/Q only on these n_+ matching instances, with zero minima excluded. Gain/n counts $Q_{\text{joint}} < Q_{\cup,\text{sep}}$ among n closed union comparisons. Historical resource-limited attempts are retained separately. Candidate-test score entries are the cost unit.

60 seconds. An earlier archived audit recorded 151 closed and 74 unresolved targets; the later summary above reports 159 and 66, respectively. Later retries allowed 60 seconds for the joint objective and, after its closure, 15 seconds per union solver call within a 30-second total budget. These call limits do not allocate an independent budget to every safety or exclusion component. These are nominal limits, with measured times and unresolved bounds recorded. Current bounds are checked against preserved solver calls.

To examine the role of intermediate contributions, we measure the fraction of active entries satisfying $0 < f_{ir} < p_i$. We retain all 1,920 targets and stratify by panel, depth and stopping criterion. Thirteen of the 48 cells contain both strict-gain and no-gain instances. The strict-gain subgroup has a larger mean intermediate fraction in eight cells and a smaller one in five. Thus the presence of intermediate contributions permits shared evidence, but its prevalence alone does not explain the observed gains. Split-level costs, initial gaps, exclusion slack and contribution fractions accompany these post hoc descriptive summaries. Overlapping splits do not supply independent population evidence.

Separating evidence costs. For each identical instance and stopping target, we compare $Q_{\text{split}} - Q_{\cup,\text{sep}}$, $Q_{\cup,\text{sep}} - Q_{\text{joint}}$ and $Q_{\text{online}} - Q_{\text{joint}}$. The first removes duplicate counting from separately optimal proofs; the second measures the benefit of choosing their evidence jointly. The last measures online discovery above the joint minimum. The first two sum to $Q_{\text{split}} - Q_{\text{joint}}$; discovery overhead is measured separately. Table 58 uses the same complete set for all three means: 960 primary and 960 exact targets. All depths and panels remain in the analysis.

The complete records include 1,920 targets and seven policies, with all joint minima and union comparisons numerically closed. Across all 960 primary targets, the mean reference excess is 809.58 entries; the exact-target mean is 796.66. Historical resource-limited attempts remain separate from these final numerically closed values. Ratios use positive numerically closed minima, and zero minima have a separate field. These overlapping-split summaries describe the recorded panels, not a population effect or an online competitive bound. Files in `outputs/evidence-cost-contrasts/` include all

Panel	q	Mean Q	Closed	n_+	Ref.	Width	Frontier	Gain/n
RewardBench 2	0.05	696.4	40	40	1.092	1.137	1.110	0/40
RewardBench 2	0.10	1212.0	40	40	1.107	1.147	1.119	0/40
RewardBench 2	0.20	1699.8	40	40	1.182	1.181	1.123	0/40
RewardBench 2	0.30	1579.6	40	40	1.362	1.207	1.152	0/40
WMT23, 13	0.05	1789.9	40	40	1.097	1.173	1.110	5/40
WMT23, 13	0.10	3387.7	40	40	1.077	1.142	1.099	0/40
WMT23, 13	0.20	1911.4	40	40	1.378	1.231	1.181	0/40
WMT23, 13	0.30	1631.7	40	40	1.721	1.296	1.257	0/40
WMT23, 12	0.05	1694.4	40	40	1.081	1.125	1.093	8/40
WMT23, 12	0.10	2930.0	40	40	1.077	1.129	1.097	0/40
WMT23, 12	0.20	1959.6	40	40	1.344	1.249	1.214	0/40
WMT23, 12	0.30	2174.8	40	40	1.531	1.299	1.255	0/40
WMT24 en-de	0.05	5370.8	40	40	1.086	1.140	1.106	3/40
WMT24 en-de	0.10	6180.4	40	40	1.116	1.137	1.113	0/40
WMT24 en-de	0.20	8856.5	40	40	1.171	1.183	1.165	0/40
WMT24 en-de	0.30	9272.1	40	40	1.304	1.225	1.189	0/40
WMT24 en-es	0.05	4785.9	40	40	1.069	1.117	1.090	2/40
WMT24 en-es	0.10	5655.5	40	40	1.122	1.147	1.120	0/40
WMT24 en-es	0.20	7825.5	40	40	1.160	1.190	1.171	0/40
WMT24 en-es	0.30	7295.5	40	40	1.345	1.239	1.205	0/40
WMT24 ja-zh	0.05	4298.4	40	40	1.079	1.130	1.119	16/40
WMT24 ja-zh	0.10	7693.5	40	40	1.069	1.122	1.104	16/40
WMT24 ja-zh	0.20	11954.3	40	40	1.114	1.161	1.148	7/40
WMT24 ja-zh	0.30	13096.4	40	40	1.168	1.217	1.177	0/40

Table 55: Full-batch mean evidence from empty history, $\alpha = .10$, $\varepsilon = 0$. All four depths and 40 original splits are included. Q averages the matching minimum joint entry count over all 40 splits; all targets are numerically closed. Closed counts minima whose feasible value matches the numerical global lower bound; n_+ counts positive matching minima. Ref., Width and Frontier average Q_{online}/Q only on these n_+ matching instances, with zero minima excluded. Gain/n counts $Q_{\text{joint}} < Q_{\cup,\text{sep}}$ among n closed union comparisons. Historical resource-limited attempts are retained separately. Candidate-test score entries are the cost unit.

Stop	Strict gains / targets	Median	Mean	Mean if positive	Max. relative reduction
primary	32 / 960	0	5.34	160.34	2.61%
exact	57 / 960	0	1.19	19.96	0.87%

Table 56: Distribution of $Q_{\cup,\text{sep}} - Q_{\text{joint}}$ across all 960 targets per stopping setting, with the same panels, depths, splits, empty histories and $\alpha = .10$ as Table 58. Median and mean are across all targets; the conditional mean includes only strict improvements. Absolute savings count score entries. Relative reduction divides by $Q_{\cup,\text{sep}} > 0$, and the last column reports its maximum, not its average.

policy-level records and bounds.

Matched inspection costs across pooling rules. Table 59 puts the median gap witness and mean joint minimum on the same entry-cost scale as reference acquisition. Within each pooling rule, each difference uses the same panel, split, depth, maps, cutoff, budget and stopping target. The supplement reports all four common policies, separate zero-query and nonzero-query summaries, and empirical quantiles. Undefined ratios at zero minima remain missing. A zero median excess is compatible with positive mean query counts: for WMT24 en-es median primary stopping, 159 of 160 instances stop without queries, while one requires inspection. These descriptive summaries reuse the saved paths; they add no scorer or optimizer runs.

Initial geometry and discovery overhead. We stratify the same saved paths by the initial gap $(V_0^+ - V_0^-)/|S_0|$, computed before any candidate-test query. The strata are descriptive and were not used to select a query policy. Writing $g_0 = (V_0^+ - V_0^-)/|S_0|$, the 960 primary mean targets fall into $0 < g_0 \le .10$, $.10 < g_0 \le .50$ and $g_0 > .50$, with 10, 466 and 484 targets, respectively. Their mean reference query-to-minimum ratios are 2.711, 1.337 and 1.096; mean excess counts are 752.9, 1,206.7 and 428.4 entries. Thus a smaller initial gap does not identify a uniformly smaller multiplicative overhead on these panels. The strata mix panels and depths and do not establish a causal effect or a deployable prediction rule. The supplement retains every policy, both stops and

Stop	Full retention	Targets	$d_* = 0$	Strict gain	Automatic exclusion
Exact	Yes	903	454	0	903
Exact	No	57	0	57	0
Primary	Yes	903	570	0	903
Primary	No	57	0	32	25

Table 57: No-gain explanations on all 1,920 mean targets at $\alpha = .10$. Full retention concerns the true complete scores. Automatic exclusion counts targets with an integer-checked optimal witness satisfying $V^- + \varepsilon \geq |S_0|$; it does not mean zero-query stopping. All 1,831 no-gain targets have such a witness, including 25 non-full primary targets. All 1,024 zero-intermediate optima remain in the full-retention region. These are descriptive properties of saved optimal witnesses, not pre-query predictors or independent trials.

Panel	Stop	Closed / tasks	Dedup.	Joint	Discovery
All panels	primary	960 / 960	18.69	5.34	809.58
All panels	exact	960 / 960	81.79	1.19	796.66
RewardBench 2	primary	160 / 160	0.00	0.00	274.59
RewardBench 2	exact	160 / 160	0.00	0.00	267.05
WMT23, 13	primary	160 / 160	0.60	0.75	531.77
WMT23, 13	exact	160 / 160	5.01	0.51	542.18
WMT23, 12	primary	160 / 160	3.33	1.70	525.33
WMT23, 12	exact	160 / 160	12.35	0.78	527.55
WMT24-en-de	primary	160 / 160	0.00	0.00	1298.33
WMT24-en-de	exact	160 / 160	7.86	0.23	1291.07
WMT24-en-es	primary	160 / 160	1.34	0.67	1140.28
WMT24-en-es	exact	160 / 160	7.95	0.17	1160.61
WMT24-ja-zh	primary	160 / 160	106.86	28.95	1087.16
WMT24-ja-zh	exact	160 / 160	457.57	5.43	991.53

Table 58: Complete-batch mean cost contrasts at $\alpha = .10$, all four depths. On the same closed-union tasks, columns average $Q_{\text{split}} - Q_{\cup,\text{sep}}$ (Dedup.), $Q_{\cup,\text{sep}} - Q_{\text{joint}}$ (Joint), and $Q_{\text{ref}} - Q_{\text{joint}}$ (Discovery), in score entries. Dedup. and Joint sum to $Q_{\text{split}} - Q_{\text{joint}}$; Discovery is measured separately. Primary uses $\varepsilon = \lceil .02|S_0| \rceil$; exact uses zero. Empty histories, maps, budgets and stopping targets match. Machine-readable records retain all targets, final lower and upper bounds, zero-minimum indicators and all seven policies. Resource-limited attempts remain in separate records.

zero-query cases separately.

Complete budget and tolerance grid. We additionally evaluate every original mean instance, without selecting on observed gain. Multiplying the integer budget by $.75, 1, 1.25$ and the primary tolerance by $0, 1, 2$ gives 8,640 tasks on the same 960 matrices. Full retention is feasible in 786, 903 and 941 instances at the three budgets, respectively (Table 60). These instances cannot exhibit strict joint advantage: their optimistic count remains the full selected count, so safety alone certifies the required tolerance. The remaining 174, 57 and 19 instances are eligible for, but need not exhibit, an advantage. Changing tolerance does not change the full-information optimum. For all 8,640 tasks we also enumerate whole-column query subsets and optimize joint and separate-union evidence over all admissible targets and tied component optima. Every joint minimum equals its best separate-union minimum. Independent sorted-row capacity calculations verify the witnesses and rule out all subsets with one fewer column; monotonicity then rules out every smaller subset. Each reported component is also minimum for its target, and its union attains the joint lower bound. The 1,920 original targets agree with the matched column study below. The negative result persists across this complete budget/tolerance grid, but the realizable column example above rules out universal equality. This reuses the same matrices; it supplies neither new entry minima nor independent NLP confirmation.

Previously unused WMT22 panels. Before extracting numeric scores, we froze a separate WMT22 protocol covering all three primary metric-task directions: en-de, en-ru and zh-en (Freitag et al., 2022). The cached archive had already been acquired. We admitted complete, finite, system-aligned reference-A segment-score streams, excluded reference outputs and removed exactly duplicate numeric streams. Admission and deduplication read the complete aligned panel before splitting; these preprocessing reads are outside

Panel	Pool	Stop	$\overline{Q}_{\min}$	$\overline{Q}_{\text{ref}}$	$\overline{\Delta Q}$	Med. ΔQ	Zero / n
RewardBench 2	median	primary	119.6	213.4	93.8	88.0	43 / 160
RewardBench 2	median	exact	157.5	270.2	112.7	111.0	35 / 160
RewardBench 2	mean	primary	1,216.9	1,491.5	274.6	216.5	0 / 160
RewardBench 2	mean	exact	1,297.0	1,564.0	267.1	208.5	0 / 160
WMT23, 13	median	primary	83.1	130.8	47.7	0.0	124 / 160
WMT23, 13	median	exact	104.3	156.5	52.2	0.0	123 / 160
WMT23, 13	mean	primary	2,012.1	2,543.9	531.8	437.5	0 / 160
WMT23, 13	mean	exact	2,180.2	2,722.3	542.2	475.0	0 / 160
WMT23, 12	median	primary	15.8	26.6	10.7	0.0	150 / 160
WMT23, 12	median	exact	26.4	48.7	22.3	0.0	145 / 160
WMT23, 12	mean	primary	2,024.3	2,549.7	525.3	431.0	0 / 160
WMT23, 12	mean	exact	2,189.7	2,717.2	527.5	435.5	0 / 160
WMT24 en-de	median	primary	10.6	17.4	6.8	0.0	153 / 160
WMT24 en-de	median	exact	27.3	46.7	19.4	0.0	152 / 160
WMT24 en-de	mean	primary	6,922.7	8,221.0	1,298.3	930.0	0 / 160
WMT24 en-de	mean	exact	7,420.0	8,711.0	1,291.1	909.0	0 / 160
WMT24 en-es	median	primary	3.4	8.3	4.9	0.0	159 / 160
WMT24 en-es	median	exact	8.9	19.3	10.4	0.0	154 / 160
WMT24 en-es	mean	primary	5,968.4	7,108.7	1,140.3	853.5	0 / 160
WMT24 en-es	mean	exact	6,390.6	7,551.2	1,160.6	873.0	0 / 160
WMT24 ja-zh	median	primary	53.2	111.1	57.9	0.0	145 / 160
WMT24 ja-zh	median	exact	98.4	174.1	75.7	0.0	136 / 160
WMT24 ja-zh	mean	primary	8,717.0	9,804.2	1,087.2	897.5	0 / 160
WMT24 ja-zh	mean	exact	9,260.6	10,252.2	991.5	714.5	0 / 160

Table 59: Matched entry costs for reference priority at $\alpha = .10$, full-reference right ECDF and all four depths. $Q_{\min}$ is the median gap witness or mean joint minimum for the same instance and stopping target; $\Delta Q = Q_{\text{ref}} - Q_{\min}$. Primary uses $\varepsilon = \lceil .02|S_0| \rceil$ and exact uses zero. Means include zero-query instances; the last column counts them. The supplement reports per-depth and nonzero-query summaries, quantiles and all four common policies. Ratios exclude zero minima. Overlapping splits support descriptive comparisons.

c	Full	Non-full	0	ε_0	$2\varepsilon_0$
0.75	786	174	0	0	0
1.00	903	57	0	0	2
1.25	941	19	2	3	6

Table 60: Budget geometry on all 960 original mean instances. Budgets are $\lfloor cb_0 \rfloor$ for the displayed multiplier c; maps, cutoff and candidates stay fixed. Full and Non-full partition the instances by full-information retention. The last three columns count zero-query gap certificates at each tolerance, with $\varepsilon_0 = \lceil .02|S_0| \rceil$. All 8,640 corresponding column tasks have equal joint and best separate-union minima. This grid supplies no newly optimized entry minima.

online acquisition counts. After splitting, candidate test scores are available to acquisition only through queries. Alphabetical stream order supplies seven base evaluators and all remaining candidates. The resulting panels contain 21, 20 and 21 metric streams, evaluated on 15, 16 and 18 translation systems, respectively. This tests a previously unevaluated, older benchmark year within the same research workflow, not an independent-team replication or an exclusion of benchmark contamination.

Each panel has one fixed, document-disjoint reference/calibration/test partition. With panel index $h = 0, 1, 2$ in the order above, distinct document IDs are collected in first-occurrence order and shuffled by Python `random.Random` with seed $20261008 + 1000h$. For D documents, the first $d = \lfloor D/3 \rfloor$ form reference, the next d calibration, and the remainder test: 60/60/61 documents for en-de and en-ru, and 85/85/86 for zh-en. The full test batches contain 9,675, 12,208 and 11,088 outputs. Score directions use Pearson correlation with BLEURT-20 on reference outputs only. Negative correlations reverse the score sign; zero correlations or undefined correlations caused by a constant vector retain the original sign. We then fit right ECDFs and the reference-consensus cutoff, fixing both for each instance. Calibration outputs are reserved and unused by these deterministic certificates. Four depths, $q \in \{.05, .10, .20, .30\}$, two pooling rules and two stops give 48 targets at $\alpha = .10$. Acquisition uses cached scores and no quality labels. The four original policies run throughout; mean pooling additionally uses all three frontier variants.

All 12 mean instances permit full retention. At

both stopping targets, exclusion is therefore automatic: joint, additive and best separate-union minima coincide in all 24 comparisons. The matching 24 column comparisons also have no strict gain. Table 61 reports every mean instance and the primary discovery counts. These results are consistent with the full-retention side of the automatic-exclusion prediction; they do not independently reproduce the strict joint advantage seen on part of the original grid. They also show that more panels alone need not increase the observed prevalence of sharing gains. One partition per panel does not support a population-level significance claim.

Independent code reconstructs reference mappings and replacement contributions, replays all 132 acquisition paths (888,799 revealed entries), and checks each first stopping state. All 24 mean entry comparisons are numerically closed; the 12 exact entry minima additionally match an independent integer reduction calculation. Direct enumeration checks 28,704 column predecessors, with monotonicity excluding smaller column sets. Positive-tolerance entry lower bounds still rely on numerical optimization. Protocol, admission manifest, code and all 48 summary rows are indexed under `review-revision/editorial-evidence-followthrough/confirmation/`. New raw score blocks and full traces remain local pending redistribution clearance; the package contains their hashes, protocol, audit receipts and summary, not those raw blocks. End-to-end reproduction additionally requires the source archive and reconstruction commands listed in `ACQUISITION_INDEX.md`; the package alone does not supply the raw inputs.

Partitioned optimization. Let m be the active-row count and L its full-information capacity. A feasible safe target lies in $s \in [\max(0, L - \varepsilon), L]$. For $s + \varepsilon < m$, the exclusion constraint is represented by an integer threshold, and every level in the formulation's declared domain is considered. For $s + \varepsilon \geq m$, exclusion is automatic; this branch uses the smallest admissible s, since increasing the safety target cannot lower its cost. Each branch retains the original constraints and fixes s and, where required, the exclusion threshold. The admissible branch family covers the original feasible set. An LP relaxation supplies a numerical lower bound; unresolved branches can then receive integer optimization. A branch with no usable bound retains the nonnegative lower bound zero. Solver-declared infeasibility and time limits remain distinct statuses.

Repeated valid bounds for the same branch are combined by their maximum. The global numerical lower bound is the minimum over an exhaustive branch family after removing solver-declared infeasible branches; any checked feasible witness gives an upper bound. Validity of this lower bound depends on both the branch bounds and the excluded-branch statuses. Integer witness reconstruction alone does not validate those statuses. This rule is applied at both the threshold and safe-target levels. Integer objective bounds use $\lceil d - 10^{-6} \rceil$ for the reported numerical bound d; 10^{-6} is a rounding convention, not a certified error bound on the solver. Closure requires that this global numerical lower bound reach the checked incumbent. The implementation retains branch domains, attempts and coverage checks in `external-verification-20261006/solver/`; `PARTITION_AND_ARTIFACT_INDEX.md` in the code package lists the exact files and replay commands.

Implementation checks. Saved results include entry witnesses, global numerical bounds and historical attempts, including failures, time limits, aborts and retries. Final closed-target comparisons are separate from those attempts. Reconstructed witnesses are checked against both capacities; solver-bound closure does not constitute independent human proof review. The code package indexes the composed results and saved runs in `review-revision/editorial-evidence-implementation/closing-index.csv`, including the remaining numerical-certification boundary.

Cached-score pipeline costs. We time the complete original entry policies at $q = .05$, $\alpha = .10$ and exact stopping on all six panels with mean and seven-member median pooling. The fixed timing grid uses splits 0–7, all test outputs and candidate evaluators, and three serial repetitions, giving 384 panel/pooling/split/policy configurations, each repeated three times (1,152 executions). Panel and policy orders rotate across repetitions. Every query count matches the corresponding original path; each reply and final mask is checked. We report the mean of eight splitwise median times, including zero-query stops.

Measurements use Python 3.9, an Apple M4 Pro

Panel	q	$\lvert S_0\rvert$	Q_{joint}	Reference	Width	Frontier	Exact Q_{joint}
en-de	0.05	385	3923	4032	4241	4206	4035
en-de	0.1	869	8657	9123	9778	9698	8951
en-de	0.2	1643	8202	10151	9896	9668	8741
en-de	0.3	2354	7197	10534	9307	9292	7981
en-ru	0.05	476	4472	4645	5026	4873	4602
en-ru	0.1	1084	10632	10934	11951	11664	10918
en-ru	0.2	2309	12054	13993	14242	13843	12767
en-ru	0.3	3315	8746	13980	11313	11121	9762
zh-en	0.05	663	7535	7837	8363	8258	7745
zh-en	0.1	1189	12676	13268	14209	13836	13068
zh-en	0.2	2183	9902	12099	12061	11599	10620
zh-en	0.3	3114	6231	10254	8075	7760	7260

Table 61: WMT22 additional-panel mean-pooling check, one document partition per panel. Query counts use entries. The joint, reference, width and frontier query columns use $V^+ - V^- \le \lceil .02|S_0| \rceil$; the final query column uses exact stopping. The selected count $|S_0|$ is an instance property. All results have $\alpha = .10$. Every instance permits full retention and has equal joint and optimal separate-union minima at both stops. Frontier is the joint variant; all other policies and median results remain in the 48-row supplement.

and single-thread numerical kernels on a shared research machine. Timed components include reference ECDF mapping, cutoff and base-bound setup, ridge fitting for reference priority, and acquisition through exact stopping. Capacity kernels, mask-prefix construction, cached lookup and flip calculation are also timed separately; remaining acquisition includes policy selection, updates and bookkeeping. Instrumentation overhead remains in the total. Imports, kernel compilation and raw-file preload are outside the timed interval; candidate scores are cached, so no evaluator forward pass is measured. Shared-host timings describe this execution environment rather than dedicated-hardware latency.

Table 62 gives complete pipeline times. Reference priority averages 19.9–146.5 ms across the twelve panel/pooling cells. Fewer queries need not imply lower cached runtime: fitting and maintaining a priority can outweigh the saved lookups. The supplement reports every repeat and component summary. The separate 5,760-attempt offline resource study measures certificate optimization, not these online paths. With a nominal 60-second algorithm budget, outcome MILP numerically closes 1,876 of 1,920 targets (931 primary and 945 exact), entry MILP 1,853 and row DP 86. DP also has the 100,000-state and 10-million-transition caps. These resource-limited runs are separate from the final numerical closures in Table 3.

M.10 Mixed entry and column prices

For odd-sized median pools with no initial candidate-test observations, fix entry price $\tau > 0$ and column prices $c_r > 0$. Let $\mathcal{O}_A = \mathcal{B} \times A$ denote the boundary evidence revealed by columns A. The full column still reveals all N scores; c_r prices that entire action. With additive prices independent of query order, the minimum mixed cost is

$$C_\varepsilon^{\text{mix}} = \min_{A \subseteq [T]} \left\{ \sum_{r \in A} c_r + \tau Q_\varepsilon(\mathcal{O}_A) \right\}. \quad (8)$$

Under these additive, order-independent prices, remove entry actions already covered by a purchased column and place the column actions first. The observations are unchanged and the cost cannot increase. Proposition 3 lower-bounds the remaining entry cost and constructs a witness attaining it. Pure entry and pure column access are included. The formula uses the true matrix to choose evidence offline; it supplies neither an online policy nor a batching-dependent execution model. For mean pooling, replace the median residual minimum by the joint residual minimum from Theorem 3, using active-row column observations; the same column-first argument applies.

For example, take $b = 0$, $\varepsilon = 0$ and opposition rows $(1, 0, 0)$ and $(0, 0, 1)$. Both rows must be excluded. With column prices $(1, 3, 7)$ and entry price two, pure entry access costs four and pure column access costs eight. Querying the first column and the second row's last entry costs three and certifies the same exact count. The first column shares observations, but buying the expensive last column is unnecessary. This illustrates why remaining cell evidence matters to mixed access.

Our residual implementation constructs the witnessing cells. A separate action-subset enumeration checks all 8,192 pairs of binary truth matrices

Panel	Pool	Fixed	Random	Width	Reference
RewardBench	Median	12.7	12.9	12.6	28.7
RewardBench	Mean	14.8	14.4	16.0	19.9
WMT23 (13)	Median	23.5	22.8	25.0	58.3
WMT23 (13)	Mean	36.1	33.0	36.0	49.1
WMT23 (12)	Median	15.7	15.6	15.5	46.3
WMT23 (12)	Mean	33.0	34.6	35.4	48.4
WMT24 en-de	Median	54.4	54.5	54.5	142.4
WMT24 en-de	Mean	99.1	96.2	107.5	146.5
WMT24 en-es	Median	48.7	48.8	48.8	130.4
WMT24 en-es	Mean	88.2	86.4	96.0	122.2
WMT24 ja-zh	Median	33.4	33.7	33.6	82.3
WMT24 ja-zh	Mean	79.6	76.3	84.7	113.6

Table 62: Instrumented cached-score pipeline time in milliseconds at $q = .05$, $\alpha = .10$ and exact stopping. Each cell averages eight splitwise medians of three serial runs. Full batches and original policies are used. Times include reference mapping, base setup and acquisition; only reference priority requires the timed ridge fit. Raw-file preload, imports, compilation and evaluator inference are excluded. Component times are supplied separately.

and observation masks with $(m, T) = (3, 2)$ or $(2, 3)$, budgets $b \in \{0, \dots, mT\}$ and tolerances $\varepsilon \in \{0, \dots, m\}$, giving 200,704 comparisons. It also checks 800 larger random cases and 4,032 mixed-action optima with three fixed price profiles. Empty initial states match the existing formula in 3,136 cases. These are finite implementation checks of the characterization, not additional NLP tasks.

M.11 Exact NLP column witnesses

Exact column witnesses use the six original panels with median pooling, full-reference right ECDF and $\alpha = .10$. Four depths and 40 splits give 960 instances. Candidate counts are 7 for RewardBench, 17 for each WMT23 panel, 14 for WMT24 en-de and en-es, and 15 for ja-zh; every family is complete. We replay all 2,880 original exact column stops and compare them with the independently checked witnesses. Table 63 summarizes nominal $q = .05$; the supplement's `column-witness-all` grid retains every depth and order. Positive-gap column costs in Appendix K remain a separate target.

At RewardBench nominal $q = .05$, the minimum whole-column witness averages 4.93 of seven columns and 70.4% of candidate-test scores; reference order uses 5.63 columns at exact stopping. For thirteen-stream WMT23 at the same depth and exact median target, the minimum whole-column witness reveals 45.6% of NT scores. For these RewardBench and thirteen-stream WMT23 configurations, minimum whole-column certificates still reveal a large fraction of candidate test scores even when evidence is chosen with full knowledge of the matrix. Online counts additionally include the cost of discovering suitable columns.

Panel	$K_{\mathrm{col}} > 0$	Q_{exact}	K_{col}	Min. (%)	Ref. cols	Ratio
RewardBench 2	40	196.6	4.93	70.4	5.63	1.183
WMT23, 13 streams	36	415.9	7.75	45.6	8.35	1.000
WMT23, 12 streams	4	41.1	0.83	4.9	0.95	1.100
WMT24 en-de	3	37.1	0.25	1.8	0.25	1.000
WMT24 en-es	2	19.0	0.23	1.6	0.28	1.125
WMT24 ja-zh	13	271.6	2.45	16.3	2.63	1.000

Table 63: Exact median column witnesses at $q = .05$, $\alpha = .10$, full-reference right ECDF. Q_{exact} is the entry witness; K_{col} is the minimum whole-column witness. Counts and Min. ($100K_{\mathrm{col}}/T$) average all 40 splits; Ratio is the median of per-split $K_{\mathrm{ref}}/K_{\mathrm{col}}$ over $K_{\mathrm{col}} > 0$ splits, where K_{ref} is the exact reference-order column count. The $K_{\mathrm{col}} > 0$ column reports split count. Whole-column score cost is NK_{col}. These exact costs have a different target from the positive-gap costs in Table 34.

M.12 Whole-column evidence for mean and even medians

We extend the same fixed selection protocols to whole-column access for mean pooling with seven base evaluators and even medians with six or eight. Six panels, three pooling configurations, four depths, 40 splits and two stopping targets give $6 \times 3 \times 4 \times 40 \times 2 = 5{,}760$ targets. All candidate columns and selected outputs remain in each instance. A column reveals all N scores and costs N logical entries, including scores outside the selected active rows.

For a column subset A, the observed row cost is $\ell_i(A) = \sum_{r \in A} f_{ir}$ and its upper bound is $u_i(A) = \ell_i(A) + (T - |A|)p_i$. We minimize $|A|$ subject to $V(\ell(A)) - V(u(A)) \leq \varepsilon$. Complete cardinality enumeration starts at the valid bound $\lceil Q_{\mathrm{joint}}/m \rceil$ when $m > 0$, because one column reveals m active entries, and ends at the best feasible policy count. Zero active rows require no queries. Integer witnesses and a separate predecessor-subset calculation check the minima.

The fixed and seeded-random column orders retain their original definitions. Reference column priority sorts columns by the sum of reference-predicted upper-cost reductions over active selected rows, quantized at 10^{-12} as in the original column protocol. It uses the same ridge features and penalty, with the actual mean or even-median pool and cutoff. All three orders are evaluated at their first valid stopping states; full scores are used only to evaluate observations and offline minima.

Table 64 shows nominal $q = .05$ results, including zero-query cases. Exact mean column minima reveal 89.1–99.6% of candidate-test scores across panels; reference order reveals 90.5–99.6%. For RewardBench, the corresponding exact minima for six- and eight-member medians are 85.0% and 50.0%. These pooling changes also change the selected set and replacement family. Every policy comparison uses the minimum for its own unchanged instance. Complete depth, policy and zero-minimum summaries are included in the supplement.

Joint evidence versus separate column optima. For all 1,920 mean targets, we compare the joint column minimum with the least expensive union over all separately optimal safety and exclusion column sets. The 1,806 full-retention targets have automatic exclusion. For the remaining 114 targets, enumeration of every column subset gives both capacities; subset envelopes retain the largest safe count and smallest optimistic count for each contained subset cardinality. Minimizing over admissible safe counts and all tied component minima gives the best union. Complete subset enumeration determines component minima. An independent sorted-row calculation additionally verifies the capacities of 8,540 predecessor subsets. Each resulting union attains the matching joint minimum, proving equality on every target, including all 89 strict entry-gain targets. A column costs N entries, so the equality also holds in that cost unit. Table 65 reports both query models on identical tasks.

The fixed-action bound and realizable $3/2$ separation above show that column gains are possible in the model. On the original 114 non-full-retention targets, every column is necessary. The following condition explains much of this full-revelation requirement using row contribution ranges. Independent finite checks compare the subset-envelope implementation with literal witness-pair enumeration on 400 seeded cases. A further 1,200 seeded histories, each under entry and column actions with positive integer prices, check the zero-contribution residual equality in Proposition 6. These computations support implementation consistency; the

Panel	Pool	Min., primary	Min., exact	Ref., exact	Zero, exact
RewardBench	Mean	96.1	99.6	99.6	0/40
RewardBench	Median 6	79.1	85.0	91.2	0/40
RewardBench	Median 8	42.1	50.0	60.4	1/40
WMT23 (13)	Mean	90.0	92.1	92.1	0/40
WMT23 (13)	Median 6	3.3	7.1	12.5	20/40
WMT23 (13)	Median 8	0.2	0.6	0.8	37/40
WMT23 (12)	Mean	89.6	91.5	91.5	0/40
WMT23 (12)	Median 6	6.9	10.6	15.4	27/40
WMT23 (12)	Median 8	1.6	3.8	4.8	34/40
WMT24 en-de	Mean	88.9	90.0	91.1	0/40
WMT24 en-de	Median 6	3.3	4.7	5.3	34/40
WMT24 en-de	Median 8	0.0	0.6	0.8	38/40
WMT24 en-es	Mean	87.5	89.1	90.5	0/40
WMT24 en-es	Median 6	5.5	8.7	10.0	29/40
WMT24 en-es	Median 8	1.0	1.5	1.5	38/40
WMT24 ja-zh	Mean	92.2	94.2	94.3	0/40
WMT24 ja-zh	Median 6	50.6	56.7	59.4	6/40
WMT24 ja-zh	Median 8	8.4	13.2	14.6	30/40

Table 64: Whole-column certification at $q = .05$ and $\alpha = .10$. Percentages use all NT candidate-test scores, averaging all 40 splits. Min. is the offline minimum; Ref. is the reference column order. Primary tolerance is $\lceil .02|S_0| \rceil$; exact tolerance is zero. Median 6 and 8 average their two central scores. Zero reports exact zero-query splits. Full depth and policy results accompany the supplement.

Stop	Retention	Targets	Entry	Column
Primary	Full	903	0	0
Primary	Non-full	57	32	0
Exact	Full	903	0	0
Exact	Non-full	57	57	0

Table 65: Strict joint gains over the best union of separately optimal witnesses on the original mean-pooling grid. Full means the full-information optimum retains every baseline-selected output. Entry and Column count targets with strict gain using their respective query units. Each stop uses the same 960 instances at $\alpha = .10$; the 40 splits overlap.

proposition rests on its completion argument.

When every column is necessary. Restrict all cost vectors to baseline-selected rows, retain capacity V and budget b_H, and start with empty history. For $T \geq 1$, define the gap when column r alone is missing by

$$\delta_r = V(F - f_{\cdot r}) - V(F + p - f_{\cdot r}).$$

Also let $\underline{f}_i = \min_r f_{ir}$, $\overline{f}_i = \max_r f_{ir}$ and

$$\gamma_{\text{col}} = V(F - \underline{f}) - V(F + p - \overline{f}).$$

Both quantities use the full score matrix to explain offline evidence requirements.

Proposition 12 (Full-column necessity)**.** *Under the completion model of Section 4, the joint minimum equals T unit-cost column queries if and only if $\min_r \delta_r > \varepsilon$. Whenever this holds, the best separate-union minimum also equals T. Moreover, $\gamma_{\text{col}} \leq \min_r \delta_r$, so $\gamma_{\text{col}} > \varepsilon$ is a sufficient condition that remains true under every within-row permutation of candidate scores, with base scores, cutoff, budget and tolerance fixed.*

Proof. If column r alone is unrevealed, the row bounds are $F_i - f_{ir}$ and $F_i + p_i - f_{ir}$, giving gap δ_r. If some $\delta_r \leq \varepsilon$, that omission certifies the target using $T - 1$ columns. Otherwise every one-column omission fails; any proper subset is contained in such an omission and has at least its gap by revealing monotonicity. Full revelation always succeeds, proving the equivalence. A best separate union is a joint certificate and uses at most the T available columns, proving equality of the two minima in this case.

Since $\underline{f}_i \leq f_{ir} \leq \overline{f}_i$ and capacity is nonincreasing in each cost,

$$\begin{aligned} &V(F - f_{\cdot r}) - V(F + p - f_{\cdot r}) \\ &\quad \geq V(F - \underline{f}) - V(F + p - \overline{f}). \end{aligned}$$

Within-row permutations preserve $F, \underline{f}, \overline{f}, p$ and hence the sufficient condition. □

The range condition holds for 56 of 57 exact and 45 of 57 primary non-full-retention targets in the original mean grid. The omission criterion establishes full revelation for the remaining 13. Independent enumeration checks the range condition on

66,339 finite integer cases; independent omission calculations check the equivalence on all 570 layout/target comparisons below, with full revelation in 566. These checks supplement the proof. The equivalence is a direct consequence of revealing monotonicity: it characterizes full revelation, not all instances of equality or strict advantage between joint and separate-union certificates.

Changing column layout at fixed row costs. We retain all 57 non-full-retention mean instances and both stops. Besides the original layout, we sort each row's candidate contributions increasingly; rotate those sorted rows by their row index modulo T; or independently permute each row using PCG64 seeds 104729 and 130363. Each permutation transports the corresponding normalized candidate scores, so the contributions remain realizable under the same base scores and cutoff. These synthetic columns need not represent a single deployed evaluator. Row multisets, F_i, p_i, V^*, budgets and tolerances stay fixed. The rowwise bijection preserves all entry witnesses and their minima; it need not preserve a column witness.

All 570 layout/target comparisons have equal joint and best separate-union minima. Every column is necessary in 566 comparisons. The four exceptions use the aligned layout and primary stopping, where both minima drop by one column and exclusion is automatic. All 114 original-layout targets require every column. Independent sorted-row calculations check 9,236 predecessor subsets and each joint witness. This post hoc intervention shows that the observed column equality persists under these changes of evidence location; it does not establish universal equality or a new NLP task result. Protocols, layout records and condition coverage are in `review-revision/granularity-risk-implementation/`.

M.13 Heterogeneous column-price sensitivity

Whole-evaluator certificate feasibility is unchanged when columns have different prices. For exact stopping, minimize $\sum_{r\in A} c_r$ under Proposition 2; for positive tolerance, require the two capacities to differ by at most the same ε as the original replay.

We retain the six original median panels, four depths, 40 splits, complete candidate families, full-reference right ECDF and $\alpha = .10$. In frozen candidate-index order $r = 0, \ldots, T-1$, the three price vectors are $(2T-1)\mathbf{1}$, $(T+2r)_r$ and its reverse. They share total price $T(2T-1)$; reporting cost divided by $2T-1$ makes the mean column price one. These are declared additive action costs rather than hardware measurements. Existing column orders and their stopping states remain fixed.

Full subset optimization gives 5,760 profile/target records. A separate raw-row enumeration checks every cheaper subset; all 960 uniform-price exact minima match the original column witnesses. Tables 66 and 67 summarize all panel–depth settings across 40 splits. Ties in total witness cost are resolved by minimizing column count. In 30 of the 3,840 nonuniform-price records, every minimum-cost witness uses more columns than the minimum feasible column count, including after breaking cost ties in favor of fewer columns. Enumerating all same-price subsets in these 30 records confirms this secondary minimum. For example, RewardBench split 17 at $q = .10$ and positive tolerance has a five-column ascending-price optimum with raw price 55, or $55/13$ after normalization, whereas its minimum column count is four. Thus minimizing column count and minimizing additive query cost can select different evidence, even under the same certification target. Existing-order priced cost ratios compare paid action costs against the minimum under the same prices and stopping target; they do not measure execution time.

M.14 Controlled score panels and implementation checks

We generate seven-member median pools at cutoff .5, with $m \in \{32, 128\}$ selected boundary rows, 16 selected interior rows, 16 invariant unselected rows and 16 unselected boundary rows. Base low and high values are drawn from $U(.1, .3)$ and $U(.65, .95)$. Selected boundary, selected interior, invariant unselected and unselected boundary rows have respectively four, five, two and three high votes. For designated selected boundary rows, a first-member high score is set to .5, preserving exactly four scores at or above the cutoff. Every selected boundary row has candidate opposition and agreement scores .25 and .75, with designated agreements tied at .5. The other three row types have candidate scores from $U(.1, .95)$. The complete tie indices, random-number order and separate reference seeds are specified by `make_panel` in `scripts/run_synthetic_certificates.py`.

With $T \in \{7, 12\}$, each selected boundary row has $d \in \{1, 3\}$ opposition bits. The layouts keep row costs fixed and vary opposition loca-

Panel	q	C_0	$C_\uparrow$	$C_\downarrow$	R_0	$R_\uparrow$	$R_\downarrow$	n_+	More
RewardBench	0.05	4.925	4.513	4.648	1.162	1.264	1.332	40	0/0
RewardBench	0.10	4.050	3.408	3.498	1.098	1.241	1.408	40	2/1
RewardBench	0.20	1.825	1.248	1.254	1.128	1.620	1.705	39	0/1
RewardBench	0.30	0.150	0.081	0.081	1.000	2.048	1.667	6	0/0
WMT23 (13)	0.05	7.750	7.841	6.942	1.095	1.145	1.227	36	2/0
WMT23 (13)	0.10	0.050	0.027	0.027	1.000	2.000	1.667	1	0/0
WMT23 (13)	0.20	0.000	0.000	0.000	–	–	–	0	0/0
WMT23 (13)	0.30	0.000	0.000	0.000	–	–	–	0	0/0
WMT23 (12)	0.05	0.825	0.764	0.762	1.300	1.345	1.646	4	1/1
WMT23 (12)	0.10	0.750	0.621	0.528	1.418	2.063	1.826	8	2/1
WMT23 (12)	0.20	0.150	0.100	0.117	1.000	1.561	1.234	1	0/0
WMT23 (12)	0.30	0.150	0.091	0.107	1.000	1.298	1.919	2	0/1
WMT24 en-de	0.05	0.250	0.207	0.207	1.000	1.606	1.269	3	0/0
WMT24 en-de	0.10	0.375	0.274	0.265	1.100	1.555	1.476	5	1/1
WMT24 en-de	0.20	0.000	0.000	0.000	–	–	–	0	0/0
WMT24 en-de	0.30	0.000	0.000	0.000	–	–	–	0	0/0
WMT24 en-es	0.05	0.225	0.185	0.204	1.125	1.677	1.628	2	0/0
WMT24 en-es	0.10	0.100	0.074	0.072	1.333	2.012	1.586	3	0/0
WMT24 en-es	0.20	0.025	0.017	0.013	1.000	1.000	2.571	1	0/0
WMT24 en-es	0.30	0.000	0.000	0.000	–	–	–	0	0/0
WMT24 ja-zh	0.05	2.450	2.305	2.241	1.101	1.436	1.337	13	0/0
WMT24 ja-zh	0.10	1.250	1.126	1.012	1.266	1.726	1.711	9	2/0
WMT24 ja-zh	0.20	0.100	0.072	0.067	1.500	1.915	2.602	2	0/0
WMT24 ja-zh	0.30	0.000	0.000	0.000	–	–	–	0	0/0

Table 66: Exact-stop column-cost sensitivity under the original median protocol and $\alpha = .10$. Uniform, ascending and descending prices have the same mean, normalized to one. The three C columns average minimum costs over all 40 splits. The R columns average reference-order cost divided by the matching optimum over the n_+ splits with positive minimum cost. All three price profiles have the same n_+. More reports ascending/descending split counts where even the fewest-column minimum-cost witness uses more columns than the minimum feasible count. Ties in total witness cost minimize column count. Undefined ratios remain missing. Prices are declared action-cost models, not measured scoring times or fees.

tions. Shared support assigns the same d candidate columns to every row; cyclic support rotates a seeded candidate order across rows; independent support draws a uniform d-subset per row.

Eight seeds and bit budgets $0, \lfloor mT/5 \rfloor, \lfloor mT/2 \rfloor$ give 192 panels and 576 instances. The nominal count is $k = m$, consistent with $q = m/(m + 48)$ in the general definition, and rational $\alpha = 2b/(7mT)$ gives flip budget exactly $4b$. Reference panels use separate seeds with the same layout; the cutoff is fixed by construction.

Every layout uses the complete four entry and three column methods from the NLP study, with the same ridge features, regularization, query interfaces and both stopping rules. The 192 panels yield 768 entry-acquisition orders, each reused across three budgets, and 1,728 budget-specific column paths. Two stopping targets give $768 \times 3 \times 2 + 1{,}728 \times 2 = 8{,}064$ records. Acquisition cannot access unqueried test scores. Complete matrices are used for evaluation and offline witness construction. A separate replay implementation rebuilds explicit single-member flips and every first stopping state. Complete generated base, candidate and reference matrices, paths, results and witness subsets are saved. No third-party score or quality data are required.

Table 68 reports the complete controlled comparison, including zero-budget cases. It shows how evidence positions affect column certificate size under odd-median pooling, complementing the entry/column comparison in Section 7.6. At $m = 128, T = 12, d = 1, b = 307$, the common minimum entry certificate size is 1,229; shared support requires ten columns and the other layouts eleven. The layouts preserve each row's contribution multiset and the minimum entry-certificate size. Their column minima differ because informative entries occupy different candidate columns. The experiment does not assess NLP quality or inference time.

N Retention Tolerance and Required Evidence

Let η denote the nominal count-tolerance fraction. We vary $\eta \in \{0, .005, .01, .02, .05, .10\}$, stopping when the integer gap is at most $\varepsilon_\eta = \lceil \eta |S_0| \rceil$. The original .02 remains the main configuration.

Panel	q	C_0	$C_\uparrow$	$C_\downarrow$	R_0	$R_\uparrow$	$R_\downarrow$	n_+	More
RewardBench	0.05	4.300	3.785	3.888	1.210	1.419	1.457	40	0/0
RewardBench	0.10	3.475	2.792	2.848	1.109	1.268	1.539	39	1/0
RewardBench	0.20	1.375	0.856	0.844	1.054	1.646	1.832	37	0/0
RewardBench	0.30	0.025	0.013	0.013	1.000	1.857	1.857	1	0/0
WMT23 (13)	0.05	6.350	6.431	5.561	1.098	1.155	1.303	36	3/0
WMT23 (13)	0.10	0.000	0.000	0.000	–	–	–	0	0/0
WMT23 (13)	0.20	0.000	0.000	0.000	–	–	–	0	0/0
WMT23 (13)	0.30	0.000	0.000	0.000	–	–	–	0	0/0
WMT23 (12)	0.05	0.625	0.558	0.557	1.262	1.586	1.795	3	0/0
WMT23 (12)	0.10	0.350	0.254	0.224	1.300	2.035	2.254	5	1/0
WMT23 (12)	0.20	0.100	0.061	0.065	1.000	1.775	1.419	1	0/0
WMT23 (12)	0.30	0.050	0.036	0.043	1.500	1.646	2.088	1	0/1
WMT24 en-de	0.05	0.175	0.148	0.143	1.000	1.160	1.321	2	1/0
WMT24 en-de	0.10	0.150	0.093	0.089	1.000	1.605	1.991	5	0/0
WMT24 en-de	0.20	0.000	0.000	0.000	–	–	–	0	0/0
WMT24 en-de	0.30	0.000	0.000	0.000	–	–	–	0	0/0
WMT24 en-es	0.05	0.125	0.109	0.130	1.600	1.712	1.643	1	1/0
WMT24 en-es	0.10	0.000	0.000	0.000	–	–	–	0	0/0
WMT24 en-es	0.20	0.000	0.000	0.000	–	–	–	0	0/0
WMT24 en-es	0.30	0.000	0.000	0.000	–	–	–	0	0/0
WMT24 ja-zh	0.05	1.700	1.564	1.514	1.213	1.387	1.686	9	2/0
WMT24 ja-zh	0.10	0.575	0.519	0.460	1.223	1.487	1.473	5	3/1
WMT24 ja-zh	0.20	0.025	0.016	0.013	3.000	4.789	5.533	1	0/0
WMT24 ja-zh	0.30	0.000	0.000	0.000	–	–	–	0	0/0

Table 67: Column-cost sensitivity at $V^+ - V^- \le \lceil .02|S_0| \rceil$ under the original median protocol and $\alpha = .10$. Uniform, ascending and descending prices have the same mean, normalized to one. The three C columns average minimum costs over all 40 splits. The R columns average reference-order cost divided by the matching optimum over the n_+ splits with positive minimum cost. All three price profiles have the same n_+. More reports ascending/descending split counts where even the fewest-column minimum-cost witness uses more columns than the minimum feasible count. Ties in total witness cost minimize column count. Undefined ratios remain missing. Prices are declared action-cost models, not measured scoring times or fees.

The six original panels retain the original seven-member pool, full-reference right ECDF, $\alpha = .10$, both pooling rules, four depths, four acquisition methods and forty splits. Acquisition choices do not depend on ε, so the saved paths through exact stopping determine every new first stop. This yields 7,680 paths, 46,080 stopping records and 1,152 configuration means. WMT24 retains its fixed-point priority ties. These are post hoc analyses on exposed panels.

Every cached reply is checked against the corresponding full score matrix. A separate sorted-cost checker reconstructs each returned mask and budget, both count bounds and the full-information optimum. For each stopping state reached after at least one query, the immediately preceding state must have a gap greater than the tolerance. The original .02 and exact stopping masks and positions match the saved results. All new masks meet the same churn budget, and realized retained-count regret never exceeds ε_η.

Nominal tolerance is not the proportion of incorrect returned outputs. It permits a retained-count gap without relaxing α. For $|S_0| > 0$, integer rounding gives

$$\eta \le \frac{\varepsilon_\eta}{|S_0|} < \eta + \frac{1}{|S_0|}.$$

Different nominal tolerances can therefore have identical stopping targets on small lists. Table 70 reports the effective proportion. Inspection fractions, undefined ratios and split aggregation follow Appendix B.

The supplement's `tolerance-all-*` grids show all depths and acquisition orders. At nominal $q = .05$, reference priority with median pooling on RewardBench inspects 71.1%, 61.4% and 33.6% of active entries at nominal tolerances 0, .02 and .10, with selected coverage 99.6%, 97.3% and 89.8%. Thirteen-stream WMT23 has corresponding active-inspection fractions 54.3%, 45.1% and 20.9%. These reductions relax the retained-count target rather than reduce queries for the same exact-certification problem.

With mean pooling, reference priority still queries a large fraction of active entries across the tolerance grid. At nominal $q = .05$, reference-priority active inspection across the six panels

m	T	d	b	Q_{exact}	Shared	Cyclic	Independent
32	7	1	0	32	1.00	7.00	7.00
32	7	1	44	180	6.00	7.00	7.00
32	7	1	112	112	4.00	5.00	4.00
32	7	3	0	32	1.00	3.00	3.63
32	7	3	44	150	7.00	7.00	7.00
32	7	3	112	112	4.00	7.00	6.13
32	12	1	0	32	1.00	12.00	11.50
32	12	1	76	308	10.00	11.00	11.00
32	12	1	192	192	6.00	7.00	7.00
32	12	3	0	32	1.00	4.00	5.00
32	12	3	76	319	12.00	12.00	12.00
32	12	3	192	192	6.00	8.00	8.00
128	7	1	0	128	1.00	7.00	7.00
128	7	1	179	717	6.00	7.00	7.00
128	7	1	448	448	4.00	5.00	4.13
128	7	3	0	128	1.00	3.00	4.25
128	7	3	179	618	7.00	7.00	7.00
128	7	3	448	448	4.00	7.00	6.25
128	12	1	0	128	1.00	12.00	12.00
128	12	1	307	1229	10.00	11.00	11.00
128	12	1	768	768	6.00	7.00	7.00
128	12	3	0	128	1.00	4.00	7.25
128	12	3	307	1300	12.00	12.00	12.00
128	12	3	768	768	6.00	8.00	8.00

Table 68: Controlled exact-stop column-evidence sharing for seven-member median pools. Each row represents eight seeds in each of three layouts with identical m, T, d, b and minimum entry-witness size. For each seed, we verify a certificate of size K_{col} and rule out every smaller column subset. Shared, Cyclic and Independent report mean minimum column counts across eight seeds; row costs and the entry minimum are identical across layouts. All 576 instances use the complete entry and column acquisition solvers. The synthetic inputs test query geometry, not NLP quality.

η	Fixed	Random	Width	Ref.
.02	2067.3	2111.6	1800.6	2363.8
.10	567.4	667.3	660.9	940.7

Table 69: Mean queries on thirteen-stream WMT23, mean pooling, $q = .30$, $\alpha = .10$: the same 40 splits and fixed paths, with stopping gap $\lceil \eta |S_0| \rceil$. Counts are reaggregated from saved records and rounded half up. Width has the lowest mean at .02, fixed order at .10.

ranges from 84.9–90.2% at exact stopping and from 72.4–76.2% at nominal tolerance .10. On thirteen-stream WMT23 with mean pooling at $q = .30$, reference priority requires more queries than width at all six tolerances. Table 69 makes the ranking change explicit: width has the lowest mean at tolerance .02, and fixed order at .10, among these four policies.

For median paths, Corollary 1 supplies the matching offline entry witness Q_{ε_η}. At nominal $q = .05$, its RewardBench mean falls from 196.6 entries at exact stopping to 155.0 at .02 and 77.9 at .10. The corresponding thirteen-stream WMT23 means fall from 415.9 to 332.3 and 137.4 entries. Table 70 separates intrinsic requirements from multiplicative entry discovery ratios. For each split with $Q_{\varepsilon_\eta} > 0$, we compute $Q_{\text{online}}/Q_{\varepsilon_\eta}$ and then average these ratios across included splits, reporting their number. A ratio of average query counts is a different statistic. A positive-gap query count below Q_0 does not violate a lower bound, because the matching minimum is Q_{ε_η} for target gap ε_η. These median closed forms do not characterize mean-pooling or whole-evaluator minima. Theorem 3 handles joint mean entries; Appendix M.9 reports numerically closed full-batch mean minima for the primary and exact targets. Other tolerances have no matching full-grid mean-entry optimization here.

The accompanying records also retain actual regret, certificate gap, zero-query stops, panel fractions, exact-path query savings and each stop's mask. Relative query savings, normalized by the exact-stopping count, are undefined when that count is zero. No tolerance or acquisition method is selected retrospectively to replace the main protocol.

Panel	η	$\varepsilon/\lvert S_0\rvert$ (%)	Q_ε	Online queries	Ratio	Positive Q_ε	Zero-query (%)
RewardBench 2	0	0.0	196.6	327.6	1.895	40	0.0
RewardBench 2	0.005	0.9	182.6	313.6	1.924	40	0.0
RewardBench 2	0.01	1.6	169.2	300.3	1.951	40	0.0
RewardBench 2	0.02	2.5	155.0	285.0	1.992	40	0.0
RewardBench 2	0.05	5.4	124.0	233.7	1.950	39	2.5
RewardBench 2	0.1	10.4	77.9	160.2	2.110	36	10.0
WMT23, thirteen streams	0	0.0	415.9	622.6	1.679	36	10.0
WMT23, thirteen streams	0.005	0.8	400.2	590.0	1.660	36	10.0
WMT23, thirteen streams	0.01	1.5	349.7	559.2	1.718	36	10.0
WMT23, thirteen streams	0.02	2.4	332.3	523.2	1.698	36	10.0
WMT23, thirteen streams	0.05	5.4	234.6	409.1	1.795	34	15.0
WMT23, thirteen streams	0.1	10.3	137.4	252.2	1.834	26	35.0
WMT23, twelve streams	0	0.0	41.1	56.0	1.952	4	90.0
WMT23, twelve streams	0.005	1.0	38.8	52.0	1.957	4	90.0
WMT23, twelve streams	0.01	1.4	37.1	49.1	2.018	4	90.0
WMT23, twelve streams	0.02	2.4	34.2	43.3	1.954	3	92.5
WMT23, twelve streams	0.05	5.4	28.3	33.2	1.714	2	95.0
WMT23, twelve streams	0.1	10.4	6.3	18.3	2.901	1	97.5
WMT24 en-de	0	0.0	37.1	60.2	1.775	3	92.5
WMT24 en-de	0.005	0.6	33.5	54.2	1.586	2	95.0
WMT24 en-de	0.01	1.1	31.4	51.0	1.597	2	95.0
WMT24 en-de	0.02	2.2	27.2	44.1	1.604	2	95.0
WMT24 en-de	0.05	5.1	14.3	22.6	1.642	2	95.0
WMT24 en-de	0.1	10.1	2.6	4.6	1.779	1	97.5
WMT24 en-es	0	0.0	19.0	46.4	2.172	2	95.0
WMT24 en-es	0.005	0.6	17.4	42.2	2.430	1	97.5
WMT24 en-es	0.01	1.2	16.0	38.4	2.401	1	97.5
WMT24 en-es	0.02	2.1	13.5	33.2	2.457	1	97.5
WMT24 en-es	0.05	5.1	5.5	11.6	2.123	1	97.5
WMT24 en-es	0.1	10.1	0.0	0.0	–	0	100.0
WMT24 ja-zh	0	0.0	271.6	371.8	1.887	13	67.5
WMT24 ja-zh	0.005	0.7	254.5	344.0	1.964	13	67.5
WMT24 ja-zh	0.01	1.2	242.9	320.3	1.939	11	72.5
WMT24 ja-zh	0.02	2.2	150.6	276.4	2.383	9	77.5
WMT24 ja-zh	0.05	5.2	105.4	161.2	1.915	5	87.5
WMT24 ja-zh	0.1	10.1	24.0	49.6	2.046	2	95.0

Table 70: Tolerance-specific median entry witnesses at $q = .05$, $\alpha = .10$, with reference-priority acquisition. Witness and online counts average all 40 splits. Ratio averages splitwise $Q_{\text{online}}/Q_\varepsilon$ over positive witnesses, whose count is reported separately. Zero witnesses have undefined ratios. Effective tolerance includes integer rounding. These descriptive ratios are not online competitive guarantees.

O Choosing the Selection and Information Protocol

Fix the reference mapping, consensus rule, cutoff and replacement family before certification. Changing the mapping or evaluator pool can change the baseline selected set and stability question. When it does, fewer queries under the new protocol do not establish an improvement for the original list. The existing mapping ablations support sensitivity analysis rather than a universal midrank recommendation.

Choose the query unit supported by the scoring interface: entries target individual outputs, whereas columns reveal all N scores. For an operational comparison, include reference fitting, base scores, preprocessing, calibration when used and certificate computation. The accounting boundary is specified in Appendix B; deployment claims require measured execution costs.

Choose retention tolerance in integer outputs. If at most r outputs of retained-count shortfall are acceptable, set $\varepsilon = r$. The replay rounds fractions up. To permit shortfall of at most a fraction η of $|S_0|$, use $\varepsilon = \lfloor \eta |S_0| \rfloor$ instead, a different target. Exact stopping certifies the optimal retained count without necessarily revealing all scores; a positive gap bounds retained-count shortfall. This changes the stopping target without relaxing the churn budget α. Report invariant retention separately from noninvariant outputs admitted within

that budget. Restricting the commitment mask for a subsequent content policy preserves the fixed-denominator budget but can reduce the returned count (Appendix L).

If a content or calibrated acceptance rule fixes an eligible set A_{cal} that is visible before certification, apply the capacity calculation to $S_0 \cap A_{\text{cal}}$. The same monotone-cost argument gives $V_A^- \le V_A^* \le V_A^+$ and a gap guarantee relative to this restricted full-information optimum. It does not preserve near-optimality over the original S_0. Pruning a calibrated mask preserves an expected bound only for the same fixed-denominator, nonnegative loss under its calibration assumptions. A churn guarantee supplies no semantic-quality bound. Report post-certification quality audits separately. If audit labels select a stopping point, statistical quality claims must account for that choice. Under the fixed selection protocol, deterministic churn and count-gap guarantees hold at any chosen state satisfying the certificate conditions of Proposition 1.

Fixed additive column prices can change the cheapest evidence without changing the certification target (Appendix M.13). Real batching, model loading and shared computation can instead make prices depend on the action history. Deployment claims require a verified scorer revision and interface, measured setup and inference costs, and the same target and retained-count accounting. The current priced-witness study does not replace those measurements.

P Prediction and Certificate Discovery

P.1 Prediction accuracy and the stopping objective

Reference priority uses predicted flips to choose queries, while stopping depends on retention capacities under lower and upper row costs. Low prediction error alone does not identify the observations that certify safety or exclude larger sets. For an active selected row, a queried candidate reduces upper cost by $p_i - f_{ir}$ and raises lower cost by f_{ir}. For odd-sized median pools, agreements and oppositions contribute to different sides of the certificate. Ranking rows by predicted total cost does not identify useful evidence across candidates or which rows determine the capacity bounds.

We analyze the six original NLP panels, both pooling rules, four depths and forty splits with full-reference right ECDF and $\alpha = .10$. Four original entry orders give 7,680 paths. Evaluating each at $\eta \in \{0, .02, .10\}$ and $\varepsilon_\eta = \lceil \eta |S_0| \rceil$ gives 23,040 stopping states and 5,760 matched reference-versus-width comparisons. Paths, mappings, cutoffs and stopping rules remain unchanged. Full test matrices are used only for post hoc diagnostics, following the information boundary and aggregation conventions in Appendix B.

P.2 Diagnostic definitions and checks

Let $\hat{f}_{ir}$ be the clipped ridge prediction and $\mathcal{A} = \{i \in S_0 : p_i > 0\}$. Entry prediction error is

$$E_{\text{entry}} = \frac{1}{mT} \sum_{i \in \mathcal{A}, r} \frac{|\hat{f}_{ir} - f_{ir}|}{p_i}.$$

It is undefined when $m = 0$. Initial row-total rank correlation is the Spearman correlation of $\sum_r \hat{f}_{ir}$ and F_i, using average ranks for ties over $\mathcal{A}$. Constant vectors or fewer than two active rows give an undefined correlation; tables report the number of defined splits.

For reference first-stop count $Q > 0$, evaluate error at $t = Q - 1$, before the first query meeting the stopping criterion. For $Q = 0$, use the initial state $t = 0$. Equivalently, $t = \max(0, Q - 1)$. Let $U_i(t)$ contain unqueried candidates and $\mathcal{J}(t)$ the active rows with at least one such candidate. Estimated total cost is $\hat{F}_i(t) = \ell_i(t) + \sum_{r \in U_i(t)} \hat{f}_{ir}$. Remaining-row error is

$$E_{\text{rem}}(t) = \frac{1}{|\mathcal{J}(t)|} \sum_{i \in \mathcal{J}(t)} \frac{|\hat{F}_i(t) - F_i|}{T p_i}.$$

It is undefined if no row remains. Its row set depends on the path and target. The error concerns row-total estimates on rows with unqueried entries, normalized by the full row scale Tp_i. Because the rows and prefix depend on acquisition and stopping, this diagnostic is not an independent measure of initial prediction accuracy or a causal explanation of query cost.

Paired differences use $\Delta Q = Q_{\text{reference}} - Q_{\text{width}}$ for the same panel, split, depth, pooling rule and tolerance. Negative values favor reference. For median pooling, excess entry queries are $Q_{\text{reference}} - Q_{\varepsilon_\eta}$, relative to the offline witness for the same target. The original full-panel mean ratios use the matching exact subset in Appendix M.7; the expanded grid covers both targets in Appendix M.9. The small-batch minima in Appendix M.6 remain separate denominators. Records also include fixed-order comparisons, strict wins against all

three other methods, remaining-row correlations and within-configuration prediction-error/query-difference correlations. Undefined values remain missing. Forty overlapping splits give descriptive summaries rather than independent replications for significance tests.

We check all 22,317,820 cached replies against full matrices and reconstruct the retained-count bounds at every saved stopping state. Reference tie rules are checked at 5,693 diagnostic prefixes, including WMT24 fixed-point priorities. Across 1,920 reference paths, we deduplicate the first-stop positions and their predecessors for three tolerances, retaining positions before the exact stop where a next query exists: 904 median and 4,789 mean prefixes. A separate implementation recomputes ridge predictions and average-rank correlations on 48 configurations, matches all 23,040 original stopping counts and checks 2,016 summary values.

P.3 Acquisition towards certificate frontiers

Frontier priority reuses the reference-only ridge predictions $\hat{f}_{ir}$, clipped to $[0, p_i]$. The policy operates on active selected rows. Let V_A be capacity restricted to these rows; equivalently, extend their cost vector with zero invariant costs and subtract m_0 from V. Define the priority only at a query state with $v^- < v^+$, where $v^- = V_A(u)$ and $v^+ = V_A(\ell)$ are active counts. Let $\hat{F}_i = \ell_i + \sum_{r \in U_i} \hat{f}_{ir}$. Sort $\hat{F}_i$ with the original label-blind tie order, and set

$$s = \min\{v^+, \max\{v^- + 1, V_A(\hat{F})\}\}.$$

The first s predicted rows form A_s. Because $s > v^-$, no s-row set is upper-cost feasible. Their safety deficit is therefore $D_s = \sum_{i \in A_s} u_i - b_H > 0$. The priority targets exclusion of $s+1$ active rows, independently of stopping tolerance. When $p = s+1 \le m$, set $t = \lceil \hat{F}_{(p)} \rceil$ and define the exclusion deficit

$$D_e = \sum_i (t - \ell_i)_+ - pt + b_H + 1.$$

For an unqueried entry, its priority is

$$\frac{\mathbf{1}[i \in A_s](p_i - \hat{f}_{ir})}{D_s} + \frac{\min\{\hat{f}_{ir}, (t - \ell_i)_+\}}{D_e},$$

where the exclusion term is omitted if $p > m$ or $D_e \le 0$. The two terms estimate fractional progress towards feasible safety and exclusion proofs, including when the integer count gap stays unchanged. They are point-prediction heuristics, not expected query-optimality claims. Safety-only retains the first term; exclusion-only retains the second. If all priorities are zero, use largest width with the same tie order. Predictions and priorities use nearest-even 10^{-12} ticks; ties follow the seeded label-blind row priority, then candidate index. Before acquisition, active rows are sorted by the original tie permutation generated with seed $s_{\text{panel}} + \text{split} + 700001$. Estimated costs add prediction ticks to observed integer costs converted to the same 10^{-12} scale. Acquisition is independent of the stopping tolerance, so one path supplies first primary and exact stops. Each selection recomputes the predicted ordering and scores, taking $O(m \log m + mT)$ time and $O(mT)$ working space. Only the query callback reveals a test contribution. All policies retain the same deterministic interval and stopping guarantees.

Matched acquisition and ablations. On all 960 mean instances, joint, safety-only and exclusion-only frontier paths each continue through exact stopping and record the first primary stop. No quality label or unqueried test contribution enters acquisition.

At the primary stop, joint frontier averages 5210.6 entries, compared with 5286.5 for reference and 5349.7 for width. Against reference it has 403 wins, 554 losses and three ties; the lower overall mean coexists with losses on more instances. Safety-only averages 5207.6, slightly below joint, while exclusion-only averages 5343.7. The experiment does not establish that both terms are necessary for lower query cost. Tables 72 and 73 give all depths, including unfavorable results. Unlike the original budget-independent orders, frontier uses the current budget in its deficits; cross-budget paths cannot be reused.

When exclusion is required. Table 71 stratifies these saved comparisons by the full-information optimum V^*, baseline count $|S_0|$ and tolerance. If $V^* + \varepsilon < |S_0|$, even the largest feasible safe set cannot meet the gap target while the optimistic count remains $|S_0|$, so exclusion evidence is necessary. If $V^* < |S_0| \le V^* + \varepsilon$, sufficient safety evidence can meet the target without lowering that count. The third stratum permits full retention. These conditions are fixed within each matched comparison. They use hidden full-information costs for post hoc diagnosis, not for acquisition, and differ from

whether the heuristic's exclusion term is active at a particular history.

At primary stopping, joint frontier saves an average of 20.28 entries over safety-only on the 32 targets requiring exclusion, with 25 wins, six losses and one tie. It uses 3.72 more entries on average on the 903 full-retention targets and 6.40 more on the remaining 25. At exact stopping, it saves 20.21 entries on average on the 57 targets requiring exclusion, but loses on ten. Thus the overall means conceal a conditional benefit of the exclusion term in these instances. The condition neither guarantees improvement nor supplies an online switching rule. These reused, overlapping-split comparisons establish no independent-task confirmation or stepwise causal mechanism.

Implementation checks. The 2,880 paths contain 5,760 stopping states. Replay checks 15,838,138 replies, every stopping mask and the failure to stop at the preceding state. A separate priority calculation checks 14,832 choices over 144 paths spanning all panels and depths.

Stop	Instance condition	Targets	Mean ΔQ	Median ΔQ	Wins/losses/ties
Primary	$V^* + \varepsilon < \lvert S_0 \rvert$	32	-20.28	-11	25/6/1
Primary	$V^* = \lvert S_0 \rvert$	903	+3.72	+0	2/65/836
Primary	$V^* < \lvert S_0 \rvert \leq V^* + \varepsilon$	25	+6.40	+3	10/14/1
Exact	$V^* + \varepsilon < \lvert S_0 \rvert$	57	-20.21	-7	40/10/7
Exact	$V^* = \lvert S_0 \rvert$	903	+2.63	+0	5/55/843

Table 71: Matched frontier ablation by instance condition on the original mean panels, at $\alpha = .10$ and all four depths. ΔQ is the joint-frontier query count minus the safety-only count, so negative values and wins favor joint frontier. Both counts use the same instance and stopping target. Primary uses $\varepsilon = \lceil .02 \lvert S_0 \rvert \rceil$; exact uses zero. The non-full, tolerance-covered stratum is empty at exact stopping. Conditions use full-information V^* for post hoc diagnosis only. These descriptive comparisons reuse saved paths and do not treat overlapping splits as independent samples.

Panel	q	Reference	Width	Frontier	S-only	E-only	W/L Ref.	W/L Width
RewardBench 2	0.05	727.9	770.6	752.6	743.0	770.5	0/39	36/3
RewardBench 2	0.10	1289.6	1356.2	1317.6	1316.1	1357.0	3/37	40/0
RewardBench 2	0.20	1919.6	1925.8	1839.4	1839.4	1925.8	40/0	40/0
RewardBench 2	0.30	2028.7	1781.6	1716.9	1716.9	1781.6	40/0	40/0
WMT23, 13	0.05	1873.7	2013.2	1921.8	1904.6	2012.0	5/34	38/0
WMT23, 13	0.10	3518.8	3782.5	3637.3	3637.3	3782.5	0/39	40/0
WMT23, 13	0.20	2419.2	2167.6	2063.8	2063.8	2167.6	39/1	40/0
WMT23, 13	0.30	2363.8	1800.6	1732.9	1732.9	1800.6	40/0	37/3
WMT23, 12	0.05	1757.7	1841.1	1800.2	1787.5	1840.1	4/36	36/4
WMT23, 12	0.10	3047.4	3239.9	3138.9	3138.2	3239.9	3/37	40/0
WMT23, 12	0.20	2413.8	2273.8	2203.0	2203.0	2273.8	40/0	38/2
WMT23, 12	0.30	2979.8	2520.6	2437.4	2437.4	2520.6	40/0	38/2
WMT24 en-de	0.05	5619.0	5920.0	5815.1	5805.8	5920.0	3/37	37/3
WMT24 en-de	0.10	6538.0	6785.0	6625.3	6625.2	6785.3	17/23	38/2
WMT24 en-de	0.20	9591.7	9911.6	9730.2	9730.2	9911.6	12/28	40/0
WMT24 en-de	0.30	11135.4	10569.2	10208.6	10208.6	10569.2	40/0	39/1
WMT24 en-es	0.05	4962.1	5228.2	5130.8	5109.5	5227.8	2/38	36/4
WMT24 en-es	0.10	6038.1	6272.6	6092.9	6092.9	6272.6	13/27	40/0
WMT24 en-es	0.20	8509.9	8836.9	8639.3	8639.3	8836.9	7/33	38/2
WMT24 en-es	0.30	8924.9	8309.6	8037.0	8037.0	8309.6	40/0	35/5
WMT24 ja-zh	0.05	4402.5	4590.0	4533.1	4531.0	4552.7	1/39	28/10
WMT24 ja-zh	0.10	7834.2	8182.0	8068.1	8069.8	8157.2	1/39	36/4
WMT24 ja-zh	0.20	12544.0	13133.2	12904.6	12905.7	13054.0	7/33	35/5
WMT24 ja-zh	0.30	14436.1	15180.5	14707.1	14707.1	15180.5	6/34	40/0

Table 72: Mean-pooling acquisition at $V^+ - V^- \leq \lceil .02 \lvert S_0 \rvert \rceil$, $\alpha = .10$, on complete original batches. Query columns average entry counts across all 40 splits. Frontier sums predicted fractional progress toward safety and exclusion; S-only and E-only retain the corresponding single term. W/L counts Frontier strict wins/losses against Reference or Width; remaining splits tie. All methods use the same normalized scores, bound updates and certification target. Prediction-based methods share reference-fitted models; none uses quality labels. These descriptive comparisons do not establish a universally best policy.

Panel	q	Reference	Width	Frontier	S-only	E-only	W/L Ref.	W/L Width
RewardBench 2	0.05	759.2	787.8	771.2	764.6	787.8	5/34	32/2
RewardBench 2	0.10	1341.4	1387.1	1354.7	1353.8	1387.7	7/33	40/0
RewardBench 2	0.20	2006.6	2006.4	1906.4	1906.4	2006.4	40/0	40/0
RewardBench 2	0.30	2148.8	1906.5	1819.1	1819.1	1906.5	40/0	40/0
WMT23, 13	0.05	1940.4	2053.1	1964.4	1953.6	2053.0	11/28	40/0
WMT23, 13	0.10	3647.6	3856.5	3715.9	3715.9	3856.5	2/38	40/0
WMT23, 13	0.20	2615.6	2352.5	2254.1	2254.1	2352.5	40/0	40/0
WMT23, 13	0.30	2685.8	2108.1	2046.9	2046.9	2108.1	40/0	36/4
WMT23, 12	0.05	1823.0	1879.4	1839.6	1833.6	1879.2	13/26	37/2
WMT23, 12	0.10	3160.6	3311.7	3218.7	3217.8	3311.7	4/36	39/1
WMT23, 12	0.20	2611.1	2447.8	2377.0	2377.0	2447.8	40/0	38/1
WMT23, 12	0.30	3274.4	2816.3	2715.6	2715.6	2816.3	40/0	39/1
WMT24 en-de	0.05	5809.8	6017.4	5921.8	5917.3	6017.4	6/33	35/3
WMT24 en-de	0.10	6847.4	7007.2	6885.7	6884.6	7007.2	17/22	37/3
WMT24 en-de	0.20	10201.6	10388.9	10236.8	10236.8	10388.9	20/20	34/5
WMT24 en-de	0.30	11985.3	11327.7	10967.6	10967.6	11327.7	40/0	38/2
WMT24 en-es	0.05	5126.9	5316.6	5227.0	5211.0	5316.6	4/36	34/6
WMT24 en-es	0.10	6320.6	6471.5	6337.9	6337.9	6471.5	15/25	40/0
WMT24 en-es	0.20	9054.4	9263.8	9094.9	9094.9	9263.8	17/23	37/3
WMT24 en-es	0.30	9703.0	9008.4	8735.0	8735.0	9008.4	40/0	35/5
WMT24 ja-zh	0.05	4554.2	4672.9	4628.0	4637.5	4670.3	12/28	29/10
WMT24 ja-zh	0.10	8093.6	8332.5	8246.4	8251.1	8331.0	7/33	36/3
WMT24 ja-zh	0.20	13115.0	13549.6	13380.4	13382.7	13548.5	10/29	35/5
WMT24 ja-zh	0.30	15245.9	15846.2	15291.7	15291.7	15846.2	21/19	40/0

Table 73: Mean-pooling acquisition at $V^+ = V^-$, $\alpha = .10$, on complete original batches. Query columns average entry counts across all 40 splits. Frontier sums predicted fractional progress toward safety and exclusion; S-only and E-only retain the corresponding single term. W/L counts Frontier strict wins/losses against Reference or Width; remaining splits tie. All methods use the same normalized scores, bound updates and certification target. Prediction-based methods share reference-fitted models; none uses quality labels. These descriptive comparisons do not establish a universally best policy.

P.4 Prediction accuracy and observed query cost

Tables 74 and 75 report every depth at the main tolerance. Across the three original mean-pooling panels, reference priority uses fewer entries on average than width at $q = .05, .10$, but more at $q = .30$. These are configuration means, not a claim about every split. On thirteen-stream WMT23 with mean pooling, initial rank correlation is .280 at $q = .05$, where reference beats width on all 40 splits with mean difference -139.5 queries. At $q = .30$, correlation is .747, but reference loses on all 40 splits with difference 563.2. Reference uses 2,363.8 queries versus width's 1,800.6. Entry errors are .213 and .260; remaining-row errors at the defined diagnostic states are .138 and .139. Depth changes the cutoff, selected rows and certification target, so this comparison does not isolate a cause. Initial row-total rank correlation alone does not identify the rule with the fewest queries in these configurations.

On RewardBench with median pooling at $q = .05$, initial row-total correlation is $-.260$, yet reference beats width on 26 of 40 splits and saves 7.2 queries on average. Query choices also depend on candidate-level predicted upper-cost reductions and observed replies; stopping depends on which rows determine the count bounds. These summaries do not establish that better predictions are harmful or identify an optimal priority. Table 13 remains a single-split illustration of different query choices. All three tolerances and paired records accompany the analysis code.

Panel	q	Entry error	$\rho(n)$	Remaining error	ΔQ	Wins / 40
RewardBench 2	0.05	0.508	-0.260 (40)	0.159	-7.2	26
RewardBench 2	0.10	0.501	-0.255 (40)	0.180	-20.4	32
RewardBench 2	0.20	0.490	-0.141 (40)	0.186	-6.0	19
RewardBench 2	0.30	0.491	-0.080 (40)	0.206	-0.1	1
WMT23, 13 streams	0.05	0.456	0.031 (40)	0.138	-212.2	32
WMT23, 13 streams	0.10	0.502	0.097 (40)	0.159	0.0	0
WMT23, 13 streams	0.20	0.504	0.122 (40)	0.136	0.0	0
WMT23, 13 streams	0.30	0.494	-0.042 (40)	0.118	0.0	0
WMT23, 12 streams	0.05	0.516	0.141 (40)	0.185	-0.6	2
WMT23, 12 streams	0.10	0.505	-0.018 (40)	0.160	-8.1	4
WMT23, 12 streams	0.20	0.506	0.024 (40)	0.135	0.9	0
WMT23, 12 streams	0.30	0.493	-0.125 (40)	0.121	3.0	0
WMT24 en-de	0.05	0.476	0.026 (38)	0.197	-18.3	2
WMT24 en-de	0.10	0.496	-0.047 (40)	0.174	-13.6	4
WMT24 en-de	0.20	0.501	0.024 (40)	0.152	0.0	0
WMT24 en-de	0.30	0.501	0.036 (40)	0.150	0.0	0
WMT24 en-es	0.05	0.483	-0.020 (40)	0.185	0.9	0
WMT24 en-es	0.10	0.498	-0.104 (40)	0.170	0.0	0
WMT24 en-es	0.20	0.504	0.120 (40)	0.150	0.0	0
WMT24 en-es	0.30	0.507	0.038 (40)	0.134	0.0	0
WMT24 ja-zh	0.05	0.514	-0.213 (40)	0.199	-13.4	7
WMT24 ja-zh	0.10	0.540	-0.054 (40)	0.279	-24.1	4
WMT24 ja-zh	0.20	0.523	0.018 (40)	0.259	-5.9	1
WMT24 ja-zh	0.30	0.514	-0.022 (40)	0.240	0.0	0

Table 74: Prediction and discovery diagnostics for median pooling, full-reference right ECDF, $\alpha = .10$ and $\varepsilon = \lceil .02|S_0| \rceil$. Each error statistic is defined on all 40 splits and averages them; query differences also average 40 splits. $\rho(n)$ is mean initial row-total rank correlation over the n defined splits. Remaining error is normalized by the full row scale Tp_i and uses $t = \max(0, Q - 1)$, the initial state for zero-query stops; ΔQ is the reference-minus-largest-width entry count, not a percentage. Wins count strict query reductions against width. Zero-query stops may contribute to prediction diagnostics when active rows remain.

Panel	q	Entry error	$\rho(n)$	Remaining error	ΔQ	Wins / 40
RewardBench 2	0.05	0.235	0.297 (40)	0.152	-42.6	40
RewardBench 2	0.10	0.231	0.581 (40)	0.137	-66.6	40
RewardBench 2	0.20	0.219	0.718 (40)	0.128	-6.2	21
RewardBench 2	0.30	0.225	0.746 (40)	0.144	247.1	0
WMT23, 13 streams	0.05	0.213	0.280 (40)	0.138	-139.5	40
WMT23, 13 streams	0.10	0.194	0.687 (40)	0.158	-263.7	40
WMT23, 13 streams	0.20	0.226	0.705 (40)	0.141	251.5	1
WMT23, 13 streams	0.30	0.260	0.747 (40)	0.139	563.2	0
WMT23, 12 streams	0.05	0.228	0.591 (40)	0.147	-83.3	40
WMT23, 12 streams	0.10	0.191	0.694 (40)	0.155	-192.5	40
WMT23, 12 streams	0.20	0.249	0.720 (40)	0.141	140.0	4
WMT23, 12 streams	0.30	0.264	0.753 (40)	0.134	459.2	0
WMT24 en-de	0.05	0.215	0.309 (40)	0.125	-301.0	40
WMT24 en-de	0.10	0.195	0.589 (40)	0.135	-247.0	36
WMT24 en-de	0.20	0.240	0.686 (40)	0.146	-319.9	32
WMT24 en-de	0.30	0.237	0.685 (40)	0.145	566.1	0
WMT24 en-es	0.05	0.200	0.522 (40)	0.124	-266.2	40
WMT24 en-es	0.10	0.203	0.619 (40)	0.139	-234.6	37
WMT24 en-es	0.20	0.234	0.711 (40)	0.153	-326.9	39
WMT24 en-es	0.30	0.243	0.710 (40)	0.143	615.3	0
WMT24 ja-zh	0.05	0.266	0.509 (40)	0.153	-187.5	40
WMT24 ja-zh	0.10	0.254	0.645 (40)	0.161	-347.8	40
WMT24 ja-zh	0.20	0.245	0.535 (40)	0.187	-589.2	36
WMT24 ja-zh	0.30	0.237	0.685 (40)	0.202	-744.4	40

Table 75: Prediction and discovery diagnostics for mean pooling, full-reference right ECDF, $\alpha = .10$ and $\varepsilon = \lceil .02|S_0| \rceil$. Each error statistic is defined on all 40 splits and averages them; query differences also average 40 splits. $\rho(n)$ is mean initial row-total rank correlation over the n defined splits. Remaining error is normalized by the full row scale Tp_i and uses $t = \max(0, Q - 1)$, the initial state for zero-query stops; ΔQ is the reference-minus-largest-width entry count, not a percentage. Wins count strict query reductions against width. Zero-query stops may contribute to prediction diagnostics when active rows remain.

P.5 Same-instance row and candidate interventions

Across all 1,920 original panel–pooling–depth–split configurations, we replaced the reference policy's row estimates, candidate priorities, or both with true flip contributions. True-row priority is $\ell_i + \sum_{r \in U_i} f_{ir} = F_i$; it uses full hidden row totals. True-candidate priority orders unqueried candidates by increasing f_{ir}, keeping the row rule otherwise unchanged. The Both variant combines these two interventions. Ties keep the reference policy's seeded row order and candidate-index order. These are full-information mechanism interventions, not online methods. The score matrix, maps, cutoff, eligible rows, initial bounds, budgets and certificate conditions are fixed within each instance; only query priorities change.

The reference, true-row, true-candidate and combined paths run through exact stopping and record the first primary stop, giving 7,680 paths and 15,360 states. Width comparisons use the corresponding original acquisition records. The reference method reproduces all 3,840 original exact and primary query counts. An independent checker validates 21,632,445 replies, every stopping mask and first stopping state. It also reconstructs 416,920 query choices by direct row and candidate sorting on all 48 split-zero configurations.

Table 76 reports mean-pooling query counts averaged over 40 splits at each depth. On thirteen-stream WMT23 with mean pooling at $q = .30$, true-row priority saves queries in all 40 splits relative to reference, but its mean remains 379.7 queries above width. True-candidate priority alone is worse on 36 splits, better on one and tied on three. At $q = .05$, true-row priority is worse on 31 and better on nine splits. True row costs rank outputs by realized churn; query value also depends on current safety deficits, achievable bound reductions and, when exclusion is active, observations that rule out larger sets. True contributions therefore do not make this fixed priority rule query-optimal. These interventions change priority within an instance without attributing each loss to a particular deficit or implying that better predictions are generally unhelpful. Complete results, including median and exact-stop gains and losses, are in `outputs/query-mechanism-interventions/records.csv`.

Panel	q	Reference	Width	True row	True candidate	Both
RewardBench 2	0.05	727.9	770.6	712.7	715.1	712.0
RewardBench 2	0.10	1289.6	1356.2	1261.0	1281.5	1260.2
RewardBench 2	0.20	1919.6	1925.8	1874.5	1914.8	1874.1
RewardBench 2	0.30	2028.7	1781.6	1956.5	2043.4	1956.3
WMT23, 13 streams	0.05	1873.7	2013.2	1892.9	1862.9	1892.1
WMT23, 13 streams	0.10	3518.8	3782.5	3498.5	3474.3	3497.0
WMT23, 13 streams	0.20	2419.2	2167.6	2349.9	2445.1	2349.7
WMT23, 13 streams	0.30	2363.8	1800.6	2180.3	2396.4	2180.0
WMT23, 12 streams	0.05	1757.7	1841.1	1744.7	1726.6	1743.8
WMT23, 12 streams	0.10	3047.4	3239.9	3013.7	3015.1	3012.6
WMT23, 12 streams	0.20	2413.8	2273.8	2341.4	2438.5	2340.9
WMT23, 12 streams	0.30	2979.8	2520.6	2794.5	3030.0	2793.9
WMT24 en-de	0.05	5619.0	5920.0	5582.0	5524.8	5581.3
WMT24 en-de	0.10	6538.0	6785.0	6398.8	6544.7	6398.4
WMT24 en-de	0.20	9591.7	9911.6	9341.0	9663.8	9340.2
WMT24 en-de	0.30	11135.4	10569.2	10633.1	11310.4	10633.0
WMT24 en-es	0.05	4962.1	5228.2	4923.6	4907.8	4922.6
WMT24 en-es	0.10	6038.1	6272.6	5913.7	6028.0	5913.1
WMT24 en-es	0.20	8509.9	8836.9	8319.1	8563.8	8318.7
WMT24 en-es	0.30	8924.9	8309.6	8534.1	9060.6	8533.9
WMT24 ja-zh	0.05	4402.5	4590.0	4399.6	4336.6	4398.6
WMT24 ja-zh	0.10	7834.2	8182.0	7738.2	7687.3	7736.9
WMT24 ja-zh	0.20	12544.0	13133.2	12318.3	12202.0	12316.7
WMT24 ja-zh	0.30	14436.1	15180.5	14236.8	14304.0	14235.6

Table 76: Same-instance mean-pooling interventions on complete original batches. Entries are mean candidate–output query counts over 40 splits, computed from integer sums with decimal half-up rounding. All use $\alpha = .10$ and first stop at $V^+ - V^- \leq \lceil .02|S_0| \rceil$. True row replaces predicted row totals in reference priority; True candidate uses true within-row flip contributions; Both combines them. These full-information diagnostics keep maps, cutoff, initial endpoint bounds, update rules, budget and stopping target fixed. They are not deployable policies. Median and exact-stop summaries accompany the code.

Q Summarization: evidence, discovery and measured scoring cost

This experiment tests whether the distinction between minimum evidence and its discovery extends to summarization. We use the 100-document, sixteen-system SummEval collection (Fabbri et al., 2021). Before inspecting certification outcomes, we fix a document-disjoint 33/33/34 reference/calibration/test split with seed 20261011, four depths $q \in \{.05, .10, .20, .30\}$, $\alpha = .10$, both pooling rules and both stopping tolerances. Calibration documents are reserved and unused. The test batch contains 544 outputs. All configurations are retained, including zero and negative gains. This is a new task evaluation after method development, not a claim that SummEval was absent from evaluator training.

Q.1 Frozen cached-score confirmation

Five metric families supply BARTScore CNN source-to-summary, BERTScore F, MoverScore, Prism average and ROUGE-L F scores. The cached matrix is obtained from `DAMO-NLP-SG/LLM_summeval`, revision `bd7dde15817a` (full revision in the supplement). All 1,600 document/system identities align with the original SummEval annotations. Text alignment uses case, whitespace, punctuation and Penn Treebank bracket normalization; this does not establish identical scoring preprocessing. The raw source is retained locally with its provenance.

All ten choices of three base metrics and two candidates are evaluated. Higher scores are used as better by metric definition; human labels do not orient scores, select queries or choose stops. Each composition refits reference-only right ECDF maps, cutoff and ridge priorities. Ten compositions, four depths and two pooling rules give 80 instances and 160 certification targets. Seven acquisition policies give 560 complete paths and 1,120 stopping states. The integer budget is $\lfloor 2kMT\alpha \rfloor$ with $M = 3, T = 2$. Independent replay checks 83,886 replies and every first stopping state.

Table 77 applies the same ex post capacity stratification used for the earlier NLP panels. The full-retention stratum has $V^* = |S_0|$; the exclusion stratum has $V^* + \varepsilon < |S_0|$; the remaining near-full stratum has $V^* < |S_0| \leq V^* + \varepsilon$. These conditions are computed after acquisition and are not inputs to

the policy. Under mean pooling, joint frontier saves two queries on average in the exclusion stratum at either stop. It increases the mean query count in the full-retention stratum, while median pooling shows no analogous mean advantage. Across the eight near-full mean primary targets, joint frontier uses 0.75 more queries on average: one target uses fewer, four tie and three use more than safety-only. Thus the observed mean-pooling pattern extends to this task conditionally; it is neither a universal advantage nor a per-instance guarantee. The compositions and depths share one test partition and are not independent replications.

For the 48 non-full-retention mean targets, each joint and best separately optimal union program closes its own incumbent–lower-bound gap numerically. Reconstructed integer witnesses satisfy both count constraints. Joint evidence is smaller on 27 targets and equal on 21; numerical bound closure is not an exact rational minimality proof. The 68 full-retention targets across both pooling rules admit a direct integer computation described below. Odd-median targets also use the closed-form certificates, and enumeration of all four candidate-column subsets gives the minimum column count for $T = 2$. Column enumeration does not establish entry-query minimality.

Exact computation in the full-retention subcase. If the true total selected-row cost is at most b_H, every partial lower-cost vector permits all selected rows. On m active rows, the certificate therefore requires safety for at least $m - \min(m, \varepsilon)$ rows; exclusion is automatic. For row i, sort the reductions $p_i - f_{ir}$ in descending order and let $c_i(a)$ be Tp_i minus the sum of the first a reductions. Among all a observations on that row, this is the smallest attainable upper cost. A dynamic program over processed rows, dropped-row count and total queries stores the minimum sum of retained upper costs. A retained row contributes $c_i(a)$ for each $a \in \{0, \ldots, T\}$; a dropped row contributes zero cost and zero queries. The smallest query count with at most ε drops and cost at most b_H is the exact minimum in the rectangular completion model.

To justify the recurrence, queries on dropped rows are unnecessary, and replacing a retained row's queried set by its largest reductions cannot worsen safety. Conversely, each dynamic-program transition specifies realizable observations and a retained subset. The recurrence therefore covers an optimal certificate and reconstructs a feasible one at its reported cost. Pairing the minimum safety witness with empty exclusion yields the same best separate union. This computation covers all 68 full-retention targets; exhaustive comparison on 60 small instances checks the implementation, not the general argument. It provides no characterization of arbitrary non-full-retention instances.

Q.2 Actual scorer execution

A second experiment recomputes scores from the source texts, with the same document split and one fixed execution composition: BARTScore, BERTScore and MoverScore as base metrics. Candidates are Prism and ROUGE-L. This is a fresh panel rather than a reproduction of the cached matrix. BERTScore uses unrescaled RoBERTa-large layer-17 F with IDF disabled; MoverScore uses the original BERT-MNLI unigram power-mean implementation and reference-only IDF. Prism uses the bidirectional average of model m39v1 with a single reference. ROUGE-L uses `rouge-score` 0.1.2 with stemming and a single reference. BARTScore uses `facebook/bart-large-cnn` for source-to-summary likelihood. The scorer wrappers follow `neulab/BARTScore` revision `248f511cb34a` (full revision in the supplement); the supplement records exact model revisions and dependencies.

One A40 GPU executes FP32 scoring with TF32 disabled, PyTorch 2.4.1/CUDA 12.4 and four CPU threads. Each run loads models from local weights, scores all 528 reference outputs with five metrics and all 544 test outputs with three base metrics, fits maps and priorities, and then runs the specified candidate-scoring policy. Full scoring calls each metric once per scoring phase, processing inputs internally in batches of eight. Acquisition calls the candidate scorer once per requested entry without reading a precomputed candidate-test matrix. All methods include the common preparation, including the ridge fit. Eight methods, four depths, two pooling rules and three repetitions give 192 runs in a seeded random order. Both stopping states are recorded on each online path. Model downloads, dependency installation and post-path rescore audits are outside the measured pipeline interval.

All 192 runs have identical mapped reference and base arrays, despite small floating-point differences in raw scores. Every queried entry is rescored in a batch after the path finishes, and all mapped scores agree. Saved traces reconstruct every first stopping state. One early full-scoring run lacks the

Pooling	Stop	Stratum	Targets	Less	Equal	More	Mean query difference
mean	exact	exclusion	24	7	15	2	-2.000
mean	exact	full	16	0	11	5	1.188
mean	primary	exclusion	16	7	7	2	-2.000
mean	primary	full	16	0	10	6	1.875
mean	primary	near full	8	1	4	3	0.750
median	exact	exclusion	22	5	11	6	0.136
median	exact	full	18	1	12	5	1.222
median	primary	exclusion	18	4	10	4	0.167
median	primary	full	18	2	11	5	1.722
median	primary	near full	4	2	0	2	1.750

Table 77: The final column is joint-frontier minus safety-only query count. SummEval cached-score acquisition: joint frontier versus safety-only under the same target. Less/equal/more count query differences, not certificate-minimum differences. All 160 targets are included; strata use full information only after acquisition.

full candidate-array archive; its timing and counters remain in the complete grid, while the other 23 full runs retain those arrays. A score-entry count, a metric function invocation and an internal neural-network forward pass are distinct units. Tables 78 and 79 report the first two along with elapsed time; detailed forward counters and every run accompany the analysis.

Primary reference priority uses 120.25 candidate entries on average, compared with 1,088 for full scoring, but its total score entries fall from 5,360 to 4,392.25 because reference and base scoring are shared costs. Mean total time falls from 108.18 to 89.97 seconds. At the same time, metric function calls rise from 10 to 128.25 because the on-demand interface uses single-entry calls. Each of the 24 paired reference runs is faster than its corresponding full run; this is a descriptive hardware-specific comparison, not a claim about all scorer deployments. Joint frontier and safety-only have similar total times, so the oracle sharing advantage does not establish a runtime advantage between these policies.

Allocated compute cost multiplies each pipeline duration by the RunPod A40 GPU-plus-container quote of USD 0.598 per hour, recorded on 11 October 2026. Primary full and reference runs average USD 0.01797 and USD 0.01495, respectively. These are duration-based allocations, not observed per-run invoices or total account expenditure: installation, downloads, rescore audits, process startup and idle time also consume rented resources. The complete execution queue took 19,217.75 seconds. Table 80 reports within-configuration runtime variation; the supplement retains all 128 method/pooling/depth/stop summaries. Time ratios pair runs by pooling, depth and repetition. The comparison fixes batch-eight full scoring against sequential entry acquisition; online micro-batching, persistent model services and alternative hardware may change the tradeoff.

Method	Candidate entries	Metric calls	Total seconds	Time/full	Allocated USD
Full scoring	1088.00	10.00	108.18	1.000	0.01797
Reference	120.25	128.25	89.97	0.832	0.01495
Width	123.25	131.25	90.43	0.836	0.01502
Fixed	125.25	133.25	89.51	0.827	0.01487
Random	124.12	132.12	89.95	0.832	0.01494
Joint frontier	123.50	131.50	90.24	0.834	0.01499
Safety-only	124.00	132.00	90.26	0.834	0.01499
Exclusion-only	122.88	130.88	90.37	0.835	0.01501

Table 78: Fresh SummEval scoring at primary stopping. Means cover two pooling rules, four depths and three repetitions (24 runs per method). Total time includes model loading and reference/base scoring. Time/full averages paired ratios. Metric calls include reference/base calls and may process multiple outputs; USD is duration times USD 0.598/hour, not a per-run invoice.

Method	Candidate entries	Metric calls	Total seconds	Time/full	Allocated USD
Full scoring	1088.00	10.00	108.18	1.000	0.01797
Reference	125.88	133.88	90.19	0.834	0.01498
Width	128.88	136.88	90.48	0.836	0.01503
Fixed	131.00	139.00	89.73	0.829	0.01491
Random	129.25	137.25	90.20	0.834	0.01498
Joint frontier	129.88	137.88	90.49	0.837	0.01503
Safety-only	130.25	138.25	90.47	0.836	0.01503
Exclusion-only	128.50	136.50	90.45	0.836	0.01502

Table 79: Fresh SummEval scoring at exact stopping. Means cover two pooling rules, four depths and three repetitions (24 runs per method). Total time includes model loading and reference/base scoring. Time/full averages paired ratios. Metric calls include reference/base calls and may process multiple outputs; USD is duration times USD 0.598/hour, not a per-run invoice.

Method	Primary SD range (s)	Exact SD range (s)	Primary range width (s)	Exact range width (s)
Fixed	0.34–1.61	0.32–1.61	0.66–3.13	0.62–3.13
Exclusion-only	0.36–1.34	0.36–1.34	0.64–2.55	0.64–2.56
Joint frontier	0.19–3.05	0.19–3.06	0.37–5.33	0.37–5.33
Safety-only	0.26–1.97	0.26–1.97	0.48–3.44	0.49–3.44
Full	0.33–1.39	0.33–1.39	0.59–2.76	0.59–2.76
Width	0.32–1.29	0.32–1.29	0.60–2.30	0.60–2.30
Random	0.66–1.76	0.66–1.76	1.25–3.20	1.26–3.20
Reference	0.31–1.79	0.32–1.79	0.61–3.56	0.61–3.56

Table 80: Runtime variation within fixed scoring configurations. Each sample standard deviation (SD) and range width uses three repetitions at one pooling/depth setting. The table reports their minimum and maximum across the eight settings for each method; it does not pool configurations to estimate run noise. Complete per-setting means, ranges and SDs accompany the supplement.

R Alternative Churn Objectives

The main certificates average churn uniformly over the declared replacements. This appendix treats fixed nonuniform replacement weights (R.1) and caps that every individual replacement must satisfy (R.2). Throughout, the batch, maps, cutoff, candidate family and completion model of Section 4 are unchanged.

R.1 Fixed replacement weights

Let $w_{jr} \geq 0$ with $\sum_{j,r} w_{jr} = 1$ be declared before candidate-test scores are seen, and write $w_{jr} = n_{jr}/W$ with integers $n_{jr} \geq 0$ and $W = \sum_{j,r} n_{jr}$ (for real weights, any rational approximation from below gives a conservative certificate). For candidate score x on row i, the weighted contribution of candidate r is

$$g_{ir}(x) = \sum_{j=1}^{M} n_{jr}\, \mathbf{1}\big[\sigma_P(i) \neq \sigma_{P^{j\to r}(x)}(i)\big],$$

where $P^{j\to r}(x)$ gives candidate r score x on row i, an integer in units of $1/W$, and the weighted churn is $R_w(C) = \sum_{i\in C} \sum_r g_{ir}(s_r(i))/(2kW)$. Uniform weights ($n_{jr} = 1$, $W = H$) recover Equation 2. Since every indicator is monotone in x, a selected row has the column-dependent cap $c_{ir} = g_{ir}(0)$, and $g_{ir}(1) = 0$. Write $v_{ir} = g_{ir}(s_r(i))$ for the true contribution and $b_W = \lfloor 2k\alpha W \rfloor$ for the integer budget; $R_w(C) \leq \alpha$ is equivalent to $\sum_{i\in C} \sum_r v_{ir} \leq b_W$.

Proposition 13 (Weighted families)**.** *Under fixed nonnegative rational replacement weights:*

(i) With $\ell_i = \sum_{r\in O_i} v_{ir}$ and $u_i = \ell_i + \sum_{r\notin O_i} c_{ir}$, Proposition 1 holds with budget b_W, and both bound vectors are attained by consistent completions.

(ii) Theorem 3 characterizes the minimum additional entry certificate after replacing the row state (a_i, v_i) by $(|J|, \sum_{r\in J} v_{ir}, \sum_{r\in J}(c_{ir} - v_{ir}))$, with $u_i = u_i^0 - \sum_{r\in J}(c_{ir} - v_{ir})$. The outcome-count program of Proposition 4 remains exact after grouping unobserved entries of a row by the pair (c_{ir}, v_{ir}); each row has at most T groups.

(iii) The sandwich $Q_{\text{joint}} \leq Q_{\cup,\text{sep}} \leq Q_{\text{split}} \leq 2Q_{\text{joint}}$ holds.

(iv) For odd-median pools, every contribution satisfies $v_{ir} \in \{0, c_{ir}\}$; hence $Q_{\text{joint}} = Q_{\cup,\text{sep}} = Q_{\text{split}}$ for every history and integer tolerance.

Proof. (i) Each g_{ir} is nonincreasing on a selected row, with maximum c_{ir} at $x = 0$ and zero at $x = 1$, so $\ell_i \leq \sum_r v_{ir} \leq u_i$. Without cross-entry constraints, assigning all unobserved entries zero or one realizes u or ℓ simultaneously; the sorted-prefix and capacity arguments of Proposition 1 use only nonnegative integer costs. (ii) The proof of Theorem 3 (Appendix M.5) uses three facts: a query raises its row's lower cost by its contribution and lowers its upper cost by cap minus contribution; joint endpoint attainability; and Lemma 1, which holds for any nonnegative integer lower costs. With column-dependent caps the upper-cost reduction of a subset J is no longer determined by $(|J|, \sum v)$, which is why the state records it. Entries with equal (c, v) in a row are interchangeable, which gives the grouped program. The budget-indexed dynamic program is pseudo-polynomial in the weight scale W. (iii) The proof of Proposition 5 uses only that both bounds improve monotonically under additional observations and that a minimum joint witness is feasible for each component; neither depends on the weights. (iv) A boundary row has exactly $h + 1$ voting members. A candidate score below t_0 flips the row under the replacement of every voting member and no other, so $v_{ir} = \sum_{j \text{ voting}} n_{jr} = c_{ir}$; a score at or above t_0 flips none. Take a minimum joint witness X and its safe target. Entries with $c_{ir} = 0$ affect neither bound and can be dropped. The entries with $v_{ir} = 0$ certify the same upper costs and hence safety, and the entries with $v_{ir} = c_{ir} > 0$ certify the same lower costs and hence exclusion. These sets are disjoint, so $Q_{\text{split}} \leq |X| = Q_{\text{joint}}$. This is Corollary 3 with $d^* = 0$. □

Part (iv) shows that the disjointness of odd-median evidence is structural, not a consequence of uniform weights. The closed forms of Theorems 1 and 2 do not carry over: with unequal caps the separate safety minimum for a fixed row set takes the largest available caps, but choosing the row set becomes a cardinality-constrained knapsack. We compute weighted minima with the program in (ii).

Does the factor-two separation survive nonuniform weights? The upper bound in (iii) holds for every weight vector. The next result shows that the tight family of Proposition 9 is not an artifact of exact uniformity.

Proposition 14 (Factor-two family under nearly uniform weights)**.** *For even $T \geq 2$ and $M = T+1$, take the realizable scores of Proposition 9 with $N = 2$, $k = 1$ and $\alpha_T = (T + \frac{1}{2})/(2MT)$, so that the uniform-weight flip budget is $\lfloor 2k\alpha_T H \rfloor = T$ and Proposition 9 applies. Let $\omega_{jr} = MTw_{jr}$ denote weights normalized to mean one. If $|\omega_{jr} - 1| < 1/(2MT)$ for every (j, r), then $Q_{\text{joint}} = T + 1$ and $Q_{\cup,\text{sep}} = Q_{\text{split}} = 2T$.*

Proof. Memberships, and hence which replacements flip each row, do not depend on the weights. In units where every weight is ω_{jr}, the condition $R_w(C) \leq \alpha_T$ reads $\sum_{i \in C} \sum_{(j,r) \text{ flipping}} \omega_{jr} \leq T + \frac{1}{2}$. Under uniform weights every row-cost quantity is an integer, so its comparison with $T + \frac{1}{2}$ has slack at least $\frac{1}{2}$. A single row's lower or upper cost sums at most MT weights, and the two-row lower-cost sum used for exclusion sums at most $T + 1$ weights, so their perturbed values change by less than $\frac{1}{2}$ and every one-row feasibility test and every exclusion test has the same outcome as under uniform weights. No two-row set is ever feasible: the true two-row cost sums $T + 1$ weights and exceeds $(T+1)(1 - \frac{1}{2MT}) > T + \frac{1}{2}$, and upper costs dominate true costs. Hence the same query subsets certify safety, exclusion and their combination, and all minima coincide with those of Proposition 9. □

The same argument gives local constancy in general: if every uniform-weight feasibility comparison of an instance is strict, all certificate minima are constant on a neighborhood of uniform weights whose radius is the smallest slack divided by the largest number of summed weights.

Table 81 checks Proposition 14 with exact joint, union and split programs. Inside the guaranteed radius every draw preserves the separation; the separation also survives considerably larger perturbations (2–10$\delta^\star$ depending on T) before it erodes.

Sensitivity on a controlled panel. Table 82 draws replacement weights from a symmetric Dirichlet distribution over the $H = 49$ replacements of the RB-like synthetic panel (Appendix X), mean pooling, with concentration $\kappa \in \{\infty, 10, 1, 0.3\}$ ($\kappa = \infty$ is uniform). Concentrating the weights leaves zero-query coverage almost unchanged but shrinks minimum certificates: columns whose replacements carry little weight contribute small caps, so fewer of their entries are needed. At $\alpha = .10$ every instance permits full retention, so exclusion is automatic and joint, union and split minima coincide. At $\alpha = .02$, exclusion is required in 80 of the 80 tight-budget instances (those whose full-information optimum cannot retain every selected output). Strict gains then appear in 56 of the 62 targets whose joint and union programs closed within 60 s, under every concentration, with a mean saving of 4.7 entries and at most 1.6% of the best union. On this panel, nonuniform weights change certificate sizes but neither remove nor amplify the modest sharing advantage.

T	M	$\delta/\delta^\star$	Draws	Preserved	$\overline{Q_{\cup,\text{sep}}/Q_{\text{joint}}}$
2	3	0.9	100	100	1.333
2	3	2	100	100	1.333
2	3	5	100	69	1.230
2	3	10	100	27	1.090
2	3	20	100	11	1.037
4	5	0.9	100	100	1.600
4	5	2	100	100	1.600
4	5	5	100	100	1.600
4	5	10	100	69	1.484
4	5	20	100	23	1.250
6	7	0.9	40	40	1.714
6	7	2	40	40	1.714
6	7	5	40	40	1.714
6	7	10	40	40	1.714
6	7	20	40	14	1.479

Table 81: Perturbed replacement weights on the Proposition 9 construction with $\alpha_T = (T + \frac{1}{2})/(2MT)$, exact stopping. Normalized weights are drawn as $\omega_{jr} = 1 + \delta U_{jr}$, $U_{jr} \sim \text{Unif}(-1, 1)$, with $\delta^\star = 1/(2MT)$ the radius of Proposition 14. Preserved counts draws with $Q_{\text{joint}} = T + 1$ and $Q_{\cup,\text{sep}} = Q_{\text{split}} = 2T$; the last column averages $Q_{\cup,\text{sep}}/Q_{\text{joint}}$. Uniform weights give $Q_{\cup,\text{sep}}/Q_{\text{joint}} = 2T/(T + 1)$.

R.2 Certifying every single replacement

If an operator will execute one replacement from the declared family and wants the guarantee whichever it is, the relevant constraint is a per-replacement cap

$$R_{jr}(C) = \frac{|(S_0(P) \triangle S_0(P^{j \to r})) \cap C|}{2k} \leq \alpha$$

for all (j, r), equivalently at most $c = \lfloor 2k\alpha \rfloor$ committed flips per replacement. For a selected row i write $\varphi_i^{jr}(x) = \mathbf{1}[\sigma_P(i) \neq \sigma_{P^{j \to r}(x)}(i)]$. Given a history, let $\bar{\varphi}_i^{jr}$ be the observed indicator if (i, r) is observed and $\varphi_i^{jr}(0)$ otherwise, and let $\underline{\varphi}_i^{jr}$ be the observed indicator or zero.

Proposition 15 (Sharp verification)**.** *For every mask C of selected and invariant rows, the*

		$\alpha = .10$			$\alpha = .02$, exact		
κ	q	Zero	Q_{joint} prim.	Q_{joint} exact	Full	Q_{joint}	Gains
∞	0.05	20.1	443.7	461.0	0/10	470.0	8/10
∞	0.30	75.3	804.7	891.8	0/10	1732.2	5/8
10	0.05	20.1	439.7	458.0	0/10	472.4	8/8
10	0.30	75.3	782.1	871.9	0/10	1724.8	4/4
1	0.05	20.1	421.6	444.8	0/10	463.3	8/9
1	0.30	75.4	660.0	747.4	0/10	1691.5	6/6
0.3	0.05	20.1	383.6	411.6	0/10	462.3	10/10
0.3	0.30	75.2	533.4	616.2	0/10	1641.9	7/7

Table 82: Dirichlet replacement weights on the RB-like synthetic panel (mean pooling, ten splits per row). Zero: zero-query selected coverage (%). Q_{joint}: mean minimum joint entry count. Full: splits whose full-information optimum retains every selected output. Gains: splits with $Q_{\text{joint}} < Q_{\cup,\text{sep}}$ among those whose joint and union programs closed within 60 s (denominator). All $\alpha = .10$ programs closed.

maximum over consistent completions of $R_{jr}(C)$ equals $\sum_{i\in C} \bar{\varphi}_i^{jr}/(2k)$ for all (j,r) simultaneously, attained by assigning zero to every unobserved selected-row entry; the minimum equals $\sum_{i\in C} \underline{\varphi}_i^{jr}/(2k)$, attained by assigning one. Hence

$$V^-_{\max} = \max\left\{|C| : \textstyle\sum_{i\in C} \bar{\varphi}_i^{jr} \le c\ \forall (j,r)\right\},$$
$$V^+_{\max} = \max\left\{|C| : \textstyle\sum_{i\in C} \underline{\varphi}_i^{jr} \le c\ \forall (j,r)\right\}$$

satisfy $V^-_{\max} \le V^*_{\max} \le V^+_{\max}$, a maximizing mask for $V^-_{\max}$ meets every per-replacement cap in every consistent completion, and $V^+_{\max} - V^-_{\max} \le \varepsilon$ certifies a count within ε of the per-replacement optimum. Verifying a given mask costs $O(|C|MT)$.

Proof. Each indicator is monotone in the candidate score of its own entry, and for a selected row a score of zero maximizes all M indicators of that entry at once, while a score of one makes them all zero. Without cross-entry constraints the extreme assignments can be made for all unobserved entries simultaneously, so the per-replacement worst and best cases are attained jointly by one completion each. The capacity statements follow as in Proposition 1, with integer programs replacing sorting. □

Proposition 16 (Per-replacement optimization is NP-hard)**.** *Computing the full-information optimum $V^*_{\max}$ is NP-hard, even for odd-median pools of any fixed size $M = 2h+1$ and cap $c = 1$.*

Proof. Reduce from maximum independent set (Garey and Johnson, 1979). Given a graph $G = (\mathcal{V}, \mathcal{E})$, create one output per vertex. Every output receives base scores 1 from the first $h+1$ base evaluators and 0 from the other h, so with cutoff $t_0 = \frac{1}{2}$ fixed by construction every output is a selected boundary row whose voting members are the same $h+1$ evaluators. Create one candidate per edge $e = \{u, v\}$ with score 0 on outputs u, v and 1 elsewhere. Replacing a voting member j by candidate e flips exactly outputs u and v; replacing a nonvoting member flips nothing. Take $N = |\mathcal{V}|$, $q \le 1/(2N)$ so that $k = 1$, and $\alpha = \frac{1}{2}$, so $c = 1$. A mask is feasible for every replacement if and only if it contains at most one endpoint of each edge, i.e., it is an independent set. Hence $V^*_{\max}$ equals the independence number of G. □

The average objective is solved by sorting, so the hardness is specific to requiring every replacement to meet the budget. Exact count gaps under per-replacement caps therefore require integer programs; feasibility of a proposed mask remains cheap to certify.

Proposition 17 (Relations between objectives)**.** *(i) A mask meeting every per-replacement cap at α meets the average budget α. (ii) If $\sum_{i\in C} u_i \le c = \lfloor 2k\alpha \rfloor$, then C meets every per-replacement cap in every consistent completion. Thus running the average certificate of Section 4 with budget c (that is, average budget α/H) certifies the per-replacement objective, and Theorems 1–3 apply with b_H replaced by c. (iii) For a known single replacement $j \to r$, row costs are Boolean and only column r is informative; Theorems 1 and 2 apply with one candidate, opposition budget c and boundary set the rows that $j \to r$ can flip, so an exact certificate with $c \le L < m$ needs $\max(0, L-c)+c+1+(m-L-1) = m$ entries.*

Proof. (i) An average of quantities each at most α is at most α. (ii) For each replacement, $\sum_{i\in C} \bar{\varphi}_i^{jr} \le \sum_{i\in C} \sum_{j',r'} \bar{\varphi}_i^{j'r'} = \sum_{i\in C} u_i \le c$. (iii) With one replacement, an observed entry reveals the row's single indicator and an unobserved flippable row has upper cost one; substituting $T = 1$ and $b = c$ into Theorems 1 and 2 gives the count. □

Table 83 compares the objectives on the controlled panels. The per-replacement cap costs coverage mainly at small depths, where few outputs absorb each replacement's flips (e.g., MT-like median at $q = .05$: 99.9% versus 86.1% under full information), and the sortable α/H reduction is far more conservative than the exact per-replacement

			Full information		0% revealed			50% revealed		Avg. mask
Panel	Pool	q	Avg.	Max	Avg.	Max	α/H	Avg.	Max	violates (%)
RB-like	median	0.05	100.0	97.7	95.3	88.6	59.2	99.5	92.0	70
RB-like	median	0.10	100.0	100.0	99.7	97.0	69.6	100.0	98.7	0
RB-like	median	0.20	100.0	100.0	100.0	100.0	74.0	100.0	100.0	0
RB-like	median	0.30	100.0	100.0	100.0	100.0	78.4	100.0	100.0	0
RB-like	mean	0.05	100.0	90.0	20.1	20.1	0.0	48.3	46.2	90
RB-like	mean	0.10	100.0	99.3	25.6	24.9	4.7	54.5	52.4	50
RB-like	mean	0.20	100.0	100.0	56.9	54.4	32.0	79.2	76.9	0
RB-like	mean	0.30	100.0	100.0	75.3	73.4	48.8	92.9	90.4	0
MT-like	median	0.05	99.9	86.1	82.7	73.3	46.7	94.2	78.1	100
MT-like	median	0.10	100.0	94.8	89.8	85.6	53.8	98.8	89.1	100
MT-like	median	0.20	100.0	98.8	94.8	91.4	59.8	100.0	95.4	60
MT-like	median	0.30	100.0	100.0	99.8	94.6	68.0	100.0	98.3	0
MT-like	mean	0.05	100.0	92.0	21.7	21.7	0.0	47.6	43.2	100
MT-like	mean	0.10	100.0	95.8	26.6	24.9	2.8	52.0	47.7	100
MT-like	mean	0.20	100.0	99.9	51.9	47.2	24.5	73.1	68.0	20
MT-like	mean	0.30	100.0	100.0	69.4	65.1	39.8	87.1	82.3	0

Table 83: Average versus per-replacement (Max) objectives on the synthetic panels at $\alpha = .10$: certified selected coverage (%), mean over ten splits. Revealed fractions use a seeded random order of active entries. α/H is the sortable sufficient reduction of Proposition 17(ii). The last column reports the percentage of splits whose full-information average-optimal mask exceeds α for at least one single replacement. All integer programs closed.

certificate. Average-optimal masks violate some individual cap in 43% of synthetic splits, consistent with the NLP diagnostic in Appendix I, where 6,985 of 15,360 average-certified masks exceed .10 for some replacement. Operators who need per-replacement guarantees should therefore certify that objective directly rather than reuse an average certificate.

S Constrained Score Completions

Section 4 lets unobserved scores range independently over $[0, 1]$. Here we ask which verifiable cross-entry constraints keep the certificate computable and sharp. Fix a history $\mathcal{O}$ and a constraint set $\mathcal{K}$ that the true candidate-test matrix is known to satisfy. Let $\Omega_{\mathcal{K}}(\mathcal{O})$ be the set of score matrices consistent with $\mathcal{O}$ and $\mathcal{K}$, ordered coordinatewise.

Proposition 18 (Lattice completions)**.** *Suppose $\Omega_{\mathcal{K}}(\mathcal{O})$ has a coordinatewise greatest element $s^{\uparrow}$ and least element $s^{\downarrow}$. For selected rows define $\ell_i^{\mathcal{K}} = \sum_{r\in O_i} f_{ir} + \sum_{r\notin O_i} f_i(s^{\uparrow}_{ir})$ and $u_i^{\mathcal{K}} = \sum_{r\in O_i} f_{ir} + \sum_{r\notin O_i} f_i(s^{\downarrow}_{ir})$. Then $\ell^{\mathcal{K}} \le F \le u^{\mathcal{K}}$, both vectors are attained in $\Omega_{\mathcal{K}}(\mathcal{O})$, and Proposition 1 holds with these bounds, which are sharp over $\Omega_{\mathcal{K}}(\mathcal{O})$. Moreover $\ell \le \ell^{\mathcal{K}}$ and $u^{\mathcal{K}} \le u$, so every certificate under the rectangular model remains valid and minimum certificate sizes cannot increase. The same holds for weighted contributions and per-replacement indicators.*

Proof. On a selected row f_i is nonincreasing, and $s^{\downarrow} \le s \le s^{\uparrow}$ for every $s \in \Omega_{\mathcal{K}}(\mathcal{O})$; summing gives the bounds. The two extreme matrices belong to the set and realize them simultaneously for all rows. Capacities are monotone in costs. The rectangular set contains $\Omega_{\mathcal{K}}(\mathcal{O})$, which gives the comparison. □

Proposition 19 (Difference constraints and envelopes)**.** *Let $\mathcal{K}$ consist of score bounds $0 \le s_r(i) \le 1$ and difference constraints $s_r(i) - s_r(i') \le \delta_r(i', i)$ for each candidate r, where $\delta_r \ge 0$ (possibly $+\infty$) satisfies the triangle inequality $\delta_r(a, c) \le \delta_r(a, b) + \delta_r(b, c)$ and $\delta_r(a, a) = 0$. If the observations O_r in column r satisfy the constraints among themselves, the column's feasible set is nonempty, with greatest and least elements*

$$U_r(i) = \min\{1,\ \min_{o\in O_r} s_r(o) + \delta_r(o, i)\},$$

$$\Lambda_r(i) = \max\{0,\ \max_{o\in O_r} s_r(o) - \delta_r(i, o)\}.$$

Columns are independent, so Proposition 18 applies with $s^{\uparrow} = U$ and $s^{\downarrow} = \Lambda$. Updating one column after a query costs $O(n)$ for n outputs.

Proof. Feasibility of U_r: it lies in $[0, 1]$ because scores and δ are nonnegative; at an observed output the term o itself gives $U_r(o) \le s_r(o)$, and consistency of the observations gives equality. For any i, i', if $U_r(i') = 1$ then $U_r(i) - U_r(i') \le 0 \le \delta_r(i', i)$; otherwise let o' attain $U_r(i')$, so $U_r(i) \le s_r(o') + \delta_r(o', i) \le s_r(o') + \delta_r(o', i') + \delta_r(i', i) = U_r(i') + \delta_r(i', i)$. Greatest: every feasible column

satisfies $s_r(i) \le s_r(o) + \delta_r(o,i)$ and $s_r(i) \le 1$. The argument for Λ_r is symmetric. □

Three useful special cases are (a) *Lipschitz constraints with slack*, $\delta_r(i,i') = L_r d(i,i') + \eta_r$ for a declared metric d on outputs (for example, an embedding distance); (b) *monotonicity* with respect to a known preorder, e.g., a candidate known to be order-consistent with a base evaluator, with $\delta_r(i',i) = 0$ when i must not score above i' and $+\infty$ otherwise; and (c) per-entry *score supports* $[a_{ir}, b_{ir}]$, whose feasible set is a product of intervals. Difference constraints between candidates on the same output, such as a declared agreement band, also keep the feasible set closed under coordinatewise maxima and minima; its extreme elements are then computed by shortest paths over the joint constraint graph. Constraints that couple scores nonmonotonically, such as low-rank structure, generally do not form a lattice.

Two consequences differ from the rectangular model. First, an observation now tightens other entries of its column, so Theorems 1–3 no longer give the minimum certificate; minima can only decrease, and queries on nonselected outputs can become informative. Second, validity rests on the constraint itself. A constant estimated from reference data yields only a conditional guarantee, and a constraint that holds with stated probability transfers that probability to the certificate only under simultaneous validity over the adaptive history (Appendix K.4).

Controlled study. We keep the ECDF-normalized base scores of the RB-like panel (Appendix X) and generate the seven candidates' normalized test scores as $g_r(z_i) + \xi_{ir}$, where $z_i \in [0,1]$ is a one-dimensional output feature, g_r is a random piecewise-linear function with slope at most L, and $\xi_{ir} \sim \mathrm{Unif}(-\eta/2, \eta/2)$. The declared constraint $|s_r(i) - s_r(i')| \le L|z_i - z_{i'}| + \eta$ is therefore valid. A misspecified run uses $(L/2, \eta/2)$. Mean pooling, $\alpha = .10$, ten splits per setting, and largest-width acquisition that queries, in the widest row, the candidate with the widest envelope. Table 84 shows that valid constraints reduce inspection substantially, most when the slack is small and the batch is deep. The misspecified constraint stops earlier still, but 14–46% of unobserved active entries lie outside its envelopes at stopping, and 12 of 80 exact-stop certificates are invalid (realized churn above α or certified count above V^*); no valid-constraint certificate failed. These synthetic scores are smoother than real evaluator outputs; the study illustrates the mechanism, not the size of NLP savings.

			Primary inspection (%)			Invalid (exact)	
L	η	q	Rect.	Valid	Missp.	Valid	Missp.
1	0.05	0.05	95.9	31.0	9.7	0	2
1	0.05	0.30	71.9	8.0	1.5	0	0
1	0.2	0.05	95.7	62.5	22.1	0	3
1	0.2	0.30	72.2	19.6	3.8	0	1
4	0.05	0.05	95.9	40.8	13.8	0	3
4	0.05	0.30	72.3	4.1	1.6	0	0
4	0.2	0.05	95.5	68.1	27.7	0	3
4	0.2	0.30	72.6	17.6	2.7	0	0

Table 84: Lipschitz-with-slack completions on a controlled panel: mean active-entry inspection at primary stopping ($\varepsilon = \lceil .02|S_0| \rceil$) under the rectangular model, the valid constraint (L, η) and a misspecified one $(L/2, \eta/2)$, ten splits per row; invalid exact-stop certificates out of ten.

T Acquisition Exploiting the Odd-Median Closed Forms

Theorems 1 and 2 describe what a minimum odd-median certificate looks like: agreements inside a feasible set of L^* rows, and oppositions capped at a common level t^*. OM-CF turns this structure into an acquisition rule. We describe it in opposition units; the implementation works in flip units, which is equivalent because every opposition contributes $h+1$ flips.

Policy. Let $\pi_{ir} \in [0,1]$ be a predicted opposition probability for an unqueried boundary entry; we use the reference ridge predictions of Section 6, $\pi_{ir} = \hat{f}_{ir}/(h+1)$ clipped to $[0,1]$. Predicted row costs are $\hat{z}_i = o_i + \sum_{r \in U_i} \pi_{ir}$, rounded and clipped to $[o_i, T - a_i]$. With current safe and optimistic boundary counts $v^- \le v^+$, set $\hat{L} = \min(v^+, \max(v^-, V_B(\hat{z})))$ and choose a target $s \in [\max(v^-, \hat{L} - \varepsilon), \hat{L}]$ minimizing the predicted residual cost of Corollary 1: the agreements needed to make the s rows with smallest $\hat{z}$ safe, plus, for $p = s + \varepsilon + 1 \le m$, the fewest predicted oppositions that make $\sum_i \min(o_i, t) \ge b + 1 + (m-p)t$ for some integer t (Lemma 1). Let A_s be those s rows and t the minimizing threshold. With deficits $D_s = \sum_{i \in A_s} u_i - b$ and $D_e = b + 1 + (m-p)t - \sum_i \min(o_i, t)$, the priority of an unqueried entry is

$$\mathbf{1}[i \in A_s] \frac{1 - \pi_{ir}}{D_s} + \mathbf{1}[o_i < t] \frac{\pi_{ir}}{D_e},$$

omitting a term whose deficit is nonpositive or whose target is already met. Ties use the seeded row priority and candidate index; if no entry has positive priority, OM-CF falls back to largest width. Each step costs $O(mT \log(mT))$ for the target search.

Proposition 20 (Consistency)**.** *For odd-median pools, empty history and exact stopping, if the predictions are exact ($\pi_{ir} = y_{ir}$), OM-CF stops after exactly Q_{exact} queries.*

Proof. With exact predictions $\hat{z} = z$. Proposition 1 gives $v^- \le L^* \le v^+$ at every history, so $\hat{L} = L^*$ and, since $\varepsilon = 0$, the target is $s = L^*$ throughout. A_s is a fixed set of L^* rows of smallest true cost, which is truly feasible. Agreements are queried only in A_s and oppositions never change upper costs, so rows outside A_s keep upper cost T and A_s minimizes $\sum_{i \in C} u_i$ over truly feasible L^*-row sets at every history; hence the safety residual of Proposition 3 equals $(\sum_{i \in A_s} u_i - b)_+$, starting at $\max(0, L^*T - b)$. The safety term gives positive priority exactly to unobserved agreements in A_s, each of which lowers this residual by one. The exclusion term gives positive priority only to unobserved oppositions in rows with $o_i < t$, where t minimizes the exact remaining count; such a query lowers D_e by one and hence that minimum by one. The minimum cannot drop by more than one per query, since otherwise a smaller certificate would exist, so the exclusion residual of Proposition 3, which starts at $B_0 + (m - p)t^*$ (Theorem 2), falls by exactly one. No query with zero priority is made while a residual is positive, because a positive residual implies an unobserved useful entry: $\sum_{A_s}(T - a_i) > b \ge \sum_{A_s} z_i$ leaves an agreement, and a positive exclusion deficit with a finite count leaves an opposition below level t. When both residuals vanish, $V^- \ge m_0 + L^* \ge V^+$ and the policy stops. The total is $\max(0, L^*T - b) + B_0 + (m - p)t^* = Q_{\text{exact}}$; for $L^* = m$ only the safety part is needed. □

Exhaustive enumeration confirms the proposition and also covers positive tolerance: on 2,262 random odd-median instances ($m \le 6$, $T \le 5$, caps $h + 1 \le 3$, every budget, every tolerance $0 \le \varepsilon \le m$), oracle OM-CF attains Q_ε of Corollary 1 exactly. In the terminology of learning-augmented algorithms (Lykouris and Vassilvitskii, 2018; Purohit et al., 2018), OM-CF is consistent. It cannot also be robust, because Propositions 7 and 8 lower-bound every always-valid policy linearly in batch size.

Panel	Stop	Fixed	Rand.	Width	Ref.	OM-CF	Oracle
RB-like	exact	2.03	2.32	2.13	1.91	2.17	1.00
RB-like	primary	2.08	2.23	2.13	1.84	2.19	1.00
MT-like	primary	2.04	2.17	2.24	1.85	2.13	1.00
MT-like	exact	2.07	2.12	2.18	1.87	2.10	1.00

Table 85: Median entry acquisition on the synthetic panels, $\alpha = .10$, four depths and ten splits: mean splitwise ratio of queries to the matching closed-form minimum Q_ε (Corollary 1), over targets with positive minima; primary uses $\varepsilon = \lceil .02|S_0| \rceil$. Oracle is OM-CF with exact predictions.

Learned predictions. Table 85 evaluates OM-CF with reference-ridge predictions on the synthetic median panels (Appendix X), against the original policies and oracle OM-CF, as splitwise ratios to the matching closed-form minimum. Reference priority attains ratios of 1.84–1.91, close to the NLP ratios in Section 7.3 (1.662–1.968). OM-CF with learned predictions is worse (2.10–2.19): with noisy opposition probabilities, targeted exclusion queries are often wasted, whereas reference priority's safety-first ordering is more forgiving. OM-CF beats reference priority on only 9 of 76 of the targets with positive minima. We therefore do not replace reference priority, and report OM-CF as a consistency result whose practical value depends on prediction quality. NLP-panel runs use the released code (Appendix X).

U When Does Joint Optimization Help?

Section 7.3 shows that strict gains beyond optimal deduplication are rare and confined to targets that need exclusion evidence. This appendix states which of these conditions can be checked before candidate-test scoring and evaluates a simple pre-scoring screen.

Proposition 21 (Pre-scoring conditions)**.** *Fix a history $\mathcal{O}$ and tolerance ε. (i) If $|S_0| - V^-(\mathcal{O}) \le \varepsilon$, then $Q_{\text{joint}} = Q_{\cup,\text{sep}} = Q_{\text{split}} = 0$. (ii) Call an active row a* staircase row *if f_i takes a value strictly between 0 and p_i on $[0, 1]$; for mean pooling this holds exactly when the replacement thresholds $\tau_{ij} = Mt_0 - \sum_{j'} s_{j'}(i) + s_j(i)$ take at least two distinct values in $(0, 1]$. If no unobserved entry lies in a staircase row, then $Q_{\text{joint}} = Q_{\cup,\text{sep}} = Q_{\text{split}}$. Both conditions use only base scores, the cutoff, the budget and the history.*

Proof. (i) The empty set of additional queries certifies the gap. (ii) Every unobserved contribution is 0 or p_i, so every minimum joint witness has $I(X) = \emptyset$ and $d^* = 0$; Corollary 3 gives equality. For mean pooling $f_i(x) = |\{j : x < \tau_{ij}\}|$ with all $\tau_{ij} \leq 1$ on a selected row, which takes an intermediate value if and only if two positive thresholds differ. □

Odd medians have no staircase rows, which is another view of Theorem 2's disjointness; mean pools usually do. The decisive ex post condition is whether exclusion is needed at all. If $V^* + \varepsilon \geq |S_0|$, all NLP targets in Appendix M.9 had an optimal witness with automatic exclusion, and full retention ($V^* = |S_0|$) provably removes any gain. V^* is unknown before scoring, but reference-fitted predictions give a plug-in screen

$$\hat{E} = \mathbf{1}\big[\hat{V}^* + \varepsilon < |S_0|\big],$$

where $\hat{V}^* = m_0 + V(\hat{F})$ and $\hat{F}_i = \sum_r \hat{f}_{ir}$ are computed from base scores and reference fits alone.

Controlled evaluation. On the two synthetic panel families with mean pooling, four depths, ten splits and $\alpha \in \{.02, .05, .10\}$, we compute Q_{joint} and $Q_{\cup,\text{sep}}$ for every target (Table 86). Of 480 targets, 396 are numerically closed and 130 of these show a strict gain. All 130 gains occur where exclusion is needed ($V^* + \varepsilon < |S_0|$), and no full-retention target has a gain. No target stops without queries. Staircase rows are present in 480 of 480 targets, so condition (ii) of Proposition 21 essentially never applies to mean pools; for odd medians it holds always and recovers the disjointness behind Theorem 2. The plug-in screen $\hat{E}$ flags 301 of 480 targets. On closed targets it misses 0 of 130 gains (recall 1.00), and 0.60 of the flagged closed targets have a gain; none of the 179 unflagged closed targets has one. The reference ridge overestimates costs (mean $\hat{V}^*$ is 223.5 against mean V^* 262.5), so the screen is conservative. It is useful to decide when a minimum safety witness suffices (unflagged targets), not to predict gains precisely. Gains are far more frequent here than on the NLP grid because the tight budgets $\alpha \in \{.02, .05\}$ make exclusion necessary; at $\alpha = .10$ every synthetic instance permits full retention and no gain occurs, as in 903 of the 960 NLP instances.

Panel	Stop	Targets	Excl.	Gains	$\hat{E}$ recall	$\hat{E}$ prec.
RB-like	exact	120	54	48/120	1.00	0.67
RB-like	primary	120	51	14/83	1.00	0.44
MT-like	exact	120	66	57/119	1.00	0.72
MT-like	primary	120	56	11/74	1.00	0.32

Table 86: Pre-scoring screen for joint gains on synthetic mean panels. Excl.: targets with $V^* + \varepsilon < |S_0|$ (exclusion needed, computed ex post). Gains: targets with $Q_{\text{joint}} < Q_{\cup,\text{sep}}$ among numerically closed targets. Recall: fraction of gain targets flagged by $\hat{E}$; precision: fraction of flagged targets with a gain.

V Choosing a Reference Mapping

Section 7.6 and Appendix K show that switching from right to midrank ECDF can change zero-query coverage and inspection dramatically. The reason is that mapping changes the selection task itself. Appendix K shows which transformations preserve the task: a common strictly increasing map of all scores and the cutoff for odd medians, and a common positive affine map for means. Separate evaluator-wise ECDF refits are not of this form, so they change the base list (Jaccard 0.77 on thirteen-stream WMT23) and the share of selected outputs at the cutoff (28.8% versus 7.3%), which in turn changes the boundary mass that certificates must handle.

No mapping can make certification difficulty invariant across datasets, because difficulty depends on the test batch: its boundary mass, its budget and its true costs. Replicable calibration (Papamichalis et al., 2026) addresses a different property: it can make the fitted cutoff stable across reference draws, which the reference bootstrap in Appendix K.2 quantifies (on thirteen-stream WMT23 median, the cutoff standard deviation is .0115 when only the cutoff is resampled). We therefore recommend the following protocol.

1. Treat the mapping, the reference-size rule and the tie convention as part of the selection protocol, and fix them before any candidate-test score is seen.

2. Choose them for selection validity, e.g., to respect each metric's score resolution and tie behavior (Deutsch et al., 2023), not to reduce certification cost, since a cheaper certificate for a different list is not an improvement for the original one.

3. Report, from reference data only, the share of selected outputs exactly at the cutoff and on the odd-median boundary, and the bootstrap stability of the cutoff and base list (Appendix K.2).

4. Report the initial gap $|S_0| - V^-(0)$ and the invariant/noninvariant decomposition of zero-query retention (Appendix G.1) on the test batch, which together determine whether queries are needed.

We also checked whether a reference-only boundary share could serve as a protocol-selection score. On the synthetic panels of Appendix X, across six reference protocols (full, half and quarter reference groups $\times$ right and midrank ECDF), median pooling, $q \in \{.05, .10\}$ and ten splits, its Spearman correlation with the test-batch exact minimum, zero-query coverage and reference-priority inspection was at most 0.27 in absolute value. Because these panels have few cutoff ties, they cannot reproduce the WMT23 effect; we report this only to explain why we recommend reporting, not optimizing, such diagnostics.

W Solver Sensitivity of the Joint Program

The full-batch minima in Appendix M.9 were computed with HiGHS through SciPy. To test sensitivity to the solver and its settings, we export the exact-stop outcome-count program (Proposition 4, safe target L^*) for identical instances to five configurations: HiGHS with presolve on and off, HiGHS with random seeds 1 and 2 (one thread), and COIN-OR CBC 2.10 (one thread), plus the OR-Tools CP-SAT solver on the equivalent pure-integer model (one worker). Each configuration has a nominal 60-second limit. Instances are the synthetic panels (two families, $\alpha \in \{.02, .10\}$, $q \in \{.05, .30\}$, splits 0–2) and a constructed uniform-score grid in the spirit of Appendix M.7 (mean pooling, cutoff .5, $q = .30$, $\alpha = .10$, $N \in \{512, 2048\}$, $M \in \{7, 15\}$, $T \in \{17, 33\}$). Across 32 instances, every configuration that closes an instance returns the same optimum (no disagreements). One constructed instance ($N = 2048$, $M = 7$, $T = 33$) closes under no configuration, and there the final incumbents differ by up to 27 entries (23,679–23,706), so unresolved values must be reported as intervals, as Appendix M.9 does. Closure and run time vary much more than optima (Table 87): on the constructed grid, one instance exhausts the limit under three HiGHS settings but closes in 16.8 s with seed 2 and in 1.8 s with CBC, and presolve alone changes another closure time from 18.6 s to 41.4 s. CP-SAT is markedly slower on this big-M formulation. Reported numerical minima are therefore insensitive to the backend, whereas closure counts under a time budget should be reported with the backend and its settings.

Configuration	Closed	Agree	Median s	Max s
HiGHS, presolve on	30/32	30/30	0.3	60.2
HiGHS, presolve off	30/32	30/30	0.2	60.1
HiGHS, seed 1	30/32	30/30	0.2	60.0
HiGHS, seed 2	31/32	31/31	0.2	60.4
CBC	30/32	30/30	0.5	61.9
CP-SAT	18/32	18/18	3.2	60.0

Table 87: Solver sensitivity of the exact-stop joint program. Closed: instances solved to proven optimality within 60 s. Agree: closed instances whose optimum equals the consensus value of all closing configurations. CBC runs are counted as closed only if they finish before the limit, because PuLP labels time-limited incumbents as optimal. Times include unresolved attempts.

X Synthetic Panels and Reproducibility of the Revision Experiments

The new experiments in Appendices R–W use controlled synthetic panels because they test mechanisms of the certificate model (weights, caps, constraints, acquisition and solvers) rather than NLP effects. Each panel draws a latent quality $q_i = g_{\mathrm{grp}(i)} + e_i$ with group effect $g \sim \mathcal{N}(0, 0.6^2)$ and $e \sim \mathcal{N}(0, 1)$, a shared nuisance factor $\nu_i \sim \mathcal{N}(0, 1)$, and evaluator scores $\lambda_e q_i + \gamma_e \nu_i + \sigma_e \epsilon_{ie}$ with $\lambda_e \sim \mathrm{Unif}(.6, 1)$, $\gamma_e \sim \mathcal{N}(0, s^2)$ and $\sigma_e \sim \mathrm{Unif}(.35, .9)$; a fraction of evaluators is rounded to half-units to create ties. *RB-like*: 1,200 groups of four outputs, 14 evaluators, $s = .35$, no ties, $M = T = 7$. *MT-like*: 450 groups of twelve outputs, 24 evaluators, $s = .5$, 30% rounded evaluators, $M = 7$, $T = 17$. Panel seeds are 101 and 202. The protocol is that of Appendix B: group splits by Python `random.Random(1000+r).shuffle` into thirds, right ECDF maps and reference-consensus cutoff fitted on the reference third, $k = \max(1, \mathrm{round}(qN))$, ridge priorities with penalty one, seeded label-blind tie order, and the same stopping rules.

The code package `experiments/` contains `certlib.py` (bounds, capacities, exact joint, separate and union programs, odd-median closed forms), `acquire.py` (all acquisition policies including OM-CF), `maxcap.py`, and one script per experiment, with JSON outputs and logs. Before use, the library was checked against exhaustive query-subset enumeration on 1,205 random instances with histories, column-dependent caps and

all tolerances, and on 300 odd-median instances against Corollary 1, with no mismatch. The Proposition 13(iv) check enumerates 8,987 weighted endpoint-only cases. All scripts also accept cached NLP score matrices in place of the synthetic generator, so the same analyses can be rerun on the panels of Section 7. Experiments ran on a four-core CPU machine with Python 3.13, NumPy 2.5, SciPy 1.18 (HiGHS), PuLP 2.9 (CBC) and OR-Tools 9.15.

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